Simplex results, explained

Five single-cell EMT datasets · three gene sets · pooled and within each unit

What is checked: (1) is there a significant simplex, and do its corners correspond to tasks, when the fit uses A. the entire gene set, B. the KS EMT genes, C. the Hallmark EMT genes; (2) how many hybrid cells are generalists, and how many generalists are hybrid, under three rules.

Thesis project, IISc · 22 October 2026

Key ideas in five lines

Read this once; every later slide uses these words

Simplex. If cells must trade off k tasks, they fill a shape (a line, triangle, tetrahedron...) whose k corners are cells specialised in one task each. We fit that shape to the leading PCA axes of the gene-expression data.

Significant simplex. The fitted shape is compared with the same data after shuffling each PCA axis independently, 100 to 200 times. If the real shape is more triangle-like than nearly all shuffles ($p \leq 0.05$), it is not just a random cloud. A small p alone does not prove a trade-off: samples that differ from each other can also give a significant shape.

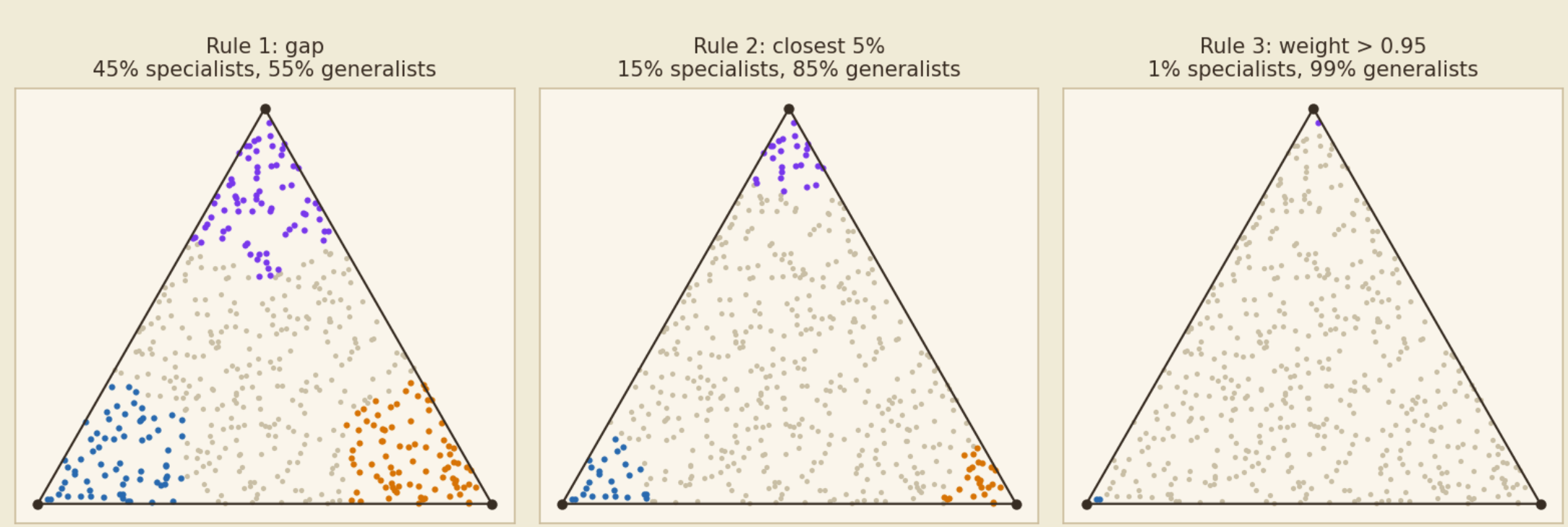
Corner = task? The 5% of cells nearest each corner are scored on the 50 MSigDB Hallmark gene sets (effect in SD versus all other cells). The top set names the programme (e.g. E2F Targets = cycling). Cell-cycle phase, EMT-state mix, dominant sample and marker genes support the name. A corner that is almost all one sample ($\geq 70\%$) is an identity corner, not a task.

Hybrid cells. The middle EMT states (Early-Hybrid, Hybrid, Late-Hybrid). Cook has no state labels, so hybrids are the middle two quartiles of the KS EMT score.

Specialist vs generalist. A cell claimed by one corner is a specialist; every other cell is a generalist, an interior mixture. The key comparison: what % of hybrid cells are generalists, versus what % of all other cells are generalists. Same % = hybrids are not special.

The three specialist rules on a toy triangle

500 simulated cells mixing three corners; the same rules are applied to the real fits



Rule 1, gap (Pal work): a cell is a specialist if its largest corner weight exceeds the second largest by at least 0.35.

Rule 2, closest 5% (Groves 2022): for each corner, the 5% of cells nearest to it are its specialists; everything else is a generalist. Caps specialists at about 15% for a triangle.

Rule 3, weight > 0.95 (Groves 2022, methods): specialist only if one weight exceeds 0.95. Because the fitted corners lie inside the cloud (delta = 0), almost no cell qualifies, so about 100% are generalists. It is the strictest and least informative.

How the numbers on each slide were produced

Everything is read from saved fits; no result is refitted for the rules

1. Fit. Each fit = PCA on the unit's cells (genes centred, not scaled) then archetypal analysis on the first k−1 PCs (15 starts); the permutation test shuffles each PC and refits (200 shuffles; 100 for the new Hallmark, cleaned-pooled and SCC scRNA-only / six-tumour pooled fits, marked in tables).

2. Best simplex. For each dataset × unit × gene set × arm the saved k values are compared; best = lowest p, ties go to the smaller k; pooled fits ignore k beyond the ESV elbow. On each slide the arm shown is the one with the lowest p (ties: cleaned arm).

3. Weights and rules. Each cell's barycentric weights on the k corners give the three rules (previous slide). Statistics use all cells of the fit; the plots draw at most 4,000 random cells. Within-unit fits are capped at 4,000 cells.

4. Corner names. 5% nearest each corner → Hallmark scores (mean log-expression per set; effect = difference in SD) + S/G2M share + state mix + dominant sample + top genes. For Hallmark-EMT fits the EMT set is excluded from naming (circular).

5. Arms. Light QC = standard cell QC; cleaned = at least 10,000 UMI (plus dataset-specific technical-gene / contamination filters). SCC has no cleaned arm. For k ≥ 4 the simplex is drawn by projecting the corners on PC1/PC2 and joining their convex hull.

Which fit is shown on each slide

One main fit per unit and gene set, so nothing is picked for having the lowest p

Mouse SCC. Main = the pooled fit on scRNA cells only, and the eight scRNA tumours. The four multiome tumours (MU3, MU4, MU7, MU8) were sequenced as nuclei, a different technology: mixing them in creates a technical corner, and inside a multiome tumour the signal is nuclear RNA. The all-cells pooled fit and the multiome tumours are kept in the appendix.

Cook, PDAC, 4T1, GSE114687. Main = the cleaned arm: cells with at least 10,000 UMI, plus each dataset's technical filter (Cook: mitochondrial fraction ≤ 0.07 and MALAT1 / NEAT1 / haemoglobin genes dropped; PDAC: liver ambient-RNA and haemoglobin genes dropped; 4T1: immune / stromal contamination removed; GSE114687: MALAT1 / NEAT1 / haemoglobin dropped). 4T1 clone E has only 142 cells after cleaning, so its light-QC fit is shown.

Why. Light QC and cleaning disagree on whether a simplex is significant in many units (Cook 2 of 12 unit × gene-set pairs, PDAC 8 of 27, 4T1 7 of 15, GSE114687 8 of 18). Picking whichever arm has the lower p would inflate significance. The light-QC p is printed on every slide as a comparison and tabulated in the appendix.

Counting. All summary numbers below count units (one main fit each), not fits.

At a glance: where is a simplex significant?

Main fits. Pooled: k and p. Within: number of units with a significant simplex

dataset	A. pooled	A. within	B. pooled	B. within	C. pooled	C. within
Mouse SCC	k=3, p=0.00498 *	4 of 8	k=5, p=0.0099 *	3 of 8	k=3, p=0.0495 *	3 of 8
Cook	k=3, p=0.0099 *	1 of 4	k=4, p=0.0099 *	1 of 4	k=4, p=0.0099 *	3 of 4
Mouse PDAC	k=4, p=0.0396 *	9 of 9	k=4, p=0.931	3 of 9	k=3, p=0.208	4 of 9
4T1 clones	k=3, p=0.00995 *	1 of 6	k=4, p=0.0299 *	1 of 6	k=4, p=0.119	1 of 6
MCF10A/HuMEC	k=4, p=0.0099 *	6 of 6	k=3, p=0.0099 *	4 of 6	k=3, p=0.0099 *	5 of 6

* significant ($p \leq 0.05$). Units: SCC 8 scRNA tumours, Cook 4 lines, PDAC 9 (7 libraries, mouse 1, mouse 2), 4T1 6 clones, GSE114687 6 samples.

A. Entire gene set: 26 of 38 main fits significant (pooled 5 of 5, within 21 of 33 units).

B. KS EMT genes: 16 of 38 main fits significant (pooled 4 of 5, within 12 of 33 units).

C. Hallmark EMT genes: 19 of 38 main fits significant (pooled 3 of 5, within 16 of 33 units).

Mouse SCC

What the dataset is, and what to keep in mind when reading its slides

What the dataset means

Mouse skin squamous cell carcinoma: 12 real tumours (4 multiome, 8 scRNA), 23,959 cells, with the authors' four EMT states. Tumour, assay and EMT state are tangled.

Units in the main analysis

SC15, SC32, SC46, SC33, SC2b1, SC2b3, SC35, SC43

Gene sets

A. all genes (11,000 to 14,500); B. KS EMT genes (Tan 2014; about 160 to 190 present); C. Hallmark Epithelial-Mesenchymal-Transition set (200 genes, those present in the data).

Keep in mind

• Hybrids = Early-Hybrid + Hybrid (the authors' labels); Late-EMT is not counted as hybrid.

• The four multiome tumours (MU3, MU4, MU7, MU8) were read as nuclei, not whole cells: a corner dominated by them is a technical axis, not biology.

• Within-tumour fits were run for all 12 tumours; most tumours contain one or two EMT states only.

This dataset in numbers (main fits)

• Entire gene set: 5 of 9 fits significant; hybrids 53% / 87% / 100% generalist (median, rules 1/2/3).

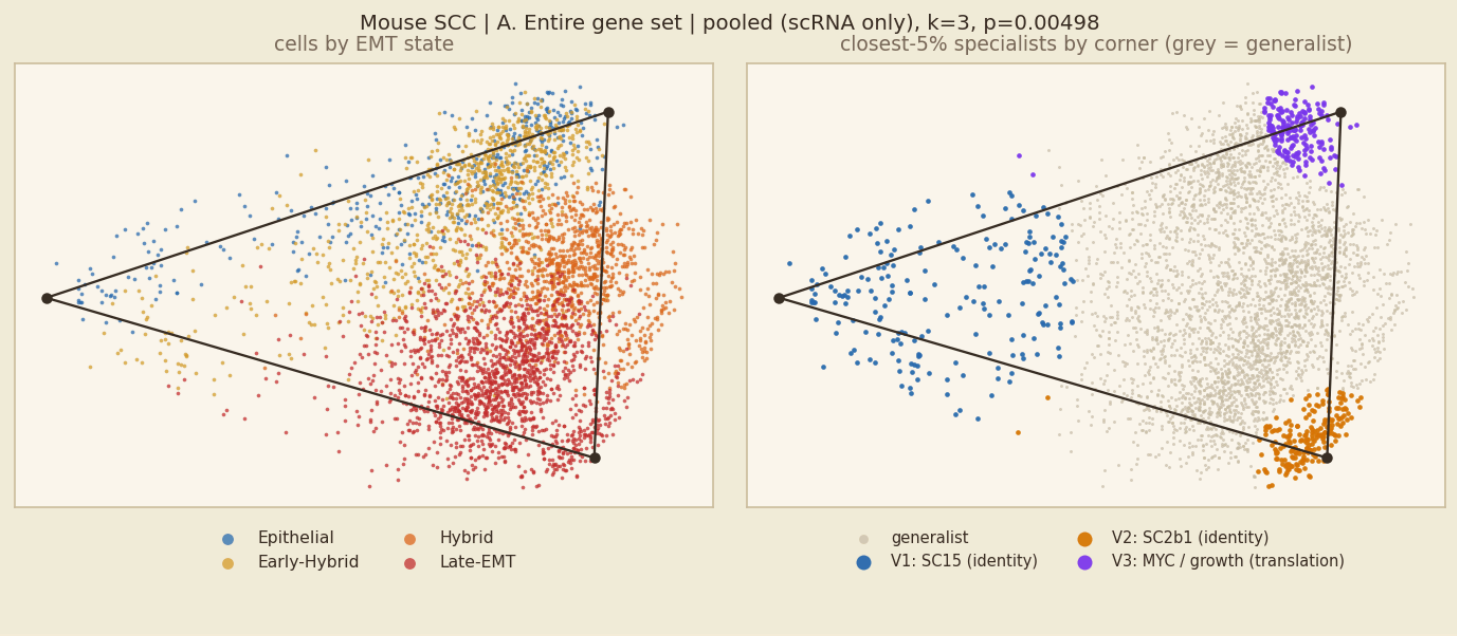
• KS EMT genes: 4 of 9 fits significant; hybrids 64% / 80% / 100% generalist (median, rules 1/2/3).

• Hallmark EMT genes: 4 of 9 fits significant; hybrids 57% / 87% / 100% generalist (median, rules 1/2/3).

Question 1 slides come first, then the Question 2 slides for this dataset.

Mouse SCC: A. Entire gene set, pooled fit

Fit shown: scRNA only · k = 3 · p = 0.00498 · 14,059 cells · 200 shuffles | comparison, light QC: k=3, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	61%	35%
2 closest 5% of cells to each corner	89%	81%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 49% of all cells here; 77% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=3, p=0.00498), but 2 of 3 corners are almost entirely one sample (SC15, SC2b1). It separates samples; it is not a shared EMT trade-off.

Each corner is essentially one tumour. The programme names (Hallmark) only describe what distinguishes that tumour, and the EMT-state colours follow the tumours. These corners are not tasks.

Hybrid vs other cells: 61% of hybrid cells and 35% of other cells are generalists (rule 1; rule 2: 89% vs 81%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 identity corner: SC15 (programme: cycling) — Mitotic Spindle +1.8 SD; 34% S/G2M; 44% Early-Hybrid; 82% SC15; Adgrl3 Ext1 Atxn1

V2 identity corner: SC2b1 (programme: coagulation / matrix-like) — Coagulation +1.04 SD; 11% S/G2M; 99% Late-EMT; 92% SC2b1; Postn Col1a1 Col3a1

V3 MYC / growth (translation) — Myc Targets V1 +1.57 SD; 80% S/G2M; 57% Early-Hybrid; Krt14 Krt5 Krt6a

Mouse SCC: A. Entire gene set, all units at a glance

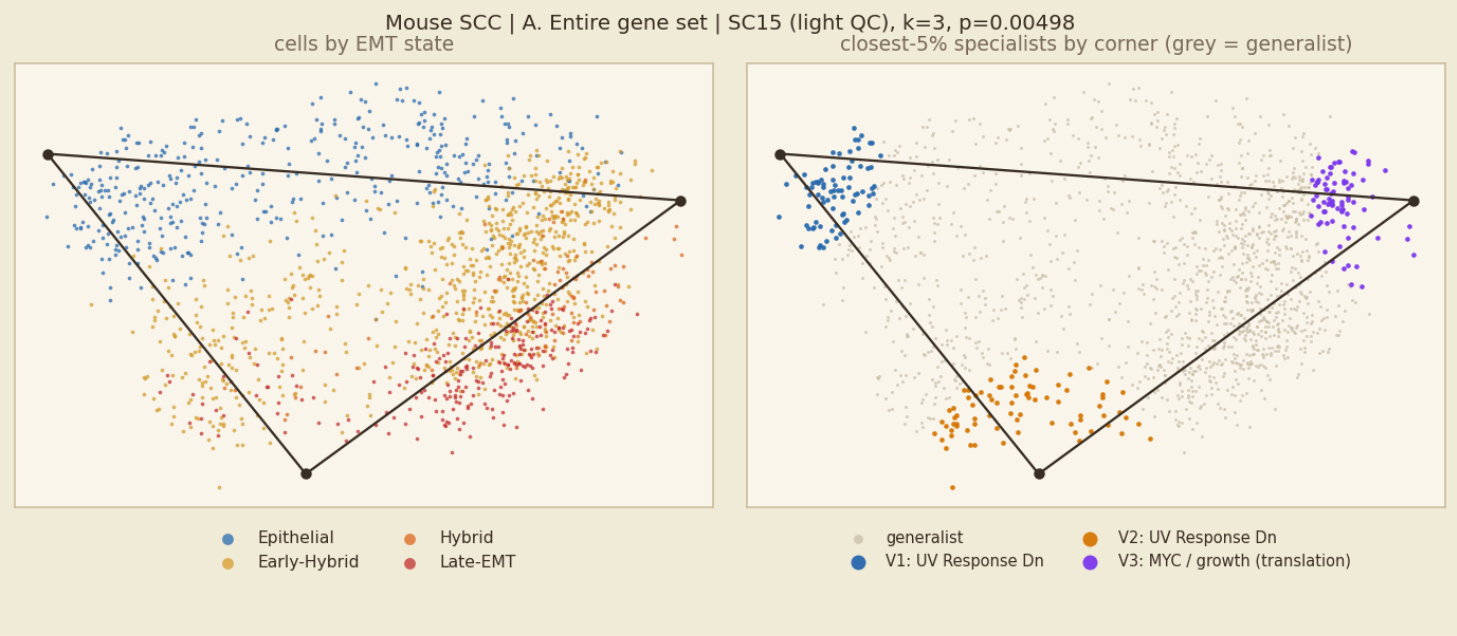
Main fit of each unit: 4 of 8 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
SC15	light QC	3	0.00498 *	–	58%	53 / 87 / 100%	57 / 59 / 58%
SC32	light QC	3	0.0149 *	–	57%	66 / 93 / 100%	68 / 62 / 58%
SC46	light QC	4	0.0796	–	50%	61 / 84 / 100%	65 / 53 / 50%
SC33	light QC	4	0.318	–	79%	70 / 83 / 100%	85 / 82 / 79%
SC2b1	light QC	3	0.00498 *	–	31%	51 / 83 / 99%	36 / 30 / 31%
SC2b3	light QC	4	0.00498 *	–	38%	42 / 82 / 100%	29 / 39 / 38%
SC35	light QC	4	0.169	–	40%	67 / 73 / 100%	44 / 37 / 40%
SC43	light QC	3	0.751	–	53%	50 / 86 / 100%	50 / 54 / 53%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

Mouse SCC: A. Entire gene set, unit SC15

Fit shown: light QC · k = 3 · p = 0.00498 · 1,576 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	53%	56%
2 closest 5% of cells to each corner	87%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 58% of all cells here; 54% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC15 alone, cells fill a 3-corner simplex (p=0.00498). 4 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: UV Response Dn; MYC / growth (translation).

Hybrid vs other cells: 53% of hybrid cells and 56% of other cells are generalists (rule 1; rule 2: 87% vs 83%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

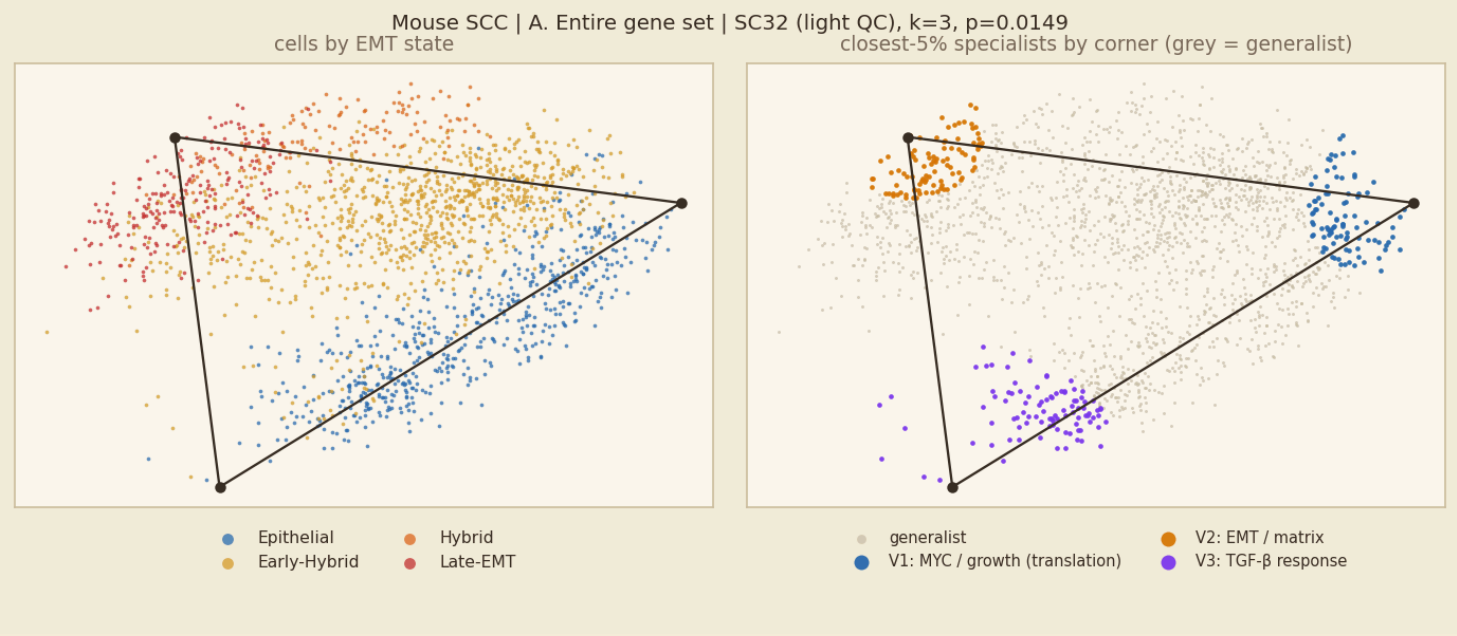
V1 UV Response Dn — UV Response Dn +1.35 SD; 22% S/G2M; 100% Epithelial; 100% SC15; Trps1 Frmd4b Adgrl2

V2 UV Response Dn — UV Response Dn +0.99 SD; 38% S/G2M; 54% Early-Hybrid; 100% SC15; Col5a2 Fn1 Col1a1

V3 MYC / growth (translation) — Myc Targets V1 +1.5 SD; 62% S/G2M; 78% Early-Hybrid; 100% SC15; S100a4 Lgals1 Ly6a

Mouse SCC: A. Entire gene set, unit SC32

Fit shown: light QC · k = 3 · p = 0.0149 · 1,910 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	66%	41%
2 closest 5% of cells to each corner	93%	75%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 57% of all cells here; 61% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC32 alone, cells fill a 3-corner simplex (p=0.0149). 4 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: MYC / growth (translation); EMT / matrix; TGF-β response.

Hybrid vs other cells: 66% of hybrid cells and 41% of other cells are generalists (rule 1; rule 2: 93% vs 75%), so hybrids are clearly more often generalists.

Corners and their tasks

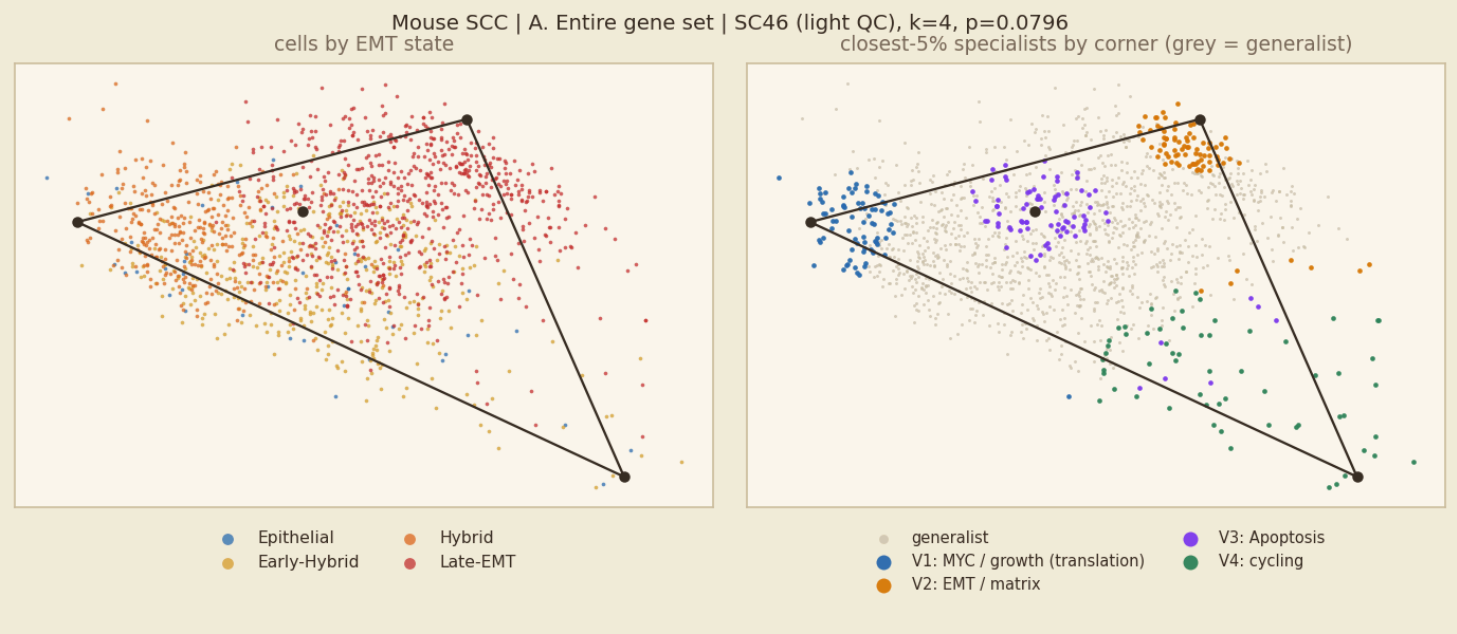
V1 MYC / growth (translation) — Myc Targets V2 +1.76 SD; 98% S/G2M; 66% Epithelial; 100% SC32; Fabp5 Krt6a Krt14

V2 EMT / matrix — Epithelial Mesenchymal Transition +1.49 SD; 1% S/G2M; 73% Late-EMT; 100% SC32; Spp1 Vim Fn1

V3 TGF-β response — TGF-beta Signaling +1.03 SD; 24% S/G2M; 76% Epithelial; 100% SC32; Thbs1 Tmem176b Krt42

Mouse SCC: A. Entire gene set, unit SC46

Fit shown: light QC · k = 4 · p = 0.0796 · 1,519 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	61%	33%
2 closest 5% of cells to each corner	84%	76%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 50% of all cells here; 28% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC46: the fit (k=4, p=0.0796) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 61% of hybrid cells and 33% of other cells are generalists (rule 1; rule 2: 84% vs 76%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 MYC / growth (translation) — Myc Targets V2 +1.98 SD; 95% S/G2M; 76% Hybrid; 100% SC46; Ccnd1 H2afz Birc5

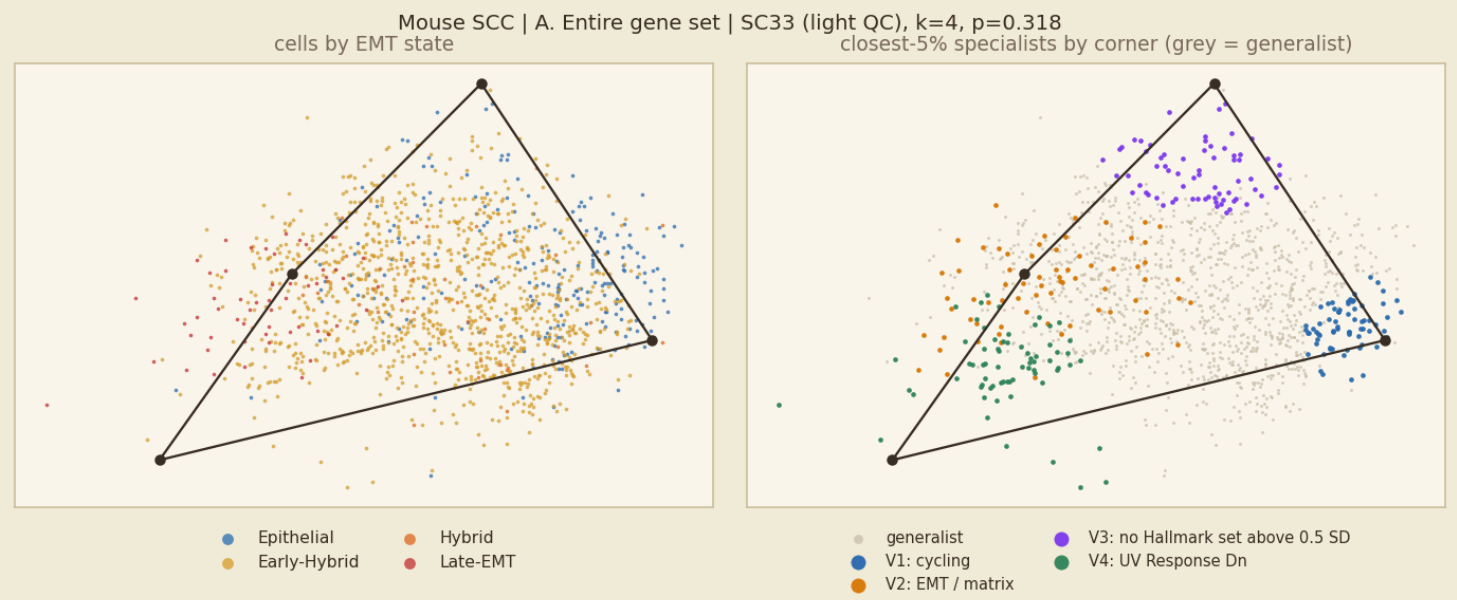
V2 EMT / matrix — Epithelial Mesenchymal Transition +1.85 SD; 25% S/G2M; 100% Late-EMT; 100% SC46; Col3a1 Eln Bgn

V3 Apoptosis — Apoptosis +1.11 SD; 12% S/G2M; 95% Late-EMT; 100% SC46; Chil4 Gsn Nupr1

V4 cycling — Mitotic Spindle +1.15 SD; 51% S/G2M; 62% Early-Hybrid; 100% SC46; Neat1 Pcdh7 Hivep2

Mouse SCC: A. Entire gene set, unit SC33

Fit shown: light QC · k = 4 · p = 0.318 · 1,379 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	70%	47%
2 closest 5% of cells to each corner	83%	68%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 79% of all cells here; 44% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC33: the fit (k=4, p=0.318) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 70% of hybrid cells and 47% of other cells are generalists (rule 1; rule 2: 83% vs 68%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 cycling — E2F Targets +1.72 SD; 100% S/G2M; 57% Early-Hybrid; 100% SC33; Hmgb2 Rrm2 Hist1h1b

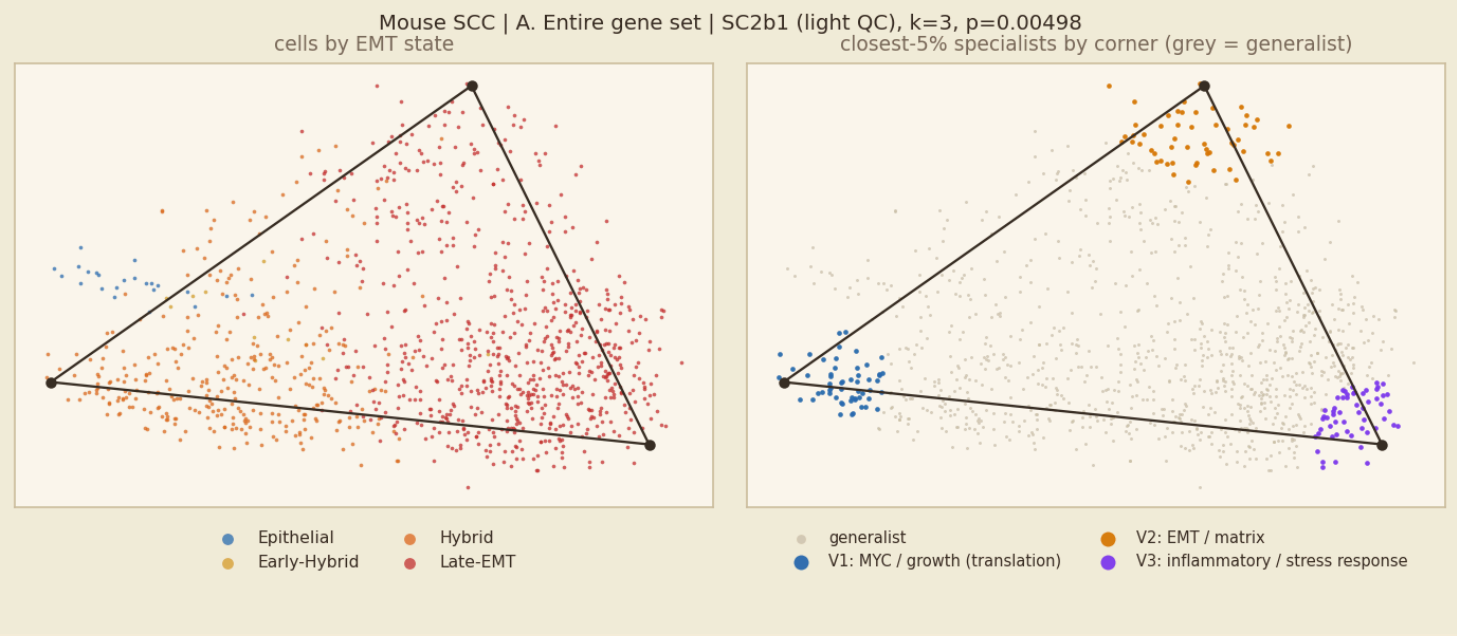
V2 EMT / matrix — Epithelial Mesenchymal Transition +1.66 SD; 9% S/G2M; 54% Late-EMT; 100% SC33; Spp1 Fn1 Dcn

V3 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 7% S/G2M; 68% Early-Hybrid; 100% SC33; Fabp5 Krtdap Ndufa4l2

V4 UV Response Dn — UV Response Dn +1.91 SD; 14% S/G2M; 96% Early-Hybrid; 100% SC33; Thbs1 Nr4a1 Fosb

Mouse SCC: A. Entire gene set, unit SC2b1

Fit shown: light QC · k = 3 · p = 0.00498 · 1,099 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	51%	41%
2 closest 5% of cells to each corner	83%	86%
3 weight > 0.95 on one corner	99%	99%

Hybrids are 31% of all cells here; 66% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC2b1 alone, cells fill a 3-corner simplex (p=0.00498). 4 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: MYC / growth (translation); EMT / matrix; inflammatory / stress response.

Hybrid vs other cells: 51% of hybrid cells and 41% of other cells are generalists (rule 1; rule 2: 83% vs 86%), so hybrids are clearly more often generalists.

Corners and their tasks

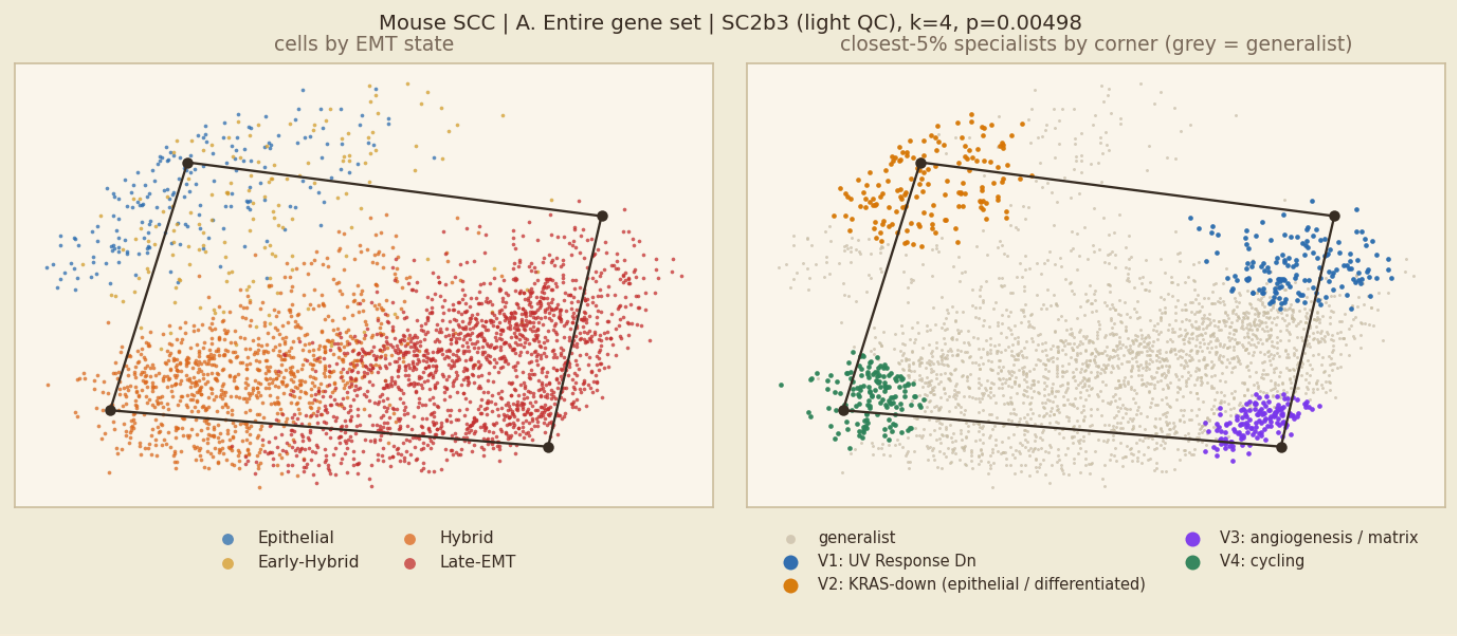
V1 MYC / growth (translation) — Myc Targets V2 +2.1 SD; 96% S/G2M; 100% Hybrid; 100% SC2b1; Hmgb2 Hist1h2ap Tuba1b

V2 EMT / matrix — Epithelial Mesenchymal Transition +1.53 SD; 5% S/G2M; 98% Late-EMT; 100% SC2b1; Mmp13 Col6a3 Mmp3

V3 inflammatory / stress response — TNF-alpha Signaling via NF-kB +1.31 SD; 31% S/G2M; 100% Late-EMT; 100% SC2b1; Cebpd Id3 Cst3

Mouse SCC: A. Entire gene set, unit SC2b3

Fit shown: light QC · k = 4 · p = 0.00498 · 2,961 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	42%	62%
2 closest 5% of cells to each corner	82%	79%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 38% of all cells here; 20% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC2b3 alone, cells fill a 4-corner simplex (p=0.00498). 4 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: UV Response Dn; KRAS-down (epithelial / differentiated); angiogenesis / matrix; cycling.

Hybrid vs other cells: 42% of hybrid cells and 62% of other cells are generalists (rule 1; rule 2: 82% vs 79%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 UV Response Dn — UV Response Dn +1.37 SD; 19% S/G2M; 97% Late-EMT; 100% SC2b3; Malat1 Col5a2 Ahnak

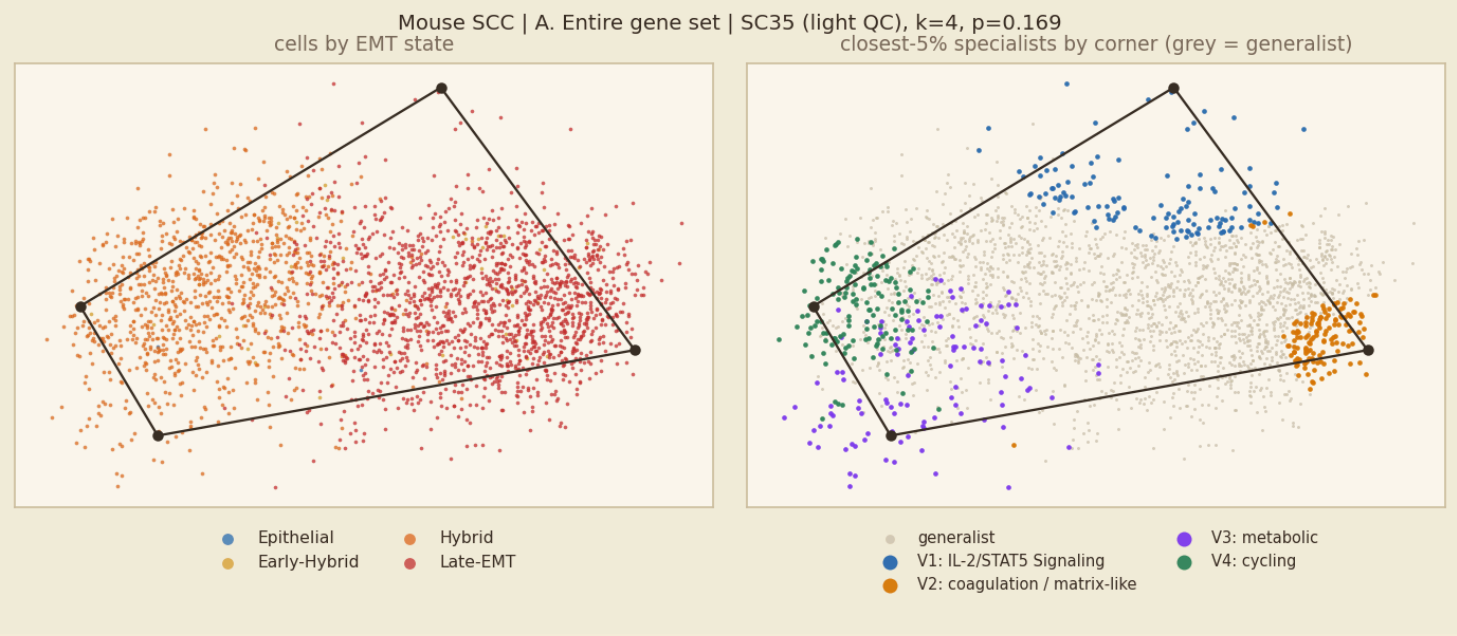
V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.5 SD; 62% S/G2M; 68% Epithelial; 100% SC2b3; Krt14 Krt5 Krt17

V3 angiogenesis / matrix — Angiogenesis +1.62 SD; 4% S/G2M; 100% Late-EMT; 100% SC2b3; Mmp3 Col3a1 Spp1

V4 cycling — E2F Targets +2.05 SD; 97% S/G2M; 100% Hybrid; 100% SC2b3; Hmgb2 H2afz Ran

Mouse SCC: A. Entire gene set, unit SC35

Fit shown: light QC · k = 4 · p = 0.169 · 2,574 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	67%	58%
2 closest 5% of cells to each corner	73%	84%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 40% of all cells here; 33% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC35: the fit (k=4, p=0.169) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 67% of hybrid cells and 58% of other cells are generalists (rule 1; rule 2: 73% vs 84%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 IL-2/STAT5 Signaling — IL-2/STAT5 Signaling +0.86 SD; 5% S/G2M; 82% Late-EMT; 100% SC35; Ereg Ptgs2 Neat1

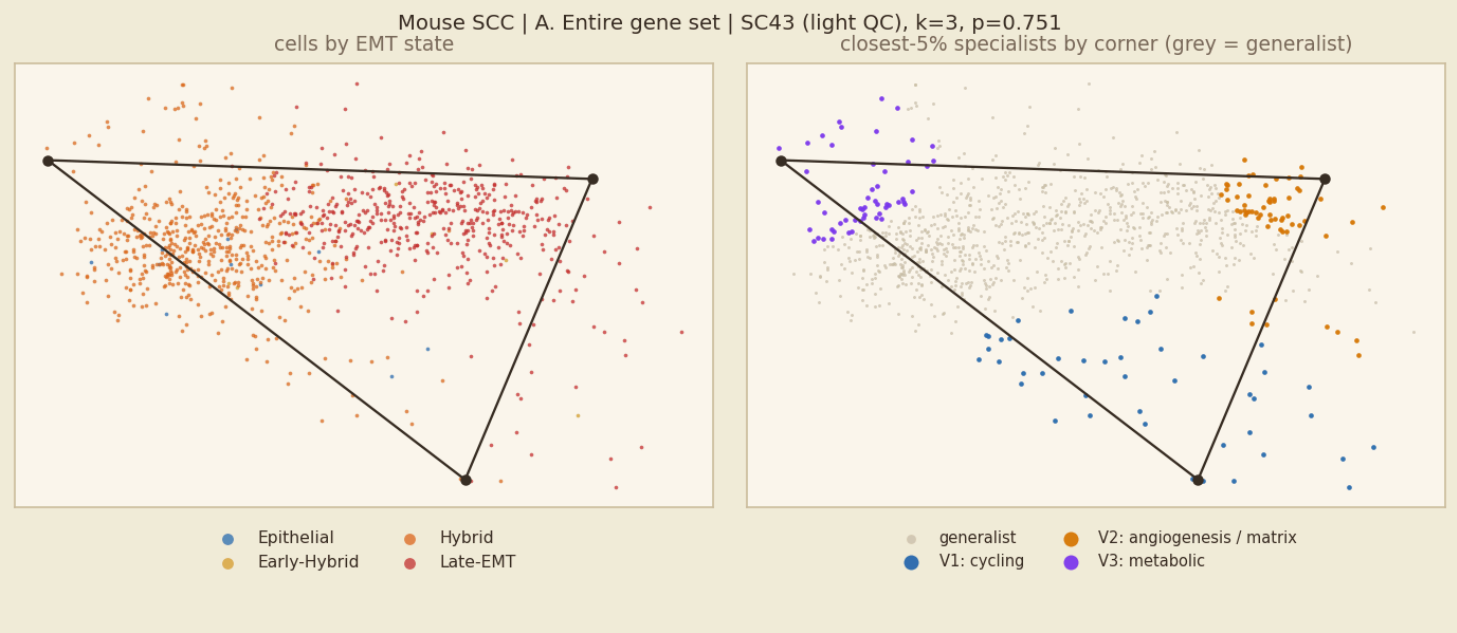
V2 coagulation / matrix-like — Coagulation +1.26 SD; 1% S/G2M; 100% Late-EMT; 100% SC35; Postn Mmp13 Col3a1

V3 metabolic — Oxidative Phosphorylation +1.51 SD; 41% S/G2M; 91% Hybrid; 100% SC35; Akr1b3 S100a4 Upp1

V4 cycling — E2F Targets +2.24 SD; 100% S/G2M; 100% Hybrid; 100% SC35; Hmgb2 Ccnd1 H2afz

Mouse SCC: A. Entire gene set, unit SC43

Fit shown: light QC · k = 3 · p = 0.751 · 1,041 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	50%	57%
2 closest 5% of cells to each corner	86%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 53% of all cells here; 74% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC43: the fit (k=3, p=0.751) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 50% of hybrid cells and 57% of other cells are generalists (rule 1; rule 2: 86% vs 83%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

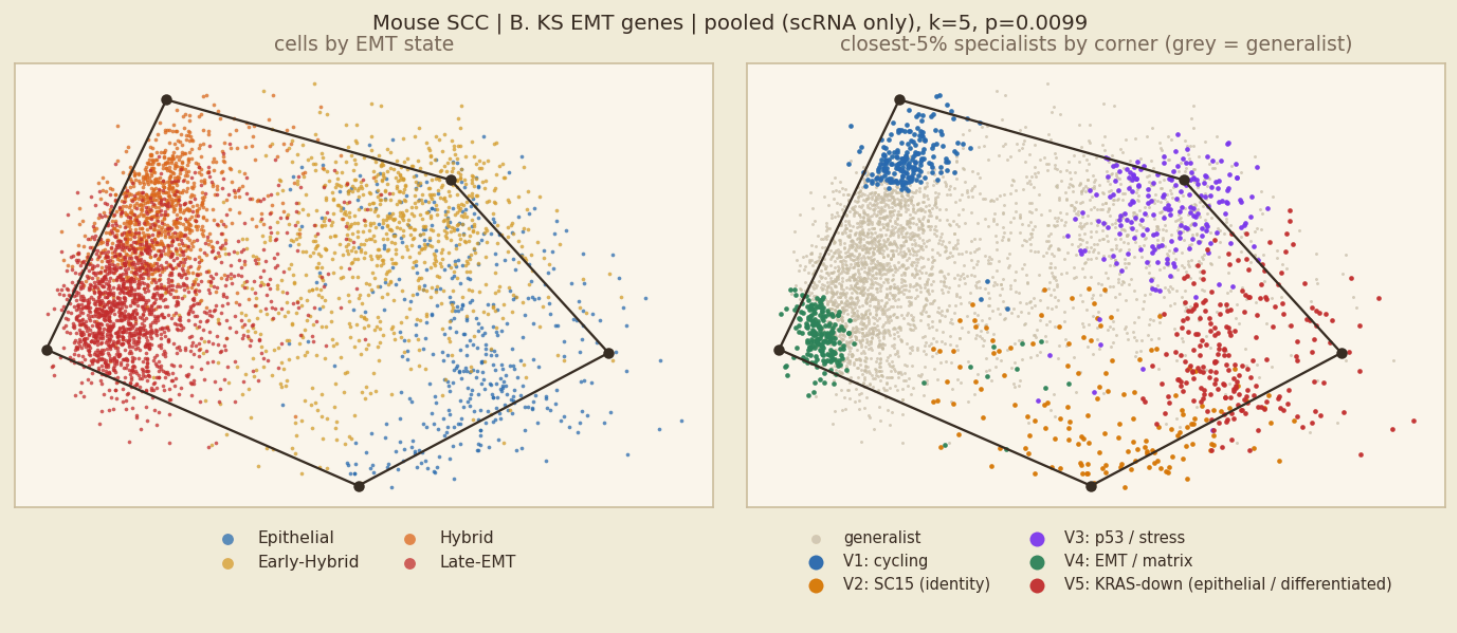
V1 cycling — Mitotic Spindle +0.79 SD; 79% S/G2M; 51% Late-EMT; 100% SC43; Malat1 AY036118 Colec12

V2 angiogenesis / matrix — Angiogenesis +1.56 SD; 0% S/G2M; 100% Late-EMT; 100% SC43; Postn Spp1 Igfbp7

V3 metabolic — Oxidative Phosphorylation +1.19 SD; 98% S/G2M; 100% Hybrid; 100% SC43; Selenoh Crisp1 H2afz

Mouse SCC: B. KS EMT genes, pooled fit

Fit shown: scRNA only · k = 5 · p = 0.0099 · 14,059 cells · 100 shuffles l comparison, light QC: k=5, p=0.0945



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	68%	56%
2 closest 5% of cells to each corner	75%	75%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 49% of all cells here; 8% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=5, p=0.0099) whose corners mix samples (no corner is 70% or more one sample), so it is not a pure sample-identity simplex.

Corners are programmes: cycling; p53 / stress; EMT / matrix; KRAS-down (epithelial / differentiated).

Hybrid vs other cells: 68% of hybrid cells and 56% of other cells are generalists (rule 1; rule 2: 75% vs 75%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 cycling — E2F Targets +1.02 SD; 72% S/G2M; 95% Hybrid; Krt8 Lgals1 Vim

V2 identity corner: SC15 (programme: cycling) — Mitotic Spindle +1.38 SD; 33% S/G2M; 51% Epithelial; 74% SC15; Dsp Rabgap1l Ank3

V3 p53 / stress — p53 Pathway +1.38 SD; 34% S/G2M; 96% Early-Hybrid; Slpi Sfn Aqp3

V4 EMT / matrix — Epithelial Mesenchymal Transition +1.66 SD; 10% S/G2M; 98% Late-EMT; Col5a2 Sparc Fstl1

V5 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.15 SD; 44% S/G2M; 84% Epithelial; 68% SC32; Fxyd3 Sfn Epcam

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse SCC: B. KS EMT genes, all units at a glance

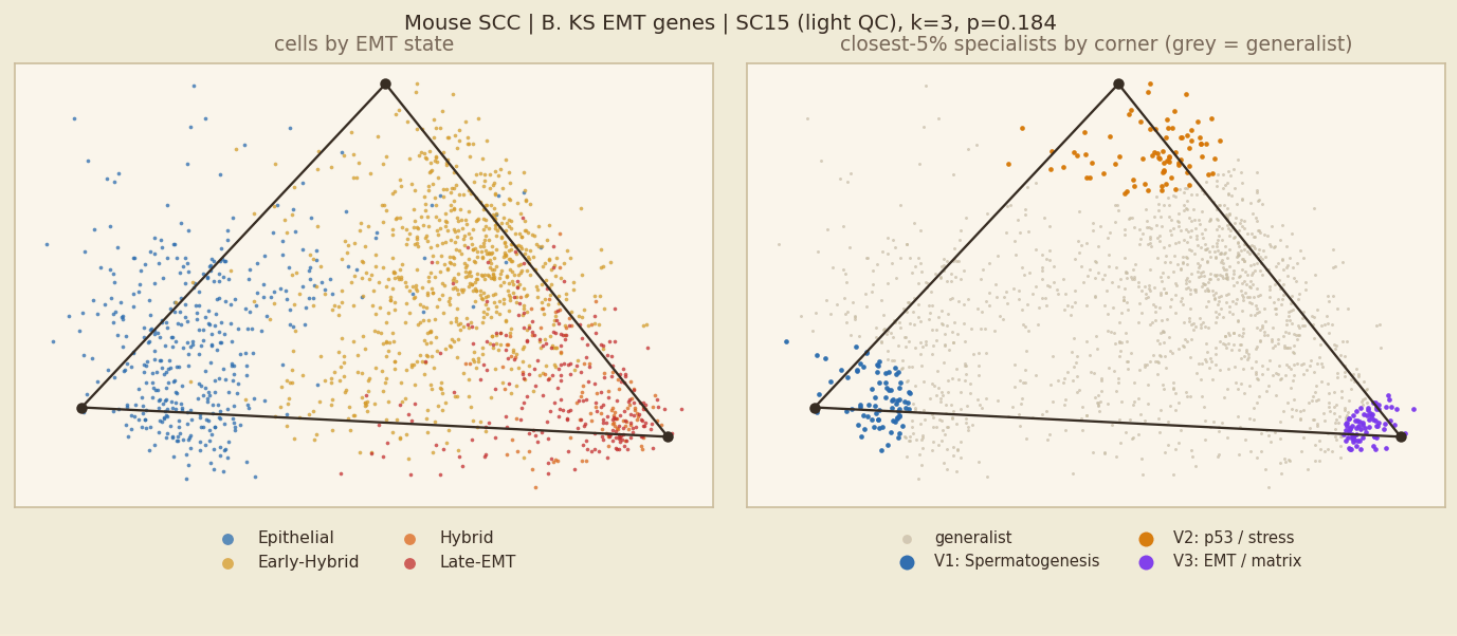
Main fit of each unit: 3 of 8 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
SC15	light QC	3	0.184	–	58%	73 / 89 / 100%	77 / 61 / 58%
SC32	light QC	3	0.00498 *	–	57%	61 / 86 / 100%	72 / 58 / 57%
SC46	light QC	3	0.164	–	50%	65 / 82 / 100%	59 / 48 / 50%
SC33	light QC	3	0.0746	–	79%	67 / 88 / 100%	80 / 82 / 79%
SC2b1	light QC	4	0.0547	–	31%	42 / 76 / 99%	24 / 29 / 30%
SC2b3	light QC	3	0.00498 *	–	38%	43 / 82 / 100%	39 / 37 / 38%
SC35	light QC	4	0.00995 *	–	40%	75 / 78 / 100%	43 / 39 / 40%
SC43	light QC	3	0.109	–	53%	45 / 88 / 99%	45 / 55 / 53%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

Mouse SCC: B. KS EMT genes, unit SC15

Fit shown: light QC · k = 3 · p = 0.184 · 1,576 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	73%	30%
2 closest 5% of cells to each corner	89%	80%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 58% of all cells here; 71% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC15: the fit (k=3, p=0.184) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 73% of hybrid cells and 30% of other cells are generalists (rule 1; rule 2: 89% vs 80%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 Spermatogenesis — Spermatogenesis +1.15 SD; 19% S/G2M; 100% Epithelial; 100% SC15; Ank3 Arap2 Epcam

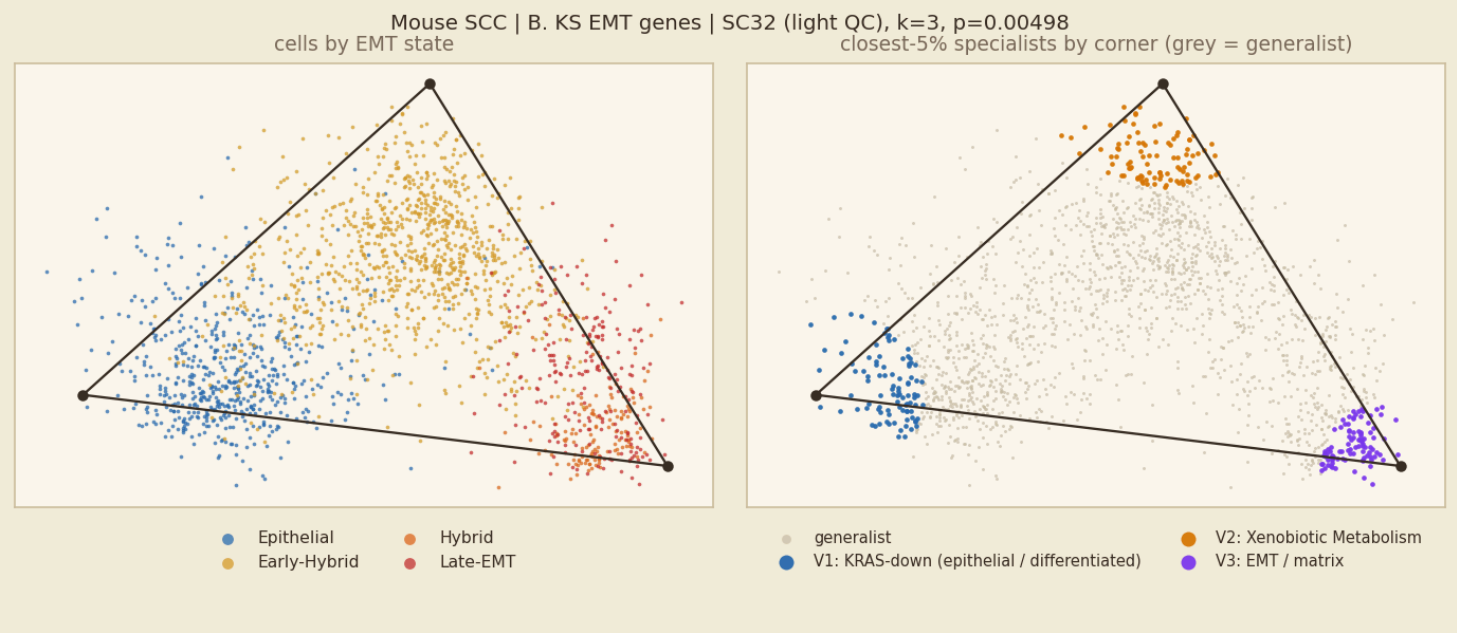
V2 p53 / stress — p53 Pathway +1.15 SD; 23% S/G2M; 97% Early-Hybrid; 100% SC15; Slpi Aqp3 Areg

V3 EMT / matrix — Epithelial Mesenchymal Transition +2.03 SD; 59% S/G2M; 66% Late-EMT; 100% SC15; Lgals1 Vim Col5a2

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse SCC: B. KS EMT genes, unit SC32

Fit shown: light QC · k = 3 · p = 0.00498 · 1,910 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	61%	31%
2 closest 5% of cells to each corner	86%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 57% of all cells here; 82% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC32 alone, cells fill a 3-corner simplex (p=0.00498). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); Xenobiotic Metabolism; EMT / matrix.

Hybrid vs other cells: 61% of hybrid cells and 31% of other cells are generalists (rule 1; rule 2: 86% vs 83%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +0.64 SD; 48% S/G2M; 98% Epithelial; 100% SC32; S100a14 Gpx2 Epcam

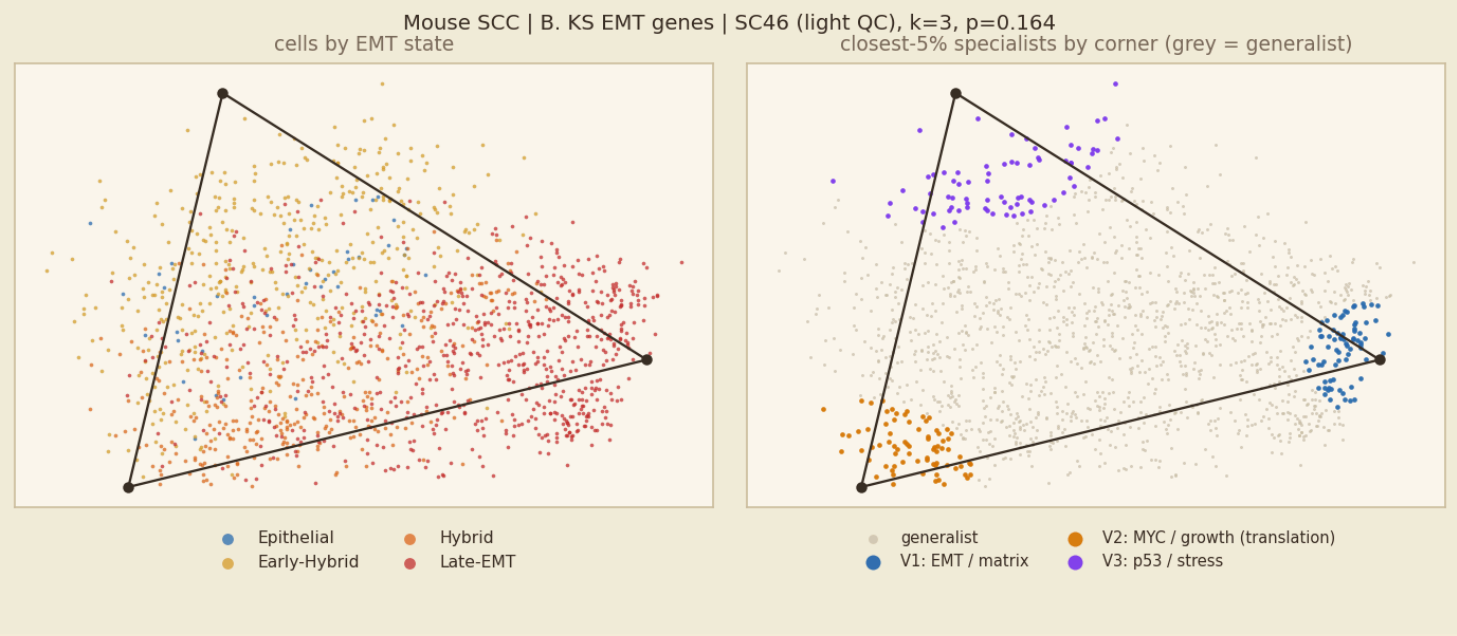
V2 Xenobiotic Metabolism — Xenobiotic Metabolism +0.92 SD; 19% S/G2M; 100% Early-Hybrid; 100% SC32; Slpi Areg Aqp3

V3 EMT / matrix — Epithelial Mesenchymal Transition +1.18 SD; 32% S/G2M; 56% Hybrid; 100% SC32; Vim Lgals1 Krt8

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse SCC: B. KS EMT genes, unit SC46

Fit shown: light QC · k = 3 · p = 0.164 · 1,519 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	65%	44%
2 closest 5% of cells to each corner	82%	88%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 50% of all cells here; 66% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC46: the fit (k=3, p=0.164) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 65% of hybrid cells and 44% of other cells are generalists (rule 1; rule 2: 82% vs 88%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.82 SD; 20% S/G2M; 100% Late-EMT; 100% SC46; Col5a2 Fstl1 Sparc

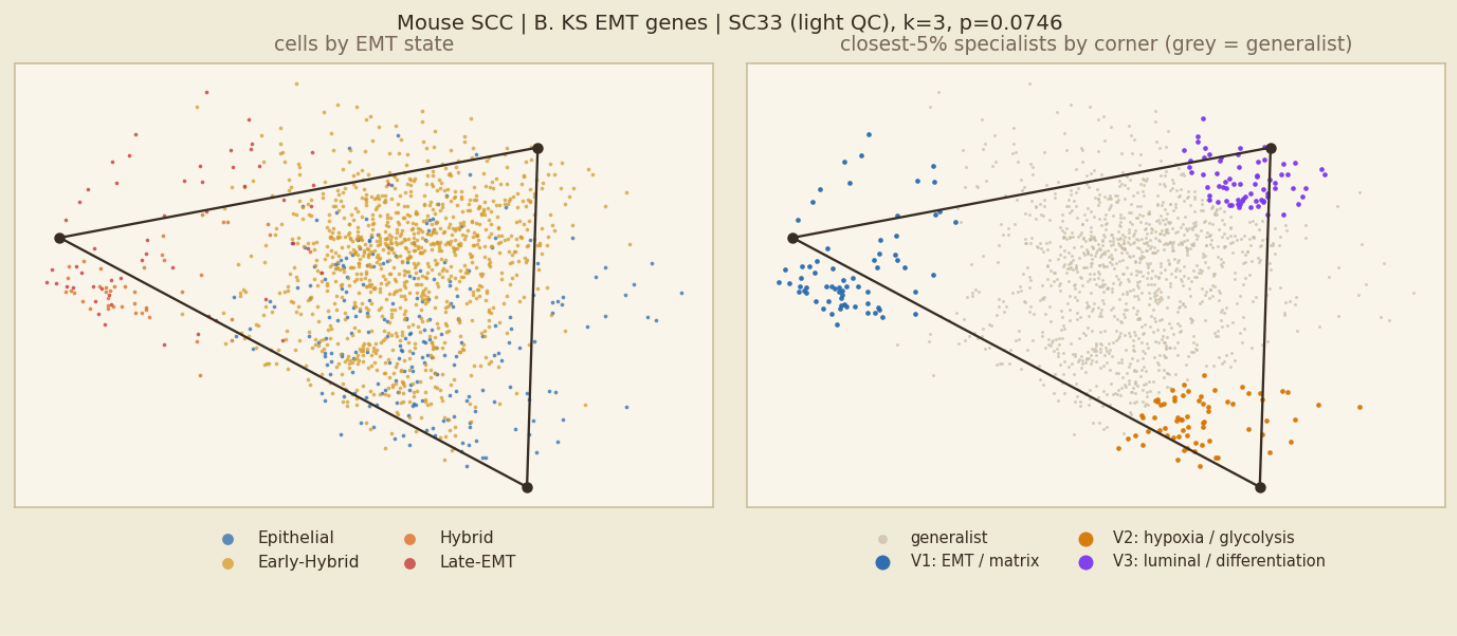
V2 MYC / growth (translation) — Myc Targets V2 +1.25 SD; 68% S/G2M; 61% Hybrid; 100% SC46; Krt18 Areg Krt8

V3 p53 / stress — p53 Pathway +1.04 SD; 25% S/G2M; 93% Early-Hybrid; 100% SC46; Fxyd3 Slpi Sfn

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse SCC: B. KS EMT genes, unit SC33

Fit shown: light QC · k = 3 · p = 0.0746 · 1,379 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	67%	65%
2 closest 5% of cells to each corner	88%	74%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 79% of all cells here; 83% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC33: the fit (k=3, p=0.0746) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 67% of hybrid cells and 65% of other cells are generalists (rule 1; rule 2: 88% vs 74%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.66 SD; 30% S/G2M; 51% Hybrid; 100% SC33; Col5a2 Vim Krt8

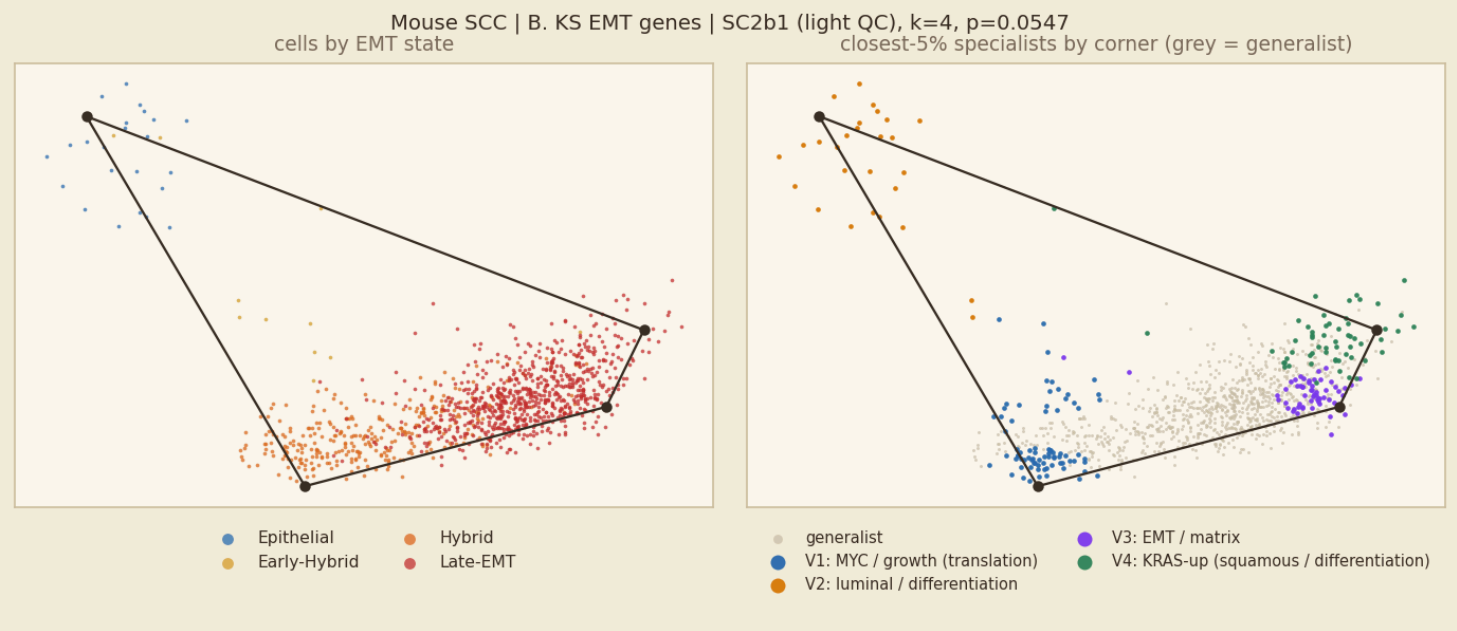
V2 hypoxia / glycolysis — Glycolysis +0.57 SD; 46% S/G2M; 58% Epithelial; 100% SC33; S100a14 Aqp3 Sfn

V3 luminal / differentiation — Estrogen Response Early +0.73 SD; 14% S/G2M; 96% Early-Hybrid; 100% SC33; Slpi Krt18 Aqp3

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse SCC: B. KS EMT genes, unit SC2b1

Fit shown: light QC · k = 4 · p = 0.0547 · 1,099 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	42%	59%
2 closest 5% of cells to each corner	76%	82%
3 weight > 0.95 on one corner	99%	100%

Hybrids are 31% of all cells here; 50% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC2b1: the fit (k=4, p=0.0547) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 42% of hybrid cells and 59% of other cells are generalists (rule 1; rule 2: 76% vs 82%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 MYC / growth (translation) — Myc Targets V1 +1.55 SD; 75% S/G2M; 96% Hybrid; 100% SC2b1; Krt8 Map1b Emp3

V2 luminal / differentiation — Estrogen Response Late +1.66 SD; 47% S/G2M; 42% Epithelial; 100% SC2b1; Sfn Areg Perp

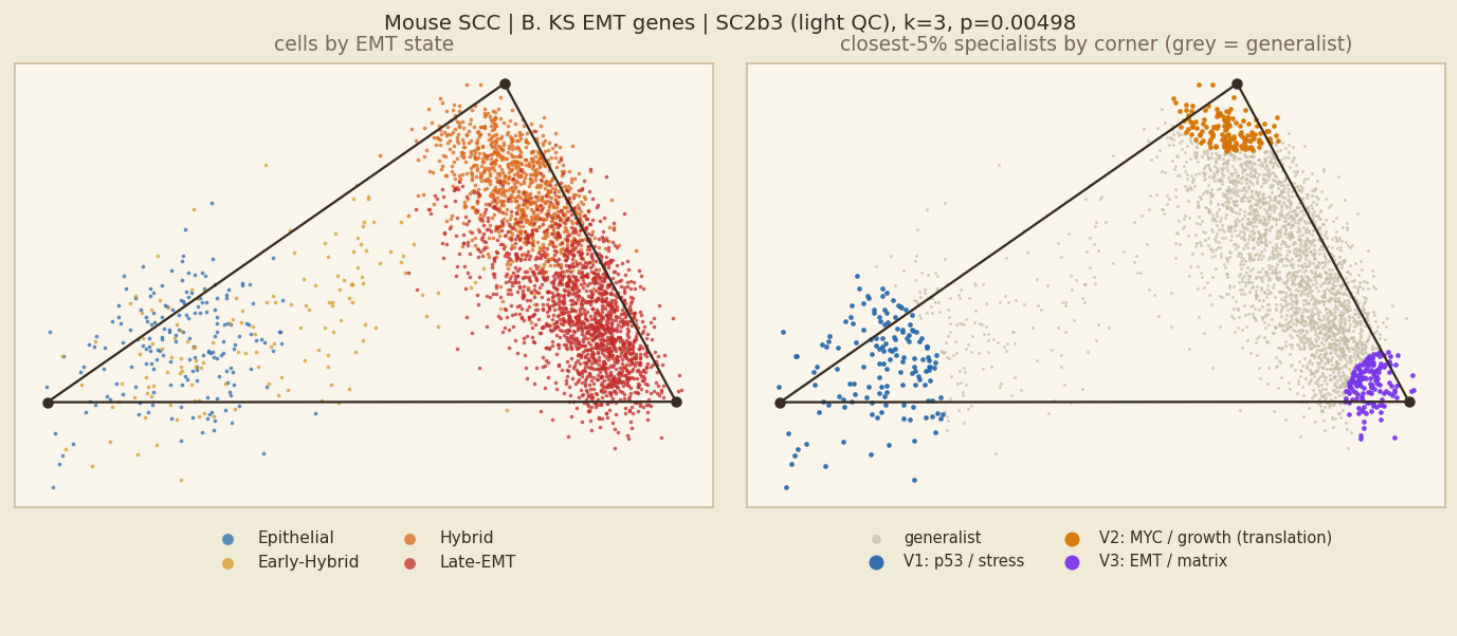
V3 EMT / matrix — Epithelial Mesenchymal Transition +1.05 SD; 16% S/G2M; 98% Late-EMT; 100% SC2b1; Col5a2 Sparc Hes1

V4 KRAS-up (squamous / differentiation) — KRAS Signaling Up +0.89 SD; 15% S/G2M; 98% Late-EMT; 100% SC2b1; Slpi Col5a2 Fstl1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse SCC: B. KS EMT genes, unit SC2b3

Fit shown: light QC · k = 3 · p = 0.00498 · 2,961 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	43%	42%
2 closest 5% of cells to each corner	82%	87%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 38% of all cells here; 86% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC2b3 alone, cells fill a 3-corner simplex (p=0.00498). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: p53 / stress; MYC / growth (translation); EMT / matrix.

Hybrid vs other cells: 43% of hybrid cells and 42% of other cells are generalists (rule 1; rule 2: 82% vs 87%), so no real difference.

Corners and their tasks

V1 p53 / stress — p53 Pathway +1.78 SD; 46% S/G2M; 65% Epithelial; 100% SC2b3; Sfn Perp Sdc1

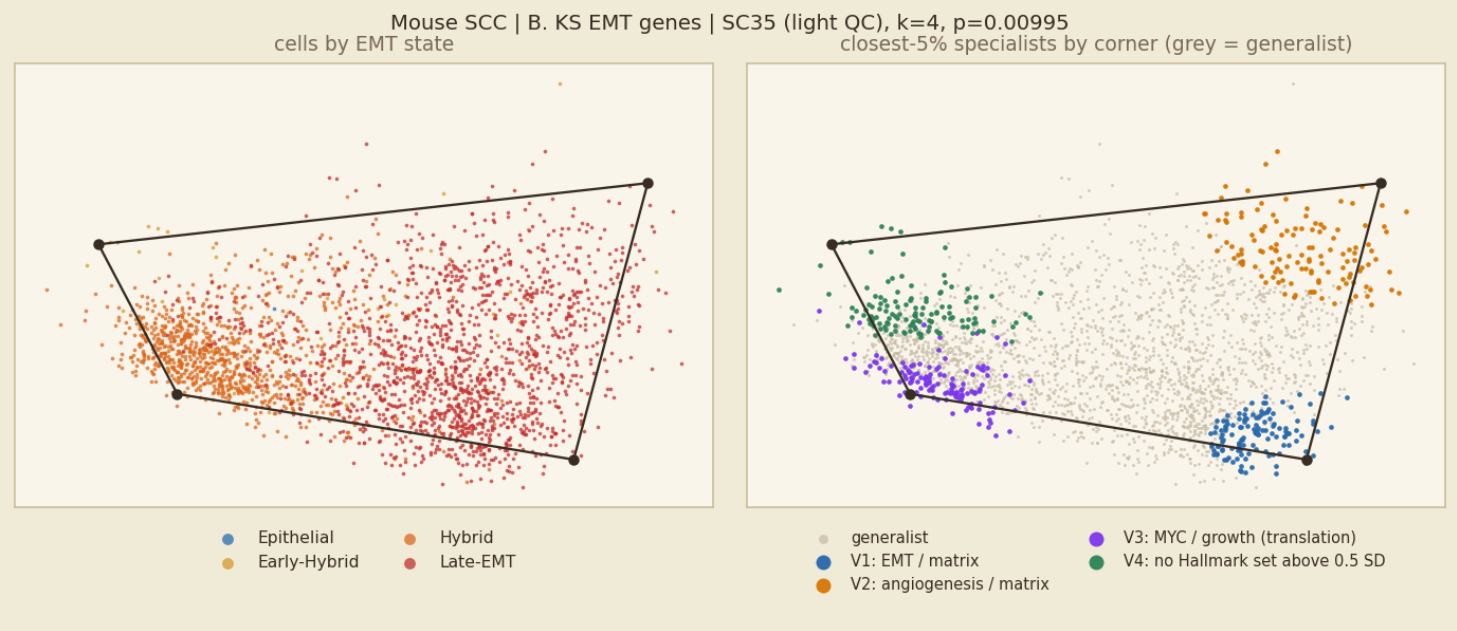
V2 MYC / growth (translation) — Myc Targets V1 +1.47 SD; 81% S/G2M; 99% Hybrid; 100% SC2b3; Krt8 Krt18 Lgals1

V3 EMT / matrix — Epithelial Mesenchymal Transition +1.59 SD; 11% S/G2M; 100% Late-EMT; 100% SC2b3; Col5a2 Sparc Fstl1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse SCC: B. KS EMT genes, unit SC35

Fit shown: light QC · k = 4 · p = 0.00995 · 2,574 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	75%	66%
2 closest 5% of cells to each corner	78%	81%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 40% of all cells here; 42% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC35 alone, cells fill a 4-corner simplex (p=0.00995). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: EMT / matrix; angiogenesis / matrix; MYC / growth (translation); no Hallmark set above 0.5 SD.

Hybrid vs other cells: 75% of hybrid cells and 66% of other cells are generalists (rule 1; rule 2: 78% vs 81%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.37 SD; 2% S/G2M; 99% Late-EMT; 100% SC35; Col5a2 Sparc Fstl1

V2 angiogenesis / matrix — Angiogenesis +1.04 SD; 2% S/G2M; 98% Late-EMT; 100% SC35; Slpi Col5a2 Sparc

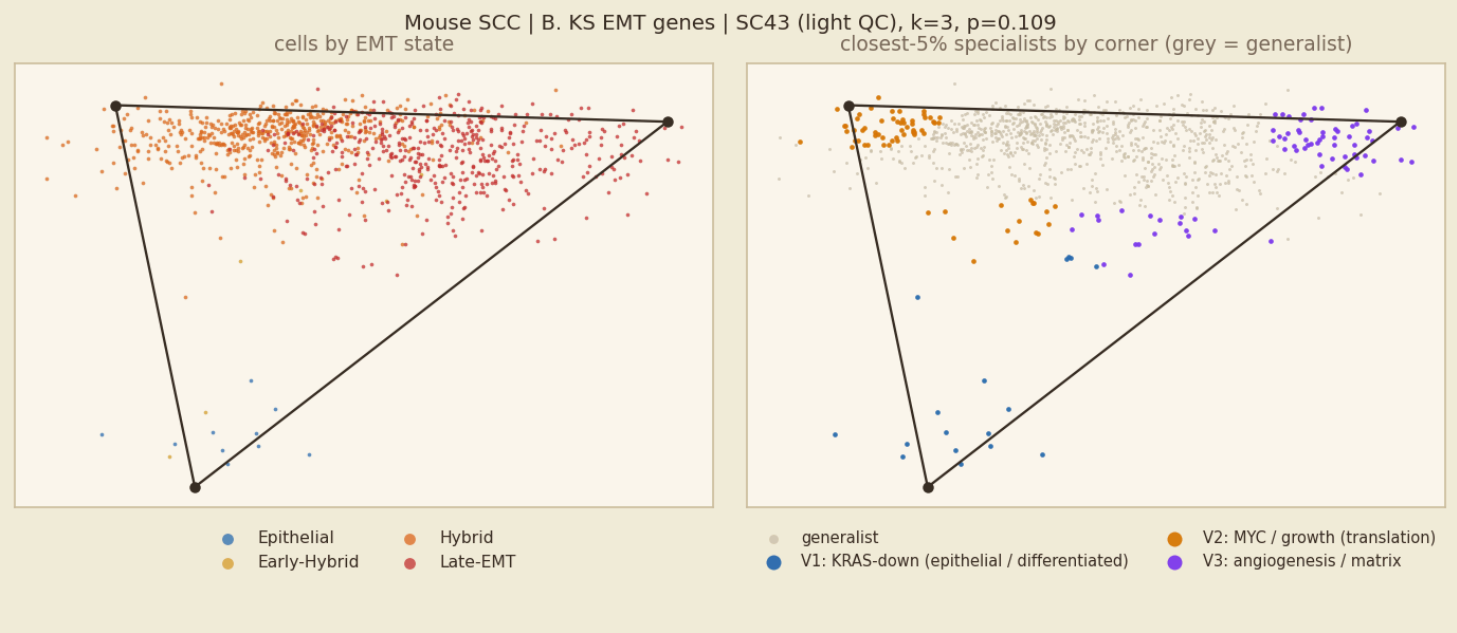
V3 MYC / growth (translation) — Myc Targets V1 +1.29 SD; 67% S/G2M; 96% Hybrid; 100% SC35; Tuba1a Map1b Krt8

V4 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 29% S/G2M; 64% Hybrid; 100% SC35; Areg Cd9 Krt18

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse SCC: B. KS EMT genes, unit SC43

Fit shown: light QC · k = 3 · p = 0.109 · 1,041 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	45%	62%
2 closest 5% of cells to each corner	88%	81%
3 weight > 0.95 on one corner	99%	99%

Hybrids are 53% of all cells here; 89% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC43: the fit (k=3, p=0.109) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 45% of hybrid cells and 62% of other cells are generalists (rule 1; rule 2: 88% vs 81%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +0.79 SD; 49% S/G2M; 55% Late-EMT; 100% SC43; Areg Perp Epcam

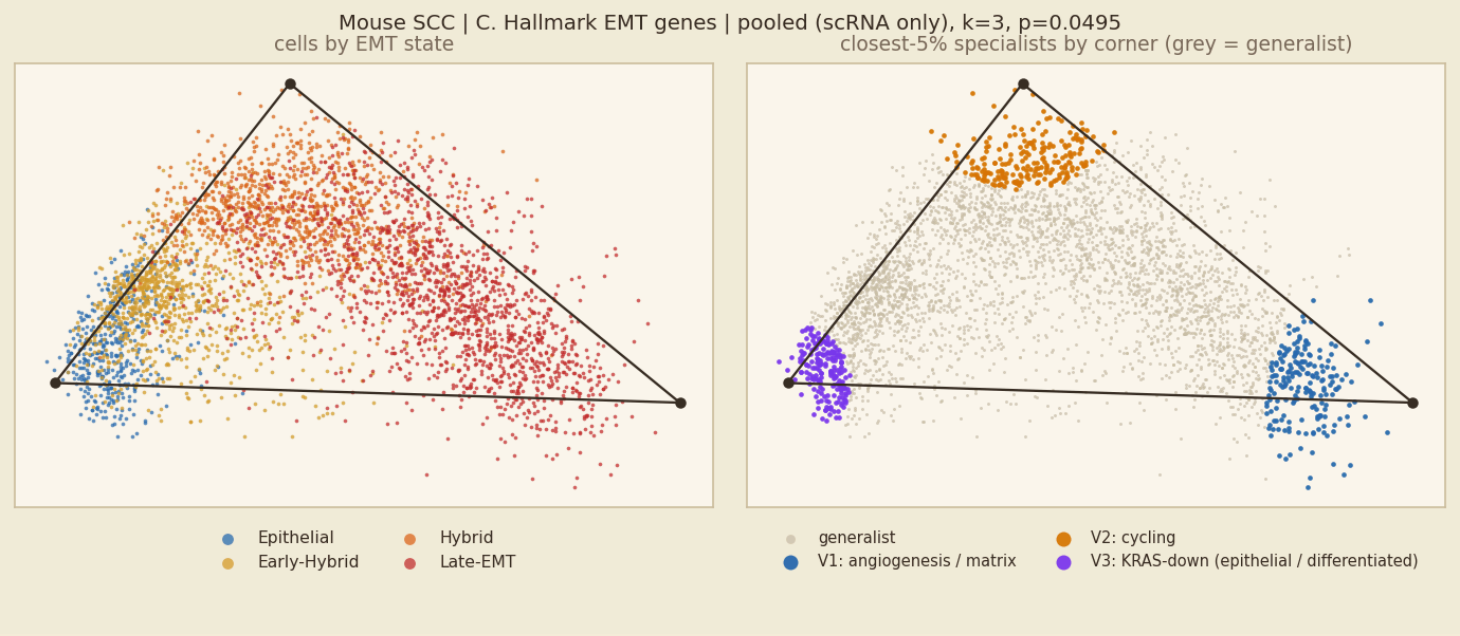
V2 MYC / growth (translation) — Myc Targets V1 +0.88 SD; 85% S/G2M; 100% Hybrid; 100% SC43; Krt8 Krt18 Lgals1

V3 angiogenesis / matrix — Angiogenesis +2.09 SD; 11% S/G2M; 98% Late-EMT; 100% SC43; Col5a2 Sparc Loxl2

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse SCC: C. Hallmark EMT genes, pooled fit

Fit shown: scRNA only · k = 3 · p = 0.0495 · 14,059 cells · 100 shuffles | comparison, light QC: k=5, p=0.277



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	57%	52%
2 closest 5% of cells to each corner	89%	81%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 49% of all cells here; 83% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=3, p=0.0495) whose corners mix samples (no corner is 70% or more one sample), so it is not a pure sample-identity simplex.

Corners are programmes: angiogenesis / matrix; cycling; KRAS-down (epithelial / differentiated).

Hybrid vs other cells: 57% of hybrid cells and 52% of other cells are generalists (rule 1; rule 2: 89% vs 81%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 angiogenesis / matrix — Angiogenesis +1.76 SD; 7% S/G2M; 99% Late-EMT; Col3a1 Postn Col1a1

V2 cycling — E2F Targets +0.94 SD; 67% S/G2M; 86% Hybrid; Mgp Timp1 Lgals1

V3 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.22 SD; 41% S/G2M; 79% Epithelial; Lama3 Sfrp1 Lamc2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse SCC: C. Hallmark EMT genes, all units at a glance

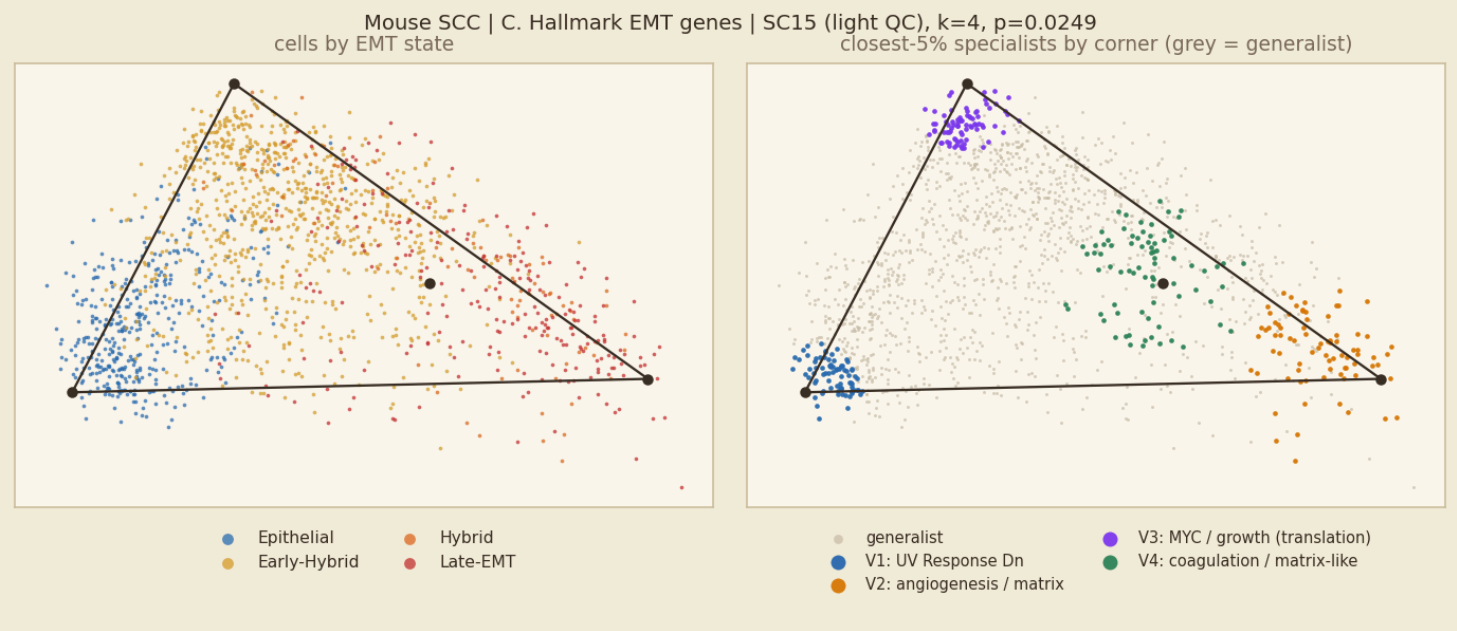
Main fit of each unit: 3 of 8 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
SC15	light QC	4	0.0249 *	–	58%	58 / 82 / 100%	64 / 60 / 58%
SC32	light QC	3	0.0299 *	–	57%	61 / 89 / 100%	70 / 60 / 58%
SC46	light QC	4	0.348	–	50%	54 / 82 / 100%	47 / 52 / 50%
SC33	light QC	4	0.692	–	79%	64 / 82 / 100%	75 / 81 / 79%
SC2b1	light QC	3	0.00995 *	–	31%	54 / 85 / 100%	33 / 31 / 31%
SC2b3	light QC	3	0.154	–	38%	62 / 88 / 100%	37 / 39 / 38%
SC35	light QC	4	0.522	–	40%	75 / 78 / 100%	41 / 39 / 40%
SC43	light QC	3	0.219	–	53%	59 / 91 / 100%	57 / 57 / 53%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

Mouse SCC: C. Hallmark EMT genes, unit SC15

Fit shown: light QC · k = 4 · p = 0.0249 · 1,576 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	58%	46%
2 closest 5% of cells to each corner	82%	77%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 58% of all cells here; 28% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC15 alone, cells fill a 4-corner simplex (p=0.0249). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: UV Response Dn; angiogenesis / matrix; MYC / growth (translation); coagulation / matrix-like.

Hybrid vs other cells: 58% of hybrid cells and 46% of other cells are generalists (rule 1; rule 2: 82% vs 77%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 UV Response Dn — UV Response Dn +1.25 SD; 25% S/G2M; 100% Epithelial; 100% SC15; Cadm1 Edil3 Sntb1

V2 angiogenesis / matrix — Angiogenesis +1.89 SD; 56% S/G2M; 68% Late-EMT; 100% SC15; Col1a1 Col1a2 Col3a1

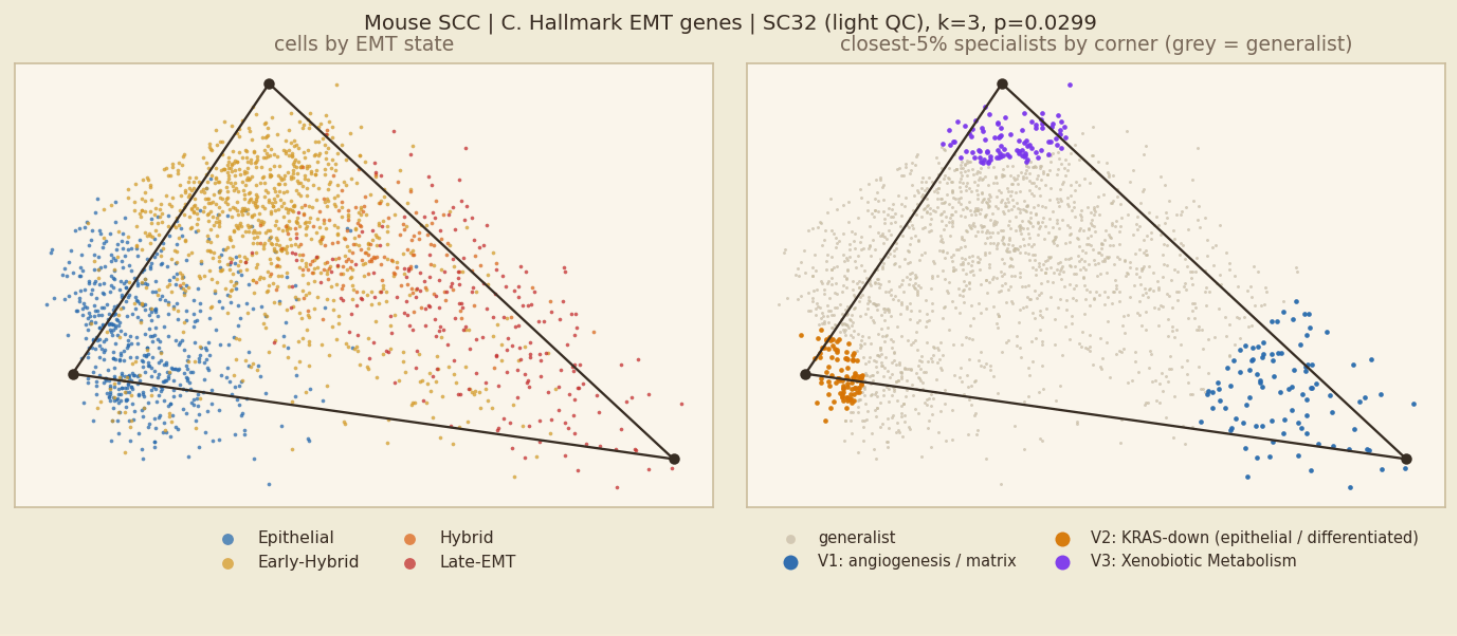
V3 MYC / growth (translation) — Myc Targets V2 +1.57 SD; 65% S/G2M; 99% Early-Hybrid; 100% SC15; Lgals1 Areg Cxcl1

V4 coagulation / matrix-like — Coagulation +1.14 SD; 6% S/G2M; 78% Early-Hybrid; 100% SC15; Fn1 Col5a2 Fbln2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse SCC: C. Hallmark EMT genes, unit SC32

Fit shown: light QC · k = 3 · p = 0.0299 · 1,910 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	61%	35%
2 closest 5% of cells to each corner	89%	80%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 57% of all cells here; 73% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC32 alone, cells fill a 3-corner simplex (p=0.0299). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: angiogenesis / matrix; KRAS-down (epithelial / differentiated); Xenobiotic Metabolism.

Hybrid vs other cells: 61% of hybrid cells and 35% of other cells are generalists (rule 1; rule 2: 89% vs 80%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 angiogenesis / matrix — Angiogenesis +2.35 SD; 5% S/G2M; 79% Late-EMT; 100% SC32; Bgn Col5a2 Spp1

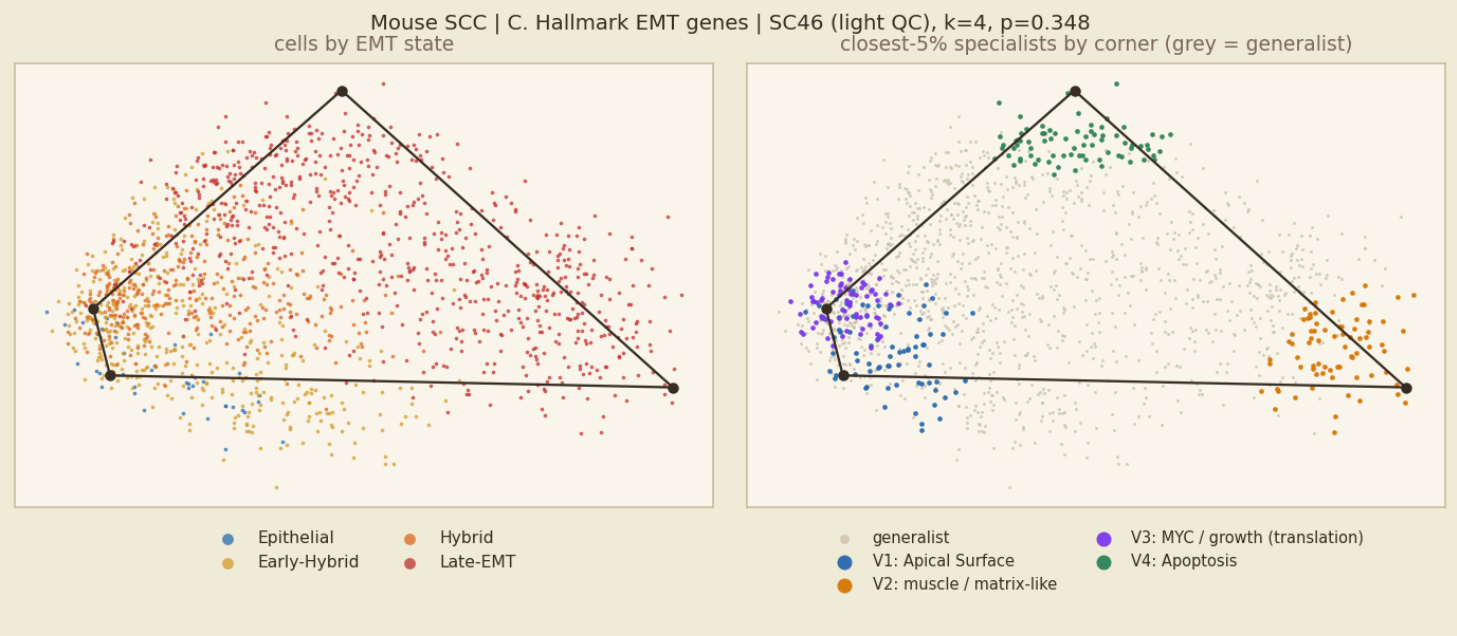
V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +0.64 SD; 34% S/G2M; 91% Epithelial; 100% SC32; Sfrp1 Thbs1 Igfbp3

V3 Xenobiotic Metabolism — Xenobiotic Metabolism +0.98 SD; 26% S/G2M; 98% Early-Hybrid; 100% SC32; Areg Ecm1 Vim

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse SCC: C. Hallmark EMT genes, unit SC46

Fit shown: light QC · k = 4 · p = 0.348 · 1,519 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	54%	62%
2 closest 5% of cells to each corner	82%	78%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 50% of all cells here; 25% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC46: the fit (k=4, p=0.348) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 54% of hybrid cells and 62% of other cells are generalists (rule 1; rule 2: 82% vs 78%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 Apical Surface — Apical Surface +1.0 SD; 28% S/G2M; 79% Early-Hybrid; 100% SC46; Cadm1 Sfrp1 Tgfbi

V2 muscle / matrix-like — Myogenesis +1.74 SD; 26% S/G2M; 100% Late-EMT; 100% SC46; Col3a1 Eln Bgn

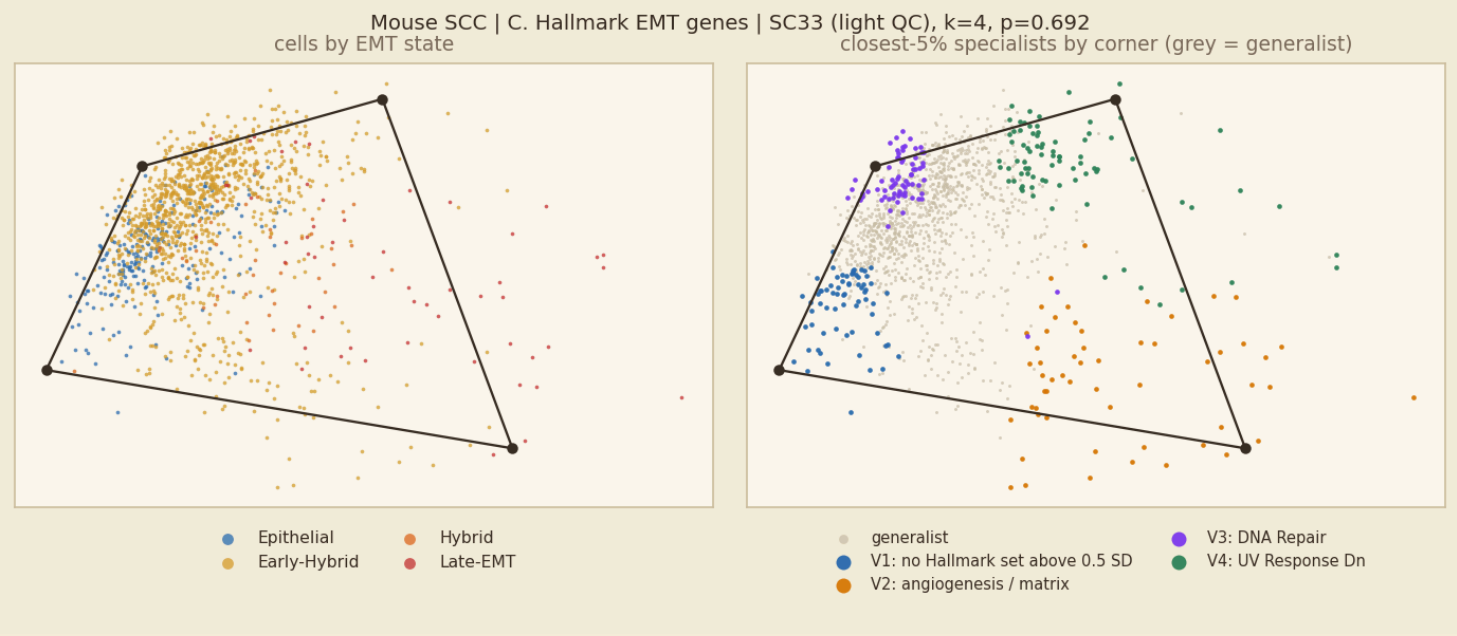
V3 MYC / growth (translation) — Myc Targets V2 +1.57 SD; 79% S/G2M; 72% Hybrid; 100% SC46; Areg Lgals1 Pthlh

V4 Apoptosis — Apoptosis +1.18 SD; 13% S/G2M; 100% Late-EMT; 100% SC46; Spp1 Dcn Fn1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse SCC: C. Hallmark EMT genes, unit SC33

Fit shown: light QC · k = 4 · p = 0.692 · 1,379 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	64%	79%
2 closest 5% of cells to each corner	82%	73%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 79% of all cells here; 50% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC33: the fit (k=4, p=0.692) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 64% of hybrid cells and 79% of other cells are generalists (rule 1; rule 2: 82% vs 73%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 38% S/G2M; 65% Epithelial; 100% SC33; Lama3 Lamc2 Gja1

V2 angiogenesis / matrix — Angiogenesis +2.27 SD; 7% S/G2M; 51% Early-Hybrid; 100% SC33; Fn1 Spp1 Col6a2

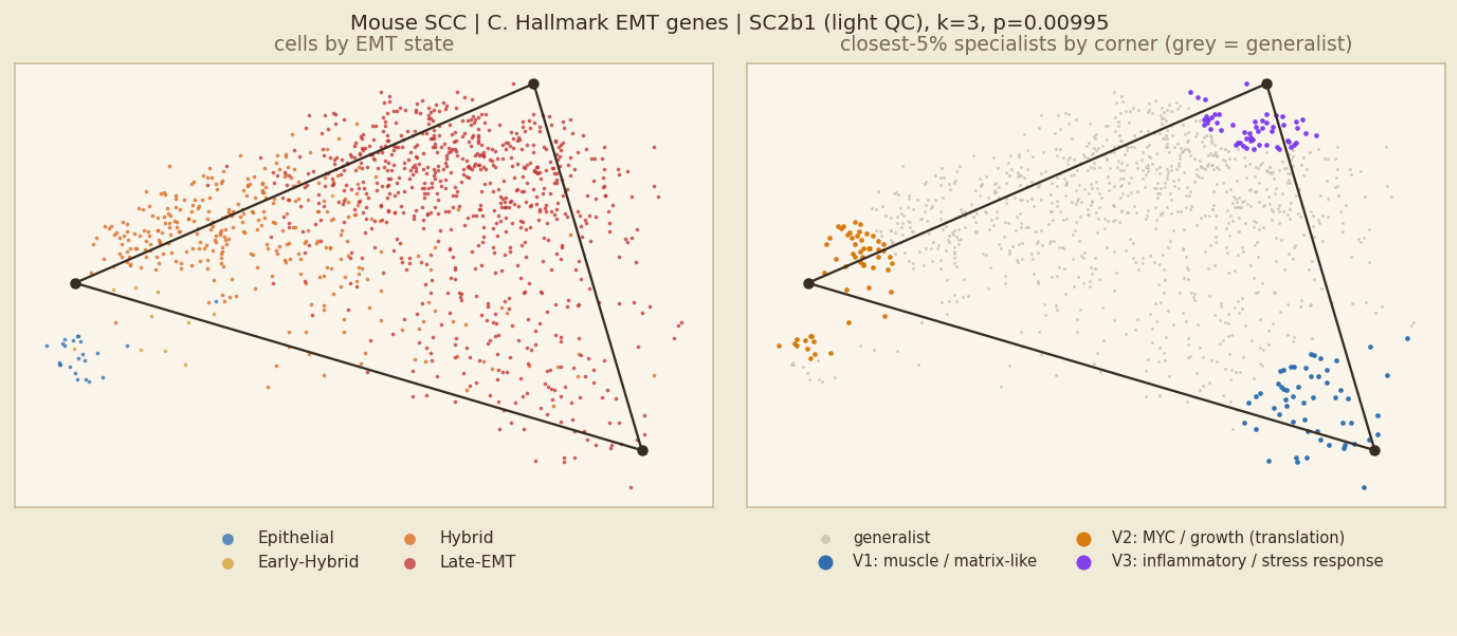
V3 DNA Repair — DNA Repair +0.65 SD; 61% S/G2M; 99% Early-Hybrid; 100% SC33; Areg Cxcl1 Mmp3

V4 UV Response Dn — UV Response Dn +1.59 SD; 16% S/G2M; 94% Early-Hybrid; 100% SC33; Thbs1 Col12a1 Acta2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse SCC: C. Hallmark EMT genes, unit SC2b1

Fit shown: light QC · k = 3 · p = 0.00995 · 1,099 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	54%	48%
2 closest 5% of cells to each corner	85%	85%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 31% of all cells here; 67% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside SC2b1 alone, cells fill a 3-corner simplex (p=0.00995). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: muscle / matrix-like; MYC / growth (translation); inflammatory / stress response.

Hybrid vs other cells: 54% of hybrid cells and 48% of other cells are generalists (rule 1; rule 2: 85% vs 85%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 muscle / matrix-like — Myogenesis +1.41 SD; 15% S/G2M; 91% Late-EMT; 100% SC2b1; Col6a3 Mmp3 Col1a1

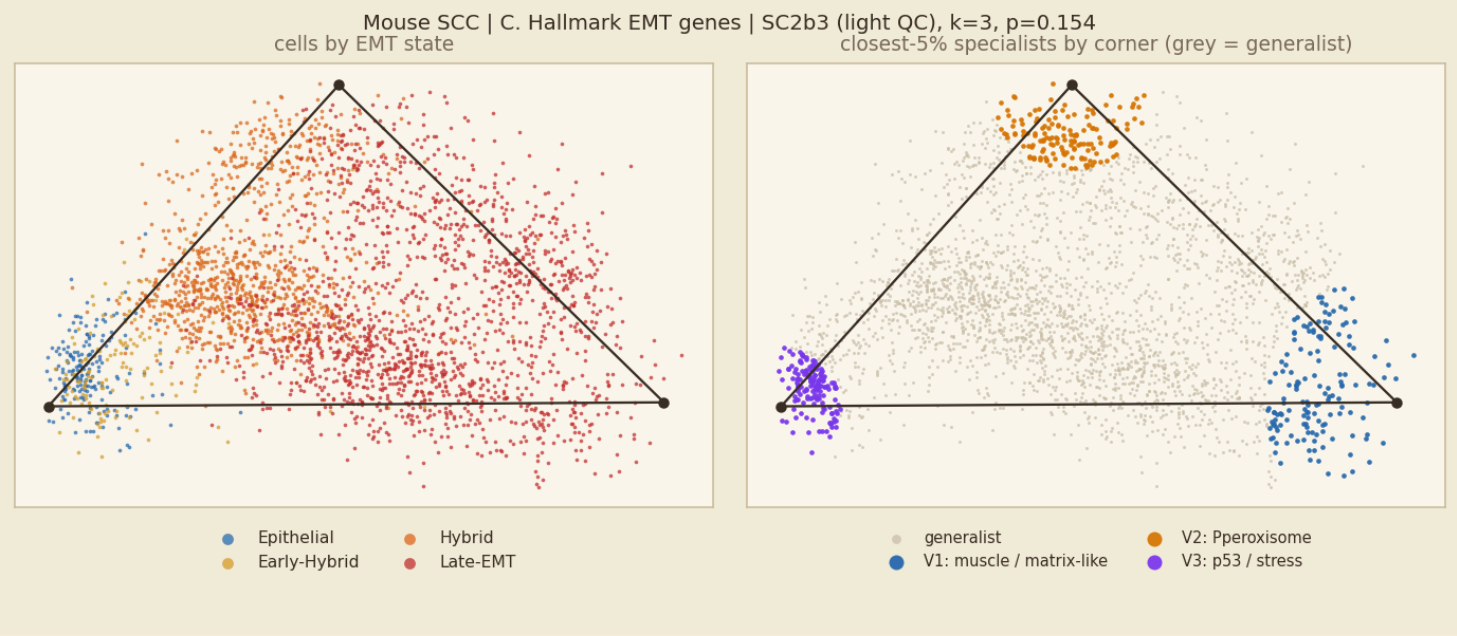
V2 MYC / growth (translation) — Myc Targets V1 +1.71 SD; 73% S/G2M; 71% Hybrid; 100% SC2b1; Areg Tnfrsf12a Basp1

V3 inflammatory / stress response — TNF-alpha Signaling via NF-kB +1.27 SD; 15% S/G2M; 100% Late-EMT; 100% SC2b1; Postn Col3a1 Lox

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse SCC: C. Hallmark EMT genes, unit SC2b3

Fit shown: light QC · k = 3 · p = 0.154 · 2,961 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	62%	63%
2 closest 5% of cells to each corner	88%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 38% of all cells here; 72% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC2b3: the fit (k=3, p=0.154) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 62% of hybrid cells and 63% of other cells are generalists (rule 1; rule 2: 88% vs 83%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 muscle / matrix-like — Myogenesis +1.53 SD; 10% S/G2M; 100% Late-EMT; 100% SC2b3; Col3a1 Postn Bgn

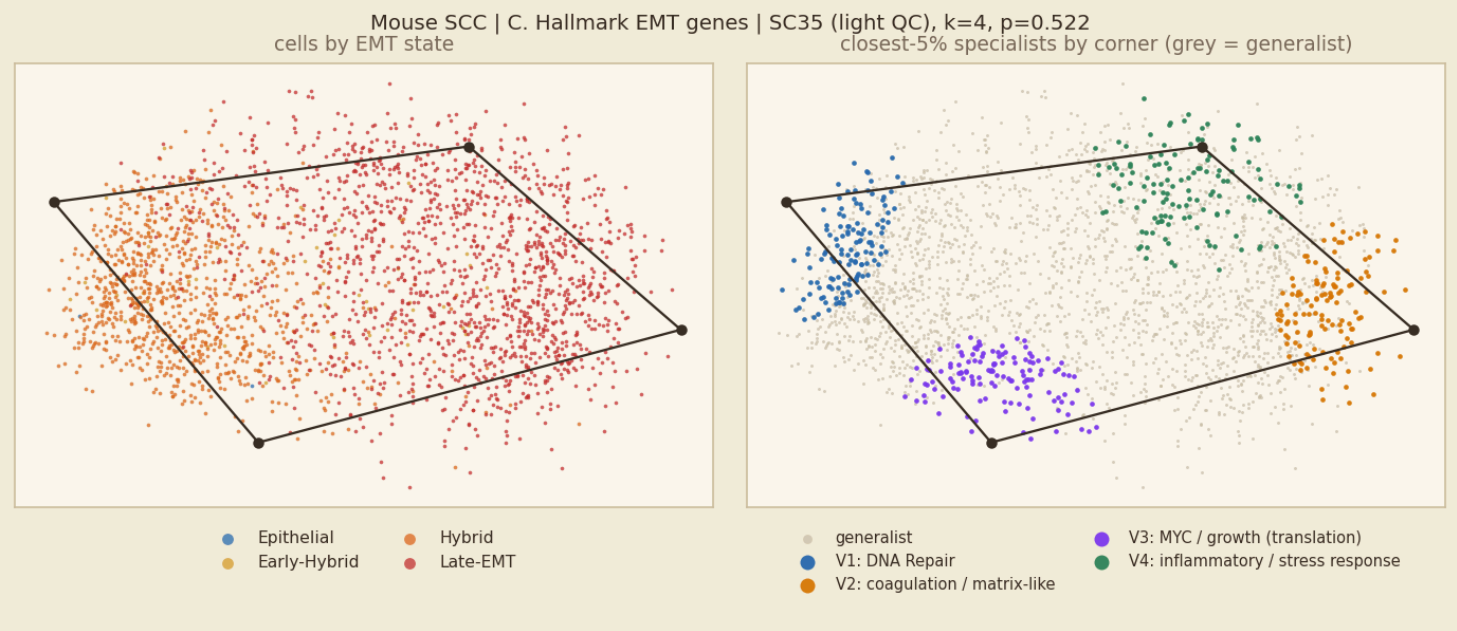
V2 Pperoxisome — Pperoxisome +0.84 SD; 51% S/G2M; 57% Hybrid; 100% SC2b3; Tagln Acta2 Spp1

V3 p53 / stress — p53 Pathway +1.55 SD; 54% S/G2M; 68% Epithelial; 100% SC2b3; Sdc1 Areg Lamc2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse SCC: C. Hallmark EMT genes, unit SC35

Fit shown: light QC · k = 4 · p = 0.522 · 2,574 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	75%	70%
2 closest 5% of cells to each corner	78%	81%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 40% of all cells here; 28% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC35: the fit (k=4, p=0.522) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 75% of hybrid cells and 70% of other cells are generalists (rule 1; rule 2: 78% vs 81%), so hybrids are slightly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 DNA Repair — DNA Repair +0.76 SD; 41% S/G2M; 86% Hybrid; 100% SC35; Spp1 Areg Sat1

V2 coagulation / matrix-like — Coagulation +1.42 SD; 0% S/G2M; 100% Late-EMT; 100% SC35; Postn Col3a1 Mmp3

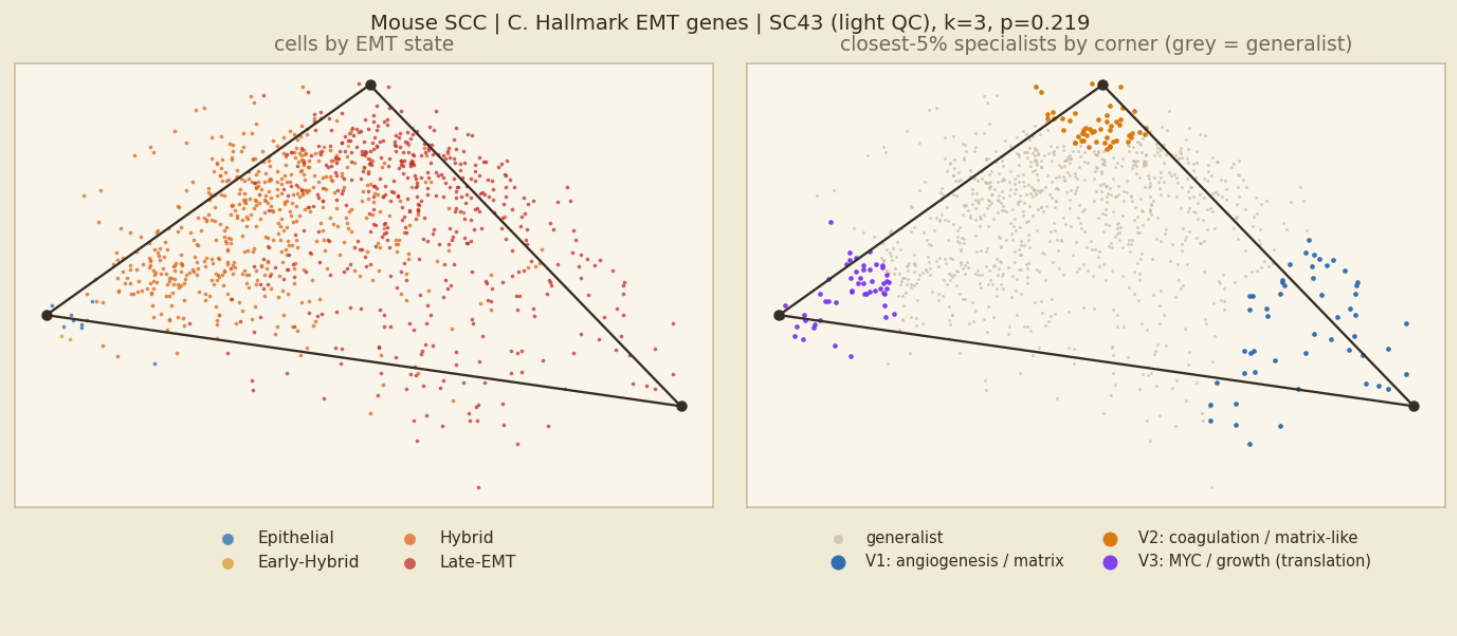
V3 MYC / growth (translation) — Myc Targets V1 +1.01 SD; 61% S/G2M; 81% Hybrid; 100% SC35; Timp3 Acta2 Dpysl3

V4 inflammatory / stress response — Inflammatory Response +0.93 SD; 2% S/G2M; 99% Late-EMT; 100% SC35; Timp1 Col3a1 Fn1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse SCC: C. Hallmark EMT genes, unit SC43

Fit shown: light QC · k = 3 · p = 0.219 · 1,041 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	59%	50%
2 closest 5% of cells to each corner	91%	78%
3 weight > 0.95 on one corner	100%	98%

Hybrids are 53% of all cells here; 79% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside SC43: the fit (k=3, p=0.219) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 59% of hybrid cells and 50% of other cells are generalists (rule 1; rule 2: 91% vs 78%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 angiogenesis / matrix — Angiogenesis +2.16 SD; 11% S/G2M; 100% Late-EMT; 100% SC43; Postn Col3a1 Col5a2

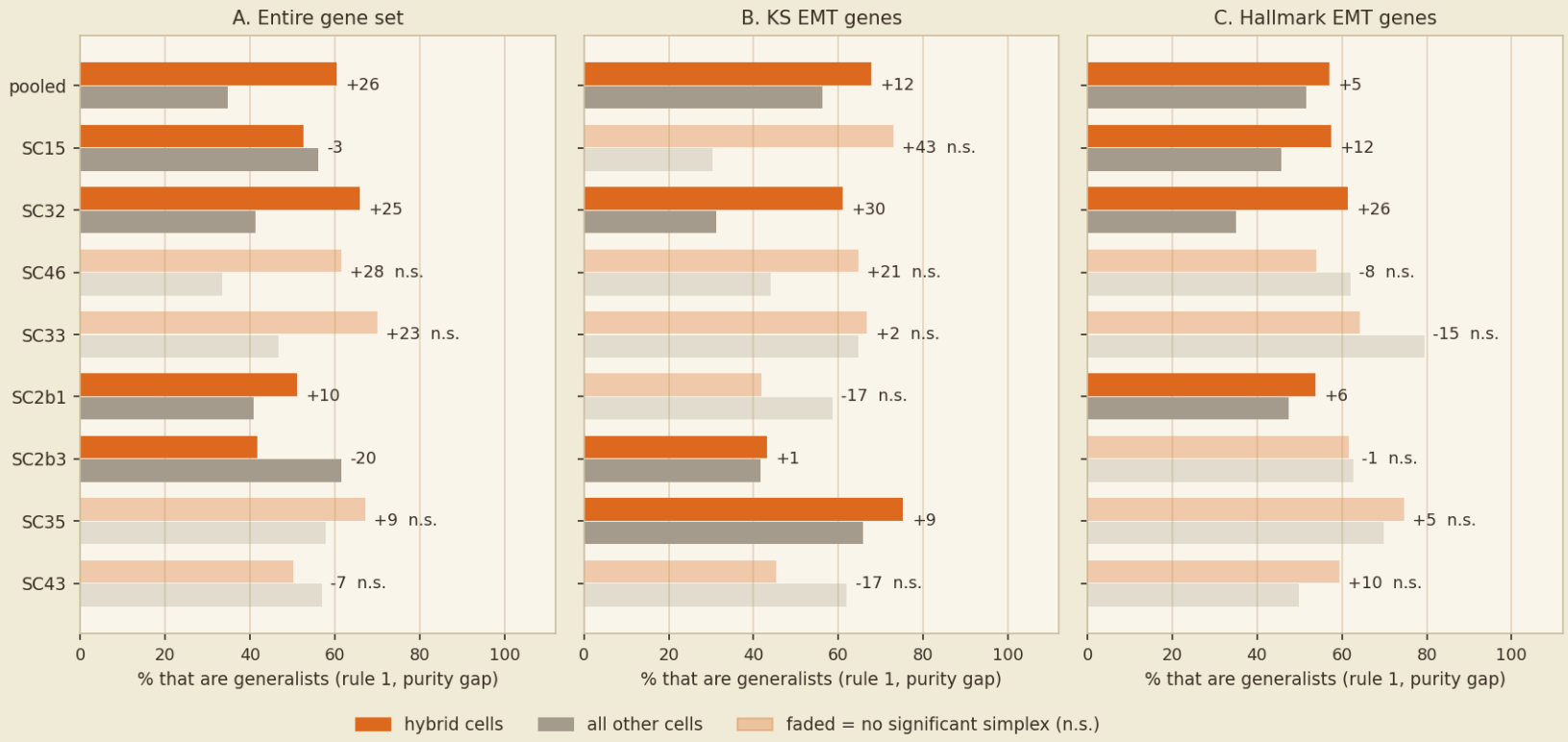
V2 coagulation / matrix-like — Coagulation +0.62 SD; 17% S/G2M; 85% Late-EMT; 100% SC43; Mgp Bgn Fbln2

V3 MYC / growth (translation) — Myc Targets V1 +1.13 SD; 83% S/G2M; 77% Hybrid; 100% SC43; Areg Pthlh Mcm7

Hallmark EMT set left out when naming (circular: fitted on those genes).

Question 2 · Mouse SCC: are hybrid cells more often generalists than other cells?

Orange = hybrid cells, grey = all other cells; one pair of bars per unit; the number = orange minus grey (percentage points)



Reading

- A. Entire gene set (5 significant): hybrids 53% vs others 41% generalist under rule 1 (rule 2: 87% vs 81%) → hybrids are clearly more often generalists.
- B. KS EMT genes (4 significant): hybrids 64% vs others 49% generalist under rule 1 (rule 2: 80% vs 82%) → hybrids are clearly more often generalists.
- C. Hallmark EMT genes (4 significant): hybrids 57% vs others 47% generalist under rule 1 (rule 2: 87% vs 81%) → hybrids are clearly more often generalists.

Rule 3 (weight > 0.95) is not drawn: it calls about 100% of every group generalist, so it cannot show a difference.

Cook

What the dataset is, and what to keep in mind when reading its slides

What the dataset means

Four human cancer cell lines given TGF-β and sampled over time. Time is an EMT coordinate independent of our score, but each line has its own identity.

Units in the main analysis

A549, DU145, MCF7, OVCA420

Gene sets

A. all genes (11,000 to 14,500); B. KS EMT genes (Tan 2014; about 160 to 190 present); C. Hallmark Epithelial-Mesenchymal-Transition set (200 genes, those present in the data).

Keep in mind

- Cook has no state labels, so hybrids are defined from the KS EMT score (middle two quartiles).
- The cut-offs are pooled over the four lines, so a line's hybrid share mostly reflects the line (A549 about 71%, MCF7 about 11%).
- The removal arm (TGF-β withdrawn) is projected as guests and left out of the cells analysed.

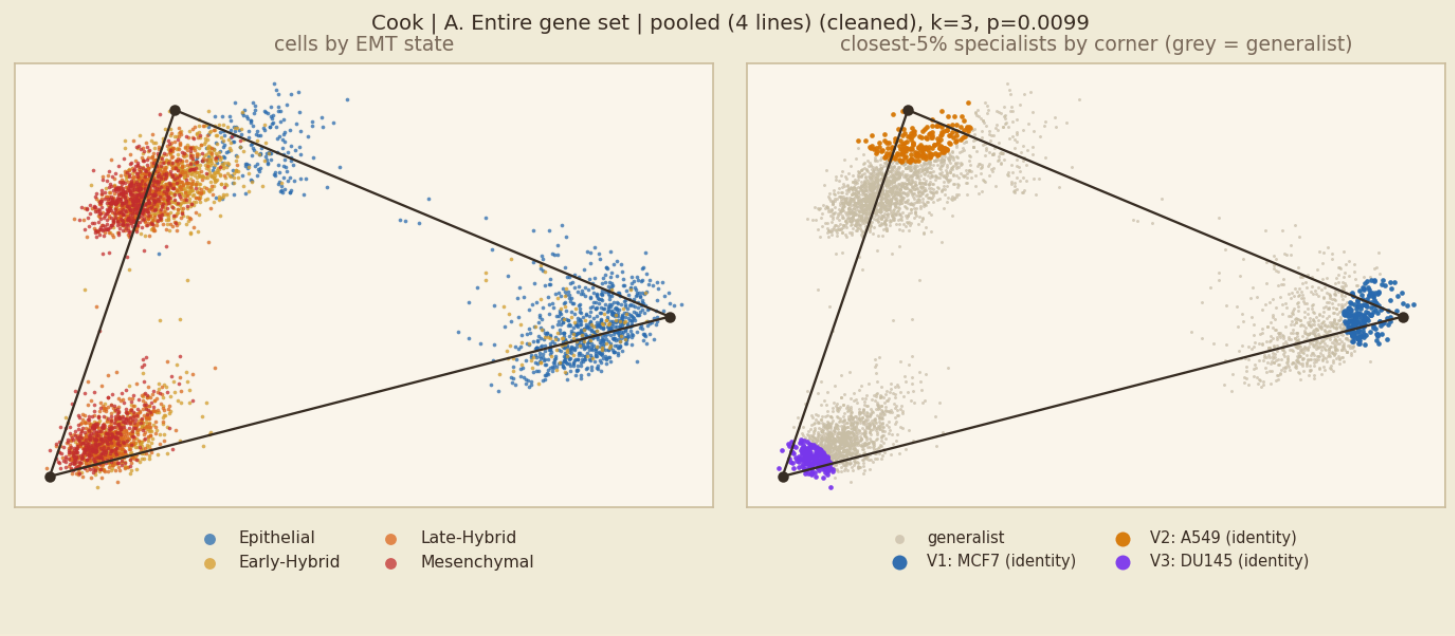
This dataset in numbers (main fits)

- Entire gene set: 2 of 5 fits significant; hybrids 38% / 86% / 100% generalist (median, rules 1/2/3).
- KS EMT genes: 2 of 5 fits significant; hybrids 33% / 84% / 100% generalist (median, rules 1/2/3).
- Hallmark EMT genes: 4 of 5 fits significant; hybrids 59% / 86% / 100% generalist (median, rules 1/2/3).

Question 1 slides come first, then the Question 2 slides for this dataset.

Cook: A. Entire gene set, pooled fit

Fit shown: cleaned · k = 3 · p = 0.0099 · 6,218 cells · 100 shuffles l comparison, light QC: k=3, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	2%	3%
2 closest 5% of cells to each corner	87%	83%
3 weight > 0.95 on one corner	99%	98%

Hybrids are 54% of all cells here; 65% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=3, p=0.0099), but 3 of 3 corners are almost entirely one sample (MCF7, A549, DU145). It separates samples; it is not a shared EMT trade-off.

Each corner is essentially one cell line. The programme names (Hallmark) only describe what distinguishes that line, and the EMT-state colours follow the lines (MCF7 scores epithelial, A549 mesenchymal). These corners are not tasks.

Hybrid vs other cells: 2% of hybrid cells and 3% of other cells are generalists (rule 1; rule 2: 87% vs 83%), so no real difference.

Corners and their tasks

V1 identity corner: MCF7 (programme: luminal / differentiation) — Estrogen Response Early +1.96 SD; 21% S/G2M; 94% Epithelial; 100% MCF7; TFF1 DSCAM-AS1 BCAS3

V2 identity corner: A549 (programme: angiogenesis / matrix) — Angiogenesis +1.32 SD; 15% S/G2M; 49% Early-Hybrid; 85% A549; KRT81 S100A4 ALDH1A1

V3 identity corner: DU145 (programme: Allograft Rejection) — Allograft Rejection +1.47 SD; 69% S/G2M; 42% Late-Hybrid; 100% DU145; MT1E CDKN2A MYL9

Cook: A. Entire gene set, all units at a glance

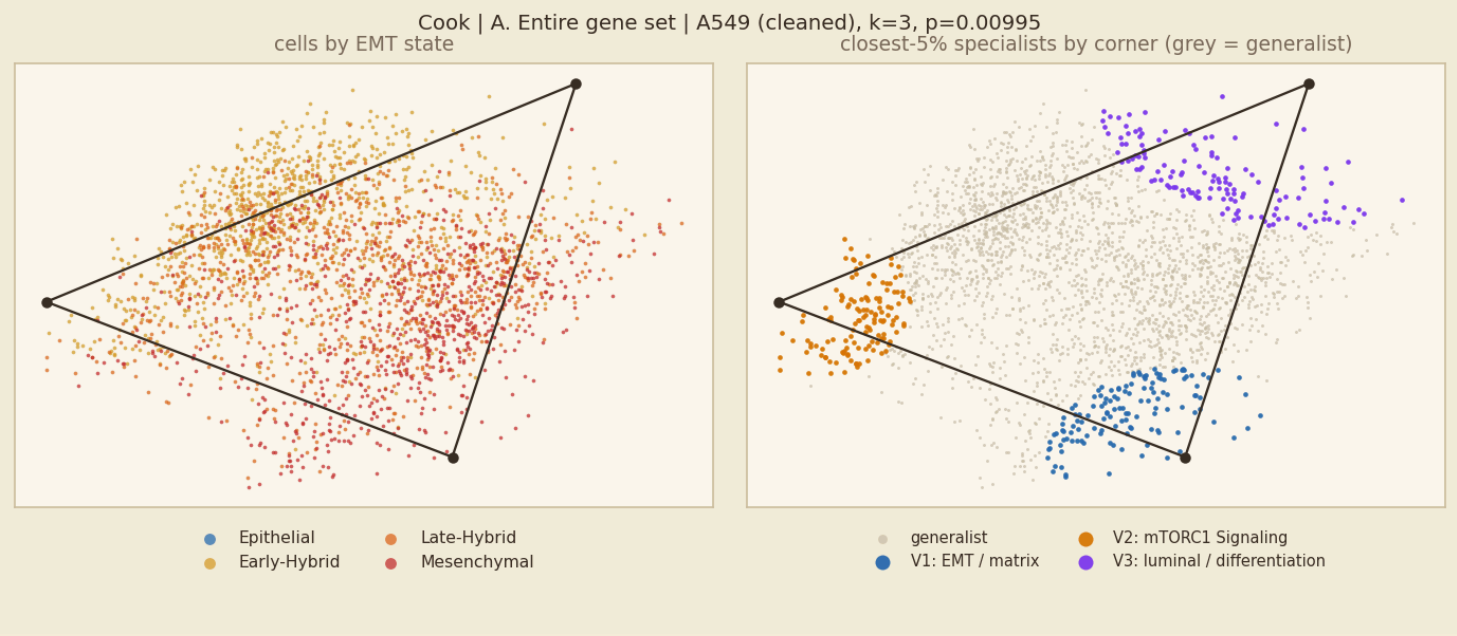
Main fit of each unit: 1 of 4 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
A549	cleaned	3	0.00995 *	0.0647	71%	75 / 86 / 100%	73 / 72 / 71%
DU145	cleaned	3	0.254	0.0995	69%	69 / 85 / 100%	68 / 69 / 69%
MCF7	cleaned	3	0.463	0.0896	11%	54 / 82 / 99%	10 / 11 / 11%
OVCA420	cleaned	3	0.756	0.453	14%	71 / 84 / 100%	15 / 14 / 14%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

Cook: A. Entire gene set, unit A549

Fit shown: cleaned · k = 3 · p = 0.00995 · 2,646 cells · 200 shuffles | comparison, light QC: k=4, p=0.0647



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	75%	69%
2 closest 5% of cells to each corner	86%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 71% of all cells here; 70% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside A549 alone, cells fill a 3-corner simplex (p=0.00995). 1 of 4 units of this dataset have a significant simplex for this gene set.

Corners are programmes: EMT / matrix; mTORC1 Signaling; luminal / differentiation.

Hybrid vs other cells: 75% of hybrid cells and 69% of other cells are generalists (rule 1; rule 2: 86% vs 83%), so hybrids are slightly more often generalists.

Corners and their tasks

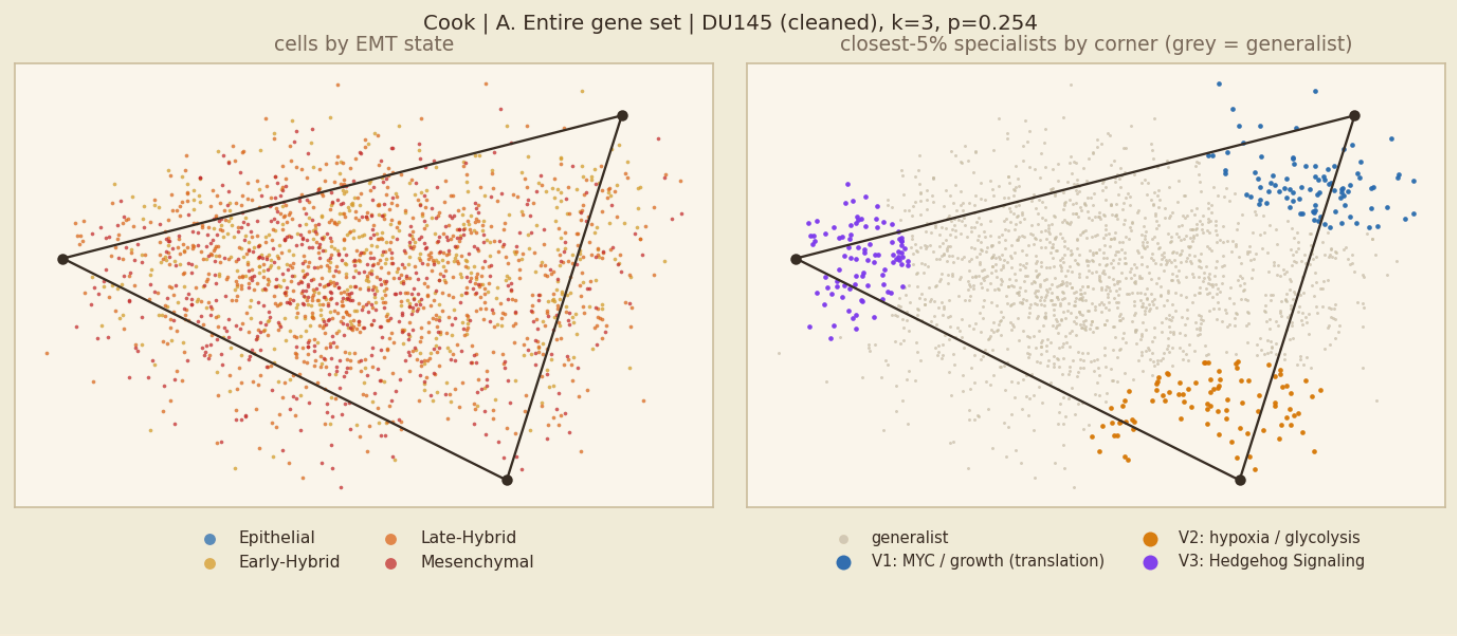
V1 EMT / matrix — Epithelial Mesenchymal Transition +1.49 SD; 70% S/G2M; 77% Mesenchymal; 100% A549; SERPINE2 IGFBP7 CCDC80

V2 mTORC1 Signaling — mTORC1 Signaling +1.82 SD; 92% S/G2M; 55% Early-Hybrid; 100% A549; AKR1B10 HSP90AA1 CCNB1

V3 luminal / differentiation — Estrogen Response Early +0.71 SD; 12% S/G2M; 66% Early-Hybrid; 100% A549; GPX2 NEAT1 SAT1

Cook: A. Entire gene set, unit DU145

Fit shown: cleaned · k = 3 · p = 0.254 · 1,839 cells · 200 shuffles | comparison, light QC: k=3, p=0.0995



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	69%	71%
2 closest 5% of cells to each corner	85%	85%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 69% of all cells here; 75% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside DU145: the fit (k=3, p=0.254) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 4 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 69% of hybrid cells and 71% of other cells are generalists (rule 1; rule 2: 85% vs 85%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

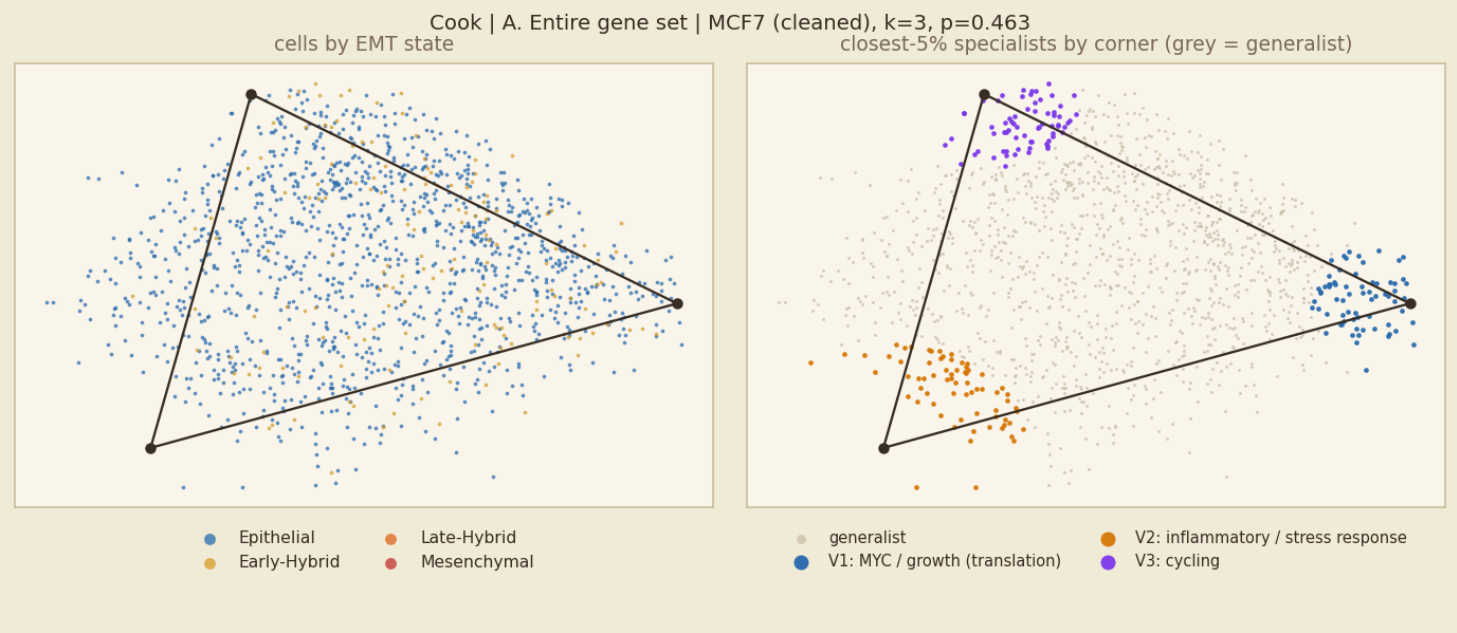
V1 MYC / growth (translation) — Myc Targets V1 +1.72 SD; 90% S/G2M; 42% Early-Hybrid; 100% DU145; CCNB1 CDC20 KPNA2

V2 hypoxia / glycolysis — Hypoxia +1.19 SD; 2% S/G2M; 43% Late-Hybrid; 100% DU145; ANGPTL4 TGFBI KRT19

V3 Hedgehog Signaling — Hedgehog Signaling +0.9 SD; 21% S/G2M; 39% Mesenchymal; 100% DU145; HIST1H4C ATP5E MT2A

Cook: A. Entire gene set, unit MCF7

Fit shown: cleaned · k = 3 · p = 0.463 · 1,378 cells · 200 shuffles | comparison, light QC: k=5, p=0.0896



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	54%	59%
2 closest 5% of cells to each corner	82%	85%
3 weight > 0.95 on one corner	99%	99%

Hybrids are 11% of all cells here; 66% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside MCF7: the fit (k=3, p=0.463) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 4 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 54% of hybrid cells and 59% of other cells are generalists (rule 1; rule 2: 82% vs 85%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

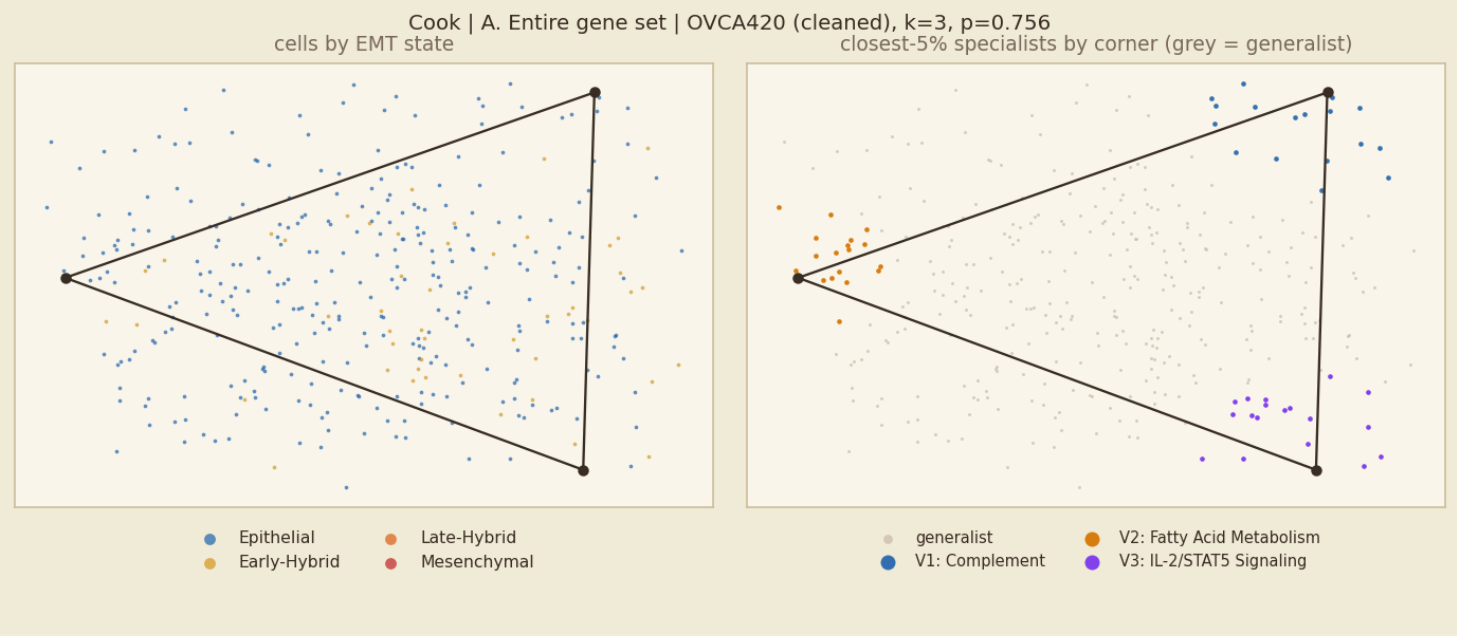
V1 MYC / growth (translation) — Myc Targets V1 +2.09 SD; 99% S/G2M; 86% Epithelial; 100% MCF7; TUBB4B TUBA1B HMGB2

V2 inflammatory / stress response — TNF-alpha Signaling via NF-kB +1.52 SD; 0% S/G2M; 93% Epithelial; 100% MCF7; NEAT1 BTG1 ANXA2

V3 cycling — Mitotic Spindle +0.61 SD; 57% S/G2M; 83% Epithelial; 100% MCF7; HIST1H4C KIAA0101 DSCAM-AS1

Cook: A. Entire gene set, unit OVCA420

Fit shown: cleaned · k = 3 · p = 0.756 · 355 cells · 200 shuffles | comparison, light QC: k=5, p=0.453



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	71%	63%
2 closest 5% of cells to each corner	84%	85%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 14% of all cells here; 62% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside OVCA420: the fit (k=3, p=0.756) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 4 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 71% of hybrid cells and 63% of other cells are generalists (rule 1; rule 2: 84% vs 85%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

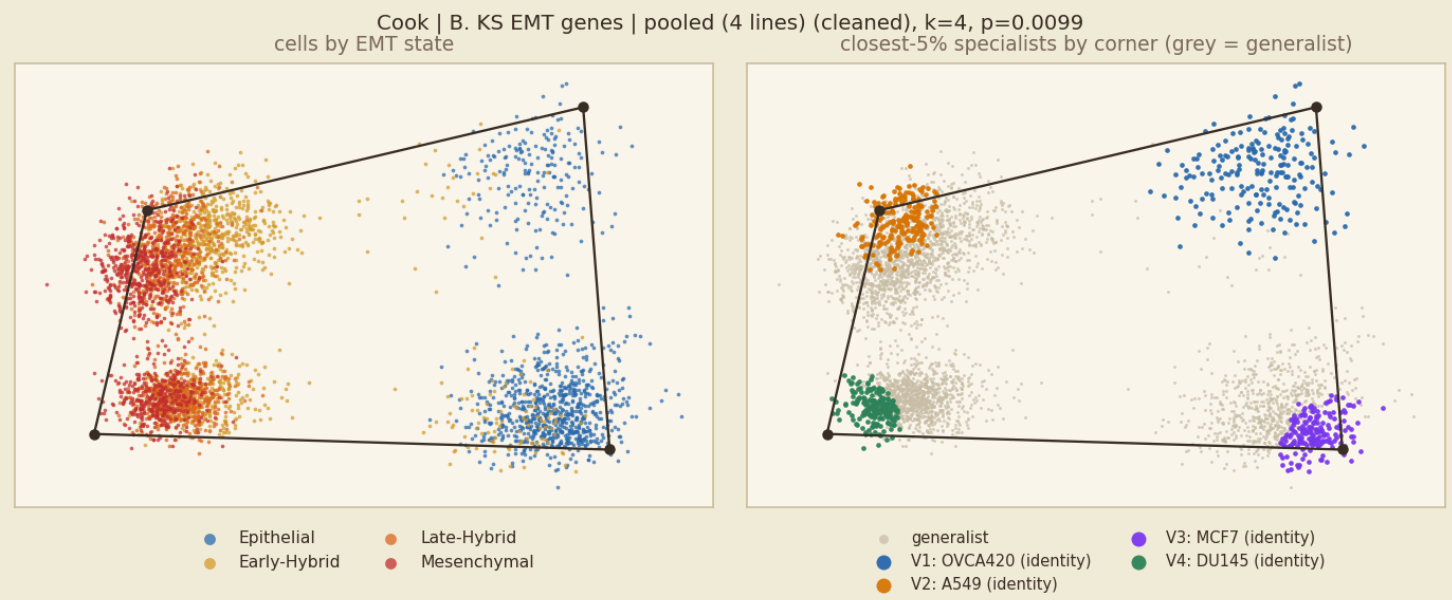
V1 Complement — Complement +1.57 SD; 6% S/G2M; 89% Epithelial; 100% OVCA420; C15orf48 KRT17 SERPINE2

V2 Fatty Acid Metabolism — Fatty Acid Metabolism +1.51 SD; 89% S/G2M; 89% Epithelial; 100% OVCA420; CBR1 AKR1B10 GSTA1

V3 IL-2/STAT5 Signaling — IL-2/STAT5 Signaling +1.45 SD; 11% S/G2M; 78% Epithelial; 100% OVCA420; MT2A LAMC2 TGM2

Cook: B. KS EMT genes, pooled fit

Fit shown: cleaned · k = 4 · p = 0.0099 · 6,218 cells · 100 shuffles l comparison, light QC: k=3, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	7%	13%
2 closest 5% of cells to each corner	88%	71%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 54% of all cells here; 47% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=4, p=0.0099), but 4 of 4 corners are almost entirely one sample (OVCA420, A549, MCF7, DU145). It separates samples; it is not a shared EMT trade-off.

Each corner is essentially one cell line. The programme names (Hallmark) only describe what distinguishes that line, and the EMT-state colours follow the lines (MCF7 scores epithelial, A549 mesenchymal). These corners are not tasks.

Hybrid vs other cells: 7% of hybrid cells and 13% of other cells are generalists (rule 1; rule 2: 88% vs 71%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 identity corner: OVCA420 (programme: p53 / stress) — p53 Pathway +2.11 SD; 51% S/G2M; 90% Epithelial; 99% OVCA420; S100A14 KRT19 FXYD3

V2 identity corner: A549 (programme: angiogenesis / matrix) — Angiogenesis +1.19 SD; 49% S/G2M; 48% Late-Hybrid; 100% A549; TPM2 KRT7 VIM

V3 identity corner: MCF7 (programme: luminal / differentiation) — Estrogen Response Late +2.08 SD; 53% S/G2M; 97% Epithelial; 100% MCF7; KRT19 XBP1 FXYD3

V4 identity corner: DU145 (programme: Allograft Rejection) — Allograft Rejection +1.19 SD; 50% S/G2M; 64% Mesenchymal; 100% DU145; MYL9 AXL LGALS1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Cook: B. KS EMT genes, all units at a glance

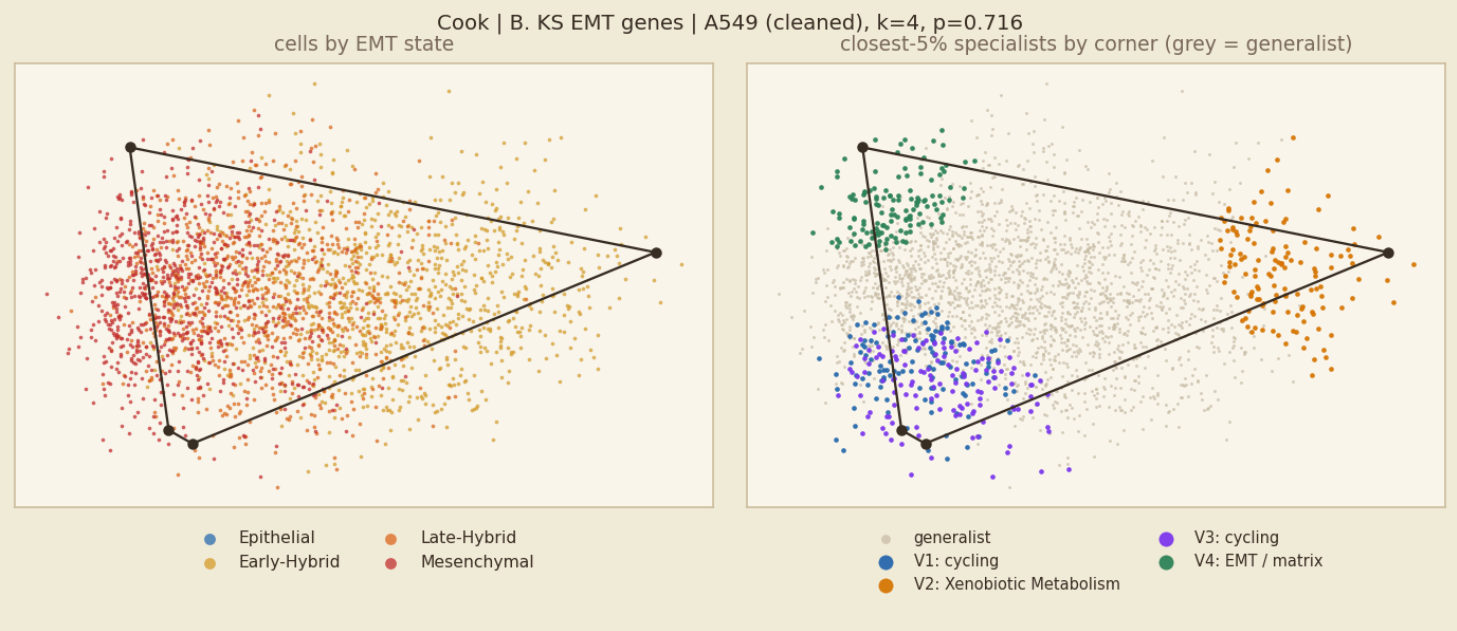
Main fit of each unit: 1 of 4 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
A549	cleaned	4	0.716	0.169	71%	83 / 84 / 100%	72 / 74 / 71%
DU145	cleaned	3	0.95	0.9	69%	78 / 84 / 100%	70 / 67 / 69%
MCF7	cleaned	3	0.0299 *	0.0149	11%	59 / 81 / 99%	10 / 10 / 11%
OVCA420	cleaned	4	0.448	0.418	14%	49 / 65 / 98%	11 / 11 / 14%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

Cook: B. KS EMT genes, unit A549

Fit shown: cleaned · k = 4 · p = 0.716 · 2,646 cells · 200 shuffles | comparison, light QC: k=5, p=0.169



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	83%	77%
2 closest 5% of cells to each corner	84%	71%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 71% of all cells here; 42% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside A549: the fit (k=4, p=0.716) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 4 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 83% of hybrid cells and 77% of other cells are generalists (rule 1; rule 2: 84% vs 71%), so hybrids are slightly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 cycling — no Hallmark set above 0.5 SD; 64% S/G2M; 47% Late-Hybrid; 100% A549; AKAP12 FHL1 SDC1

V2 Xenobiotic Metabolism — Xenobiotic Metabolism +0.78 SD; 38% S/G2M; 99% Early-Hybrid; 100% A549; GPX2 AGR2 ELF3

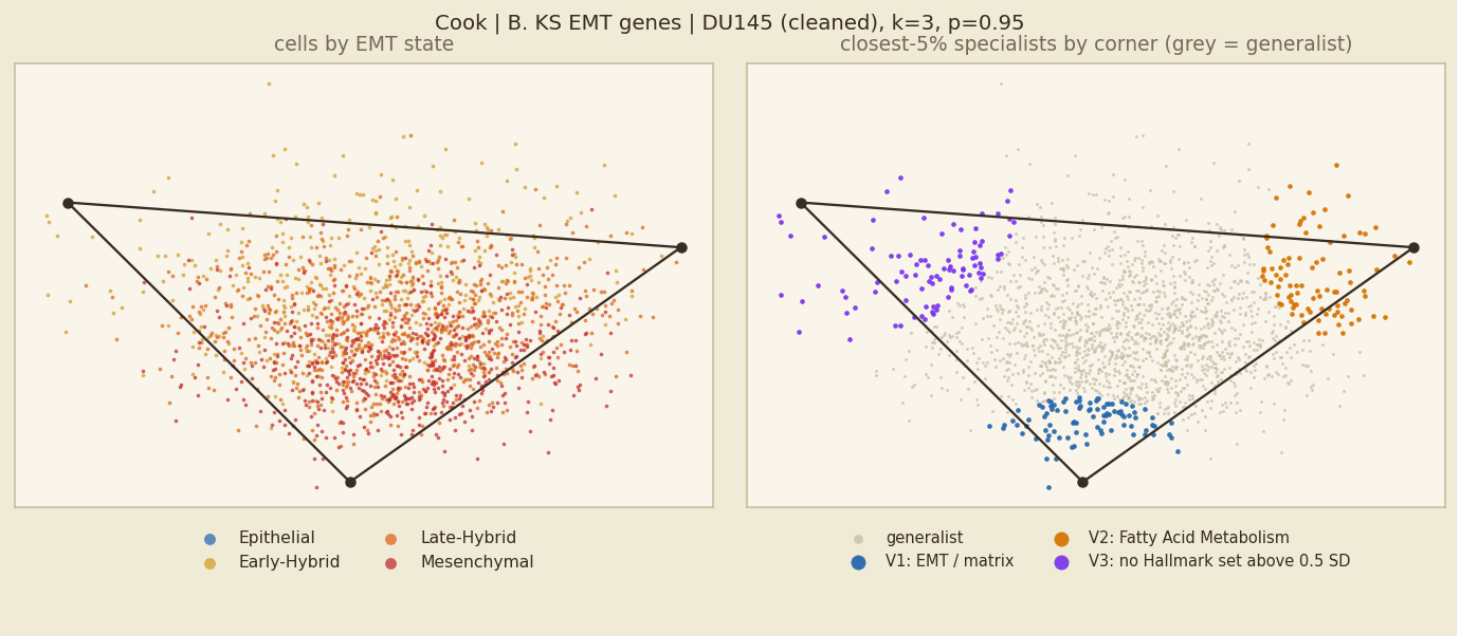
V3 cycling — Mitotic Spindle +0.6 SD; 55% S/G2M; 59% Mesenchymal; 100% A549; AXL MAP1B PTRF

V4 EMT / matrix — Epithelial Mesenchymal Transition +1.27 SD; 60% S/G2M; 68% Mesenchymal; 100% A549; KRT7 SERPINE1 TPM2

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Cook: B. KS EMT genes, unit DU145

Fit shown: cleaned · k = 3 · p = 0.95 · 1,839 cells · 200 shuffles | comparison, light QC: k=4, p=0.9



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	78%	72%
2 closest 5% of cells to each corner	84%	88%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 69% of all cells here; 85% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside DU145: the fit (k=3, p=0.95) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 4 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 78% of hybrid cells and 72% of other cells are generalists (rule 1; rule 2: 84% vs 88%), so hybrids are slightly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 EMT / matrix — Epithelial Mesenchymal Transition +0.69 SD; 32% S/G2M; 58% Mesenchymal; 100% DU145; MYL9 MSN LGALS1

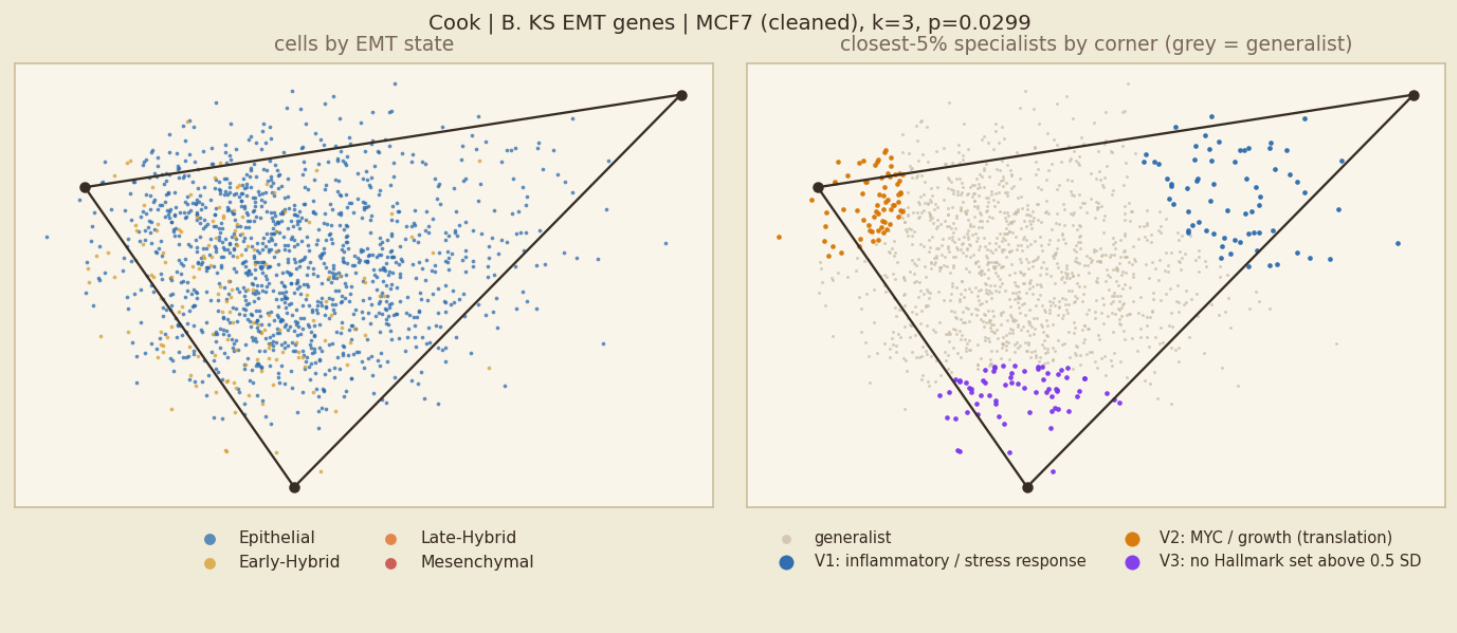
V2 Fatty Acid Metabolism — Fatty Acid Metabolism +1.26 SD; 43% S/G2M; 60% Late-Hybrid; 100% DU145; KRT19 KRT8 KRT18

V3 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 39% S/G2M; 57% Early-Hybrid; 100% DU145; AREG TOB1 ELF3

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Cook: B. KS EMT genes, unit MCF7

Fit shown: cleaned · k = 3 · p = 0.0299 · 1,378 cells · 200 shuffles l comparison, light QC: k=3, p=0.0149



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	59%	68%
2 closest 5% of cells to each corner	81%	85%
3 weight > 0.95 on one corner	99%	100%

Hybrids are 11% of all cells here; 86% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MCF7 alone, cells fill a 3-corner simplex (p=0.0299). 1 of 4 units of this dataset have a significant simplex for this gene set.

Corners are programmes: inflammatory / stress response; MYC / growth (translation); no Hallmark set above 0.5 SD.

Hybrid vs other cells: 59% of hybrid cells and 68% of other cells are generalists (rule 1; rule 2: 81% vs 85%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 inflammatory / stress response — TNF-alpha Signaling via NF-kB +1.8 SD; 6% S/G2M; 99% Epithelial; 100% MCF7; AQP3 S100A14 AREG

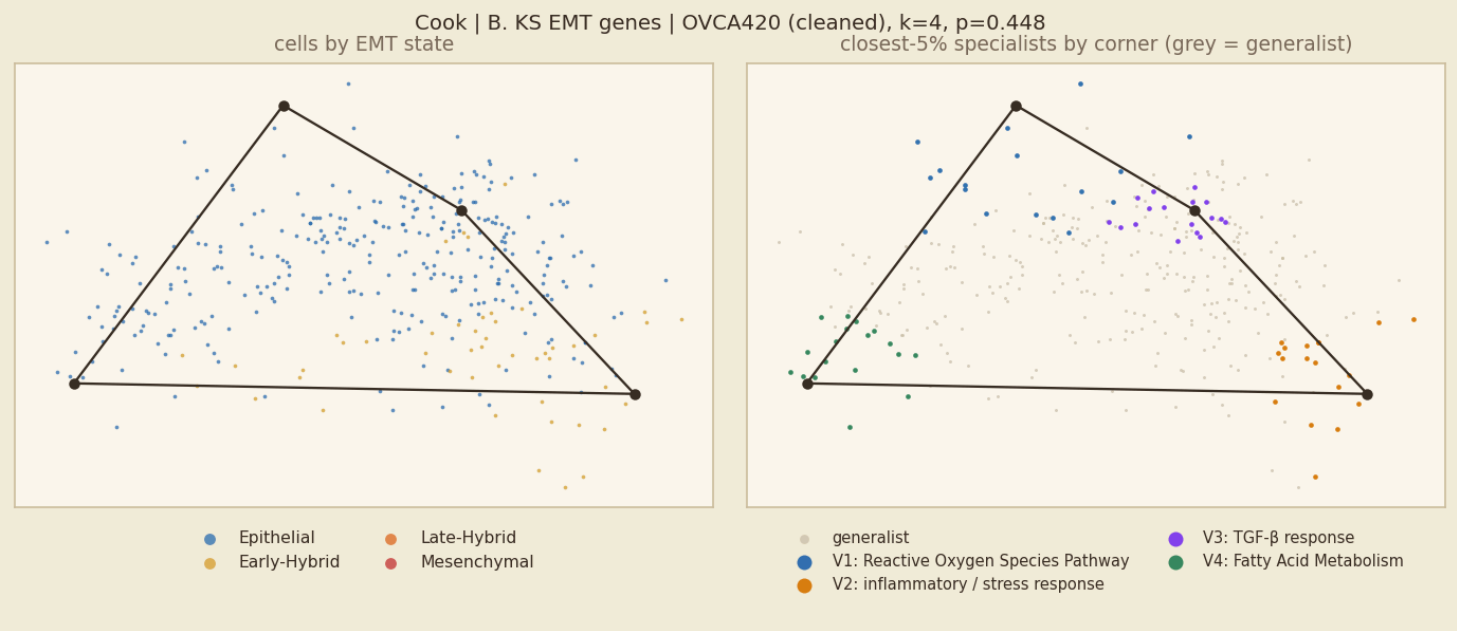
V2 MYC / growth (translation) — Myc Targets V1 +1.22 SD; 74% S/G2M; 86% Epithelial; 100% MCF7; AGR2 KRT18 KRT8

V3 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 58% S/G2M; 74% Epithelial; 100% MCF7; LGALS1 CYB561 SLC9A3R1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Cook: B. KS EMT genes, unit OVCA420

Fit shown: cleaned · k = 4 · p = 0.448 · 355 cells · 200 shuffles | comparison, light QC: k=3, p=0.418



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	49%	64%
2 closest 5% of cells to each corner	65%	82%
3 weight > 0.95 on one corner	98%	100%

Hybrids are 14% of all cells here; 34% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside OVCA420: the fit (k=4, p=0.448) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 4 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 49% of hybrid cells and 64% of other cells are generalists (rule 1; rule 2: 65% vs 82%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 Reactive Oxygen Species Pathway — Reactive Oxygen Species Pathway +0.86 SD; 28% S/G2M; 100% Epithelial; 100% OVCA420; AGR2 KRT19 FXYD3

V2 inflammatory / stress response — Inflammatory Response +1.86 SD; 28% S/G2M; 67% Early-Hybrid; 100% OVCA420; TACSTD2 PTRF SERPINE1

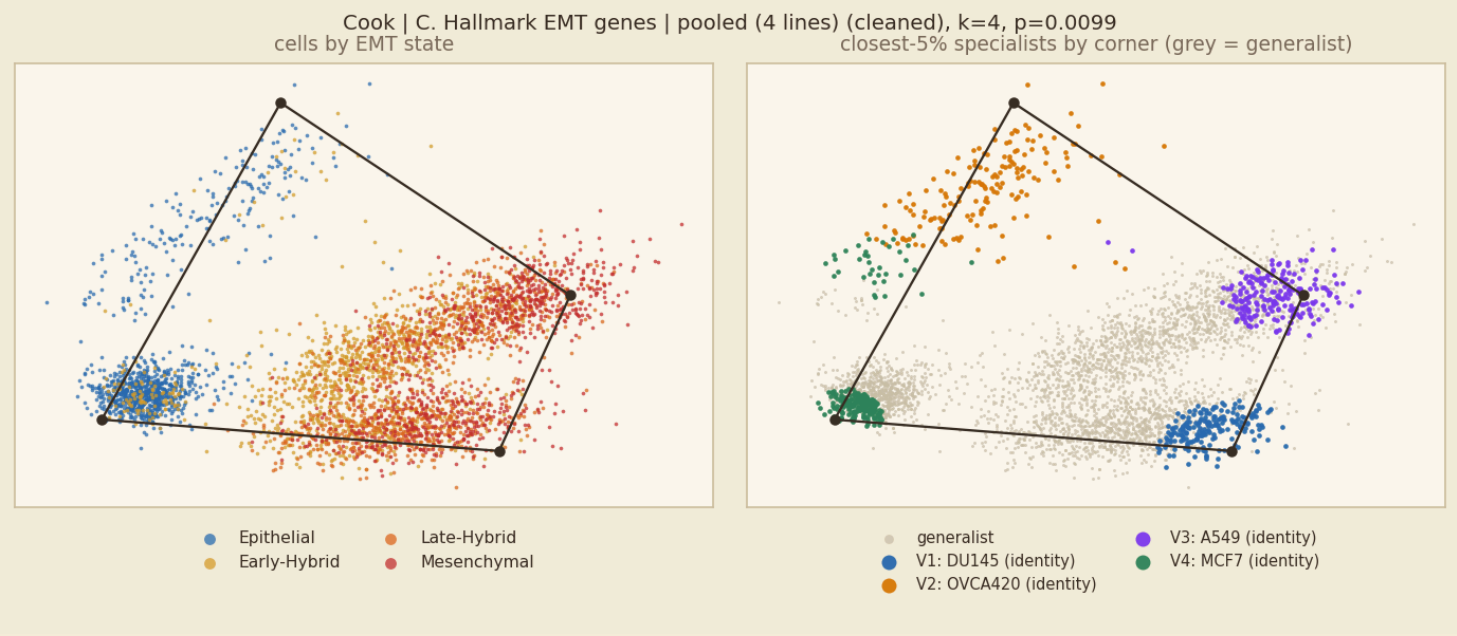
V3 TGF-β response — TGF-beta Signaling +0.54 SD; 50% S/G2M; 78% Epithelial; 100% OVCA420; SERPINE1 CTSH ITGB4

V4 Fatty Acid Metabolism — Fatty Acid Metabolism +0.82 SD; 72% S/G2M; 94% Epithelial; 100% OVCA420; ELF3 ANXA4 ABCC3

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Cook: C. Hallmark EMT genes, pooled fit

Fit shown: cleaned · k = 4 · p = 0.0099 · 6,218 cells · 100 shuffles l comparison, light QC: k=4, p=0.0099



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	53%	18%
2 closest 5% of cells to each corner	89%	70%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 54% of all cells here; 33% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=4, p=0.0099), but 4 of 4 corners are almost entirely one sample (DU145, OVCA420, A549, MCF7). It separates samples; it is not a shared EMT trade-off.

Each corner is essentially one cell line. The programme names (Hallmark) only describe what distinguishes that line, and the EMT-state colours follow the lines (MCF7 scores epithelial, A549 mesenchymal). These corners are not tasks.

Hybrid vs other cells: 53% of hybrid cells and 18% of other cells are generalists (rule 1; rule 2: 89% vs 70%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 identity corner: DU145 (programme: epithelial junction) — Apical Junction +1.41 SD; 46% S/G2M; 48% Mesenchymal; 100% DU145; TAGLN MYL9 LGALS1

V2 identity corner: OVCA420 (programme: p53 / stress) — p53 Pathway +2.15 SD; 50% S/G2M; 86% Epithelial; 100% OVCA420; SPP1 SERPINE1 SERPINE2

V3 identity corner: A549 (programme: angiogenesis / matrix) — Angiogenesis +1.24 SD; 56% S/G2M; 56% Mesenchymal; 100% A549; TGFBI TPM2 SERPINE2

V4 identity corner: MCF7 (programme: luminal / differentiation) — Estrogen Response Late +1.78 SD; 64% S/G2M; 89% Epithelial; 100% MCF7; MGP ID2 AREG

Hallmark EMT set left out when naming (circular: fitted on those genes).

Cook: C. Hallmark EMT genes, all units at a glance

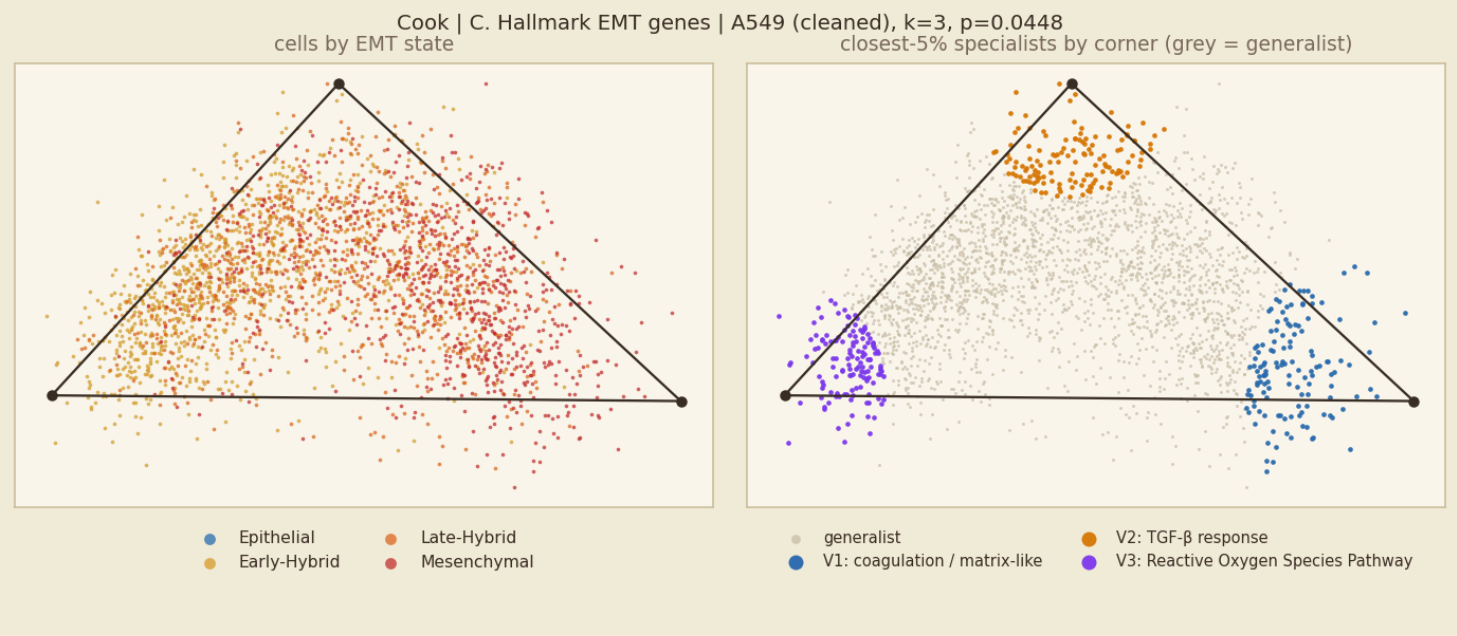
Main fit of each unit: 3 of 4 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
A549	cleaned	3	0.0448 *	0.259	71%	65 / 86 / 100%	71 / 72 / 71%
DU145	cleaned	3	0.0249 *	0.0199	69%	77 / 86 / 100%	67 / 69 / 69%
MCF7	cleaned	4	0.368	0.891	11%	68 / 83 / 100%	10 / 11 / 11%
OVCA420	cleaned	3	0.00995 *	0.00498	14%	47 / 80 / 100%	14 / 13 / 14%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

Cook: C. Hallmark EMT genes, unit A549

Fit shown: cleaned · k = 3 · p = 0.0448 · 2,646 cells · 200 shuffles l comparison, light QC: k=3, p=0.259



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	65%	65%
2 closest 5% of cells to each corner	86%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 71% of all cells here; 85% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside A549 alone, cells fill a 3-corner simplex (p=0.0448). 3 of 4 units of this dataset have a significant simplex for this gene set.

Corners are programmes: coagulation / matrix-like; TGF-β response; Reactive Oxygen Species Pathway.

Hybrid vs other cells: 65% of hybrid cells and 65% of other cells are generalists (rule 1; rule 2: 86% vs 83%), so no real difference.

Corners and their tasks

V1 coagulation / matrix-like — Coagulation +1.37 SD; 62% S/G2M; 81% Mesenchymal; 100% A549; TAGLN SERPINE2 TGFBI

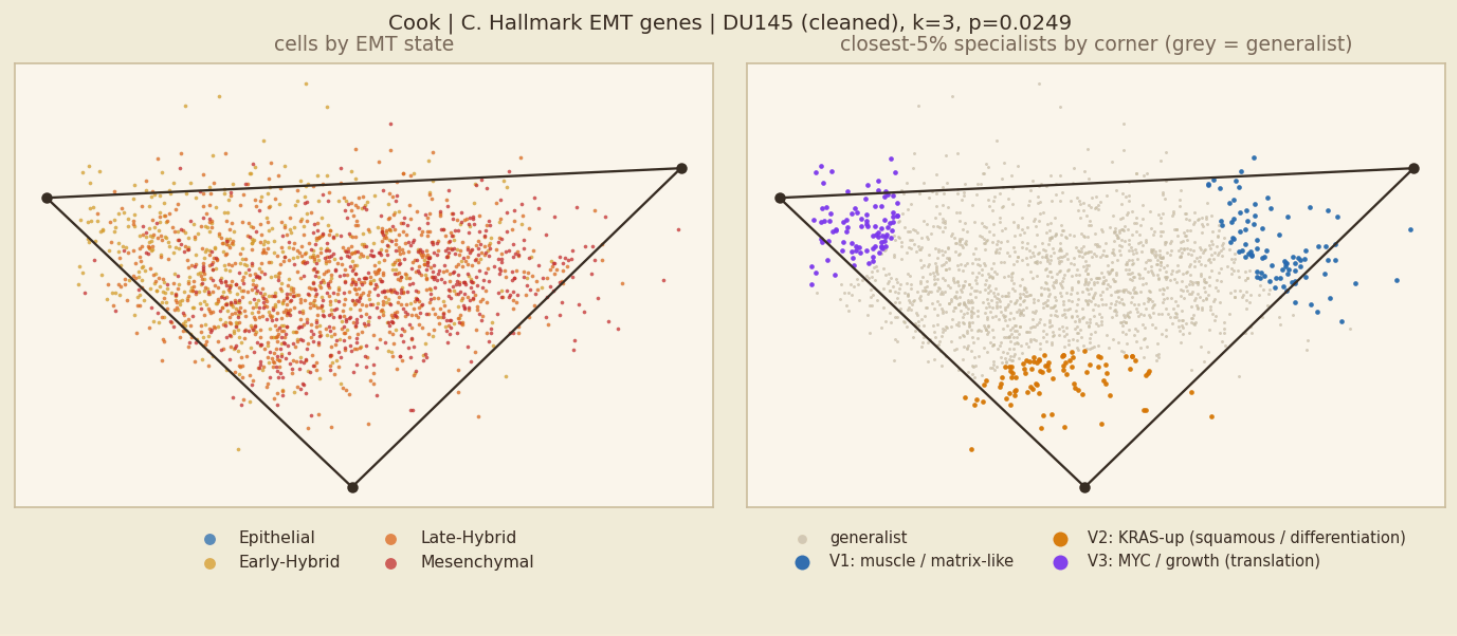
V2 TGF-β response — TGF-beta Signaling +0.85 SD; 44% S/G2M; 43% Late-Hybrid; 100% A549; IGFBP4 SERPINE1 PMEPA1

V3 Reactive Oxygen Species Pathway — Reactive Oxygen Species Pathway +0.71 SD; 51% S/G2M; 89% Early-Hybrid; 100% A549; AREG BASP1 DKK1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Cook: C. Hallmark EMT genes, unit DU145

Fit shown: cleaned · k = 3 · p = 0.0249 · 1,839 cells · 200 shuffles l comparison, light QC: k=3, p=0.0199



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	77%	83%
2 closest 5% of cells to each corner	86%	82%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 69% of all cells here; 92% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside DU145 alone, cells fill a 3-corner simplex (p=0.0249). 3 of 4 units of this dataset have a significant simplex for this gene set.

Corners are programmes: muscle / matrix-like; KRAS-up (squamous / differentiation); MYC / growth (translation).

Hybrid vs other cells: 77% of hybrid cells and 83% of other cells are generalists (rule 1; rule 2: 86% vs 82%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 muscle / matrix-like — Myogenesis +1.37 SD; 37% S/G2M; 60% Mesenchymal; 100% DU145; TAGLN MYL9 TPM1

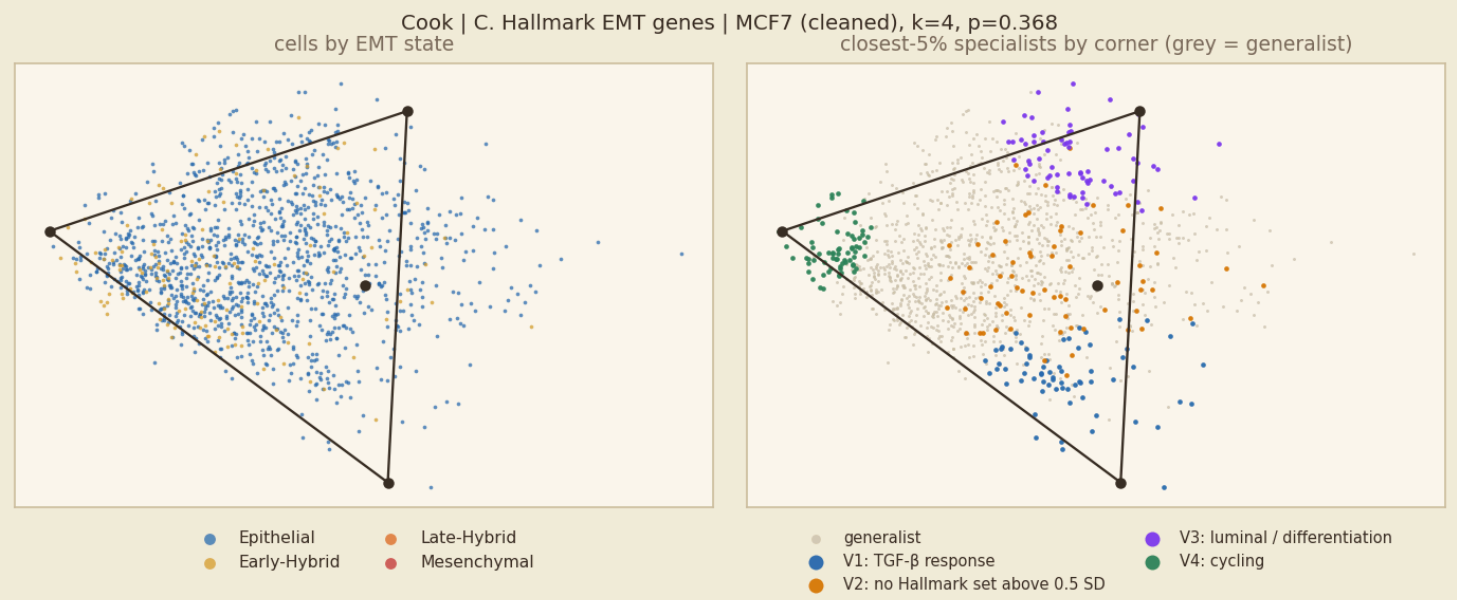
V2 KRAS-up (squamous / differentiation) — KRAS Signaling Up +1.07 SD; 30% S/G2M; 45% Mesenchymal; 100% DU145; TIMP1 TGFBI VIM

V3 MYC / growth (translation) — Myc Targets V2 +0.55 SD; 39% S/G2M; 63% Early-Hybrid; 100% DU145; AREG SAT1 GPC1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Cook: C. Hallmark EMT genes, unit MCF7

Fit shown: cleaned · k = 4 · p = 0.368 · 1,378 cells · 200 shuffles | comparison, light QC: k=4, p=0.891



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	68%	74%
2 closest 5% of cells to each corner	83%	80%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 11% of all cells here; 45% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside MCF7: the fit (k=4, p=0.368) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 4 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 68% of hybrid cells and 74% of other cells are generalists (rule 1; rule 2: 83% vs 80%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 TGF-β response — TGF-beta Signaling +1.31 SD; 13% S/G2M; 94% Epithelial; 100% MCF7; PMEPA1 CD59 LGALS1

V2 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 45% S/G2M; 93% Epithelial; 100% MCF7; MGP PMEPA1 DKK1

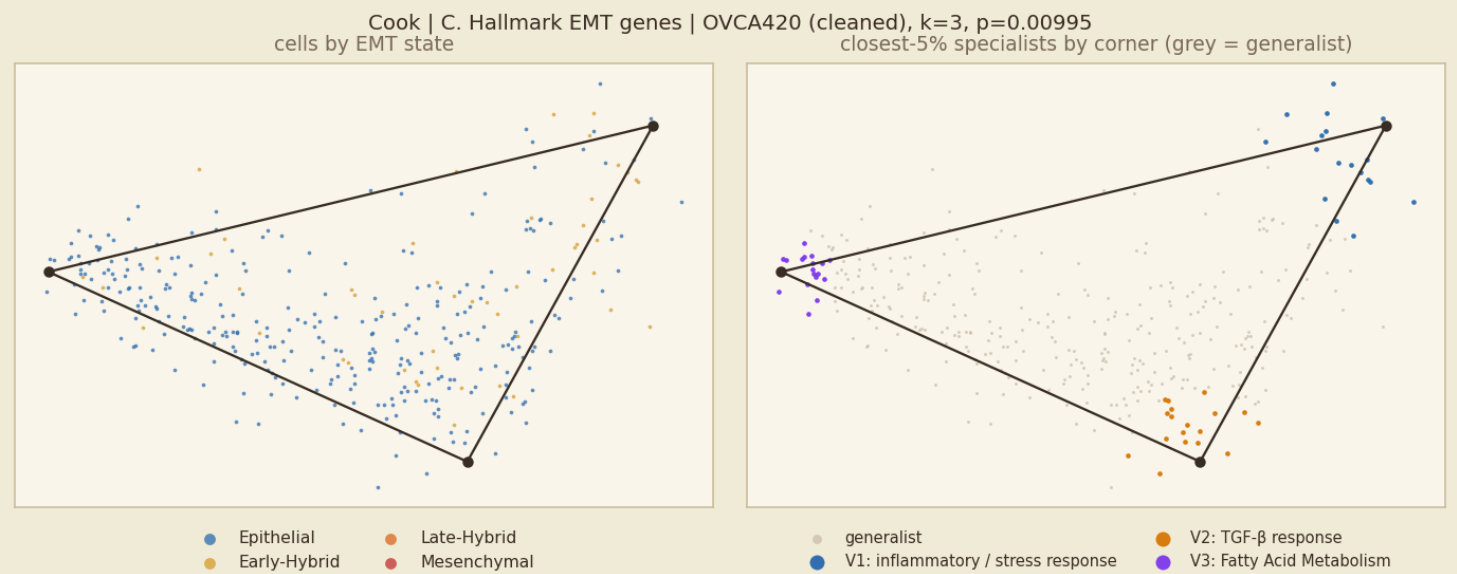
V3 luminal / differentiation — Estrogen Response Late +0.61 SD; 52% S/G2M; 96% Epithelial; 100% MCF7; AREG BASP1 DKK1

V4 cycling — no Hallmark set above 0.5 SD; 71% S/G2M; 81% Epithelial; 100% MCF7; WIPF1 PMP22 LGALS1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Cook: C. Hallmark EMT genes, unit OVCA420

Fit shown: cleaned · k = 3 · p = 0.00995 · 355 cells · 200 shuffles | comparison, light QC: k=3, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	47%	45%
2 closest 5% of cells to each corner	80%	86%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 14% of all cells here; 74% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside OVCA420 alone, cells fill a 3-corner simplex (p=0.00995). 3 of 4 units of this dataset have a significant simplex for this gene set.

Corners are programmes: inflammatory / stress response; TGF-β response; Fatty Acid Metabolism.

Hybrid vs other cells: 47% of hybrid cells and 45% of other cells are generalists (rule 1; rule 2: 80% vs 86%), so no real difference.

Corners and their tasks

V1 inflammatory / stress response — Inflammatory Response +2.2 SD; 0% S/G2M; 56% Epithelial; 100% OVCA420; TFPI2 LAMC2 IL32

V2 TGF-β response — TGF-beta Signaling +1.07 SD; 44% S/G2M; 94% Epithelial; 100% OVCA420; SERPINE1 TGM2 SERPINE2

V3 Fatty Acid Metabolism — Fatty Acid Metabolism +1.39 SD; 78% S/G2M; 94% Epithelial; 100% OVCA420; SPP1 ID2 LRP1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Question 2 · Cook: are hybrid cells more often generalists than other cells?

Orange = hybrid cells, grey = all other cells; one pair of bars per unit; the number = orange minus grey (percentage points)

Reading

- A. Entire gene set (2 significant): hybrids 38% vs others 36% generalist under rule 1 (rule 2: 86% vs 83%) → no real difference.
- B. KS EMT genes (2 significant): hybrids 33% vs others 41% generalist under rule 1 (rule 2: 84% vs 78%) → hybrids are less often generalists (they sit at corners).
- C. Hallmark EMT genes (4 significant): hybrids 59% vs others 55% generalist under rule 1 (rule 2: 86% vs 83%) → no real difference.

Rule 3 (weight > 0.95) is not drawn: it calls about 100% of every group generalist, so it cannot show a difference.

Mouse PDAC

What the dataset is, and what to keep in mind when reading its slides

What the dataset means

Mouse pancreatic tumour cells from two mice at several body sites (primary, liver, metastasis). EMT states are ours (KS-score bins), so partly circular.

Units in the main analysis

M1, M1-PT, M1-SS, M1-Liver, M1-Met, M2, M2-PT, M2-Liver, M2-Met

Gene sets

A. all genes (11,000 to 14,500); B. KS EMT genes (Tan 2014; about 160 to 190 present); C. Hallmark Epithelial-Mesenchymal-Transition set (200 genes, those present in the data).

Keep in mind

- EMT states are ours (KS-score bins), so they are partly circular with the KS-gene fits.
- Units: seven libraries, mouse 1, mouse 2. 'Cleaned' keeps cells with at least 10,000 UMI and drops liver ambient-RNA genes.
- Pooled fit = in vivo tumour cells; mouse and site are confounded.

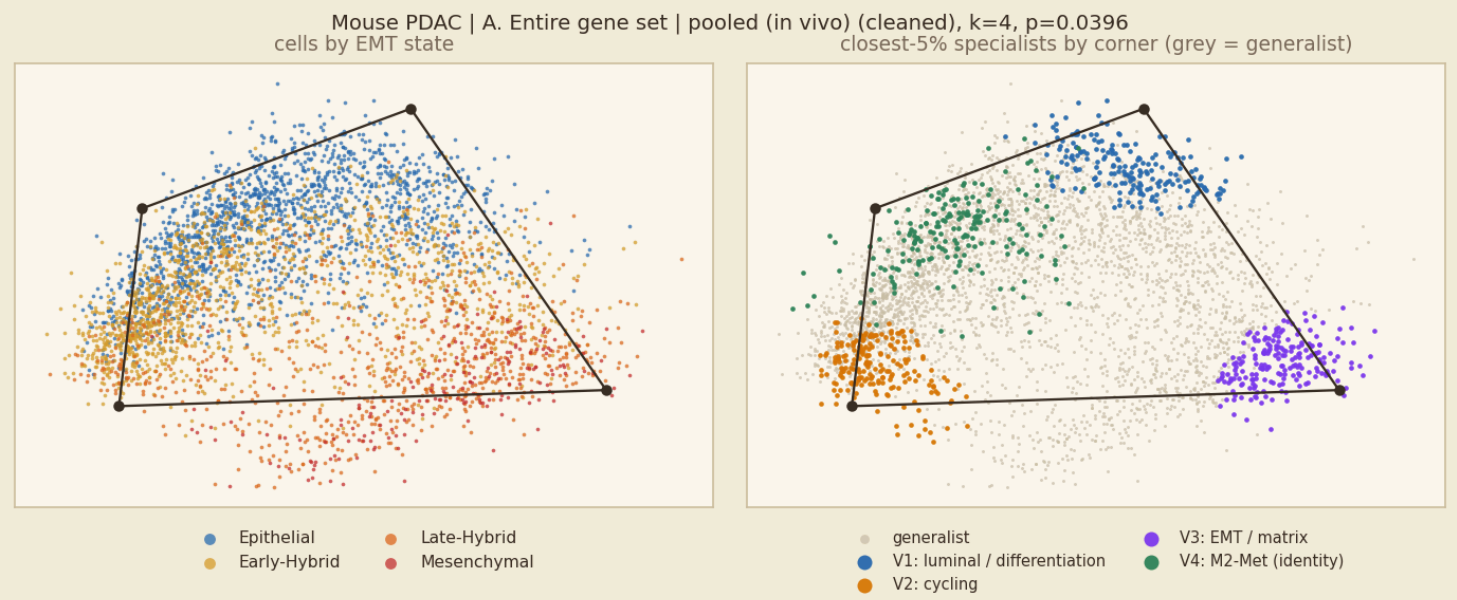
This dataset in numbers (main fits)

- Entire gene set: 10 of 10 fits significant; hybrids 61% / 84% / 100% generalist (median, rules 1/2/3).
- KS EMT genes: 3 of 10 fits significant; hybrids 75% / 80% / 100% generalist (median, rules 1/2/3).
- Hallmark EMT genes: 4 of 10 fits significant; hybrids 78% / 80% / 100% generalist (median, rules 1/2/3).

Question 1 slides come first, then the Question 2 slides for this dataset.

Mouse PDAC: A. Entire gene set, pooled fit

Fit shown: cleaned · k = 4 · p = 0.0396 · 37,387 cells · 100 shuffles l comparison, light QC: k=5, p=0.0796



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	61%	69%
2 closest 5% of cells to each corner	80%	80%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 43% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=4, p=0.0396) whose corners mix samples (no corner is 70% or more one sample), so it is not a pure sample-identity simplex.

Corners are programmes: luminal / differentiation; cycling; EMT / matrix; no Hallmark set above 0.5 SD.

Hybrid vs other cells: 61% of hybrid cells and 69% of other cells are generalists (rule 1; rule 2: 80% vs 80%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 luminal / differentiation — Estrogen Response Early +1.69 SD; 0% S/G2M; 98% Epithelial; Ttr Atp1b1 Lgals4

V2 cycling — E2F Targets +1.91 SD; 100% S/G2M; 59% Early-Hybrid; Hmgb2 H2afz Birc5

V3 EMT / matrix — Epithelial Mesenchymal Transition +1.88 SD; 1% S/G2M; 60% Late-Hybrid; Trf Gm26870 Fn1

V4 identity corner: M2-Met (programme: no Hallmark set above 0.5 SD) — no Hallmark set above 0.5 SD; 12% S/G2M; 51% Early-Hybrid; 70% M2-Met; Sprr1a S100a4 Ly6a

Mouse PDAC: A. Entire gene set, all units at a glance

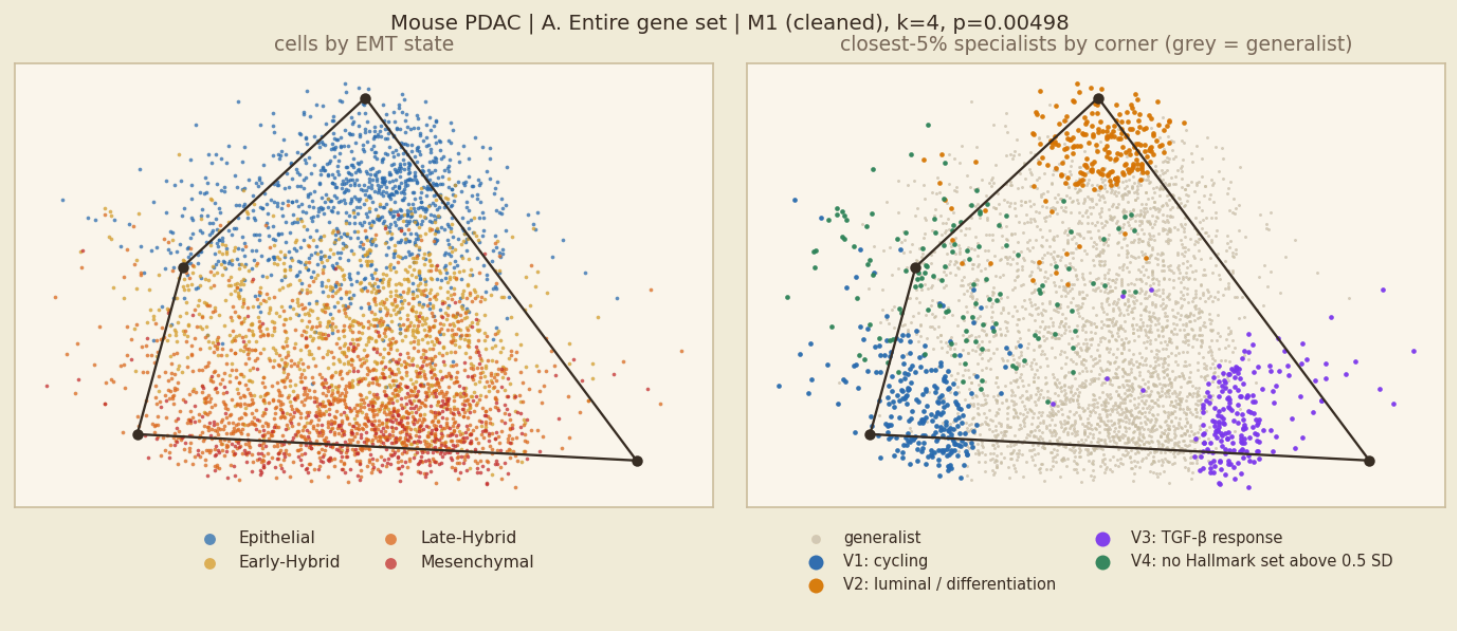
Main fit of each unit: 9 of 9 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
M1	cleaned	4	0.00498 *	0.00498	61%	70 / 81 / 100%	69 / 62 / 61%
M1-PT	cleaned	3	0.0199 *	0.00498	49%	61 / 86 / 100%	58 / 50 / 49%
M1-SS	cleaned	3	0.00498 *	0.00498	51%	66 / 86 / 100%	58 / 52 / 51%
M1-Liver	cleaned	4	0.00498 *	0.00498	67%	62 / 83 / 100%	67 / 69 / 67%
M1-Met	cleaned	4	0.00995 *	0.0249	66%	62 / 84 / 100%	73 / 69 / 66%
M2	cleaned	3	0.00498 *	0.0498	50%	49 / 84 / 99%	44 / 49 / 49%
M2-PT	cleaned	3	0.00498 *	0.00498	39%	68 / 82 / 100%	41 / 37 / 39%
M2-Liver	cleaned	3	0.00498 *	0.00498	53%	59 / 88 / 100%	56 / 55 / 53%
M2-Met	cleaned	3	0.00498 *	0.0647	56%	53 / 85 / 100%	49 / 56 / 56%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

Mouse PDAC: A. Entire gene set, unit M1

Fit shown: cleaned · k = 4 · p = 0.00498 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	70%	50%
2 closest 5% of cells to each corner	81%	78%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 61% of all cells here; 50% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M1 alone, cells fill a 4-corner simplex (p=0.00498). 9 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: cycling; luminal / differentiation; TGF-β response; no Hallmark set above 0.5 SD.

Hybrid vs other cells: 70% of hybrid cells and 50% of other cells are generalists (rule 1; rule 2: 81% vs 78%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 cycling — E2F Targets +2.39 SD; 94% S/G2M; 73% Late-Hybrid; Hmgb2 Birc5 H2afz

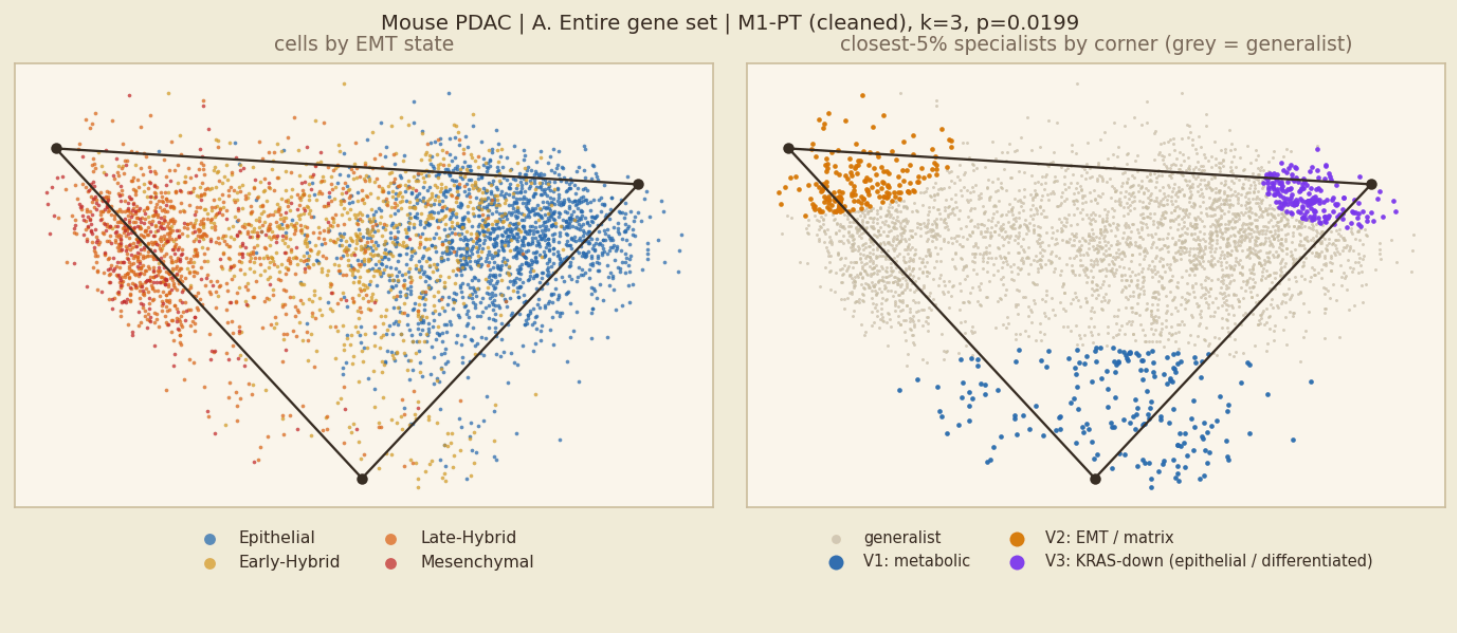
V2 luminal / differentiation — Estrogen Response Early +1.38 SD; 1% S/G2M; 100% Epithelial; Lgals4 Atp1b1 Dmbt1

V3 TGF-β response — TGF-beta Signaling +1.4 SD; 0% S/G2M; 54% Late-Hybrid; Gm26870 Neat1 Fn1

V4 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 30% S/G2M; 40% Late-Hybrid; Lgals1 Tmsb10 Tmsb4x

Mouse PDAC: A. Entire gene set, unit M1-PT

Fit shown: cleaned · k = 3 · p = 0.0199 · 3,770 cells · 200 shuffles l comparison, light QC: k=3, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	61%	43%
2 closest 5% of cells to each corner	86%	84%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 49% of all cells here; 73% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M1-PT alone, cells fill a 3-corner simplex (p=0.0199). 9 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: metabolic; EMT / matrix; KRAS-down (epithelial / differentiated).

Hybrid vs other cells: 61% of hybrid cells and 43% of other cells are generalists (rule 1; rule 2: 86% vs 84%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 metabolic — Oxidative Phosphorylation +1.02 SD; 49% S/G2M; 44% Early-Hybrid; 100% M1-PT; Lgals1 S100a6 Lgals2

V2 EMT / matrix — Epithelial Mesenchymal Transition +1.29 SD; 2% S/G2M; 59% Late-Hybrid; 100% M1-PT; Inhba Gm26870 Trf

V3 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.46 SD; 1% S/G2M; 97% Epithelial; 100% M1-PT; Ttr Atp1b1 Dmbt1

Mouse PDAC: A. Entire gene set, unit M1-SS

Fit shown: cleaned · k = 3 · p = 0.00498 · 2,397 cells · 200 shuffles | comparison, light QC: k=3, p=0.00498

Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	66%	50%
2 closest 5% of cells to each corner	86%	84%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 51% of all cells here; 70% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M1-SS alone, cells fill a 3-corner simplex (p=0.00498). 9 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: metabolic; TGF-β response.

Hybrid vs other cells: 66% of hybrid cells and 50% of other cells are generalists (rule 1; rule 2: 86% vs 84%), so hybrids are clearly more often generalists.

Corners and their tasks

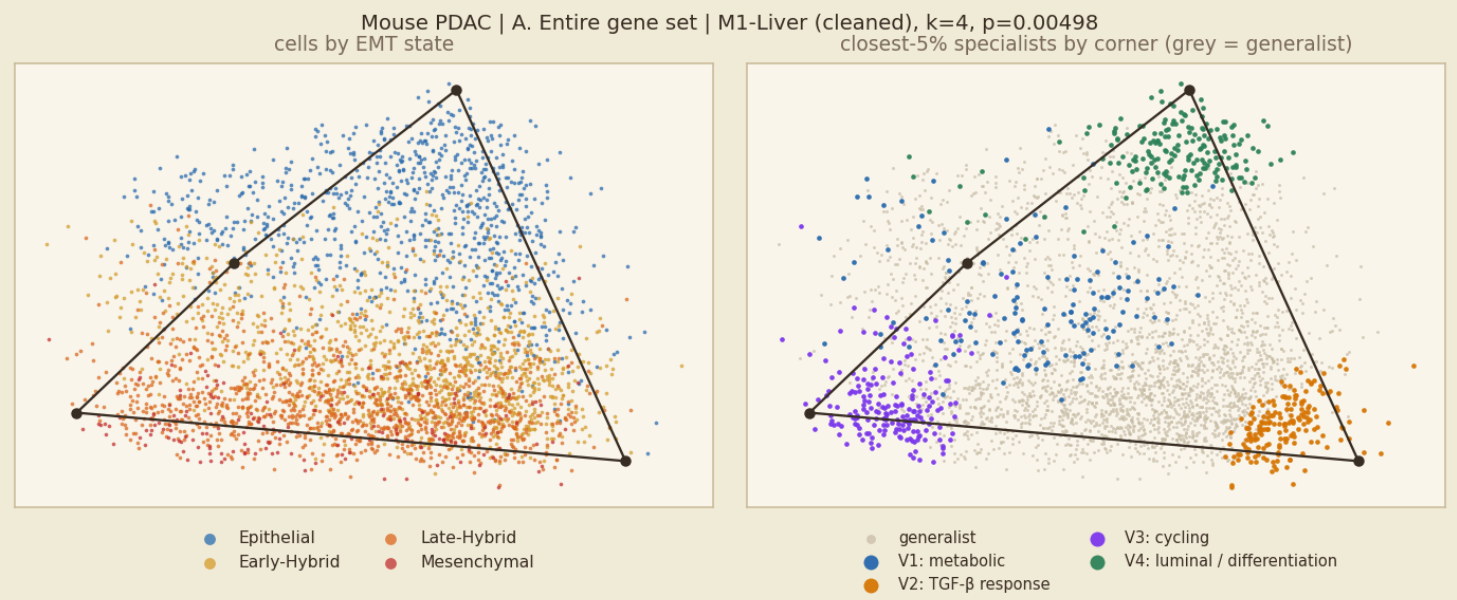
V1 metabolic — Oxidative Phosphorylation +0.62 SD; 39% S/G2M; 42% Early-Hybrid; 100% M1-SS; Lgals1 S100a6 Rplp1

V2 metabolic — Bile Acid Metabolism +1.57 SD; 0% S/G2M; 99% Epithelial; 100% M1-SS; Atp1b1 Ttr C3

V3 TGF-β response — TGF-beta Signaling +1.43 SD; 1% S/G2M; 52% Late-Hybrid; 100% M1-SS; Gm26870 Neat1 Id3

Mouse PDAC: A. Entire gene set, unit M1-Liver

Fit shown: cleaned · k = 4 · p = 0.00498 · 3,521 cells · 200 shuffles | comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	62%	62%
2 closest 5% of cells to each corner	83%	75%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 67% of all cells here; 43% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M1-Liver alone, cells fill a 4-corner simplex (p=0.00498). 9 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: metabolic; TGF-β response; cycling; luminal / differentiation.

Hybrid vs other cells: 62% of hybrid cells and 62% of other cells are generalists (rule 1; rule 2: 83% vs 75%), so no real difference.

Corners and their tasks

V1 metabolic — Oxidative Phosphorylation +0.56 SD; 19% S/G2M; 41% Early-Hybrid; 100% M1-Liver; Lgals2 Lgals1 AY036118

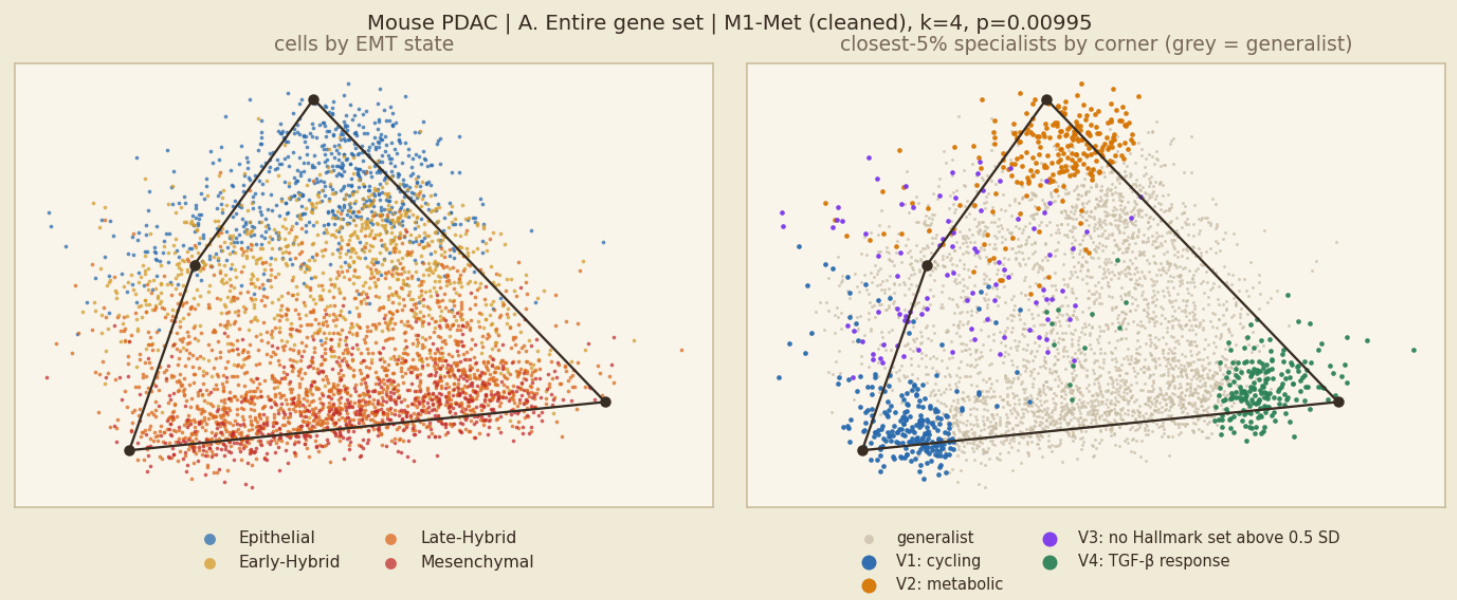
V2 TGF-β response — TGF-beta Signaling +1.43 SD; 0% S/G2M; 47% Early-Hybrid; 100% M1-Liver; Gm26870 Fn1 Neat1

V3 cycling — E2F Targets +2.33 SD; 98% S/G2M; 67% Late-Hybrid; 100% M1-Liver; Hmgb2 Birc5 Cenpa

V4 luminal / differentiation — Estrogen Response Early +1.5 SD; 0% S/G2M; 100% Epithelial; 100% M1-Liver; Dmbt1 Lgals4 Atp1b1

Mouse PDAC: A. Entire gene set, unit M1-Met

Fit shown: cleaned · k = 4 · p = 0.00995 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.0249



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	62%	45%
2 closest 5% of cells to each corner	84%	72%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 66% of all cells here; 42% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M1-Met alone, cells fill a 4-corner simplex (p=0.00995). 9 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: cycling; metabolic; no Hallmark set above 0.5 SD; TGF-β response.

Hybrid vs other cells: 62% of hybrid cells and 45% of other cells are generalists (rule 1; rule 2: 84% vs 72%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 cycling — E2F Targets +2.23 SD; 92% S/G2M; 69% Late-Hybrid; 100% M1-Met; Hmgb2 Birc5 Cenpa

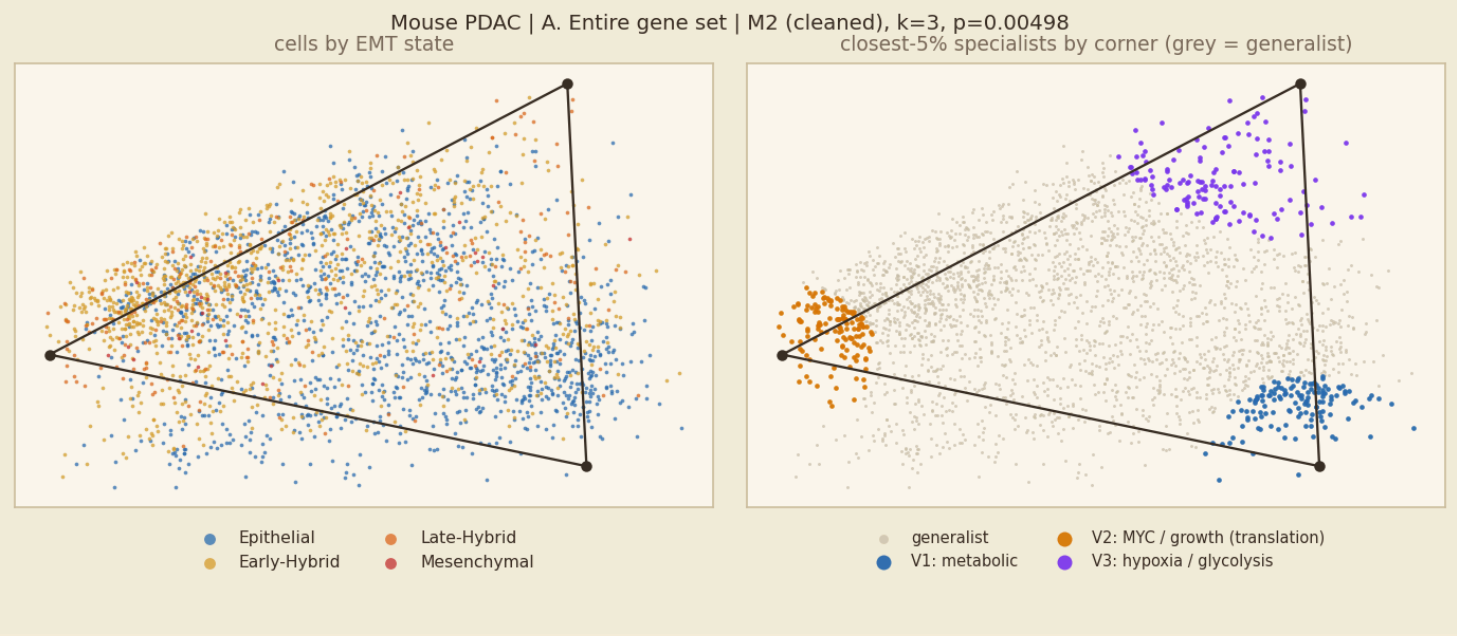
V2 metabolic — Bile Acid Metabolism +1.84 SD; 1% S/G2M; 95% Epithelial; 100% M1-Met; Lgals4 Atp1b1 Gsta4

V3 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 37% S/G2M; 44% Late-Hybrid; 100% M1-Met; Lgals1 S100a6 S100a11

V4 TGF-β response — TGF-beta Signaling +1.48 SD; 0% S/G2M; 55% Late-Hybrid; 100% M1-Met; Gm26870 Col6a1 Fn1

Mouse PDAC: A. Entire gene set, unit M2

Fit shown: cleaned · k = 3 · p = 0.00498 · 2,728 cells · 200 shuffles l comparison, light QC: k=4, p=0.0498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	49%	60%
2 closest 5% of cells to each corner	84%	86%
3 weight > 0.95 on one corner	99%	100%

Hybrids are 50% of all cells here; 70% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M2 alone, cells fill a 3-corner simplex (p=0.00498). 9 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: metabolic; MYC / growth (translation); hypoxia / glycolysis.

Hybrid vs other cells: 49% of hybrid cells and 60% of other cells are generalists (rule 1; rule 2: 84% vs 86%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

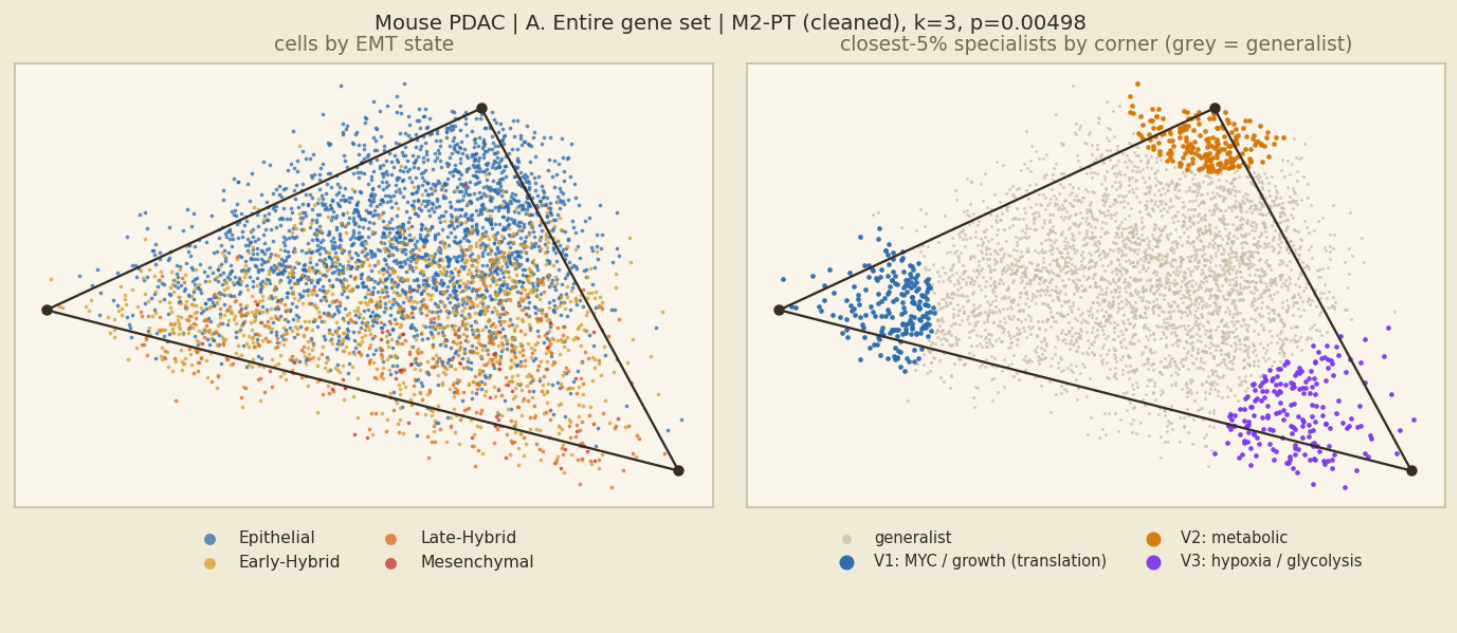
V1 metabolic — Bile Acid Metabolism +1.57 SD; 0% S/G2M; 91% Epithelial; 89% M2-PT; 2210407C18Rik Gsta4 Ttr

V2 MYC / growth (translation) — Myc Targets V1 +1.67 SD; 100% S/G2M; 64% Early-Hybrid; Hmgb2 Birc5 Cks1b

V3 hypoxia / glycolysis — Hypoxia +1.52 SD; 1% S/G2M; 46% Early-Hybrid; Mt1 Sprr1a Malat1

Mouse PDAC: A. Entire gene set, unit M2-PT

Fit shown: cleaned · k = 3 · p = 0.00498 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	68%	62%
2 closest 5% of cells to each corner	82%	87%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 39% of all cells here; 86% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M2-PT alone, cells fill a 3-corner simplex (p=0.00498). 9 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: MYC / growth (translation); metabolic; hypoxia / glycolysis.

Hybrid vs other cells: 68% of hybrid cells and 62% of other cells are generalists (rule 1; rule 2: 82% vs 87%), so hybrids are slightly more often generalists.

Corners and their tasks

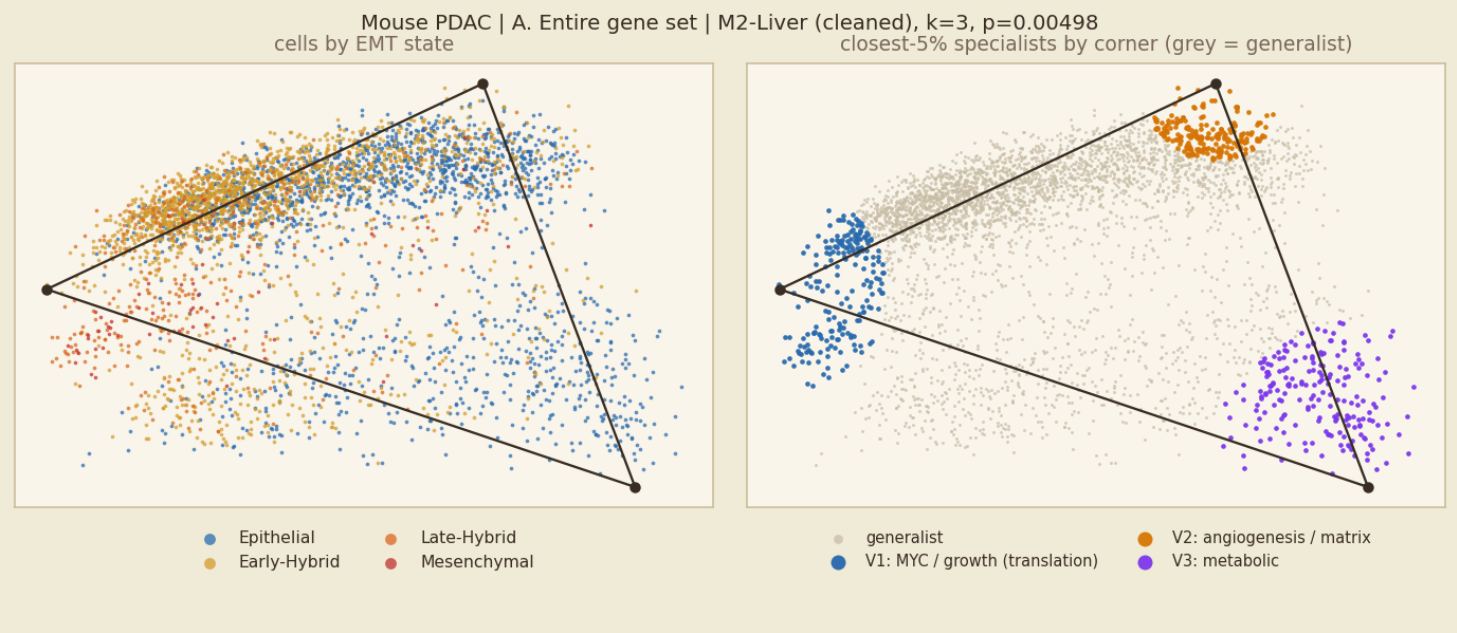
V1 MYC / growth (translation) — Myc Targets V1 +1.86 SD; 92% S/G2M; 50% Early-Hybrid; 100% M2-PT; H2afz Hmgb2 Birc5

V2 metabolic — Bile Acid Metabolism +1.73 SD; 0% S/G2M; 100% Epithelial; 100% M2-PT; 2210407C18Rik Ttr Dmbt1

V3 hypoxia / glycolysis — Hypoxia +1.87 SD; 2% S/G2M; 42% Early-Hybrid; 100% M2-PT; Ero1l Vegfa Hilpda

Mouse PDAC: A. Entire gene set, unit M2-Liver

Fit shown: cleaned · k = 3 · p = 0.00498 · 4,000 cells · 200 shuffles l comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	59%	51%
2 closest 5% of cells to each corner	88%	82%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 53% of all cells here; 56% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M2-Liver alone, cells fill a 3-corner simplex (p=0.00498). 9 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: MYC / growth (translation); angiogenesis / matrix; metabolic.

Hybrid vs other cells: 59% of hybrid cells and 51% of other cells are generalists (rule 1; rule 2: 88% vs 82%), so hybrids are clearly more often generalists.

Corners and their tasks

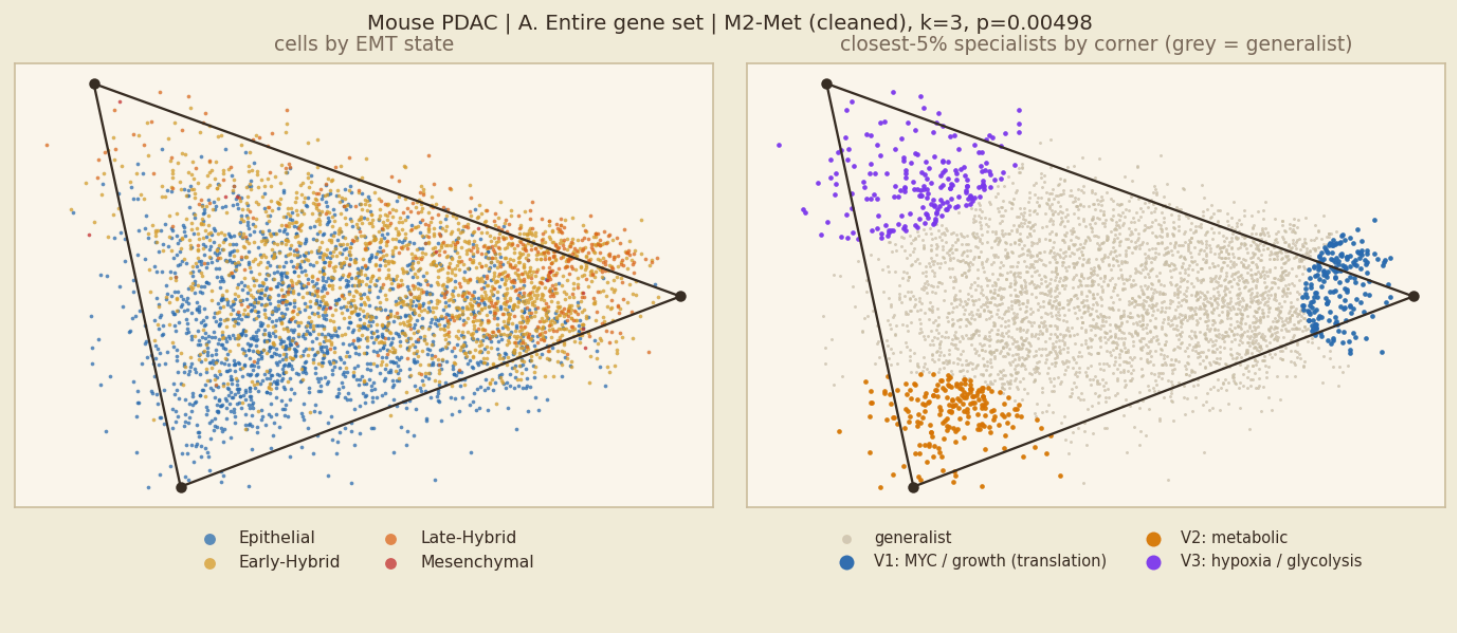
V1 MYC / growth (translation) — Myc Targets V1 +1.55 SD; 100% S/G2M; 42% Early-Hybrid; 100% M2-Liver; Hmgb2 Birc5 Phgdh

V2 angiogenesis / matrix — Angiogenesis +1.06 SD; 2% S/G2M; 62% Epithelial; 100% M2-Liver; Clu Sprr1a Spp1

V3 metabolic — Bile Acid Metabolism +1.8 SD; 2% S/G2M; 94% Epithelial; 100% M2-Liver; 2210407C18Rik Gsta4 Sptssb

Mouse PDAC: A. Entire gene set, unit M2-Met

Fit shown: cleaned · k = 3 · p = 0.00498 · 4,000 cells · 200 shuffles | comparison, light QC: k=3, p=0.0647



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	53%	72%
2 closest 5% of cells to each corner	85%	85%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 88% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M2-Met alone, cells fill a 3-corner simplex (p=0.00498). 9 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: MYC / growth (translation); metabolic; hypoxia / glycolysis.

Hybrid vs other cells: 53% of hybrid cells and 72% of other cells are generalists (rule 1; rule 2: 85% vs 85%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

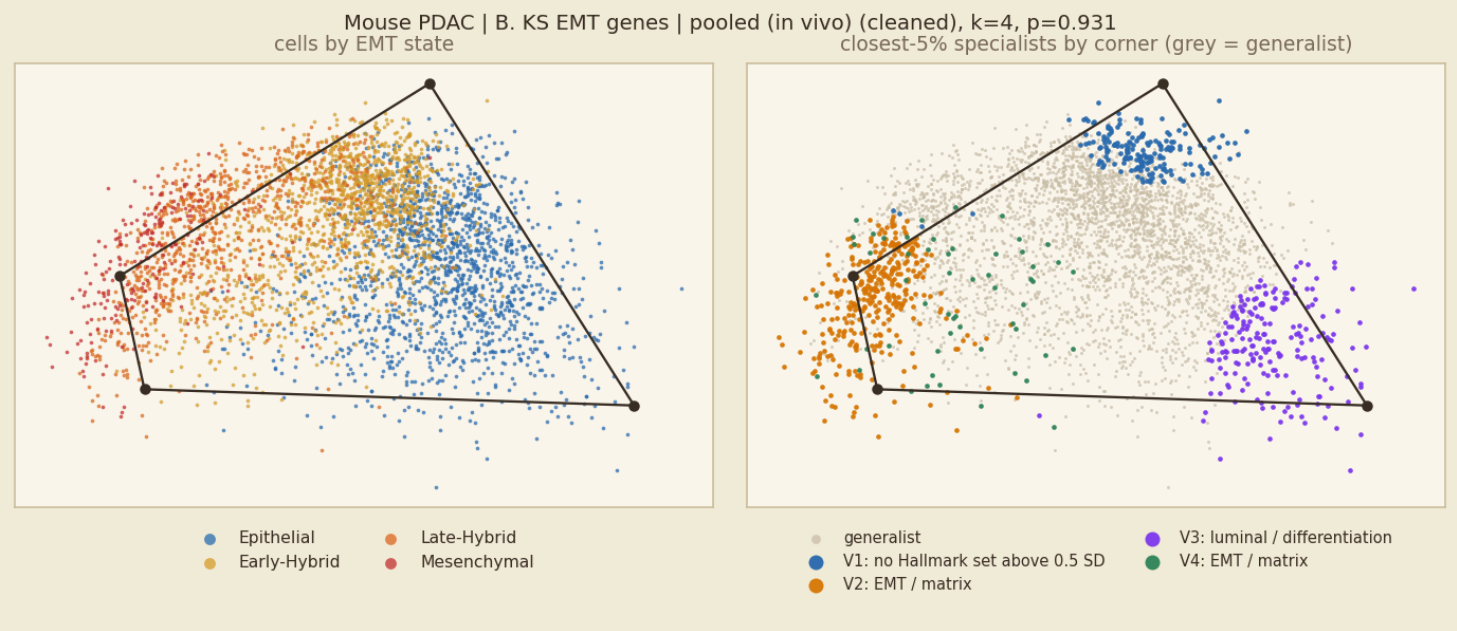
V1 MYC / growth (translation) — Myc Targets V1 +1.75 SD; 100% S/G2M; 58% Early-Hybrid; 100% M2-Met; Hmgb2 Phgdh Birc5

V2 metabolic — Bile Acid Metabolism +1.42 SD; 5% S/G2M; 94% Epithelial; 100% M2-Met; Ttr Fxyd3 Sptssb

V3 hypoxia / glycolysis — Hypoxia +2.11 SD; 4% S/G2M; 52% Early-Hybrid; 100% M2-Met; Mt1 Ero1l Ndrg1

Mouse PDAC: B. KS EMT genes, pooled fit

Fit shown: cleaned · k = 4 · p = 0.931 · 37,387 cells · 100 shuffles l comparison, light QC: k=4, p=0.97



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	54%	70%
2 closest 5% of cells to each corner	82%	78%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 38% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex: the pooled fit (k=4, p=0.931) is no more triangular than shuffled data, so its corners should not be read as tasks.

Hybrid vs other cells: 54% of hybrid cells and 70% of other cells are generalists (rule 1; rule 2: 82% vs 78%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 54% S/G2M; 59% Early-Hybrid; Krt19 Krt7 Areg

V2 EMT / matrix — Epithelial Mesenchymal Transition +1.76 SD; 9% S/G2M; 66% Late-Hybrid; Cald1 Vim Map1b

V3 luminal / differentiation — Estrogen Response Early +1.49 SD; 8% S/G2M; 100% Epithelial; Fxyd3 Atp1b1 Epcam

V4 EMT / matrix — Epithelial Mesenchymal Transition +0.73 SD; 11% S/G2M; 39% Late-Hybrid; Sparc Cald1 Map1b

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: B. KS EMT genes, all units at a glance

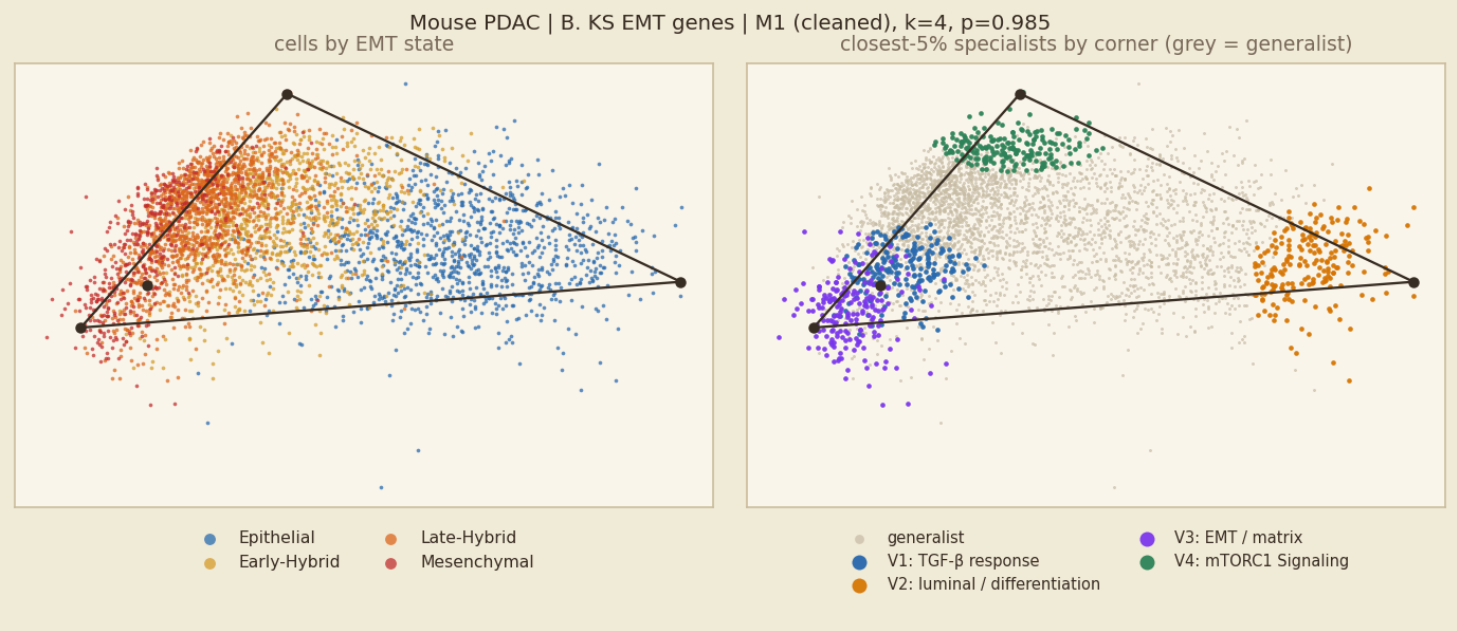
Main fit of each unit: 3 of 9 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
M1	cleaned	4	0.985	1	61%	77 / 82 / 100%	63 / 63 / 61%
M1-PT	cleaned	4	0.154	0.00498	49%	74 / 82 / 100%	45 / 50 / 49%
M1-SS	cleaned	4	0.00498 *	0.448	51%	75 / 81 / 100%	58 / 52 / 51%
M1-Liver	cleaned	4	0.97	0.945	67%	58 / 87 / 100%	63 / 73 / 67%
M1-Met	cleaned	4	0.597	0.343	66%	76 / 84 / 100%	66 / 69 / 66%
M2	cleaned	4	0.0249 *	0.363	50%	66 / 77 / 100%	46 / 47 / 50%
M2-PT	cleaned	3	0.622	0.99	39%	69 / 84 / 100%	36 / 38 / 39%
M2-Liver	cleaned	4	0.746	0.0796	53%	69 / 81 / 100%	50 / 53 / 53%
M2-Met	cleaned	4	0.0398 *	0.925	56%	81 / 80 / 100%	54 / 56 / 56%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

Mouse PDAC: B. KS EMT genes, unit M1

Fit shown: cleaned · k = 4 · p = 0.985 · 4,000 cells · 200 shuffles | comparison, light QC: k=3, p=1



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	77%	73%
2 closest 5% of cells to each corner	82%	77%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 61% of all cells here; 42% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M1: the fit (k=4, p=0.985) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 77% of hybrid cells and 73% of other cells are generalists (rule 1; rule 2: 82% vs 77%), so hybrids are slightly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 TGF-β response — TGF-beta Signaling +1.18 SD; 1% S/G2M; 58% Late-Hybrid; Hes1 Cald1 Sdc1

V2 luminal / differentiation — Estrogen Response Early +1.43 SD; 8% S/G2M; 100% Epithelial; Krt7 Fxyd3 Atp1b1

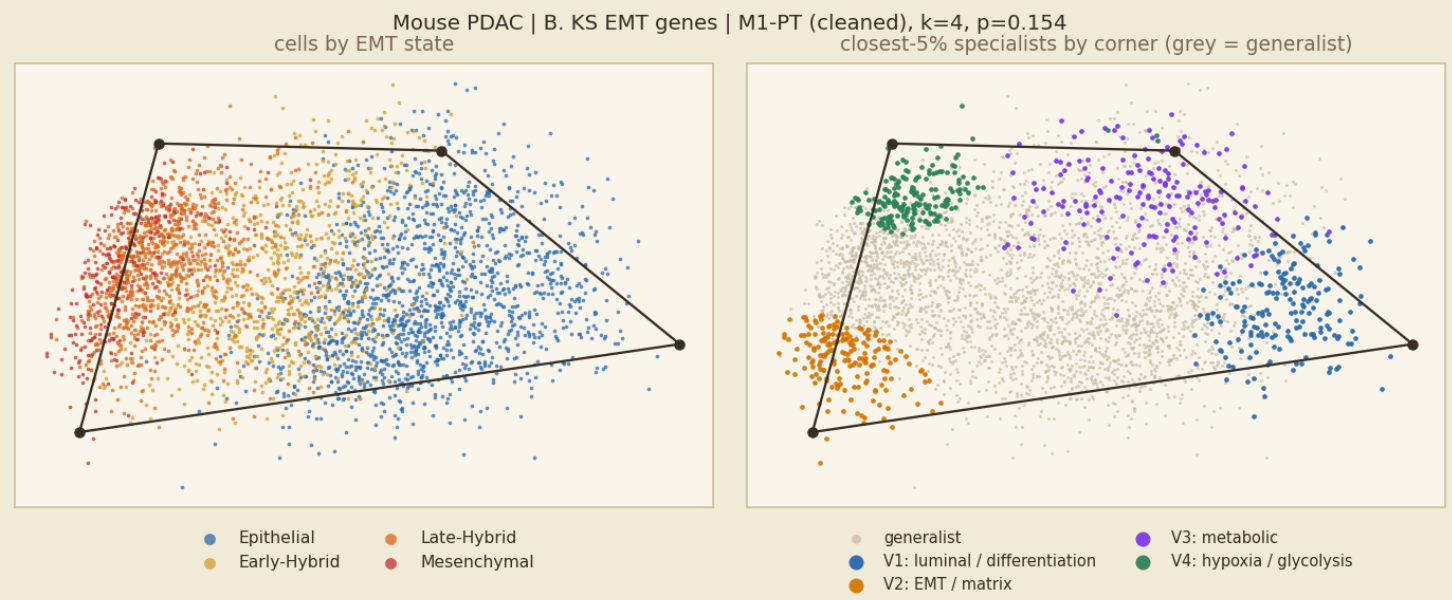
V3 EMT / matrix — Epithelial Mesenchymal Transition +1.13 SD; 8% S/G2M; 58% Mesenchymal; 66% M1-SS; Sparc Cald1 Serpine1

V4 mTORC1 Signaling — mTORC1 Signaling +1.07 SD; 48% S/G2M; 59% Late-Hybrid; Krt19 Krt8 Lgals1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: B. KS EMT genes, unit M1-PT

Fit shown: cleaned · k = 4 · p = 0.154 · 3,770 cells · 200 shuffles | comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	74%	87%
2 closest 5% of cells to each corner	82%	78%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 49% of all cells here; 42% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M1-PT: the fit (k=4, p=0.154) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 74% of hybrid cells and 87% of other cells are generalists (rule 1; rule 2: 82% vs 78%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 luminal / differentiation — Estrogen Response Early +1.49 SD; 2% S/G2M; 100% Epithelial; 100% M1-PT; Epcam Fxyd3 Atp1b1

V2 EMT / matrix — Epithelial Mesenchymal Transition +1.16 SD; 2% S/G2M; 46% Late-Hybrid; 100% M1-PT; Cald1 Sparc Sdc1

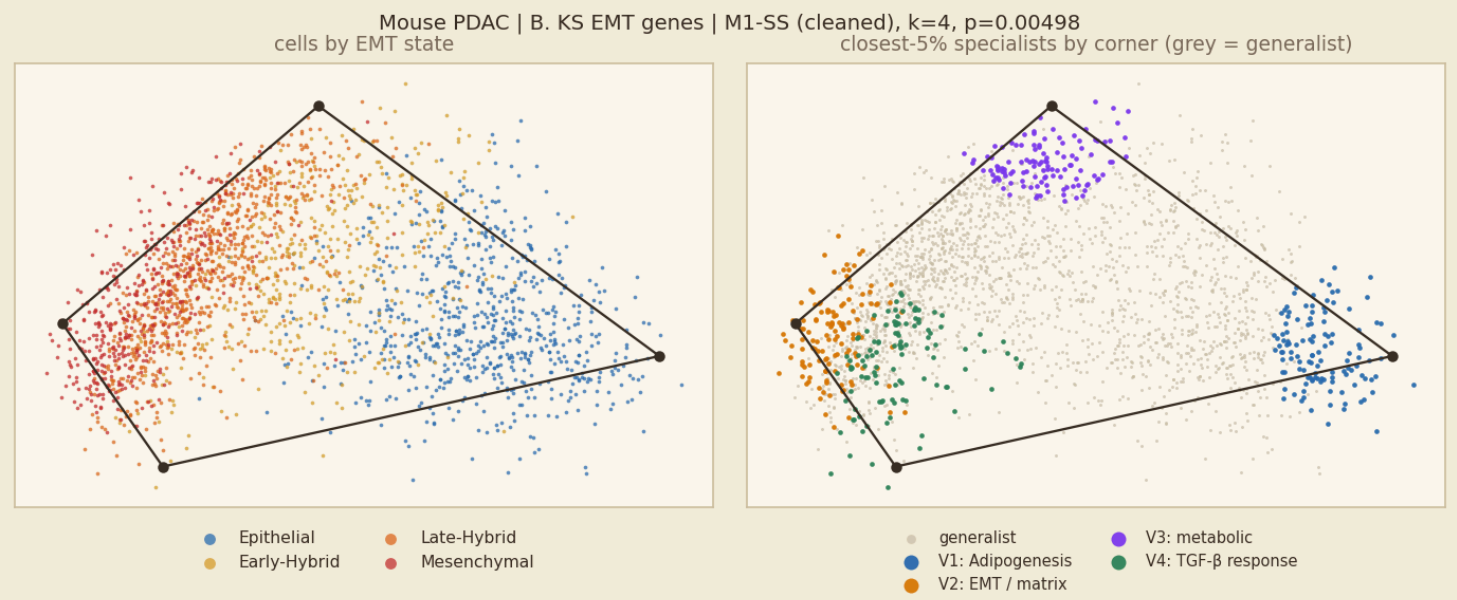
V3 metabolic — Oxidative Phosphorylation +1.18 SD; 36% S/G2M; 66% Epithelial; 100% M1-PT; Krt7 Fxyd3 Krt19

V4 hypoxia / glycolysis — Glycolysis +1.21 SD; 18% S/G2M; 72% Late-Hybrid; 100% M1-PT; Areg Vim Tuba1a

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: B. KS EMT genes, unit M1-SS

Fit shown: cleaned · k = 4 · p = 0.00498 · 2,397 cells · 200 shuffles | comparison, light QC: k=4, p=0.448



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	75%	57%
2 closest 5% of cells to each corner	81%	78%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 51% of all cells here; 43% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M1-SS alone, cells fill a 4-corner simplex (p=0.00498). 3 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: Adipogenesis; EMT / matrix; metabolic; TGF-β response.

Hybrid vs other cells: 75% of hybrid cells and 57% of other cells are generalists (rule 1; rule 2: 81% vs 78%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 Adipogenesis — Adipogenesis +1.54 SD; 12% S/G2M; 100% Epithelial; 100% M1-SS; Fxyd3 Krt19 Atp1b1

V2 EMT / matrix — Epithelial Mesenchymal Transition +1.76 SD; 10% S/G2M; 88% Mesenchymal; 100% M1-SS; Sparc Cald1 Lhfp

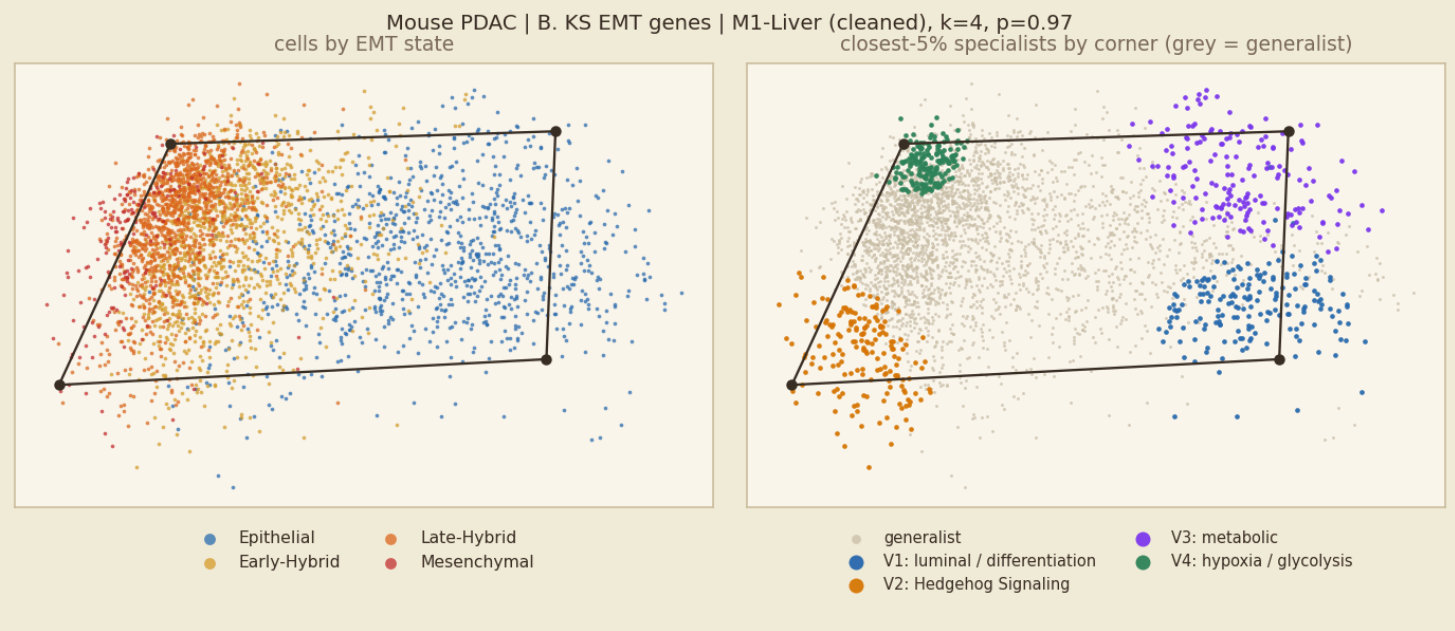
V3 metabolic — Cholesterol Homeostasis +1.0 SD; 31% S/G2M; 61% Late-Hybrid; 100% M1-SS; Krt8 Krt18 Lgals1

V4 TGF-β response — TGF-beta Signaling +1.3 SD; 1% S/G2M; 63% Late-Hybrid; 100% M1-SS; Hes1 Serpine1 Cald1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: B. KS EMT genes, unit M1-Liver

Fit shown: cleaned · k = 4 · p = 0.97 · 3,521 cells · 200 shuffles | comparison, light QC: k=4, p=0.945



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	58%	70%
2 closest 5% of cells to each corner	87%	66%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 67% of all cells here; 28% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M1-Liver: the fit (k=4, p=0.97) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 58% of hybrid cells and 70% of other cells are generalists (rule 1; rule 2: 87% vs 66%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 luminal / differentiation — Estrogen Response Early +1.33 SD; 1% S/G2M; 100% Epithelial; 100% M1-Liver; Atp1b1 Fxyd3 Krt7

V2 Hedgehog Signaling — Hedgehog Signaling +0.7 SD; 5% S/G2M; 51% Late-Hybrid; 100% M1-Liver; Sparc Cald1 Lcn2

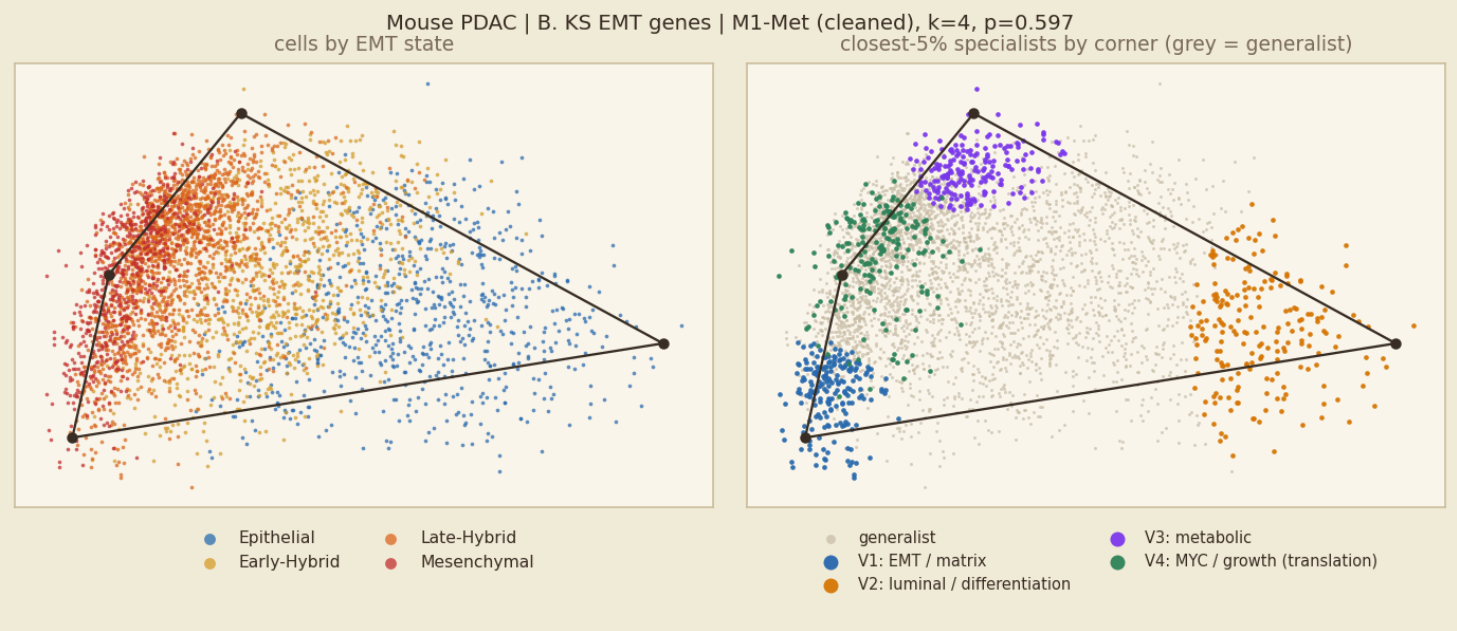
V3 metabolic — Oxidative Phosphorylation +1.12 SD; 40% S/G2M; 94% Epithelial; 100% M1-Liver; Krt7 Krt19 Fxyd3

V4 hypoxia / glycolysis — Glycolysis +0.65 SD; 36% S/G2M; 73% Late-Hybrid; 100% M1-Liver; Tuba1a Lgals1 Krt19

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: B. KS EMT genes, unit M1-Met

Fit shown: cleaned · k = 4 · p = 0.597 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.343



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	76%	76%
2 closest 5% of cells to each corner	84%	73%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 66% of all cells here; 41% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M1-Met: the fit (k=4, p=0.597) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 76% of hybrid cells and 76% of other cells are generalists (rule 1; rule 2: 84% vs 73%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.35 SD; 0% S/G2M; 52% Late-Hybrid; 100% M1-Met; Sparc Sdc1 Hes1

V2 luminal / differentiation — Estrogen Response Early +1.45 SD; 10% S/G2M; 98% Epithelial; 100% M1-Met; Krt7 Krt19 Atp1b1

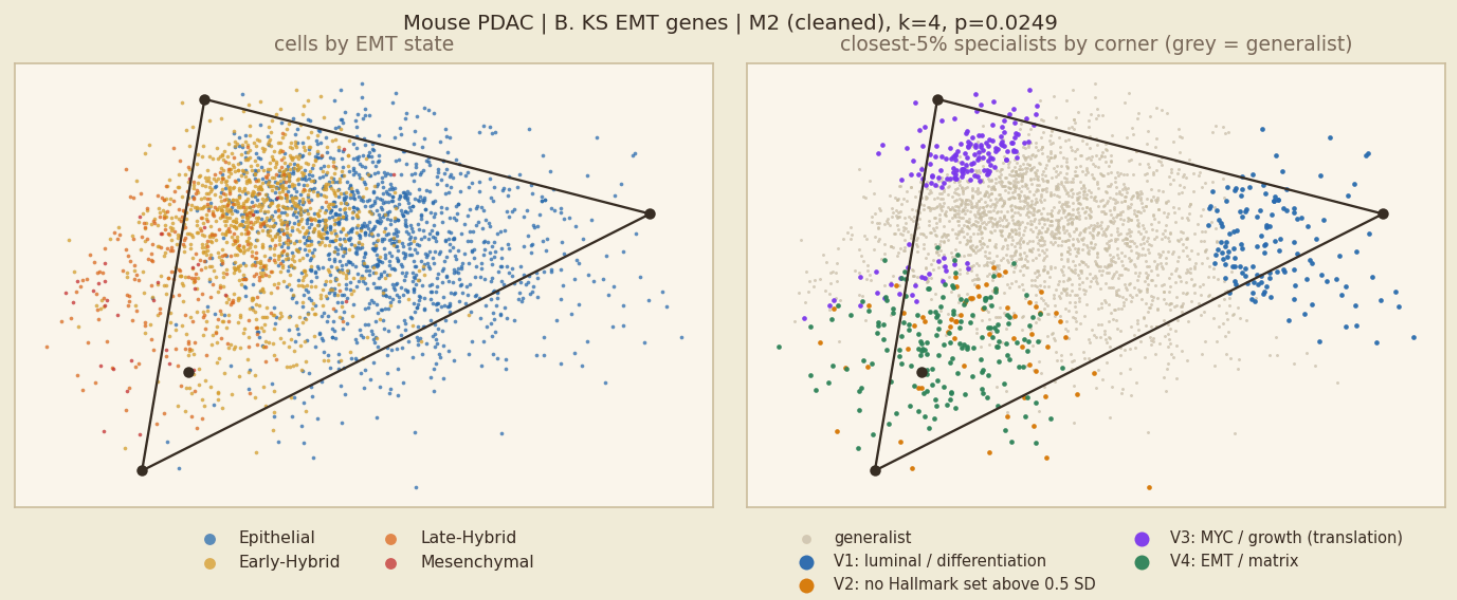
V3 metabolic — Cholesterol Homeostasis +0.87 SD; 38% S/G2M; 71% Late-Hybrid; 100% M1-Met; Krt19 Krt8 Areg

V4 MYC / growth (translation) — Myc Targets V1 +0.82 SD; 59% S/G2M; 52% Late-Hybrid; 100% M1-Met; Lgals1 Tubb6 Axl

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: B. KS EMT genes, unit M2

Fit shown: cleaned · k = 4 · p = 0.0249 · 2,728 cells · 200 shuffles l comparison, light QC: k=4, p=0.363



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	66%	77%
2 closest 5% of cells to each corner	77%	84%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 50% of all cells here; 46% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M2 alone, cells fill a 4-corner simplex (p=0.0249). 3 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: luminal / differentiation; no Hallmark set above 0.5 SD; MYC / growth (translation); EMT / matrix.

Hybrid vs other cells: 66% of hybrid cells and 77% of other cells are generalists (rule 1; rule 2: 77% vs 84%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 luminal / differentiation — Estrogen Response Early +1.47 SD; 13% S/G2M; 100% Epithelial; 69% M2-PT; Fxyd3 Epcam Krt19

V2 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 35% S/G2M; 47% Early-Hybrid; Cald1 Sparc Lcn2

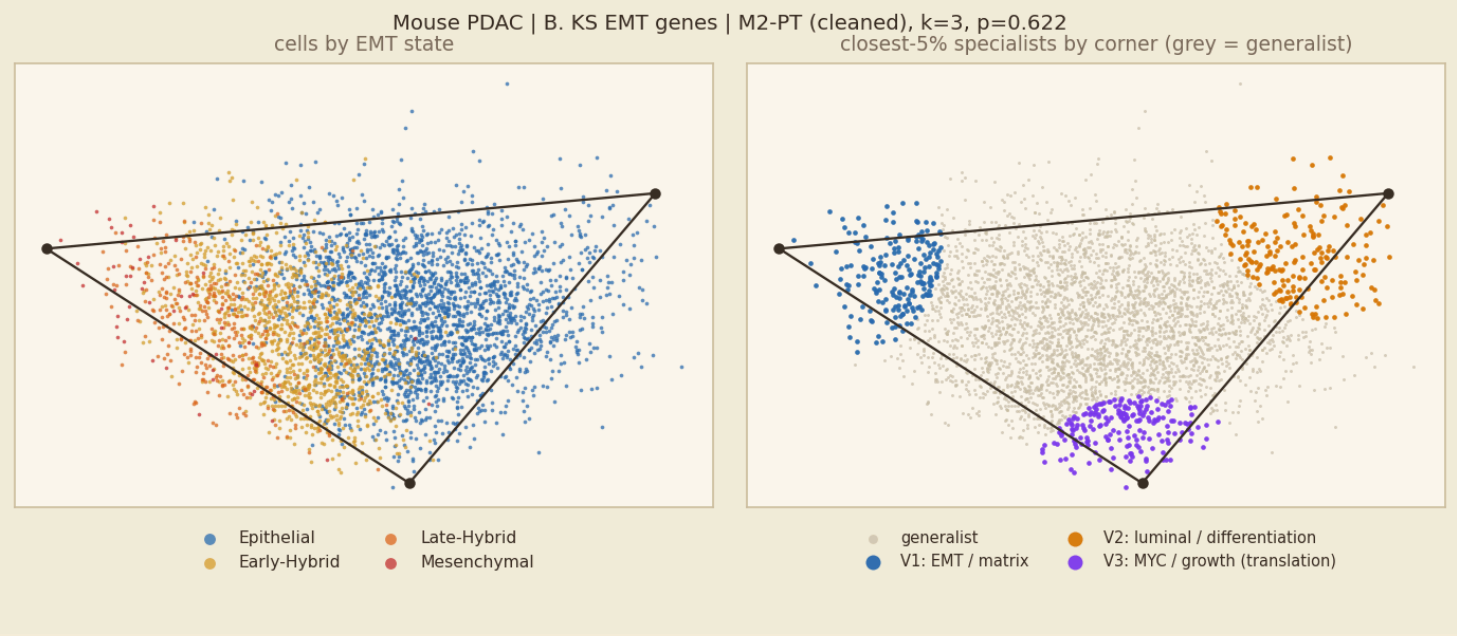
V3 MYC / growth (translation) — Myc Targets V1 +0.68 SD; 72% S/G2M; 77% Early-Hybrid; Areg Tuba1a Lgals1

V4 EMT / matrix — Epithelial Mesenchymal Transition +1.67 SD; 5% S/G2M; 51% Early-Hybrid; 81% M2-PT; Cald1 Camk2n1 Sparc

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: B. KS EMT genes, unit M2-PT

Fit shown: cleaned · k = 3 · p = 0.622 · 4,000 cells · 200 shuffles | comparison, light QC: k=3, p=0.99



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	69%	78%
2 closest 5% of cells to each corner	84%	85%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 39% of all cells here; 87% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M2-PT: the fit (k=3, p=0.622) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 69% of hybrid cells and 78% of other cells are generalists (rule 1; rule 2: 84% vs 85%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.2 SD; 8% S/G2M; 44% Late-Hybrid; 100% M2-PT; Cald1 Tuba1a Sparc

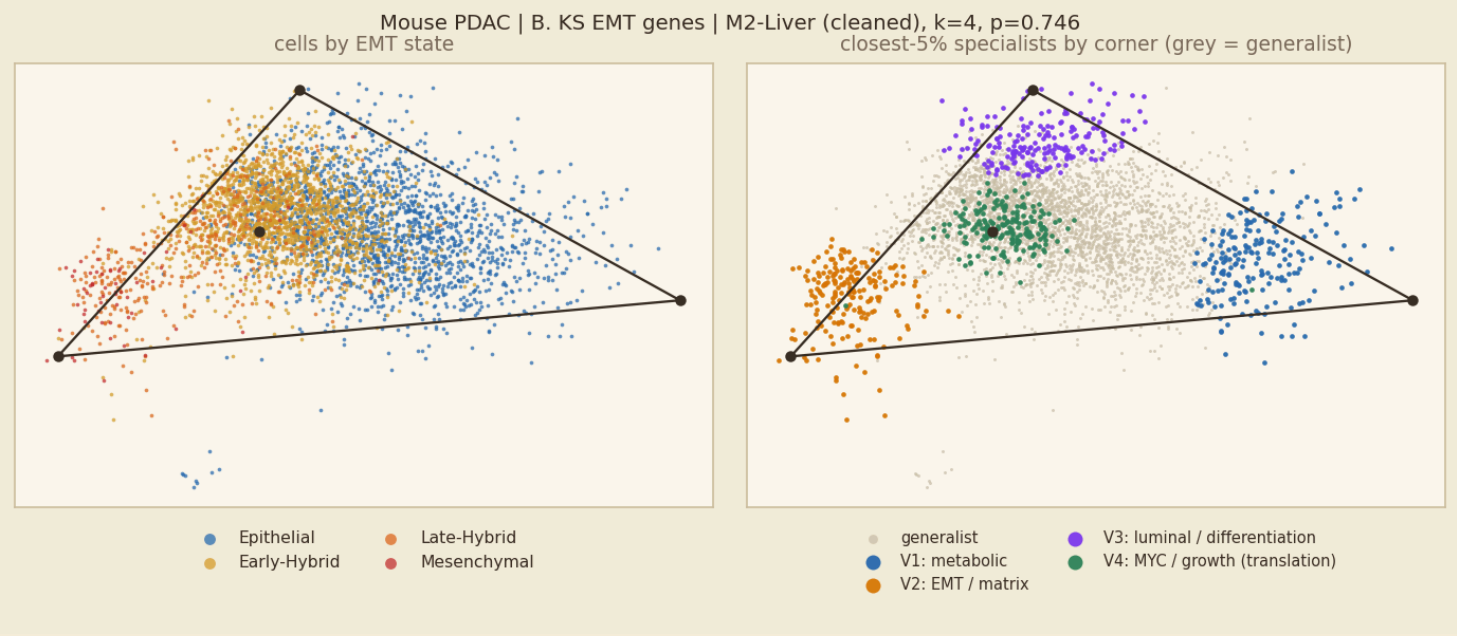
V2 luminal / differentiation — Estrogen Response Early +0.88 SD; 9% S/G2M; 100% Epithelial; 100% M2-PT; Epcam Fxyd3 Atp1b1

V3 MYC / growth (translation) — Myc Targets V1 +0.71 SD; 40% S/G2M; 60% Epithelial; 100% M2-PT; Areg Krt7 Krt19

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: B. KS EMT genes, unit M2-Liver

Fit shown: cleaned · k = 4 · p = 0.746 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.0796



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	69%	77%
2 closest 5% of cells to each corner	81%	79%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 53% of all cells here; 51% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M2-Liver: the fit (k=4, p=0.746) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 69% of hybrid cells and 77% of other cells are generalists (rule 1; rule 2: 81% vs 79%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 metabolic — Bile Acid Metabolism +1.57 SD; 6% S/G2M; 100% Epithelial; 100% M2-Liver; Fxyd3 Atp1b1 Elf3

V2 EMT / matrix — Epithelial Mesenchymal Transition +1.87 SD; 74% S/G2M; 66% Late-Hybrid; 100% M2-Liver; Cald1 Sparc Tubb6

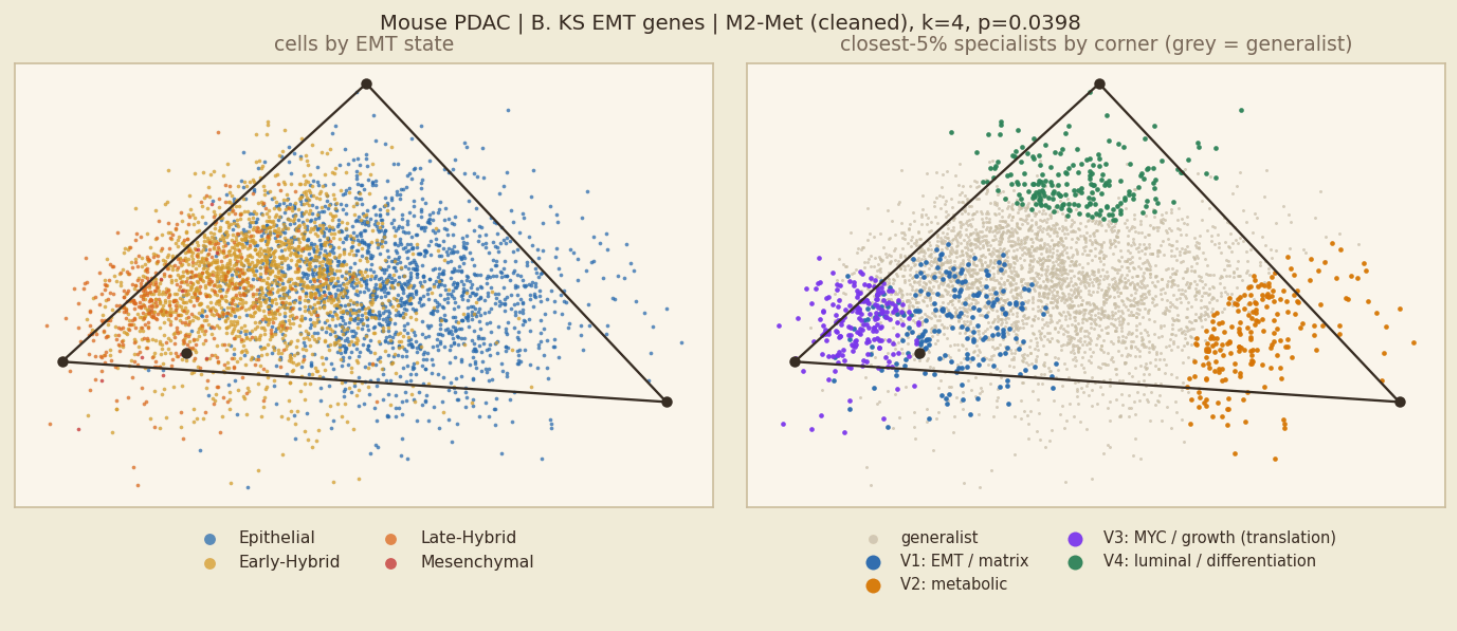
V3 luminal / differentiation — Estrogen Response Early +0.78 SD; 61% S/G2M; 50% Early-Hybrid; 100% M2-Liver; Krt19 Il1rn Areg

V4 MYC / growth (translation) — Myc Targets V1 +0.81 SD; 89% S/G2M; 65% Early-Hybrid; 100% M2-Liver; Lgals1 Krt7 Areg

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: B. KS EMT genes, unit M2-Met

Fit shown: cleaned · k = 4 · p = 0.0398 · 4,000 cells · 200 shuffles l comparison, light QC: k=3, p=0.925



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	81%	88%
2 closest 5% of cells to each corner	80%	80%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 59% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M2-Met alone, cells fill a 4-corner simplex (p=0.0398). 3 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: EMT / matrix; metabolic; MYC / growth (translation); luminal / differentiation.

Hybrid vs other cells: 81% of hybrid cells and 88% of other cells are generalists (rule 1; rule 2: 80% vs 80%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.1 SD; 44% S/G2M; 59% Early-Hybrid; 100% M2-Met; Cald1 Sparc Emp3

V2 metabolic — Bile Acid Metabolism +1.11 SD; 10% S/G2M; 96% Epithelial; 100% M2-Met; Fxyd3 Atp1b1 Camk2n1

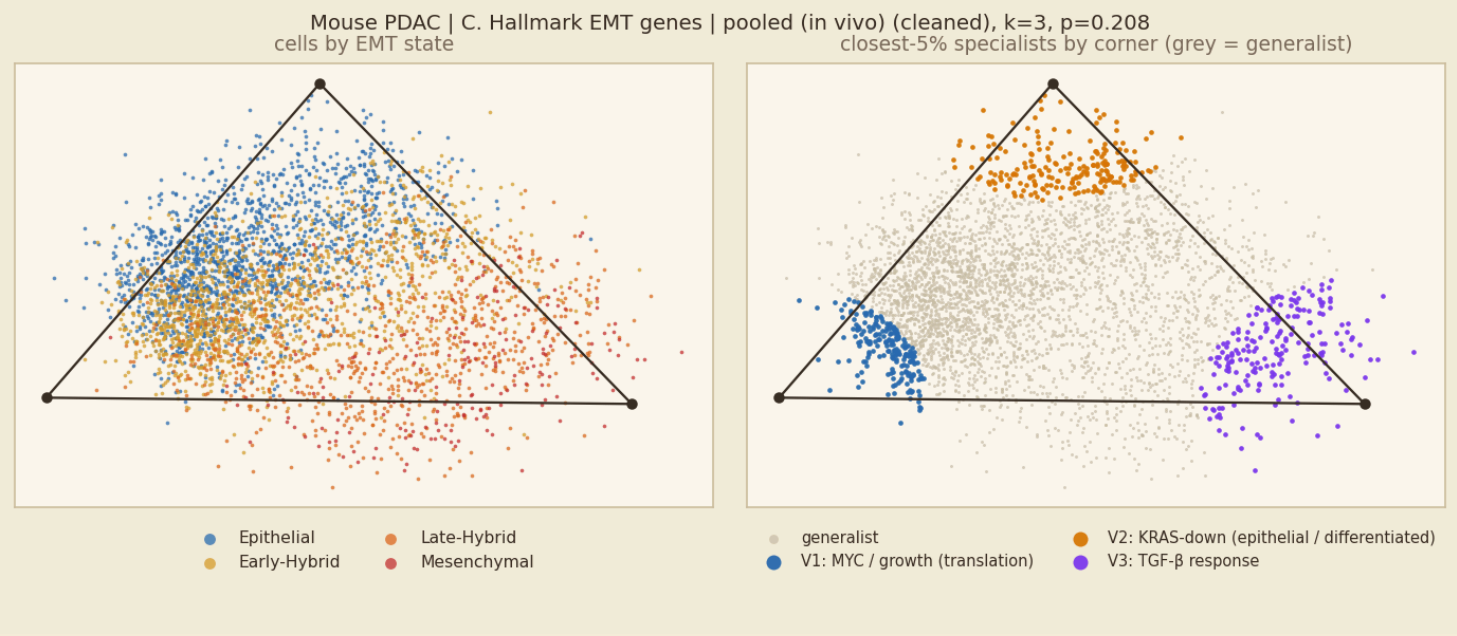
V3 MYC / growth (translation) — Myc Targets V1 +1.16 SD; 89% S/G2M; 52% Late-Hybrid; 100% M2-Met; Areg Tubb6 Lgals1

V4 luminal / differentiation — Estrogen Response Early +0.62 SD; 34% S/G2M; 63% Epithelial; 100% M2-Met; Krt19 Krt7 Krt8

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Mouse PDAC: C. Hallmark EMT genes, pooled fit

Fit shown: cleaned · k = 3 · p = 0.208 · 37,387 cells · 100 shuffles l comparison, light QC: k=3, p=0.129



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	72%	71%
2 closest 5% of cells to each corner	87%	82%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 84% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex: the pooled fit (k=3, p=0.208) is no more triangular than shuffled data, so its corners should not be read as tasks.

Hybrid vs other cells: 72% of hybrid cells and 71% of other cells are generalists (rule 1; rule 2: 87% vs 82%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 MYC / growth (translation) — Myc Targets V1 +1.04 SD; 82% S/G2M; 54% Early-Hybrid; Areg Lgals1 Mcm7

V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.13 SD; 1% S/G2M; 87% Epithelial; Fn1 Spp1 Jun

V3 TGF-β response — TGF-beta Signaling +1.56 SD; 4% S/G2M; 57% Late-Hybrid; Inhba Fn1 Fbln2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse PDAC: C. Hallmark EMT genes, all units at a glance

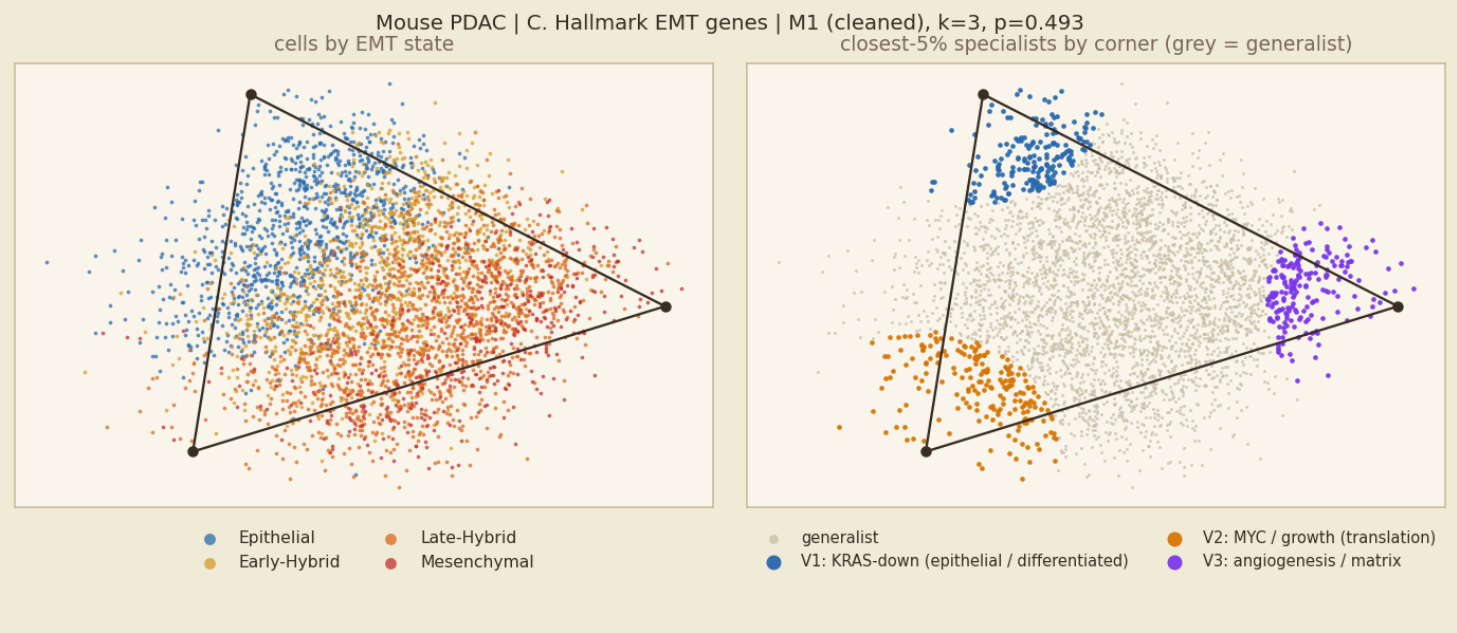
Main fit of each unit: 4 of 9 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
M1	cleaned	3	0.493	0.234	61%	76 / 90 / 100%	66 / 65 / 61%
M1-PT	cleaned	4	0.642	0.726	49%	76 / 78 / 100%	48 / 48 / 49%
M1-SS	cleaned	3	0.00498 *	0.00995	51%	79 / 93 / 100%	61 / 56 / 51%
M1-Liver	cleaned	4	0.443	0.204	67%	70 / 80 / 100%	66 / 68 / 67%
M1-Met	cleaned	4	0.0448 *	0.204	66%	70 / 83 / 100%	68 / 69 / 66%
M2	cleaned	4	0.00995 *	0.124	50%	84 / 77 / 100%	49 / 48 / 50%
M2-PT	cleaned	4	0.0299 *	0.0348	39%	77 / 75 / 100%	37 / 36 / 39%
M2-Liver	cleaned	3	0.164	0.0249	53%	75 / 86 / 100%	50 / 54 / 53%
M2-Met	cleaned	3	0.537	0.0846	56%	73 / 84 / 100%	52 / 56 / 56%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

Mouse PDAC: C. Hallmark EMT genes, unit M1

Fit shown: cleaned · k = 3 · p = 0.493 · 4,000 cells · 200 shuffles | comparison, light QC: k=3, p=0.234



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	76%	63%
2 closest 5% of cells to each corner	90%	78%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 61% of all cells here; 79% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M1: the fit (k=3, p=0.493) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 76% of hybrid cells and 63% of other cells are generalists (rule 1; rule 2: 90% vs 78%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.24 SD; 0% S/G2M; 98% Epithelial; Spp1 Tgm2 Jun

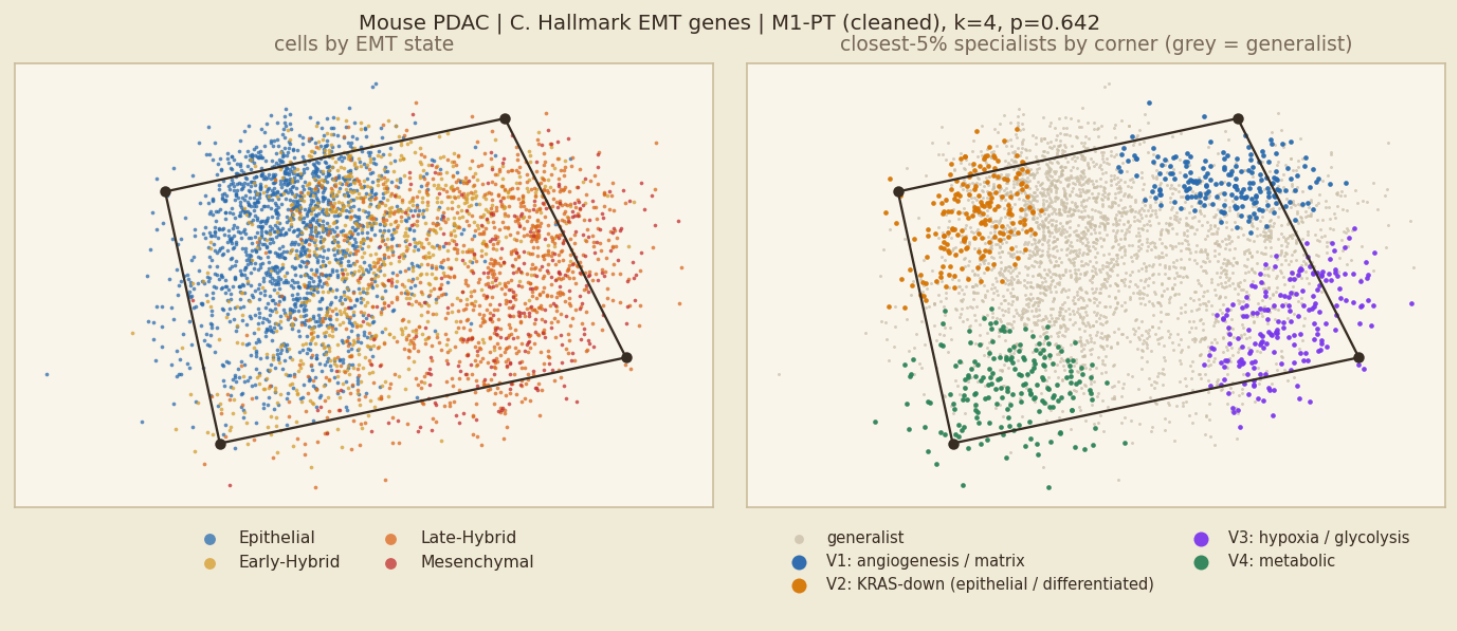
V2 MYC / growth (translation) — Myc Targets V1 +0.75 SD; 52% S/G2M; 49% Late-Hybrid; Lgals1 Tpm4 Dpysl3

V3 angiogenesis / matrix — Angiogenesis +1.38 SD; 3% S/G2M; 50% Late-Hybrid; Inhba Fn1 Fbln2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse PDAC: C. Hallmark EMT genes, unit M1-PT

Fit shown: cleaned · k = 4 · p = 0.642 · 3,770 cells · 200 shuffles | comparison, light QC: k=4, p=0.726



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	76%	80%
2 closest 5% of cells to each corner	78%	81%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 49% of all cells here; 40% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M1-PT: the fit (k=4, p=0.642) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 76% of hybrid cells and 80% of other cells are generalists (rule 1; rule 2: 78% vs 81%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 angiogenesis / matrix — Angiogenesis +1.01 SD; 1% S/G2M; 48% Late-Hybrid; 100% M1-PT; Spp1 Fbln2 Ecm1

V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.12 SD; 1% S/G2M; 84% Epithelial; 100% M1-PT; Jun Id2 Tgm2

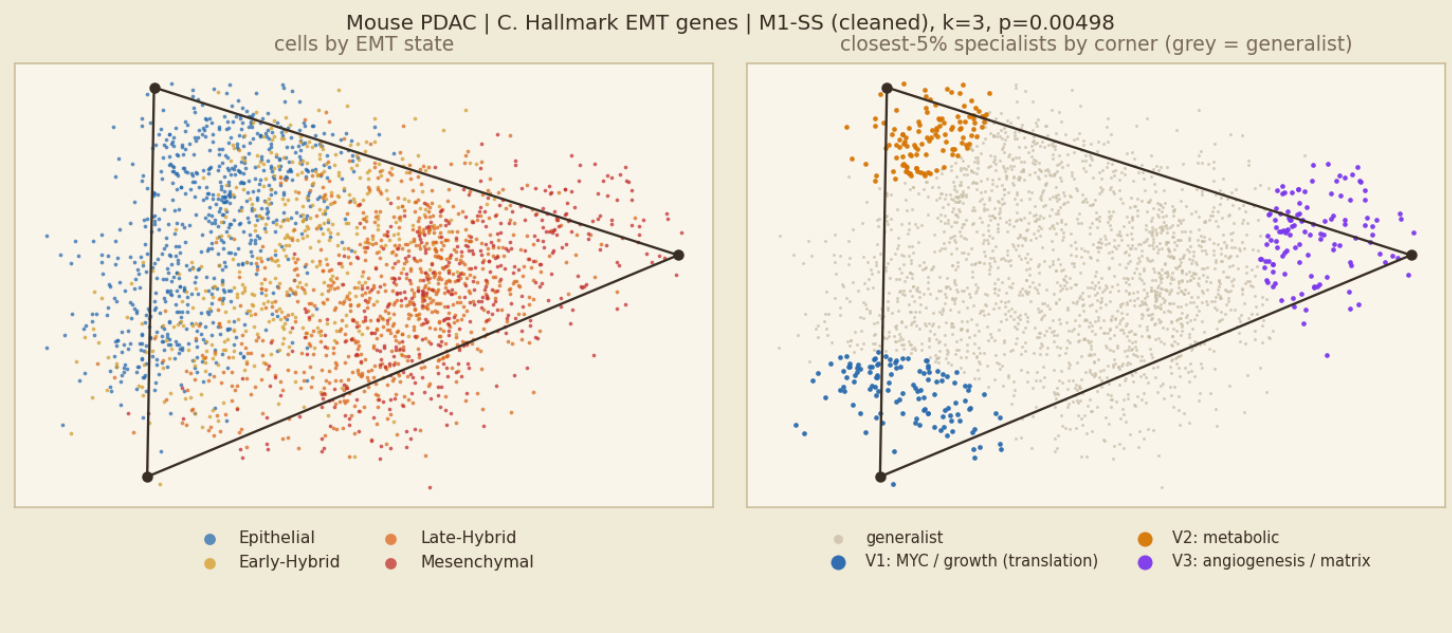
V3 hypoxia / glycolysis — Glycolysis +1.24 SD; 6% S/G2M; 60% Late-Hybrid; 100% M1-PT; Inhba Igfbp3 Fbln2

V4 metabolic — Oxidative Phosphorylation +1.19 SD; 49% S/G2M; 50% Epithelial; 100% M1-PT; Lgals1 Capg Ppib

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse PDAC: C. Hallmark EMT genes, unit M1-SS

Fit shown: cleaned · k = 3 · p = 0.00498 · 2,397 cells · 200 shuffles l comparison, light QC: k=3, p=0.00995



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	79%	54%
2 closest 5% of cells to each corner	93%	77%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 51% of all cells here; 81% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M1-SS alone, cells fill a 3-corner simplex (p=0.00498). 4 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: MYC / growth (translation); metabolic; angiogenesis / matrix.

Hybrid vs other cells: 79% of hybrid cells and 54% of other cells are generalists (rule 1; rule 2: 93% vs 77%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 MYC / growth (translation) — Myc Targets V1 +1.13 SD; 42% S/G2M; 40% Epithelial; 100% M1-SS; Lgals1 Capg Qsox1

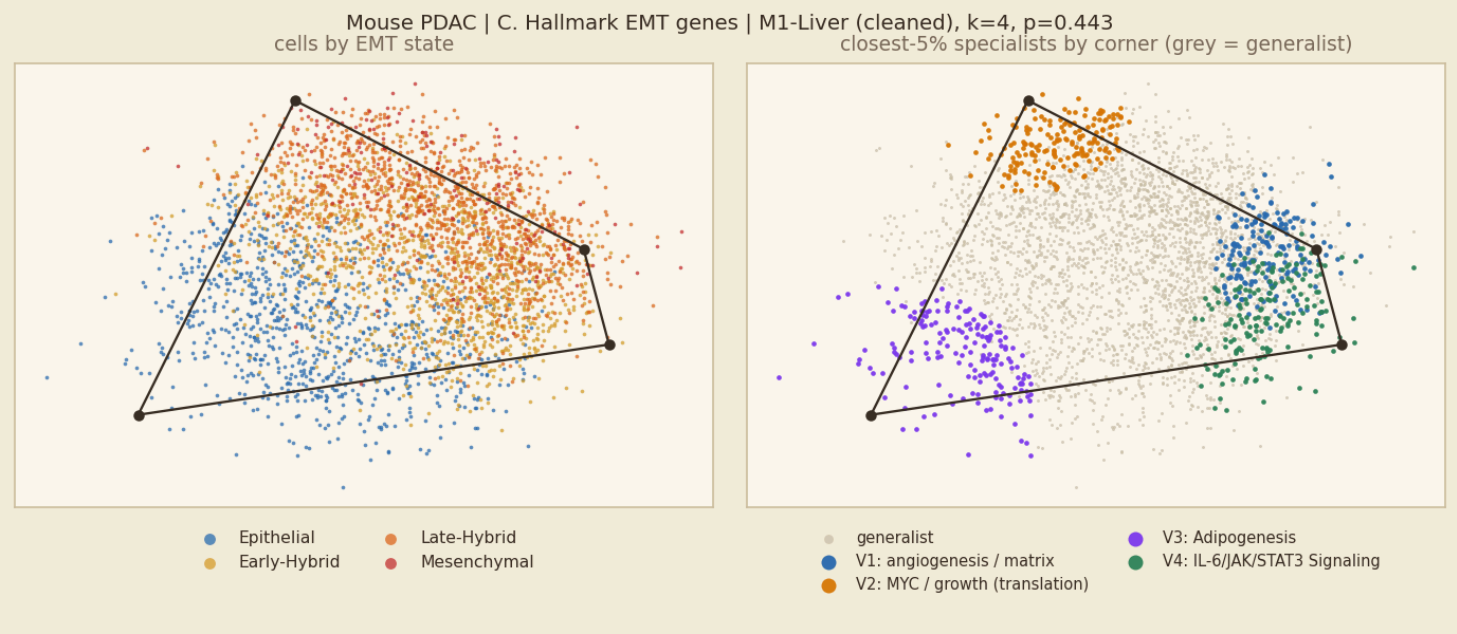
V2 metabolic — Bile Acid Metabolism +1.18 SD; 0% S/G2M; 98% Epithelial; 100% M1-SS; Spp1 Tgm2 Id2

V3 angiogenesis / matrix — Angiogenesis +1.95 SD; 1% S/G2M; 82% Mesenchymal; 100% M1-SS; Col3a1 Inhba Sparc

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse PDAC: C. Hallmark EMT genes, unit M1-Liver

Fit shown: cleaned · k = 4 · p = 0.443 · 3,521 cells · 200 shuffles | comparison, light QC: k=3, p=0.204



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	70%	75%
2 closest 5% of cells to each corner	80%	79%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 67% of all cells here; 37% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M1-Liver: the fit (k=4, p=0.443) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 70% of hybrid cells and 75% of other cells are generalists (rule 1; rule 2: 80% vs 79%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 angiogenesis / matrix — Angiogenesis +0.92 SD; 3% S/G2M; 50% Late-Hybrid; 100% M1-Liver; Spp1 Fbln2 Fn1

V2 MYC / growth (translation) — Myc Targets V1 +1.64 SD; 70% S/G2M; 62% Late-Hybrid; 100% M1-Liver; Lgals1 Igfbp3 Dpysl3

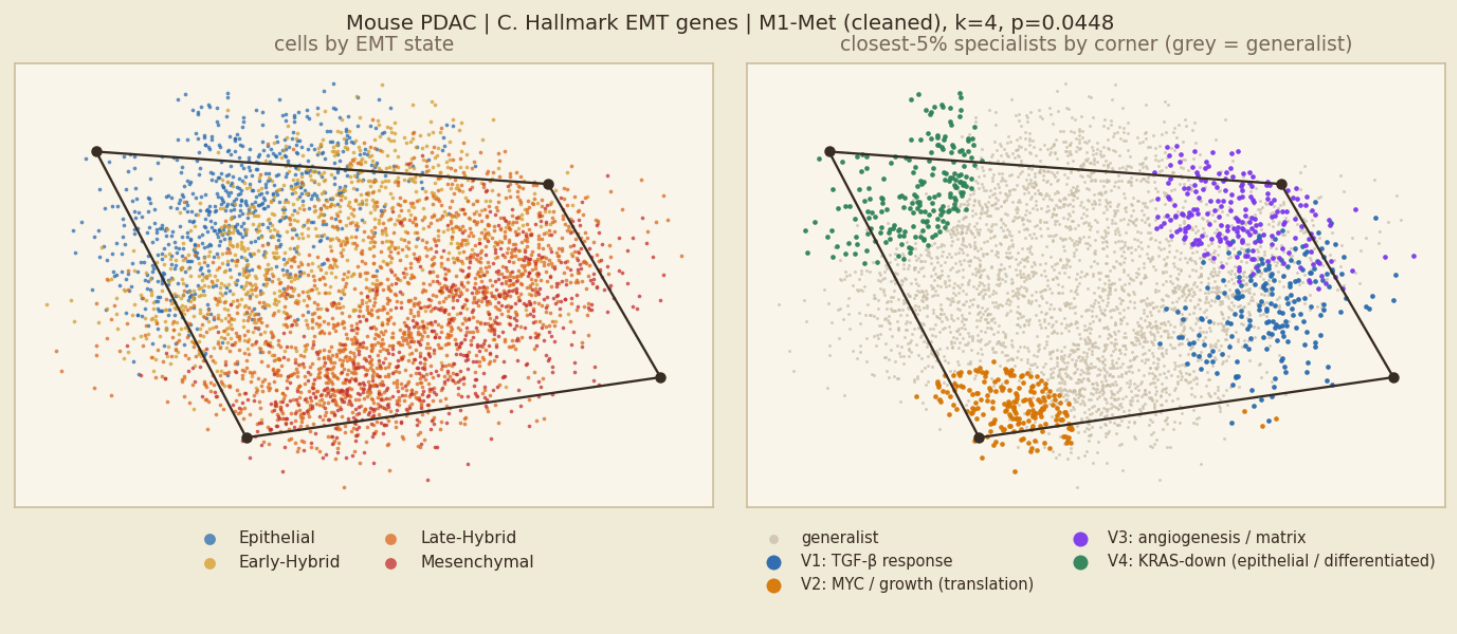
V3 Adipogenesis — Adipogenesis +1.19 SD; 6% S/G2M; 97% Epithelial; 100% M1-Liver; Qsox1 Sat1 Plaur

V4 IL-6/JAK/STAT3 Signaling — IL-6/JAK/STAT3 Signaling +1.57 SD; 1% S/G2M; 49% Early-Hybrid; 100% M1-Liver; Cxcl1 Vcam1 Fn1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse PDAC: C. Hallmark EMT genes, unit M1-Met

Fit shown: cleaned · k = 4 · p = 0.0448 · 4,000 cells · 200 shuffles l comparison, light QC: k=3, p=0.204



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	70%	64%
2 closest 5% of cells to each corner	83%	74%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 66% of all cells here; 37% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M1-Met alone, cells fill a 4-corner simplex (p=0.0448). 4 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: TGF-β response; MYC / growth (translation); angiogenesis / matrix; KRAS-down (epithelial / differentiated).

Hybrid vs other cells: 70% of hybrid cells and 64% of other cells are generalists (rule 1; rule 2: 83% vs 74%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 TGF-β response — TGF-beta Signaling +1.3 SD; 2% S/G2M; 52% Mesenchymal; 100% M1-Met; Inhba Fn1 Col6a2

V2 MYC / growth (translation) — Myc Targets V1 +1.56 SD; 62% S/G2M; 71% Late-Hybrid; 100% M1-Met; Igfbp3 Lgals1 Areg

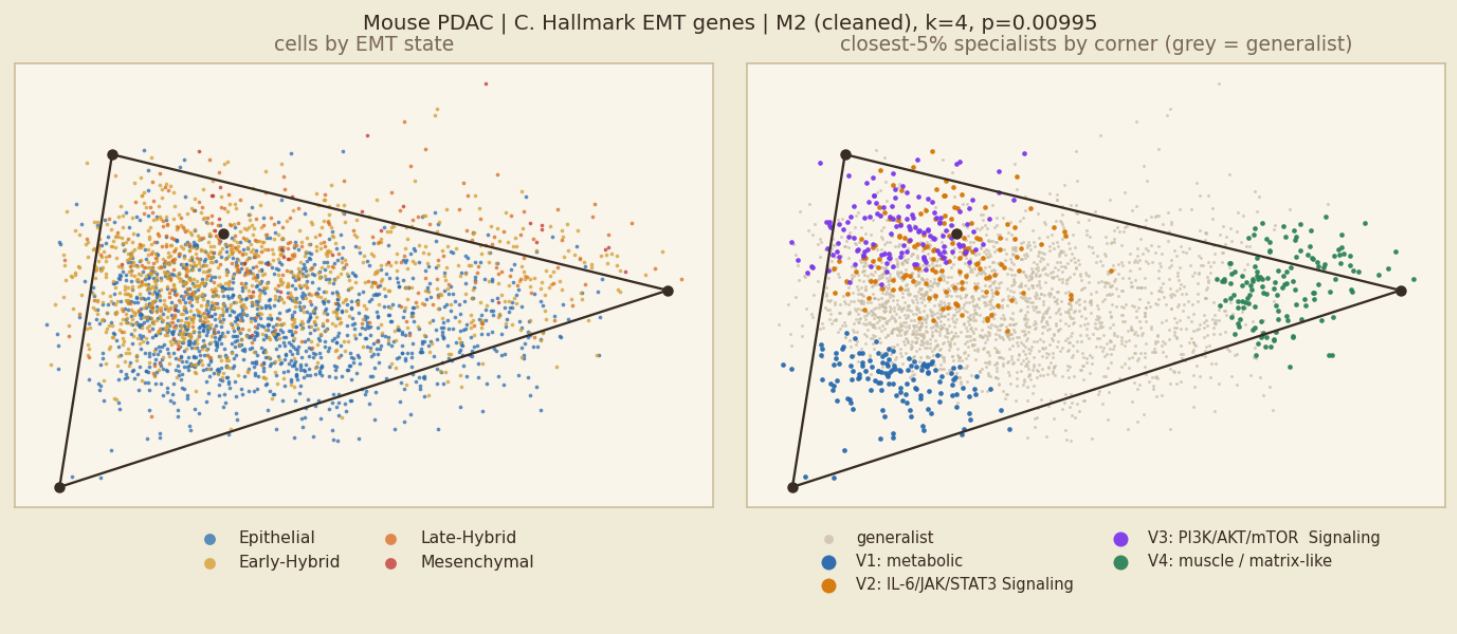
V3 angiogenesis / matrix — Angiogenesis +1.1 SD; 1% S/G2M; 63% Late-Hybrid; 100% M1-Met; Spp1 Cdh6 Ecm1

V4 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.47 SD; 6% S/G2M; 92% Epithelial; 100% M1-Met; Tgm2 Sat1 Qsox1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse PDAC: C. Hallmark EMT genes, unit M2

Fit shown: cleaned · k = 4 · p = 0.00995 · 2,728 cells · 200 shuffles | comparison, light QC: k=3, p=0.124



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	84%	86%
2 closest 5% of cells to each corner	77%	82%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 50% of all cells here; 63% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M2 alone, cells fill a 4-corner simplex (p=0.00995). 4 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: metabolic; IL-6/JAK/STAT3 Signaling; PI3K/AKT/mTOR Signaling; muscle / matrix-like.

Hybrid vs other cells: 84% of hybrid cells and 86% of other cells are generalists (rule 1; rule 2: 77% vs 82%), so no real difference.

Corners and their tasks

V1 metabolic — Oxidative Phosphorylation +0.75 SD; 58% S/G2M; 77% Epithelial; Fuca1 Mcm7 Capg

V2 IL-6/JAK/STAT3 Signaling — IL-6/JAK/STAT3 Signaling +1.22 SD; 61% S/G2M; 39% Early-Hybrid; 82% M2-Liver; Cxcl1 Vcam1 Tnfaip3

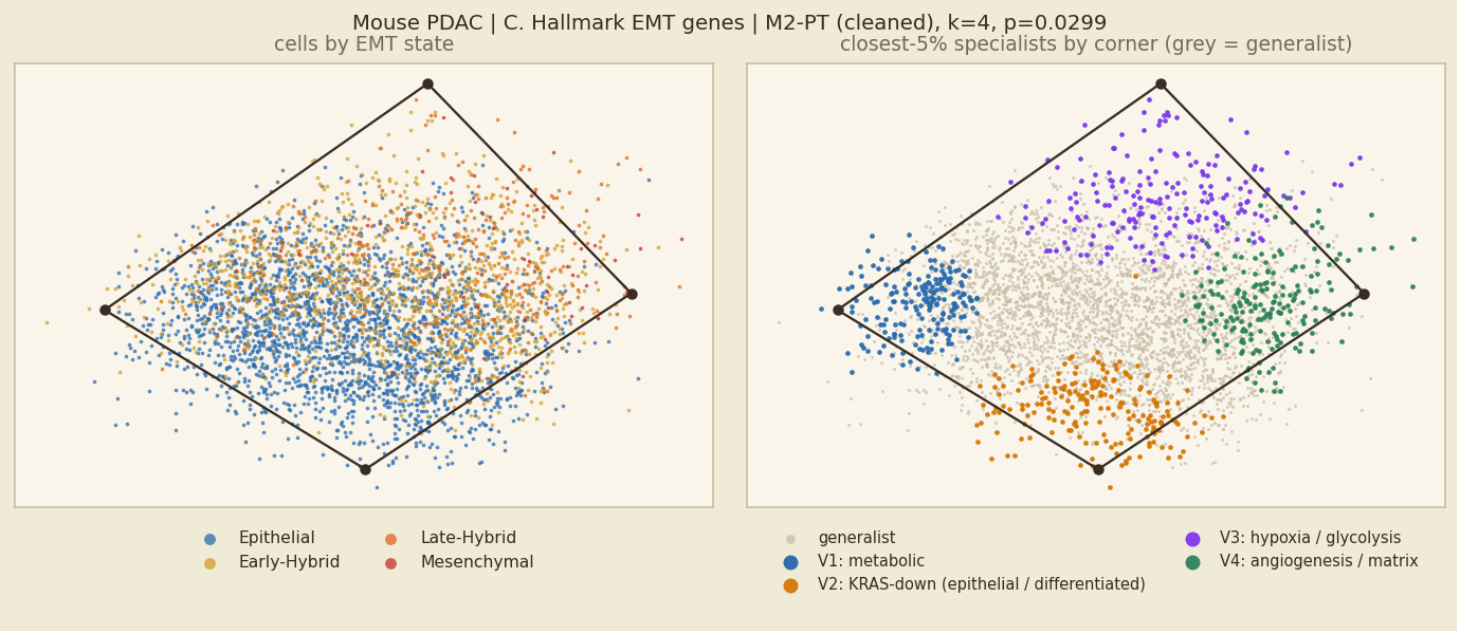
V3 PI3K/AKT/mTOR Signaling — PI3K/AKT/mTOR Signaling +0.92 SD; 66% S/G2M; 61% Early-Hybrid; Areg Tnfrsf12a Inhba

V4 muscle / matrix-like — Myogenesis +1.47 SD; 2% S/G2M; 49% Early-Hybrid; 93% M2-PT; Fn1 Spp1 Fbln2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse PDAC: C. Hallmark EMT genes, unit M2-PT

Fit shown: cleaned · k = 4 · p = 0.0299 · 4,000 cells · 200 shuffles l comparison, light QC: k=4, p=0.0348



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	77%	82%
2 closest 5% of cells to each corner	75%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 39% of all cells here; 52% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside M2-PT alone, cells fill a 4-corner simplex (p=0.0299). 4 of 9 units of this dataset have a significant simplex for this gene set.

Corners are programmes: metabolic; KRAS-down (epithelial / differentiated); hypoxia / glycolysis; angiogenesis / matrix.

Hybrid vs other cells: 77% of hybrid cells and 82% of other cells are generalists (rule 1; rule 2: 75% vs 83%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 metabolic — Oxidative Phosphorylation +1.09 SD; 67% S/G2M; 55% Epithelial; 100% M2-PT; Lgals1 Capg Tpm2

V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.01 SD; 2% S/G2M; 92% Epithelial; 100% M2-PT; Jun Tgm2 Pdlim4

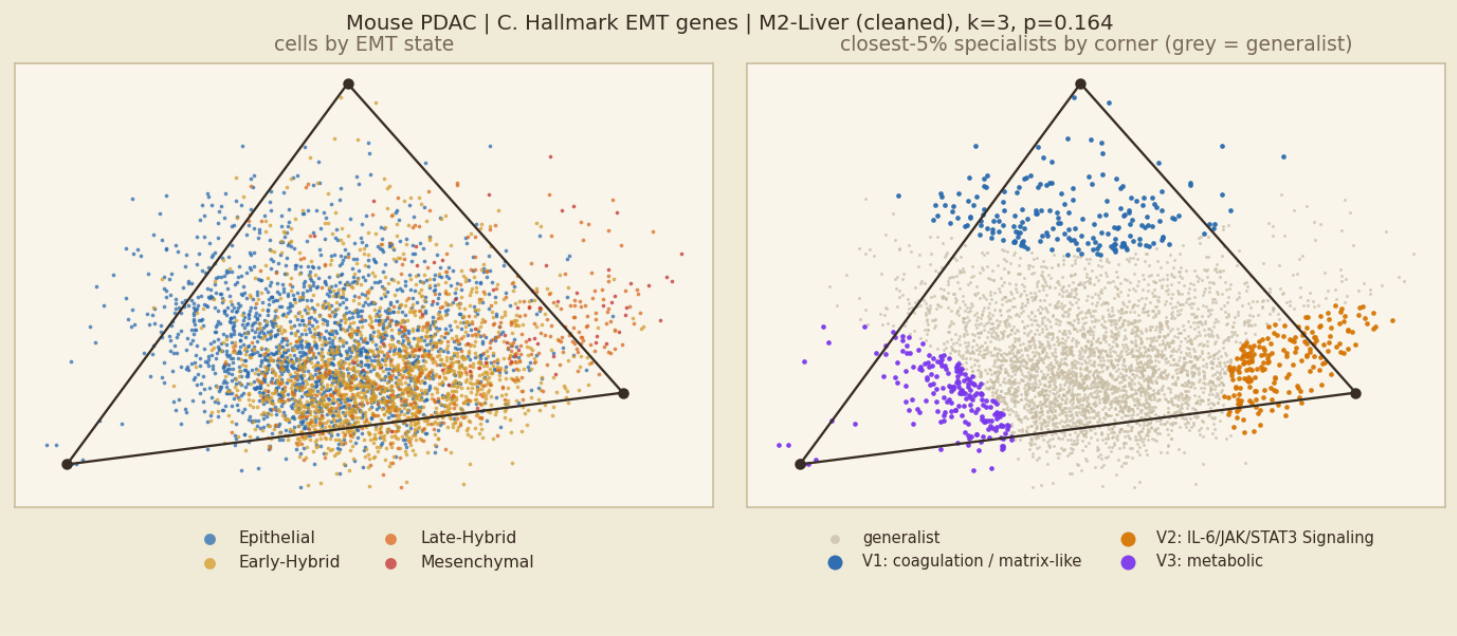
V3 hypoxia / glycolysis — Hypoxia +1.67 SD; 8% S/G2M; 41% Early-Hybrid; 100% M2-PT; Vegfa Serpine1 Igfbp3

V4 angiogenesis / matrix — Angiogenesis +1.11 SD; 4% S/G2M; 48% Early-Hybrid; 100% M2-PT; Fn1 Spp1 Fbln2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse PDAC: C. Hallmark EMT genes, unit M2-Liver

Fit shown: cleaned · k = 3 · p = 0.164 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.0249



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	75%	84%
2 closest 5% of cells to each corner	86%	84%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 53% of all cells here; 86% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M2-Liver: the fit (k=3, p=0.164) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 75% of hybrid cells and 84% of other cells are generalists (rule 1; rule 2: 86% vs 84%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 coagulation / matrix-like — Coagulation +1.55 SD; 6% S/G2M; 56% Epithelial; 100% M2-Liver; Fn1 Spp1 Fbln2

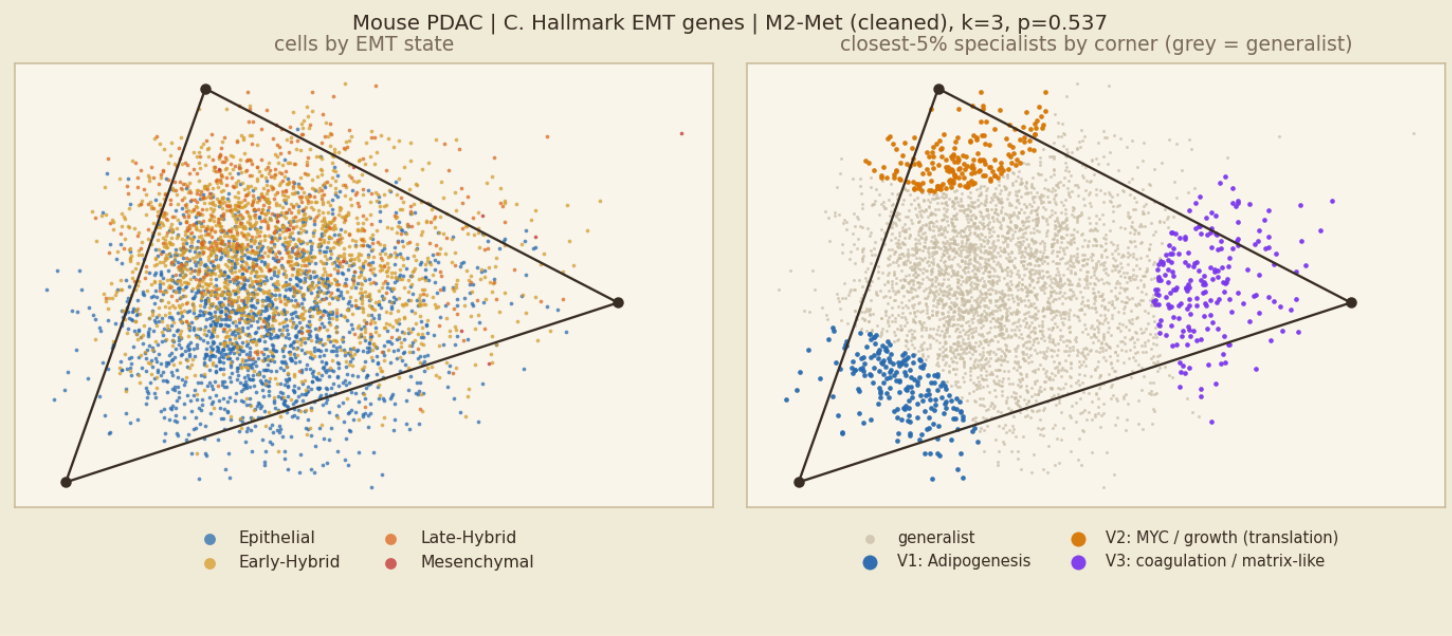
V2 IL-6/JAK/STAT3 Signaling — IL-6/JAK/STAT3 Signaling +0.95 SD; 80% S/G2M; 42% Early-Hybrid; 100% M2-Liver; Cxcl1 Areg Timp1

V3 metabolic — Oxidative Phosphorylation +0.58 SD; 59% S/G2M; 76% Epithelial; 100% M2-Liver; Mcm7 Fuca1 Pfn2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Mouse PDAC: C. Hallmark EMT genes, unit M2-Met

Fit shown: cleaned · k = 3 · p = 0.537 · 4,000 cells · 200 shuffles | comparison, light QC: k=3, p=0.0846



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	73%	85%
2 closest 5% of cells to each corner	84%	86%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 89% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M2-Met: the fit (k=3, p=0.537) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 9 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 73% of hybrid cells and 85% of other cells are generalists (rule 1; rule 2: 84% vs 86%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 Adipogenesis — Adipogenesis +0.74 SD; 44% S/G2M; 82% Epithelial; 100% M2-Met; Fuca1 Jun Lrp1

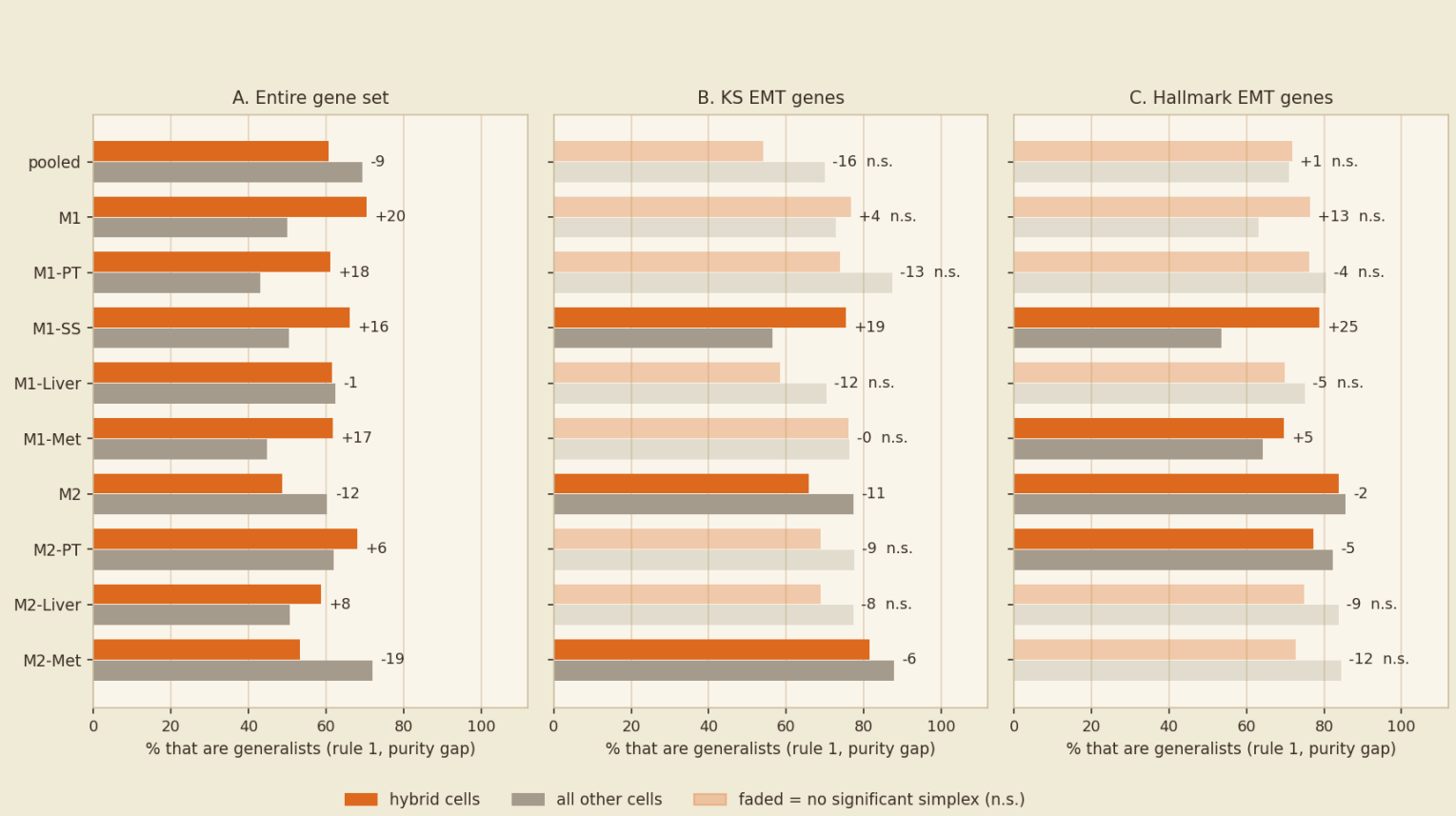
V2 MYC / growth (translation) — Myc Targets V1 +1.02 SD; 86% S/G2M; 53% Early-Hybrid; 100% M2-Met; Areg Cxcl1 Tnfrsf12a

V3 coagulation / matrix-like — Coagulation +1.54 SD; 10% S/G2M; 49% Early-Hybrid; 100% M2-Met; Fn1 Fbln2 Ecm1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Question 2 · Mouse PDAC: are hybrid cells more often generalists than other cells?

Orange = hybrid cells, grey = all other cells; one pair of bars per unit; the number = orange minus grey (percentage points)



Reading

- A. Entire gene set (10 significant): hybrids 61% vs others 55% generalist under rule 1 (rule 2: 84% vs 83%) → hybrids are slightly more often generalists.
- B. KS EMT genes (3 significant): hybrids 75% vs others 77% generalist under rule 1 (rule 2: 80% vs 80%) → hybrids are less often generalists (they sit at corners).
- C. Hallmark EMT genes (4 significant): hybrids 78% vs others 73% generalist under rule 1 (rule 2: 80% vs 79%) → no real difference.

Rule 3 (weight > 0.95) is not drawn: it calls about 100% of every group generalist, so it cannot show a difference.

4T1 clones

What the dataset is, and what to keep in mind when reading its slides

What the dataset means

Five single-cell clones of mouse 4T1 breast cancer chosen along an epithelial–mesenchymal spectrum, plus the parental line, one tumour each.

Units in the main analysis

E, EM1, EM2, EM3, M, 4T1

Gene sets

A. all genes (11,000 to 14,500); B. KS EMT genes (Tan 2014; about 160 to 190 present); C. Hallmark Epithelial-Mesenchymal-Transition set (200 genes, those present in the data).

Keep in mind

- One tumour per clone: clone, tumour and batch are the same thing.
- Clone E has only 142 cells after cleaning, so its cleaned arm is n/a.
- Only clone E is clearly separate on the EMT score; the 'spectrum' is one epithelial clone plus five similar ones.

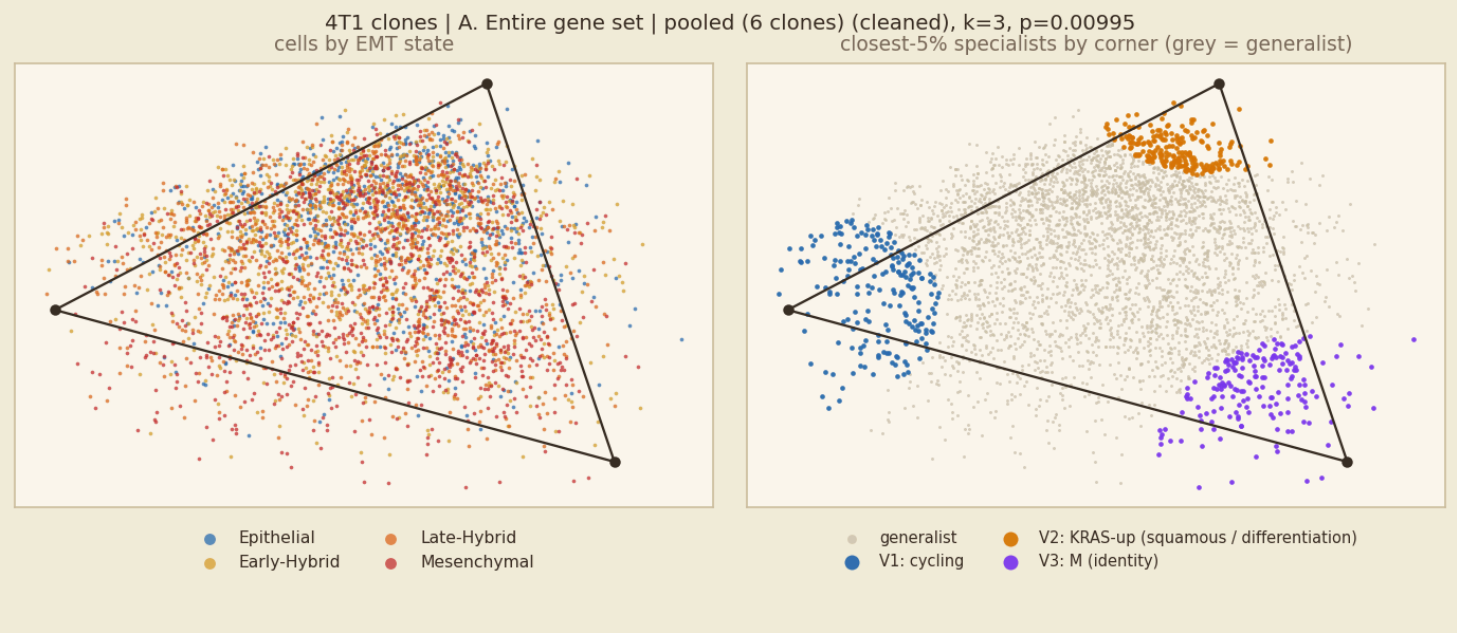
This dataset in numbers (main fits)

- Entire gene set: 2 of 7 fits significant; hybrids 62% / 90% / 100% generalist (median, rules 1/2/3).
- KS EMT genes: 2 of 7 fits significant; hybrids 72% / 85% / 100% generalist (median, rules 1/2/3).
- Hallmark EMT genes: 1 of 7 fits significant; hybrids 94% / 94% / 100% generalist (median, rules 1/2/3).

Question 1 slides come first, then the Question 2 slides for this dataset.

4T1 clones: A. Entire gene set, pooled fit

Fit shown: cleaned · k = 3 · p = 0.00995 · 13,256 cells · 200 shuffles l comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	68%	66%
2 closest 5% of cells to each corner	86%	84%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 54% of all cells here; 81% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=3, p=0.00995) whose corners mix samples (no corner is 70% or more one sample), so it is not a pure sample-identity simplex.

Corners are programmes: cycling; KRAS-up (squamous / differentiation); TGF-β response.

Hybrid vs other cells: 68% of hybrid cells and 66% of other cells are generalists (rule 1; rule 2: 86% vs 84%), so no real difference.

Corners and their tasks

V1 cycling — E2F Targets +2.19 SD; 98% S/G2M; 32% Late-Hybrid; Top2a Hist1h1e Hist1h2ap

V2 KRAS-up (squamous / differentiation) — KRAS Signaling Up +0.73 SD; 1% S/G2M; 34% Epithelial; Csf3 Plet1 Cxcl1

V3 identity corner: M (programme: TGF-β response) — TGF-beta Signaling +1.63 SD; 3% S/G2M; 48% Mesenchymal; 90% M; Ctla2a Fosb Mgp

4T1 clones: A. Entire gene set, all units at a glance

Main fit of each unit: 1 of 6 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
E	light QC	3	0.00995 *	–	2%	56 / 94 / 100%	2 / 2 / 2%
EM1	cleaned	3	0.622	1	59%	57 / 84 / 100%	58 / 58 / 59%
EM2	cleaned	3	0.627	0.642	66%	73 / 86 / 100%	66 / 67 / 66%
EM3	cleaned	4	0.403	0.95	44%	68 / 80 / 100%	45 / 44 / 44%
M	cleaned	3	0.498	0.00995	45%	59 / 86 / 100%	46 / 46 / 45%
4T1	cleaned	4	0.0597	0.00995	56%	77 / 81 / 100%	56 / 57 / 56%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

4T1 clones: A. Entire gene set, unit E

Fit shown: light QC · k = 3 · p = 0.00995 · 794 cells · 200 shuffles

Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	56%	47%
2 closest 5% of cells to each corner	94%	85%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 2% of all cells here; 68% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside E alone, cells fill a 3-corner simplex (p=0.00995). 1 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: no Hallmark set above 0.5 SD; metabolic; cycling.

Hybrid vs other cells: 56% of hybrid cells and 47% of other cells are generalists (rule 1; rule 2: 94% vs 85%), so hybrids are clearly more often generalists.

Corners and their tasks

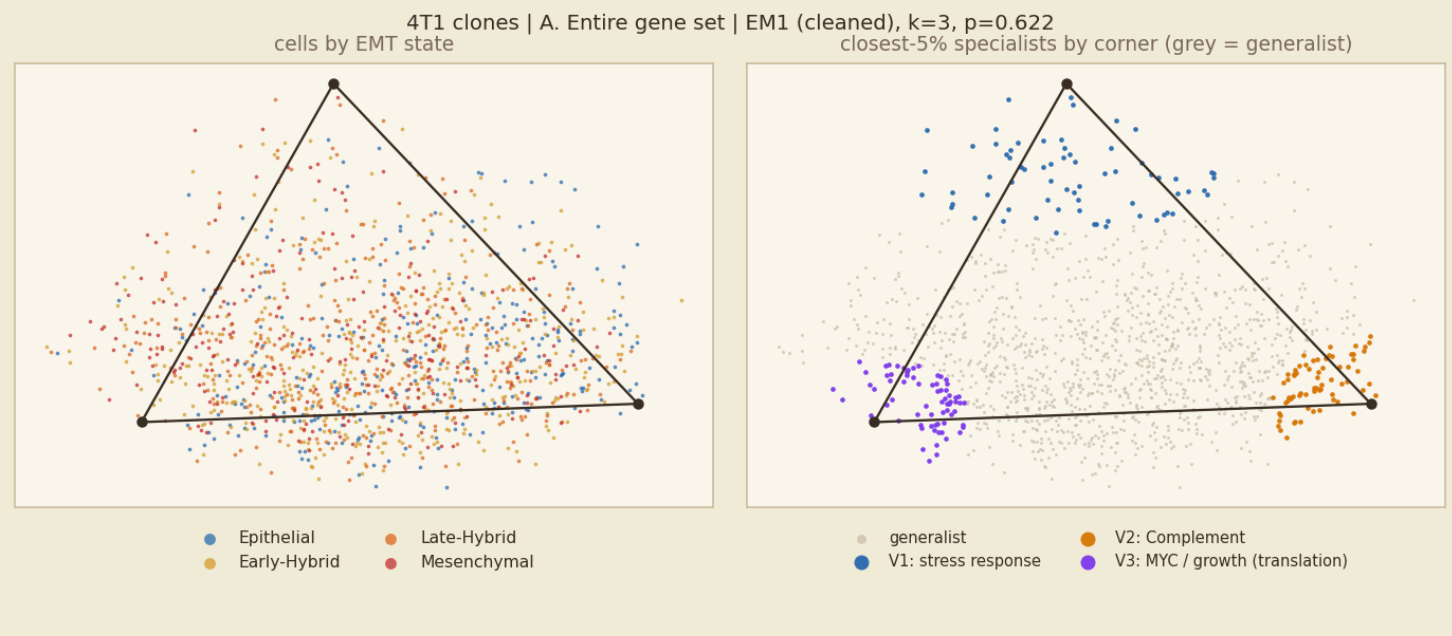
V1 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 30% S/G2M; 92% Epithelial; 100% E; S100a8 Fos H2-Aa

V2 metabolic — Oxidative Phosphorylation +1.43 SD; 28% S/G2M; 100% Epithelial; 100% E; Atf3 Fos Fosb

V3 cycling — Mitotic Spindle +0.83 SD; 8% S/G2M; 100% Epithelial; 100% E; Csf3 Itga2 Clcf1

4T1 clones: A. Entire gene set, unit EM1

Fit shown: cleaned · k = 3 · p = 0.622 · 1,365 cells · 200 shuffles | comparison, light QC: k=3, p=1



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	57%	58%
2 closest 5% of cells to each corner	84%	86%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 59% of all cells here; 69% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside EM1: the fit (k=3, p=0.622) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 57% of hybrid cells and 58% of other cells are generalists (rule 1; rule 2: 84% vs 86%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

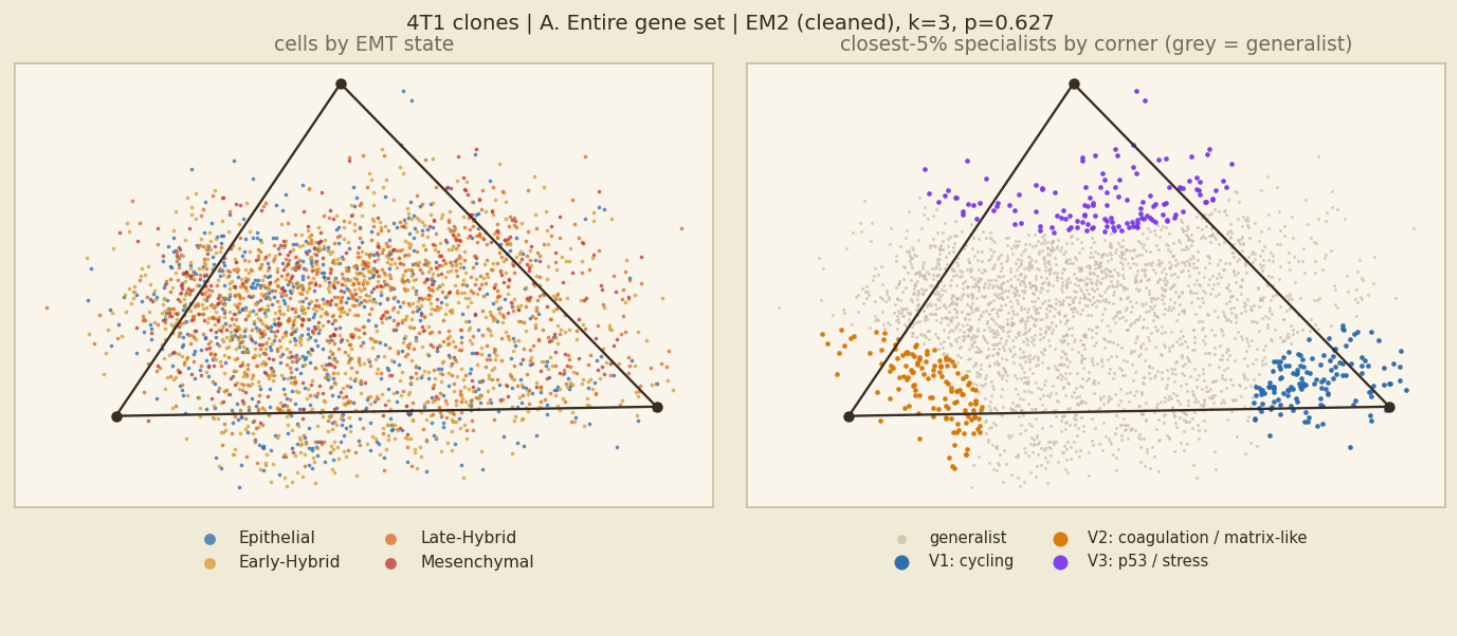
V1 stress response — UV Response Up +1.61 SD; 72% S/G2M; 32% Early-Hybrid; 100% EM1; Fos Atf3 Fosb

V2 Complement — Complement +0.85 SD; 0% S/G2M; 38% Early-Hybrid; 100% EM1; Lcn2 Mmp13 Mgp

V3 MYC / growth (translation) — Myc Targets V1 +1.13 SD; 55% S/G2M; 39% Early-Hybrid; 100% EM1; Areg Odc1 Lif

4T1 clones: A. Entire gene set, unit EM2

Fit shown: cleaned · k = 3 · p = 0.627 · 2,701 cells · 200 shuffles | comparison, light QC: k=3, p=0.642



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	73%	71%
2 closest 5% of cells to each corner	86%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 66% of all cells here; 74% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside EM2: the fit (k=3, p=0.627) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 73% of hybrid cells and 71% of other cells are generalists (rule 1; rule 2: 86% vs 83%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

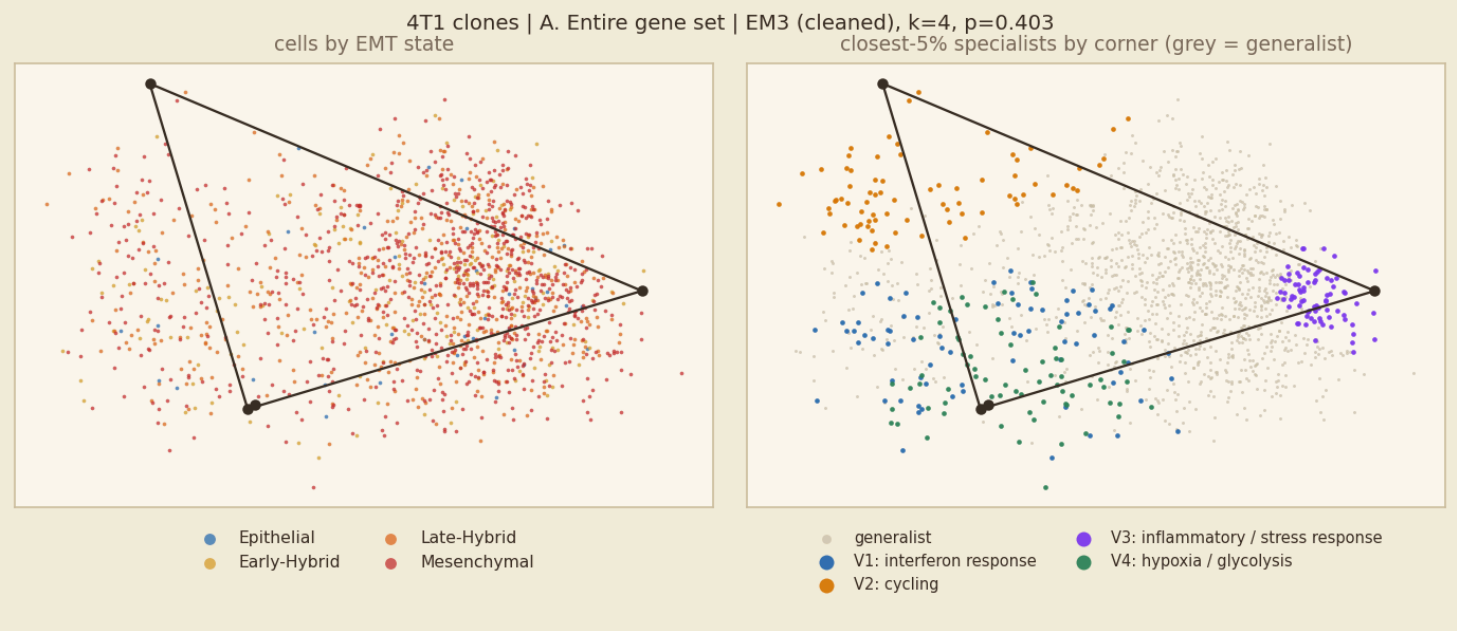
V1 cycling — G2-M Checkpoint +2.05 SD; 100% S/G2M; 41% Early-Hybrid; 100% EM2; Top2a Hist1h1e Hist1h2ap

V2 coagulation / matrix-like — Coagulation +1.05 SD; 0% S/G2M; 38% Early-Hybrid; 100% EM2; Lcn2 Serpine2 F13a1

V3 p53 / stress — p53 Pathway +1.22 SD; 73% S/G2M; 38% Early-Hybrid; 100% EM2; Fosb Sfn Atf3

4T1 clones: A. Entire gene set, unit EM3

Fit shown: cleaned · k = 4 · p = 0.403 · 1,507 cells · 200 shuffles | comparison, light QC: k=4, p=0.95



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	68%	66%
2 closest 5% of cells to each corner	80%	79%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 44% of all cells here; 23% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside EM3: the fit (k=4, p=0.403) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 68% of hybrid cells and 66% of other cells are generalists (rule 1; rule 2: 80% vs 79%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 interferon response — Interferon Alpha Response +0.96 SD; 3% S/G2M; 51% Mesenchymal; 100% EM3; Lcn2 Cp Spp1

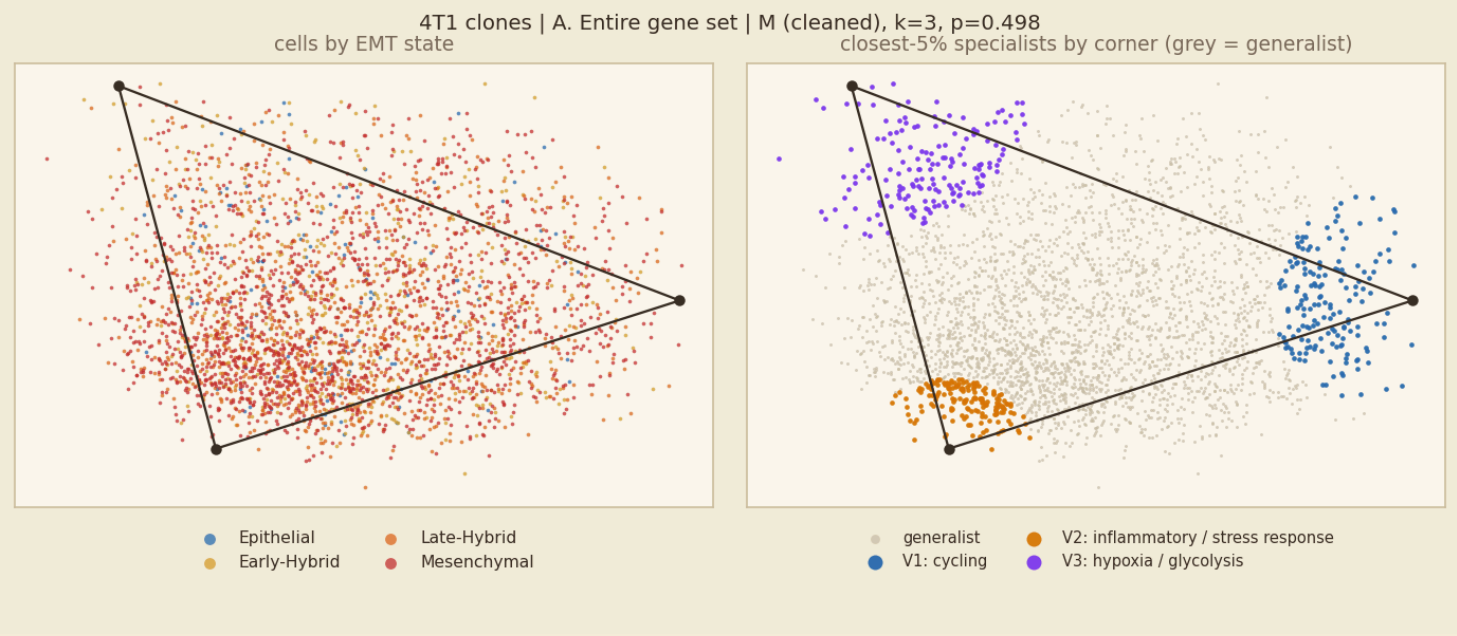
V2 cycling — E2F Targets +2.28 SD; 93% S/G2M; 57% Mesenchymal; 100% EM3; H1f0 Stmn1 Ccnd1

V3 inflammatory / stress response — Inflammatory Response +0.55 SD; 5% S/G2M; 57% Mesenchymal; 100% EM3; Areg Plet1 Sfn

V4 hypoxia / glycolysis — Hypoxia +1.71 SD; 24% S/G2M; 51% Mesenchymal; 100% EM3; Fos Atf3 Jun

4T1 clones: A. Entire gene set, unit M

Fit shown: cleaned · k = 3 · p = 0.498 · 3,324 cells · 200 shuffles | comparison, light QC: k=4, p=0.00995



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	59%	57%
2 closest 5% of cells to each corner	86%	84%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 45% of all cells here; 71% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M: the fit (k=3, p=0.498) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 59% of hybrid cells and 57% of other cells are generalists (rule 1; rule 2: 86% vs 84%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

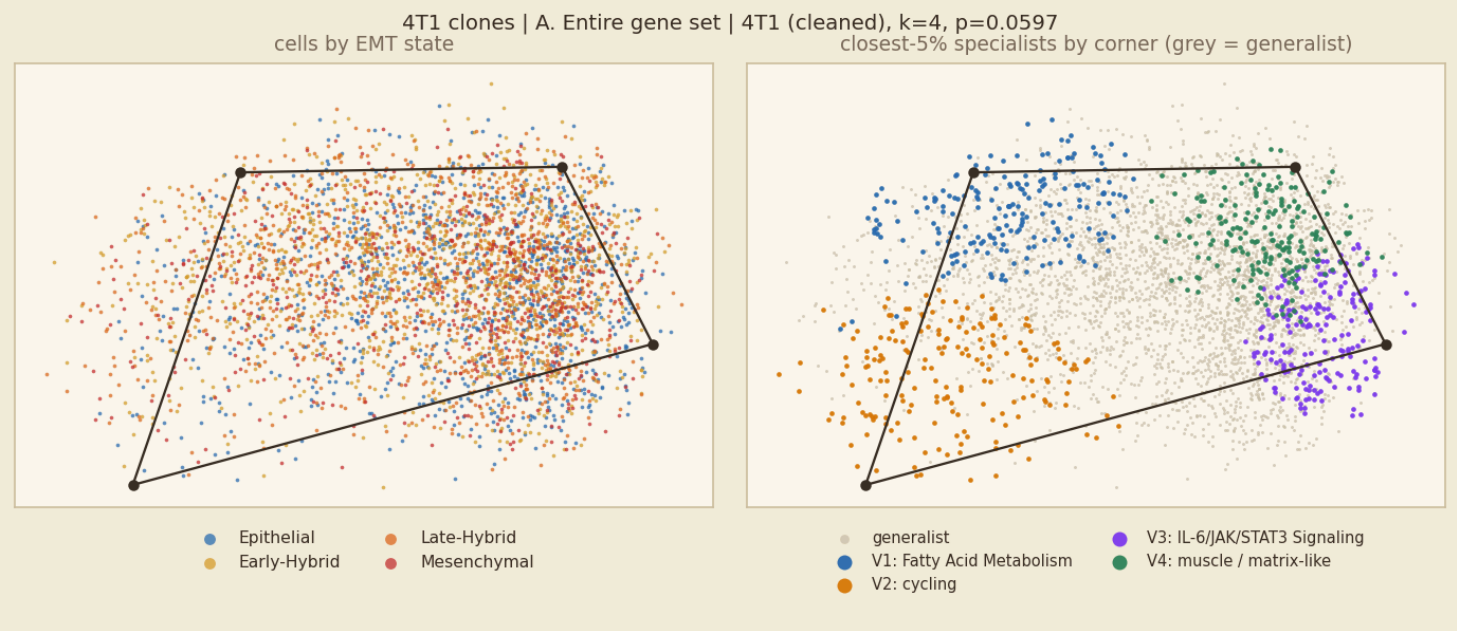
V1 cycling — E2F Targets +2.2 SD; 99% S/G2M; 57% Mesenchymal; 100% M; S100a4 Top2a Hist1h2ap

V2 inflammatory / stress response — Inflammatory Response +0.76 SD; 0% S/G2M; 63% Mesenchymal; 100% M; Ctla2a Inhba Ptgs2

V3 hypoxia / glycolysis — Hypoxia +1.65 SD; 3% S/G2M; 41% Mesenchymal; 100% M; Fos Jun Fosb

4T1 clones: A. Entire gene set, unit 4T1

Fit shown: cleaned · k = 4 · p = 0.0597 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.00995



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	77%	78%
2 closest 5% of cells to each corner	81%	79%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 39% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside 4T1: the fit (k=4, p=0.0597) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 77% of hybrid cells and 78% of other cells are generalists (rule 1; rule 2: 81% vs 79%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 Fatty Acid Metabolism — Fatty Acid Metabolism +1.29 SD; 76% S/G2M; 36% Early-Hybrid; 100% 4T1; Procr S100a4 Odc1

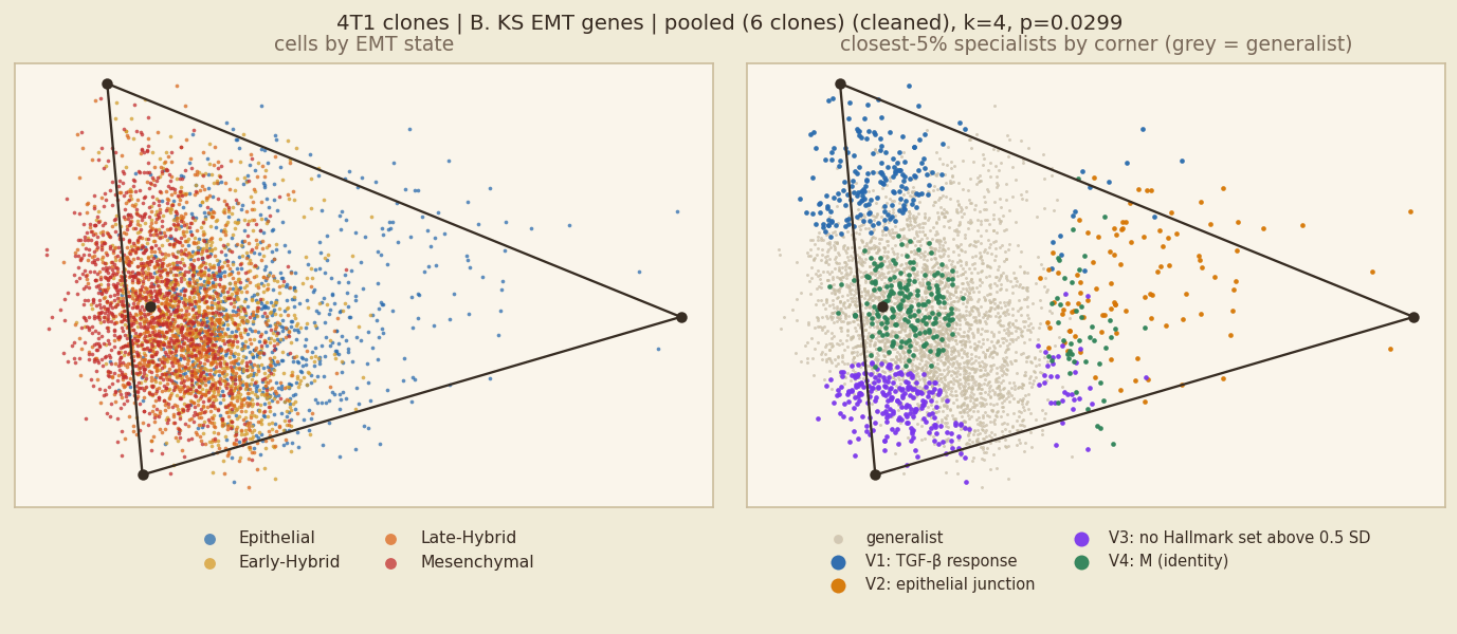
V2 cycling — E2F Targets +1.9 SD; 93% S/G2M; 36% Epithelial; 100% 4T1; Fos Atf3 Top2a

V3 IL-6/JAK/STAT3 Signaling — IL-6/JAK/STAT3 Signaling +0.89 SD; 1% S/G2M; 27% Late-Hybrid; 100% 4T1; Lcn2 Slpi Cp

V4 muscle / matrix-like — Myogenesis +0.86 SD; 2% S/G2M; 46% Epithelial; 100% 4T1; Cldn4 Jup Epcam

4T1 clones: B. KS EMT genes, pooled fit

Fit shown: cleaned · k = 4 · p = 0.0299 · 13,256 cells · 200 shuffles l comparison, light QC: k=5, p=0.174



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	75%	75%
2 closest 5% of cells to each corner	85%	74%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 54% of all cells here; 52% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=4, p=0.0299) whose corners mix samples (no corner is 70% or more one sample), so it is not a pure sample-identity simplex.

Corners are programmes: TGF-β response; epithelial junction; no Hallmark set above 0.5 SD; MYC / growth (translation).

Hybrid vs other cells: 75% of hybrid cells and 75% of other cells are generalists (rule 1; rule 2: 85% vs 74%), so no real difference.

Corners and their tasks

V1 TGF-β response — TGF-beta Signaling +0.92 SD; 10% S/G2M; 40% Mesenchymal; Lcn2 Slpi Atp1b1

V2 epithelial junction — Apical Junction +0.98 SD; 28% S/G2M; 89% Epithelial; 67% 4T1; Cldn4 Epcam Cdh1

V3 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 34% S/G2M; 36% Late-Hybrid; Areg Slpi Sfn

V4 identity corner: M (programme: MYC / growth (translation)) — Myc Targets V1 +0.74 SD; 60% S/G2M; 64% Mesenchymal; 75% M; Axl Hes1 Vim

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

4T1 clones: B. KS EMT genes, all units at a glance

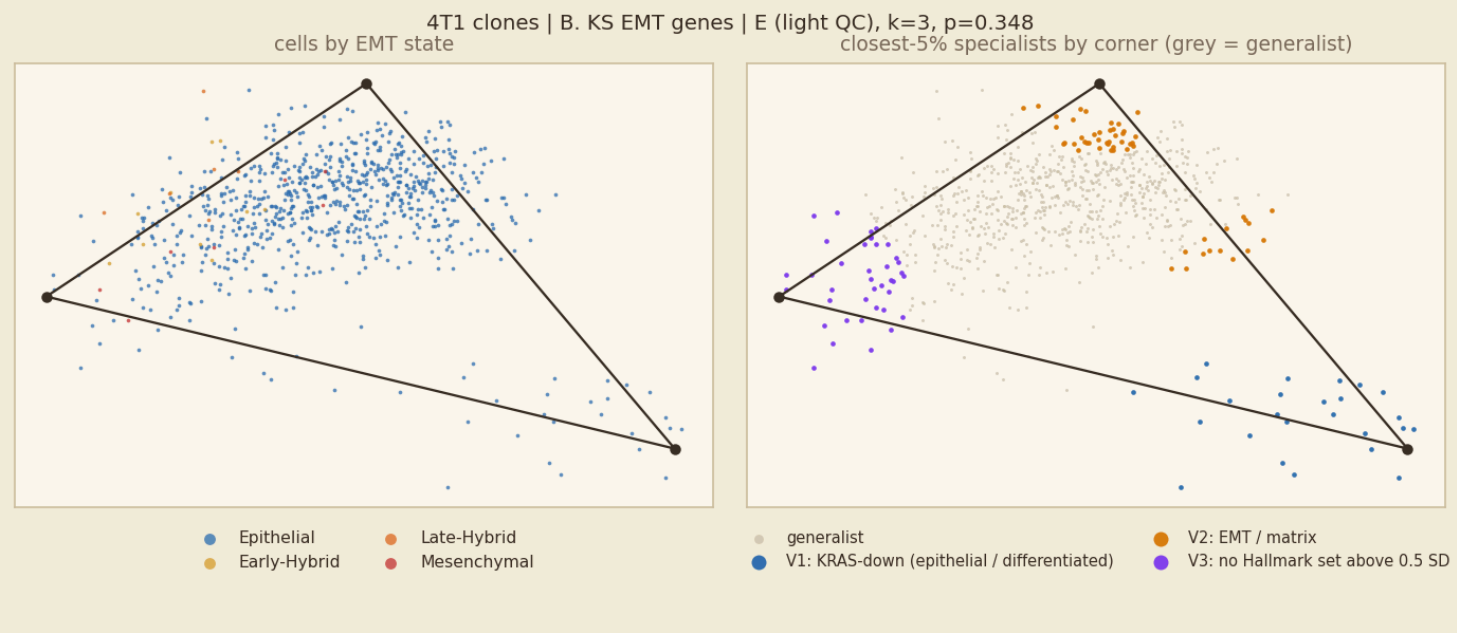
Main fit of each unit: 1 of 6 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
E	light QC	3	0.348	–	2%	75 / 81 / 100%	3 / 2 / 2%
EM1	cleaned	4	0.219	0.00498	59%	69 / 82 / 100%	58 / 60 / 59%
EM2	cleaned	3	0.0149 *	0.279	66%	69 / 86 / 100%	65 / 67 / 66%
EM3	cleaned	4	0.458	0.0746	44%	80 / 79 / 100%	43 / 43 / 44%
M	cleaned	3	0.164	0.174	45%	67 / 84 / 100%	45 / 45 / 45%
4T1	cleaned	4	0.0597	0.199	56%	72 / 82 / 100%	53 / 57 / 56%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

4T1 clones: B. KS EMT genes, unit E

Fit shown: light QC · k = 3 · p = 0.348 · 794 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	75%	52%
2 closest 5% of cells to each corner	81%	85%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 2% of all cells here; 83% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside E: the fit (k=3, p=0.348) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 75% of hybrid cells and 52% of other cells are generalists (rule 1; rule 2: 81% vs 85%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +0.81 SD; 5% S/G2M; 100% Epithelial; 100% E; Krt19 Elf3 Krt8

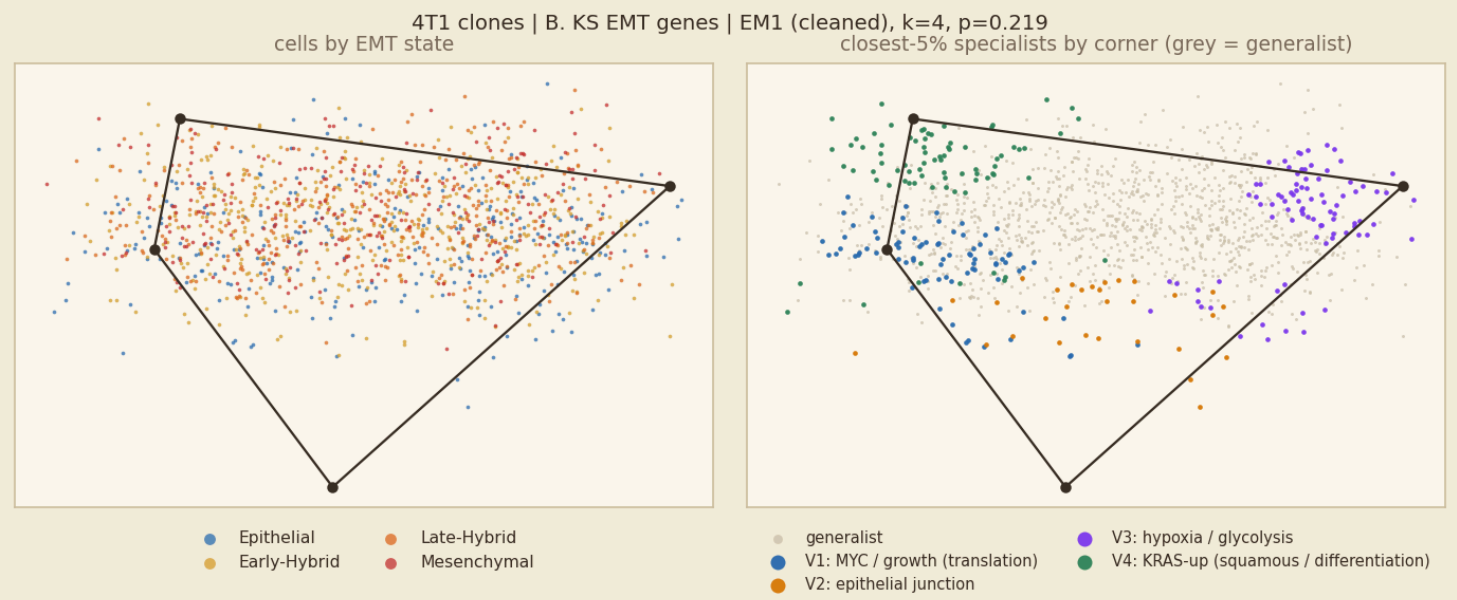
V2 EMT / matrix — Epithelial Mesenchymal Transition +1.03 SD; 30% S/G2M; 100% Epithelial; 100% E; Slpi Vim Gja1

V3 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 28% S/G2M; 88% Epithelial; 100% E; Sparc Mxra7 Chn1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

4T1 clones: B. KS EMT genes, unit EM1

Fit shown: cleaned · k = 4 · p = 0.219 · 1,365 cells · 200 shuffles | comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	69%	70%
2 closest 5% of cells to each corner	82%	76%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 59% of all cells here; 47% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside EM1: the fit (k=4, p=0.219) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 69% of hybrid cells and 70% of other cells are generalists (rule 1; rule 2: 82% vs 76%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 MYC / growth (translation) — Myc Targets V1 +1.6 SD; 65% S/G2M; 42% Mesenchymal; 100% EM1; Areg Tubb6 Sfn

V2 epithelial junction — Apical Junction +0.83 SD; 17% S/G2M; 59% Epithelial; 100% EM1; Cldn4 Epcam Krt7

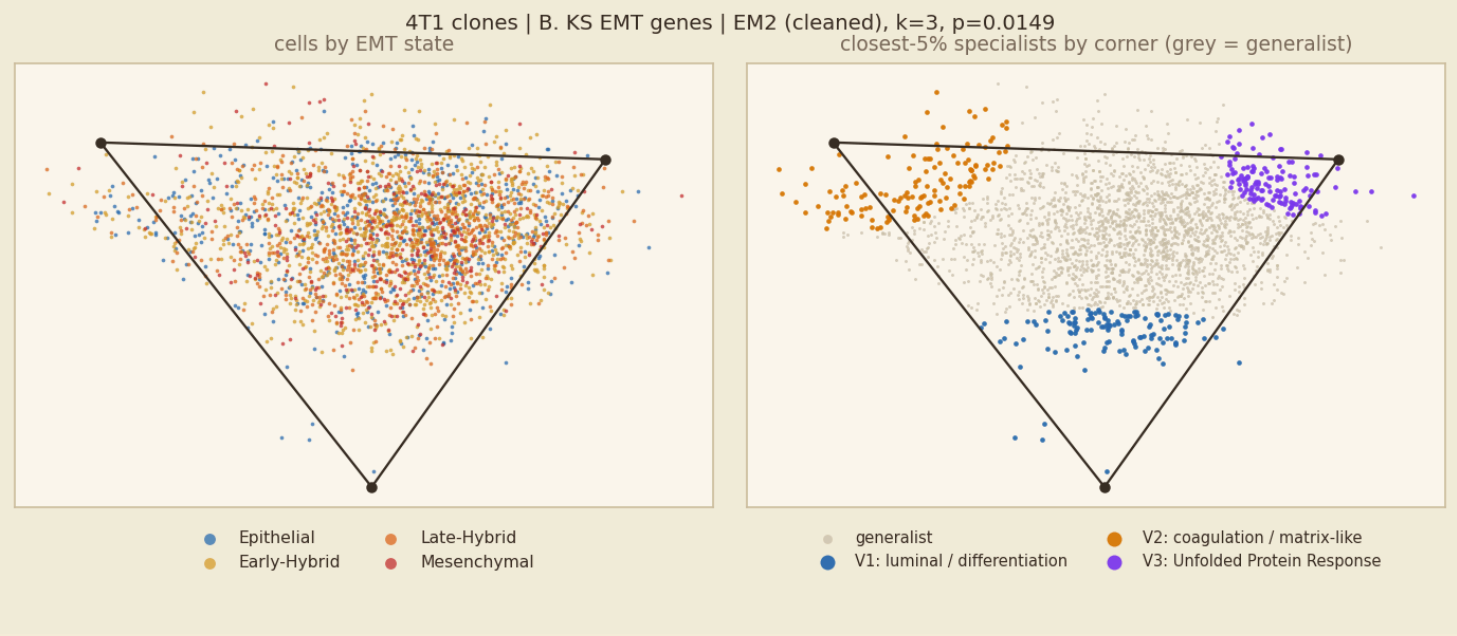
V3 hypoxia / glycolysis — Hypoxia +0.75 SD; 23% S/G2M; 30% Mesenchymal; 100% EM1; Lcn2 Atp1b1 Sparc

V4 KRAS-up (squamous / differentiation) — KRAS Signaling Up +0.6 SD; 23% S/G2M; 38% Early-Hybrid; 100% EM1; Areg Slpi Sdc1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

4T1 clones: B. KS EMT genes, unit EM2

Fit shown: cleaned · k = 3 · p = 0.0149 · 2,701 cells · 200 shuffles l comparison, light QC: k=3, p=0.279



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	69%	71%
2 closest 5% of cells to each corner	86%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 66% of all cells here; 90% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside EM2 alone, cells fill a 3-corner simplex (p=0.0149). 1 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: luminal / differentiation; coagulation / matrix-like; Unfolded Protein Response.

Hybrid vs other cells: 69% of hybrid cells and 71% of other cells are generalists (rule 1; rule 2: 86% vs 83%), so no real difference.

Corners and their tasks

V1 luminal / differentiation — Estrogen Response Late +0.9 SD; 74% S/G2M; 38% Early-Hybrid; 100% EM2; Sfn Tubb6 Epcam

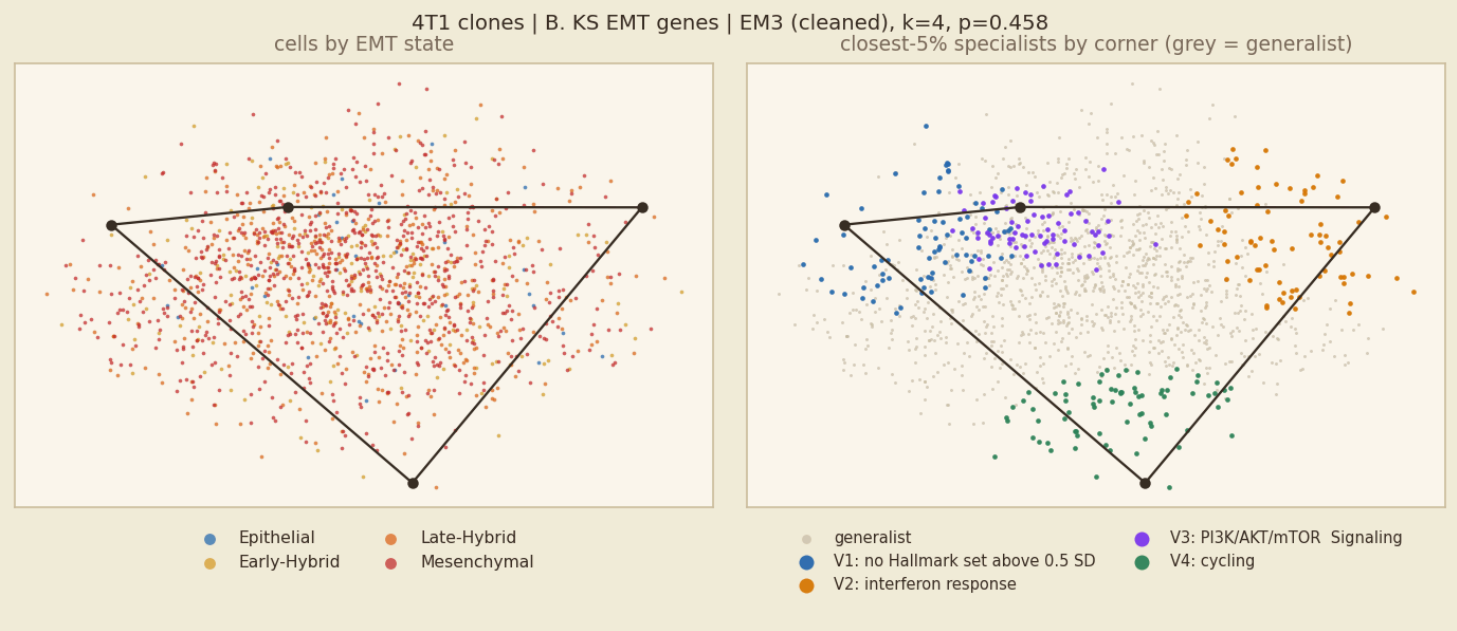
V2 coagulation / matrix-like — Coagulation +0.71 SD; 4% S/G2M; 35% Early-Hybrid; 100% EM2; Lcn2 Jup Serpine1

V3 Unfolded Protein Response — Unfolded Protein Response +0.78 SD; 50% S/G2M; 46% Early-Hybrid; 100% EM2; Slpi Areg Rbm47

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

4T1 clones: B. KS EMT genes, unit EM3

Fit shown: cleaned · k = 4 · p = 0.458 · 1,507 cells · 200 shuffles | comparison, light QC: k=4, p=0.0746



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	80%	83%
2 closest 5% of cells to each corner	79%	81%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 44% of all cells here; 39% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside EM3: the fit (k=4, p=0.458) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 80% of hybrid cells and 83% of other cells are generalists (rule 1; rule 2: 79% vs 81%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 24% S/G2M; 54% Mesenchymal; 100% EM3; Areg Klf5 Sfn

V2 interferon response — Interferon Alpha Response +0.57 SD; 7% S/G2M; 51% Mesenchymal; 100% EM3; Slpi Lcn2 Etv1

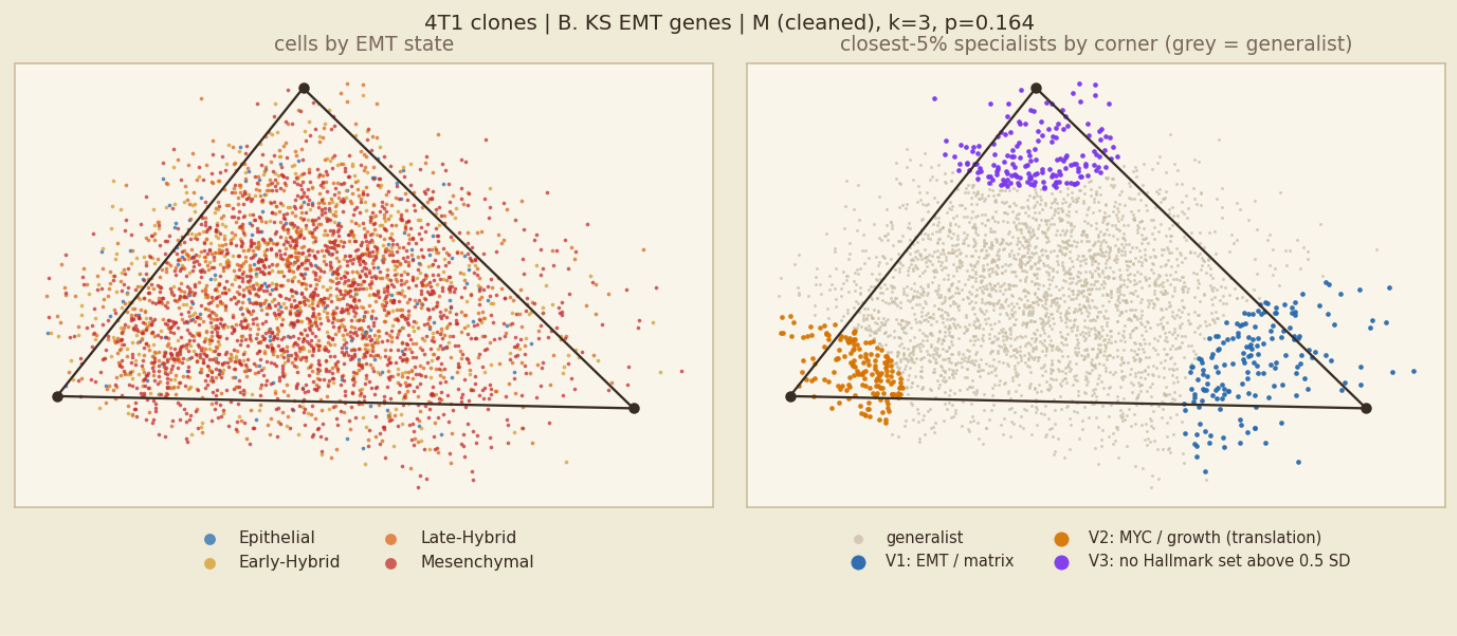
V3 PI3K/AKT/mTOR Signaling — PI3K/AKT/mTOR Signaling +0.62 SD; 26% S/G2M; 49% Mesenchymal; 100% EM3; Areg Slpi Sfn

V4 cycling — E2F Targets +1.21 SD; 58% S/G2M; 51% Mesenchymal; 100% EM3; Krt18 Hes1 Krt8

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

4T1 clones: B. KS EMT genes, unit M

Fit shown: cleaned · k = 3 · p = 0.164 · 3,324 cells · 200 shuffles | comparison, light QC: k=4, p=0.174



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	67%	69%
2 closest 5% of cells to each corner	84%	85%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 45% of all cells here; 83% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M: the fit (k=3, p=0.164) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 67% of hybrid cells and 69% of other cells are generalists (rule 1; rule 2: 84% vs 85%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.1 SD; 2% S/G2M; 61% Mesenchymal; 100% M; Lcn2 Atp1b1 Serpine1

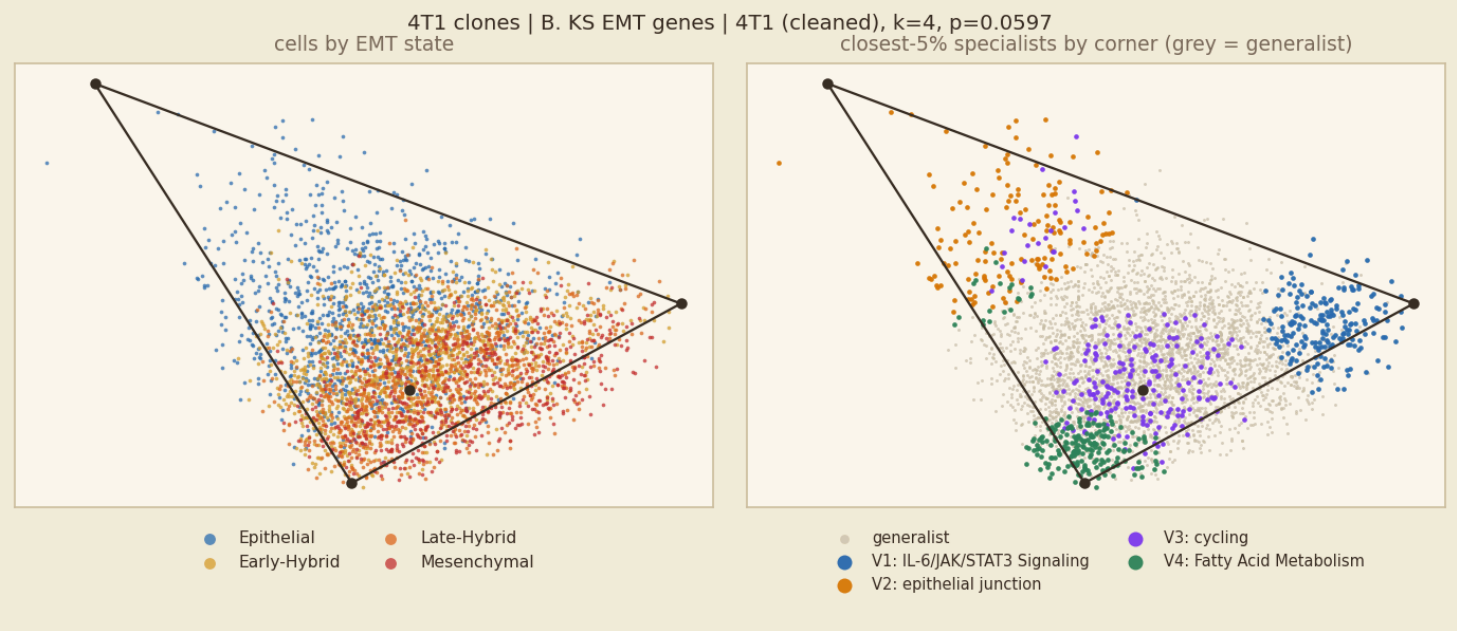
V2 MYC / growth (translation) — Myc Targets V1 +1.3 SD; 80% S/G2M; 50% Mesenchymal; 100% M; Krt8 Krt18 Krt7

V3 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 29% S/G2M; 38% Mesenchymal; 100% M; Slpi Areg Sdc1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

4T1 clones: B. KS EMT genes, unit 4T1

Fit shown: cleaned · k = 4 · p = 0.0597 · 4,000 cells · 200 shuffles l comparison, light QC: k=4, p=0.199



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	72%	80%
2 closest 5% of cells to each corner	82%	78%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 53% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside 4T1: the fit (k=4, p=0.0597) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 72% of hybrid cells and 80% of other cells are generalists (rule 1; rule 2: 82% vs 78%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 IL-6/JAK/STAT3 Signaling — IL-6/JAK/STAT3 Signaling +0.59 SD; 5% S/G2M; 34% Early-Hybrid; 100% 4T1; Slpi Lcn2 Etv1

V2 epithelial junction — Apical Junction +1.28 SD; 27% S/G2M; 96% Epithelial; 100% 4T1; Cldn4 Epcam Cdh1

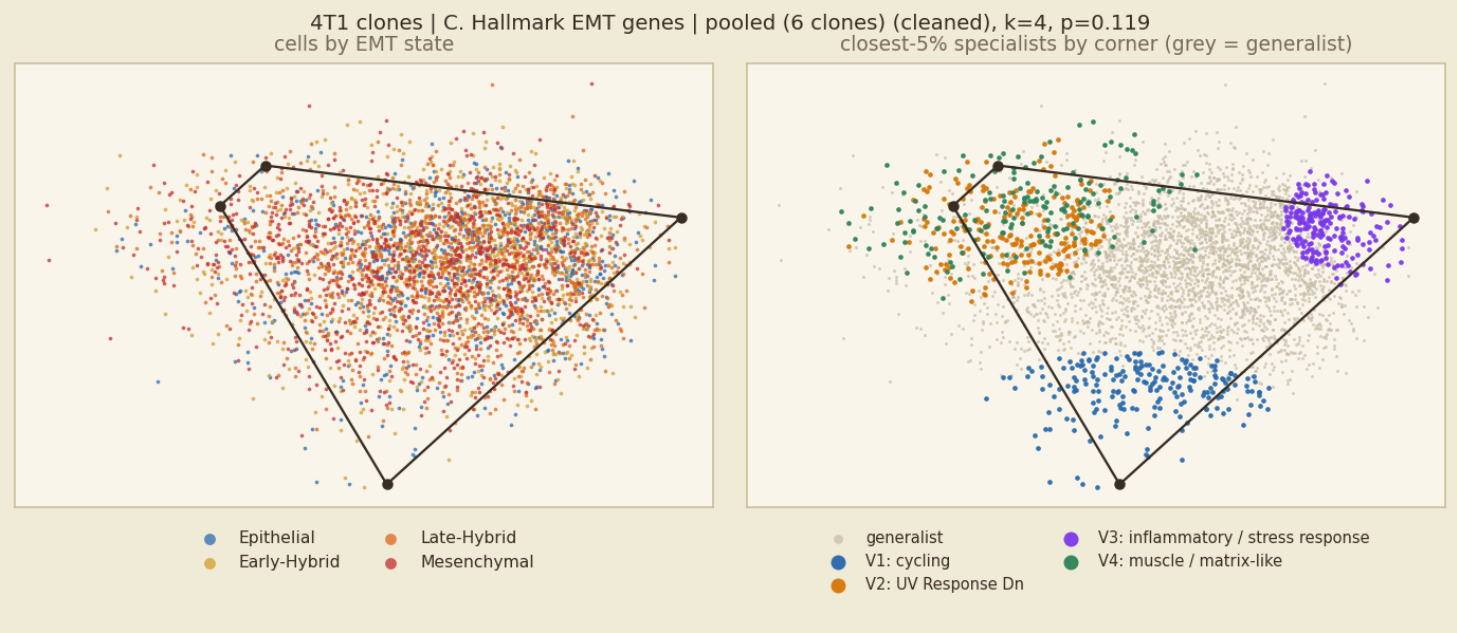
V3 cycling — E2F Targets +0.58 SD; 45% S/G2M; 37% Late-Hybrid; 100% 4T1; Hes1 Sh2b3 Tob1

V4 Fatty Acid Metabolism — Fatty Acid Metabolism +0.99 SD; 67% S/G2M; 40% Late-Hybrid; 100% 4T1; Sfn Areg Pmp22

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

4T1 clones: C. Hallmark EMT genes, pooled fit

Fit shown: cleaned · k = 4 · p = 0.119 · 13,256 cells · 100 shuffles | comparison, light QC: k=4, p=0.0198



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	75%	78%
2 closest 5% of cells to each corner	81%	79%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 54% of all cells here; 47% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex: the pooled fit (k=4, p=0.119) is no more triangular than shuffled data, so its corners should not be read as tasks.

Hybrid vs other cells: 75% of hybrid cells and 78% of other cells are generalists (rule 1; rule 2: 81% vs 79%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 cycling — E2F Targets +1.2 SD; 81% S/G2M; 29% Mesenchymal; Jun Mcm7 Rhob

V2 UV Response Dn — UV Response Dn +0.63 SD; 7% S/G2M; 39% Mesenchymal; Serpine2 Mgp Gadd45a

V3 inflammatory / stress response — Inflammatory Response +0.7 SD; 21% S/G2M; 31% Early-Hybrid; 64% 4T1; Areg Cxcl1 Inhba

V4 muscle / matrix-like — Myogenesis +1.31 SD; 15% S/G2M; 35% Mesenchymal; 64% EM1; Ccn1 Mgp Thbs1

Hallmark EMT set left out when naming (circular: fitted on those genes).

4T1 clones: C. Hallmark EMT genes, all units at a glance

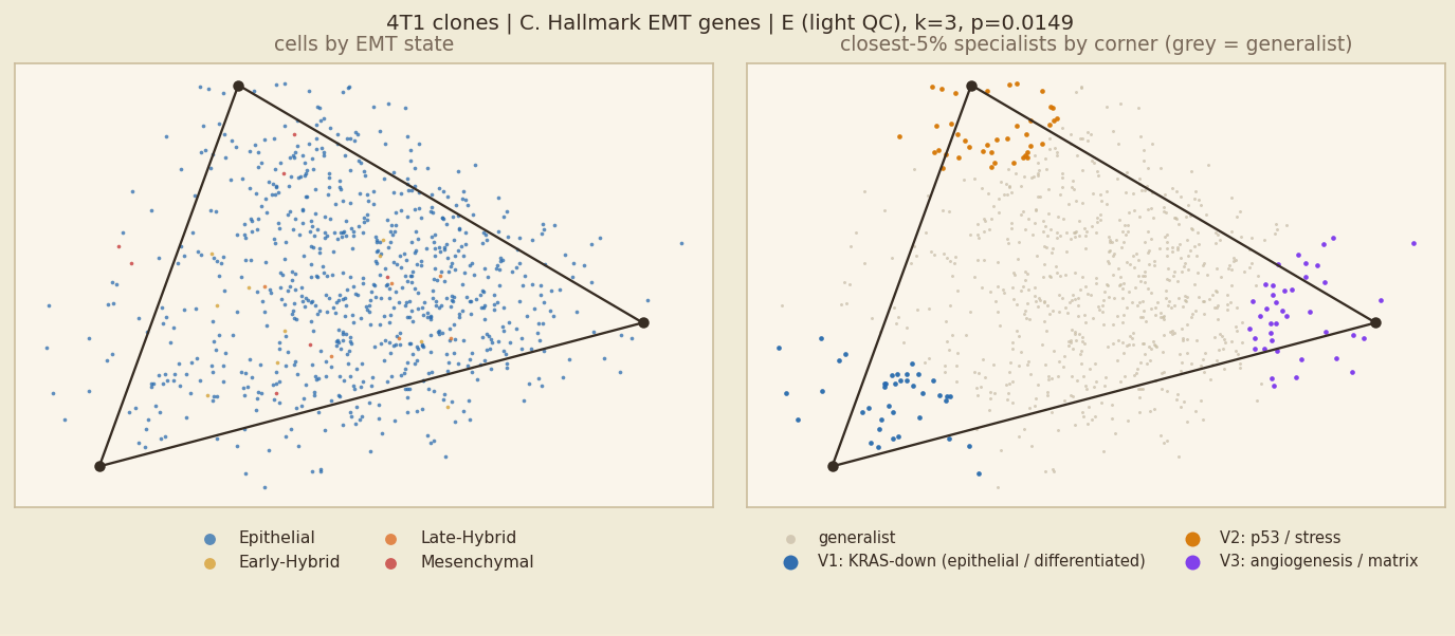
Main fit of each unit: 1 of 6 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
E	light QC	3	0.0149 *	–	2%	94 / 94 / 100%	3 / 2 / 2%
EM1	cleaned	4	0.403	0.00498	59%	70 / 80 / 100%	58 / 58 / 59%
EM2	cleaned	4	0.139	0.209	66%	73 / 79 / 100%	65 / 66 / 66%
EM3	cleaned	3	0.124	0.134	44%	74 / 85 / 100%	43 / 44 / 44%
M	cleaned	4	0.398	0.408	45%	77 / 80 / 100%	45 / 45 / 45%
4T1	cleaned	4	0.562	0.905	56%	78 / 80 / 100%	56 / 56 / 56%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

4T1 clones: C. Hallmark EMT genes, unit E

Fit shown: light QC · k = 3 · p = 0.0149 · 794 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	94%	63%
2 closest 5% of cells to each corner	94%	85%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 2% of all cells here; 80% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside E alone, cells fill a 3-corner simplex (p=0.0149). 1 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); p53 / stress; angiogenesis / matrix.

Hybrid vs other cells: 94% of hybrid cells and 63% of other cells are generalists (rule 1; rule 2: 94% vs 85%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 KRAS-down (epithelial / differentiated) — no Hallmark set above 0.5 SD; 18% S/G2M; 98% Epithelial; 100% E; Sdc4 Pdlim4 Timp3

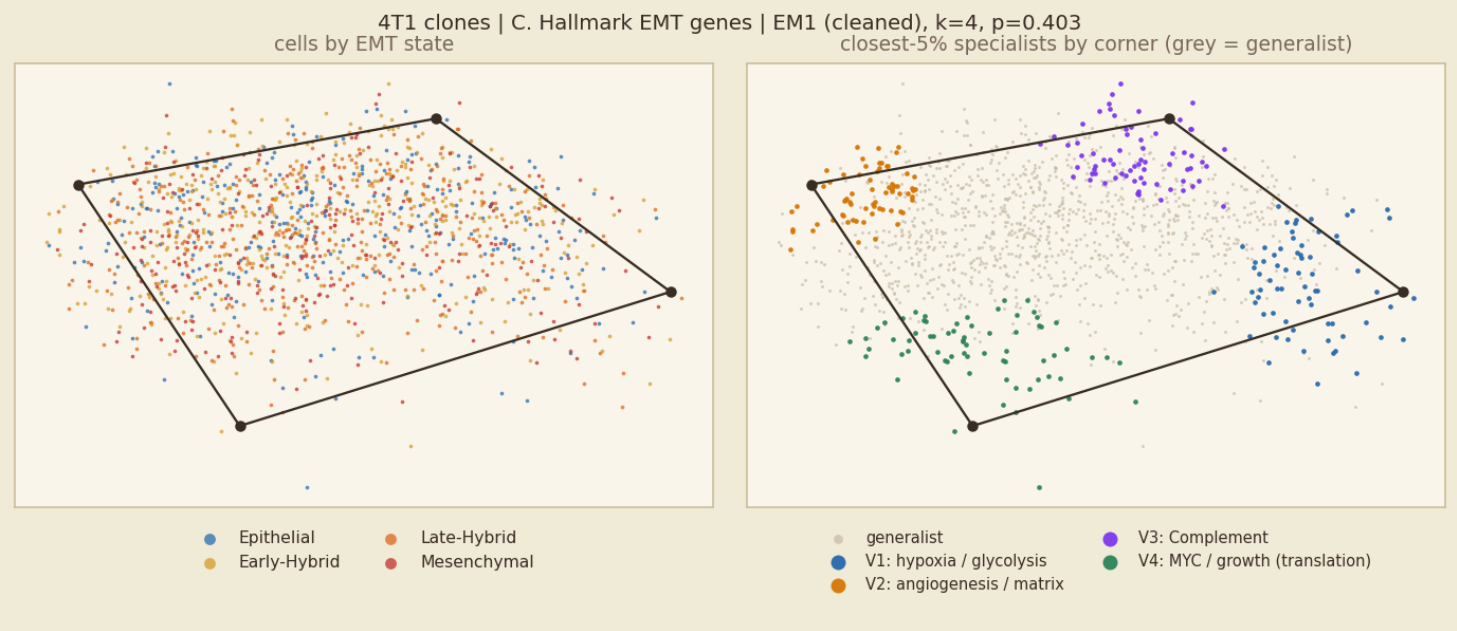
V2 p53 / stress — p53 Pathway +1.18 SD; 15% S/G2M; 98% Epithelial; 100% E; Jun Gadd45a Rhob

V3 angiogenesis / matrix — Angiogenesis +1.31 SD; 15% S/G2M; 100% Epithelial; 100% E; Inhba Lama3 Areg

Hallmark EMT set left out when naming (circular: fitted on those genes).

4T1 clones: C. Hallmark EMT genes, unit EM1

Fit shown: cleaned · k = 4 · p = 0.403 · 1,365 cells · 200 shuffles | comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	70%	71%
2 closest 5% of cells to each corner	80%	80%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 59% of all cells here; 43% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside EM1: the fit (k=4, p=0.403) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 70% of hybrid cells and 71% of other cells are generalists (rule 1; rule 2: 80% vs 80%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 hypoxia / glycolysis — Hypoxia +1.33 SD; 33% S/G2M; 33% Late-Hybrid; 100% EM1; Ccn1 Thbs1 Bgn

V2 angiogenesis / matrix — Angiogenesis +1.03 SD; 23% S/G2M; 36% Early-Hybrid; 100% EM1; Areg Pthlh Cxcl1

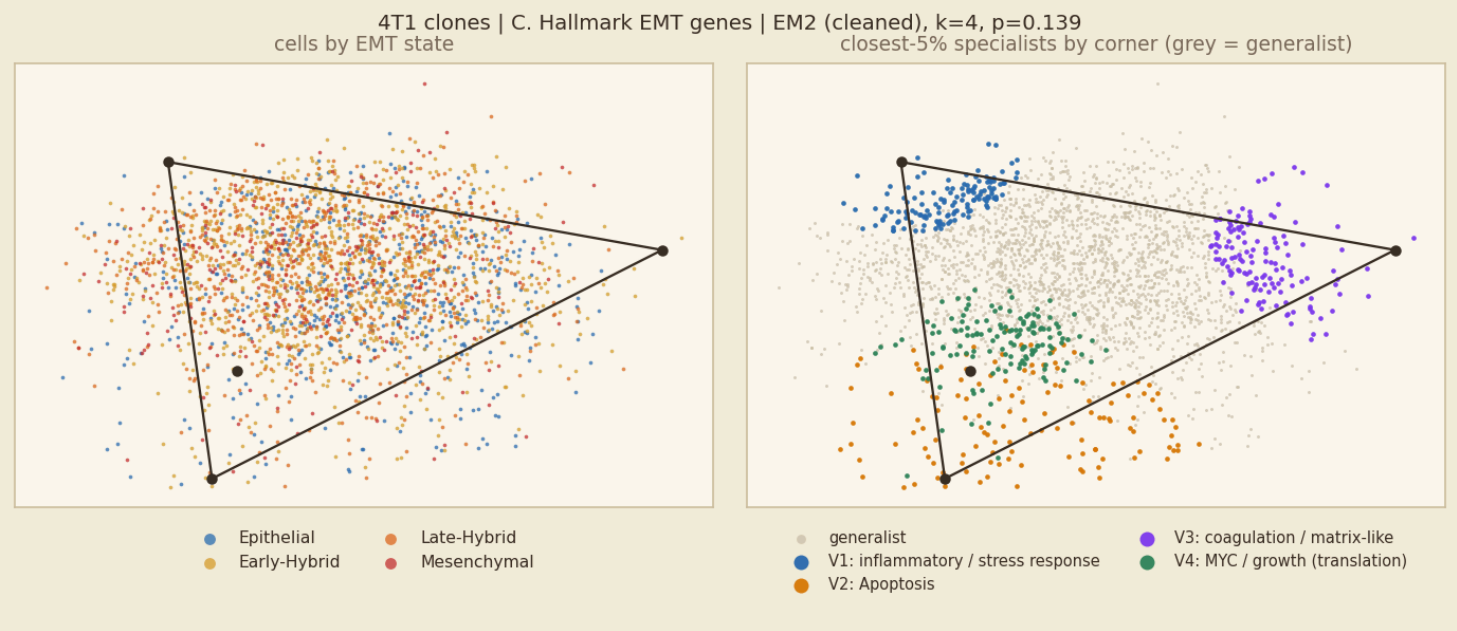
V3 Complement — Complement +0.82 SD; 0% S/G2M; 41% Early-Hybrid; 100% EM1; Mgp Crlf1 Serpine2

V4 MYC / growth (translation) — Myc Targets V1 +1.11 SD; 65% S/G2M; 29% Early-Hybrid; 100% EM1; Jun Mcm7 Gadd45b

Hallmark EMT set left out when naming (circular: fitted on those genes).

4T1 clones: C. Hallmark EMT genes, unit EM2

Fit shown: cleaned · k = 4 · p = 0.139 · 2,701 cells · 200 shuffles | comparison, light QC: k=3, p=0.209



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	73%	77%
2 closest 5% of cells to each corner	79%	81%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 66% of all cells here; 44% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside EM2: the fit (k=4, p=0.139) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 73% of hybrid cells and 77% of other cells are generalists (rule 1; rule 2: 79% vs 81%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 inflammatory / stress response — Inflammatory Response +0.89 SD; 64% S/G2M; 43% Early-Hybrid; 100% EM2; Cxcl1 Areg Inhba

V2 Apoptosis — Apoptosis +1.66 SD; 74% S/G2M; 35% Early-Hybrid; 100% EM2; Jun Ccn1 Gadd45b

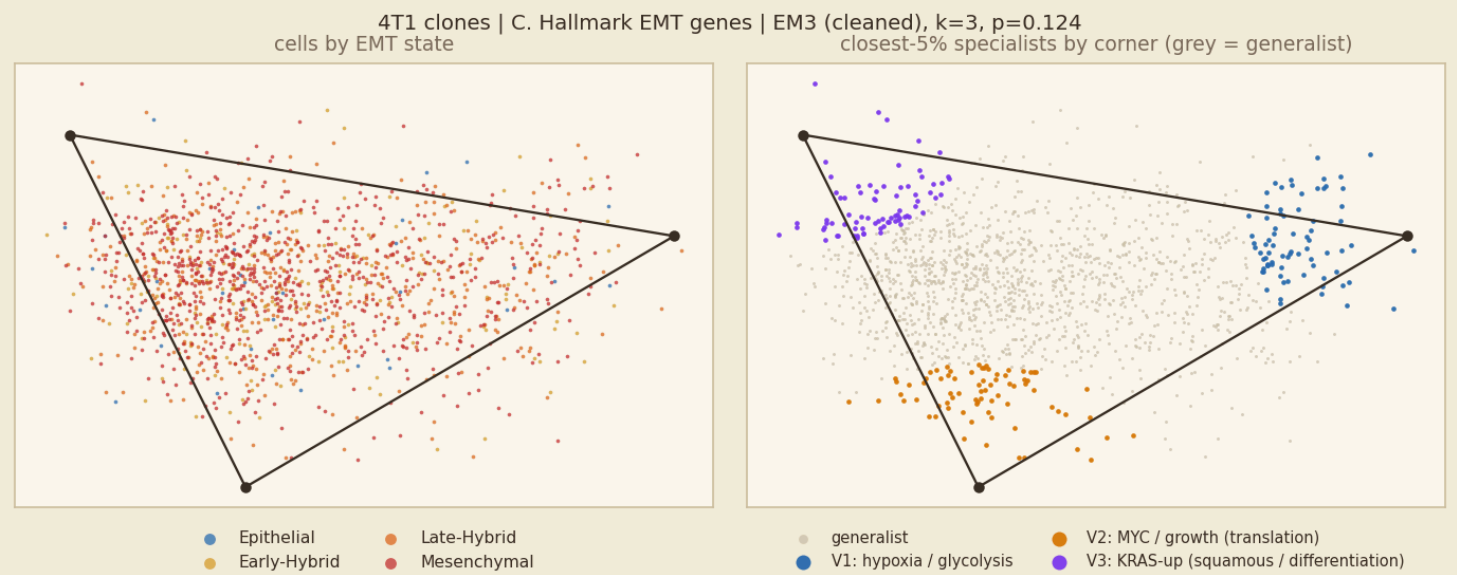
V3 coagulation / matrix-like — Coagulation +1.05 SD; 3% S/G2M; 38% Early-Hybrid; 100% EM2; Serpine2 Crlf1 Spp1

V4 MYC / growth (translation) — Myc Targets V1 +1.03 SD; 77% S/G2M; 36% Early-Hybrid; 100% EM2; Col4a1 Col4a2 Mcm7

Hallmark EMT set left out when naming (circular: fitted on those genes).

4T1 clones: C. Hallmark EMT genes, unit EM3

Fit shown: cleaned · k = 3 · p = 0.124 · 1,507 cells · 200 shuffles | comparison, light QC: k=3, p=0.134



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	74%	76%
2 closest 5% of cells to each corner	85%	85%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 44% of all cells here; 85% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside EM3: the fit (k=3, p=0.124) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 74% of hybrid cells and 76% of other cells are generalists (rule 1; rule 2: 85% vs 85%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 hypoxia / glycolysis — Hypoxia +1.39 SD; 9% S/G2M; 53% Mesenchymal; 100% EM3; Jun Ccn1 Gadd45a

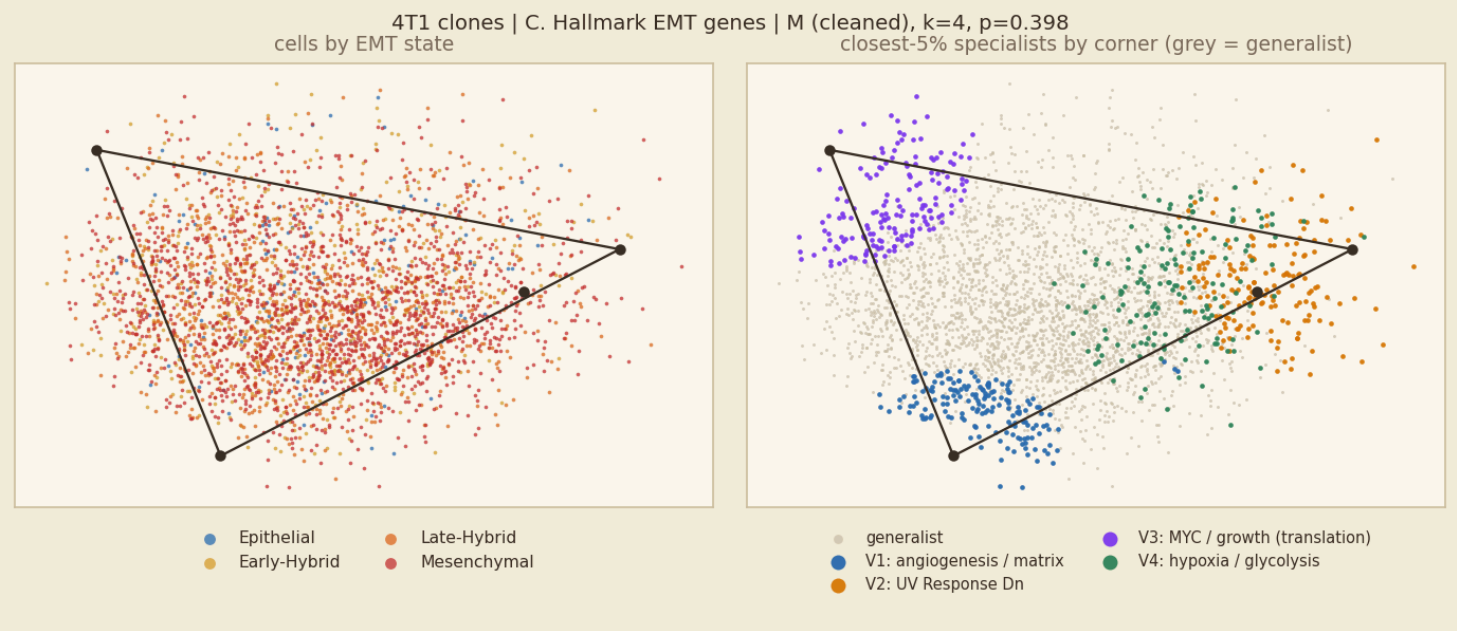
V2 MYC / growth (translation) — Myc Targets V1 +1.05 SD; 51% S/G2M; 47% Mesenchymal; 100% EM3; Pthlh Lgals1 Vim

V3 KRAS-up (squamous / differentiation) — KRAS Signaling Up +1.23 SD; 7% S/G2M; 55% Mesenchymal; 100% EM3; Inhba Areg Cxcl1

Hallmark EMT set left out when naming (circular: fitted on those genes).

4T1 clones: C. Hallmark EMT genes, unit M

Fit shown: cleaned · k = 4 · p = 0.398 · 3,324 cells · 200 shuffles | comparison, light QC: k=3, p=0.408



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	77%	78%
2 closest 5% of cells to each corner	80%	80%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 45% of all cells here; 45% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside M: the fit (k=4, p=0.398) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 77% of hybrid cells and 78% of other cells are generalists (rule 1; rule 2: 80% vs 80%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 angiogenesis / matrix — Angiogenesis +0.56 SD; 41% S/G2M; 47% Mesenchymal; 100% M; Areg Inhba Cxcl1

V2 UV Response Dn — UV Response Dn +1.45 SD; 6% S/G2M; 63% Mesenchymal; 100% M; Ccn2 Ccn1 Mgp

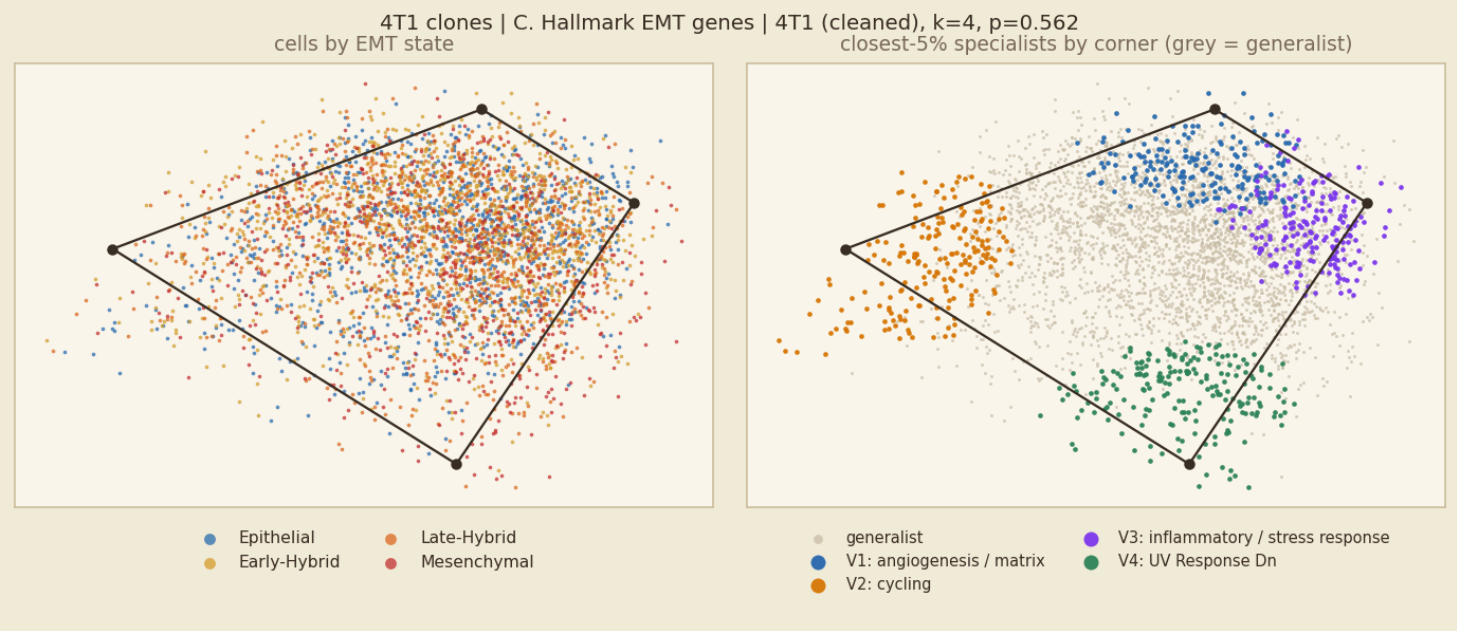
V3 MYC / growth (translation) — Myc Targets V1 +1.26 SD; 82% S/G2M; 44% Mesenchymal; 100% M; Jun Lgals1 Vim

V4 hypoxia / glycolysis — Hypoxia +0.82 SD; 7% S/G2M; 49% Mesenchymal; 100% M; Mgp Serpine2 Spp1

Hallmark EMT set left out when naming (circular: fitted on those genes).

4T1 clones: C. Hallmark EMT genes, unit 4T1

Fit shown: cleaned · k = 4 · p = 0.562 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.905



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	78%	78%
2 closest 5% of cells to each corner	80%	80%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 43% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside 4T1: the fit (k=4, p=0.562) is no more triangular than shuffled data, so its corners should not be read as tasks. 1 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 78% of hybrid cells and 78% of other cells are generalists (rule 1; rule 2: 80% vs 80%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 angiogenesis / matrix — Angiogenesis +0.77 SD; 36% S/G2M; 38% Epithelial; 100% 4T1; Inhba Lamc2 Itga5

V2 cycling — E2F Targets +1.28 SD; 74% S/G2M; 30% Late-Hybrid; 100% 4T1; Jun Htra1 Mcm7

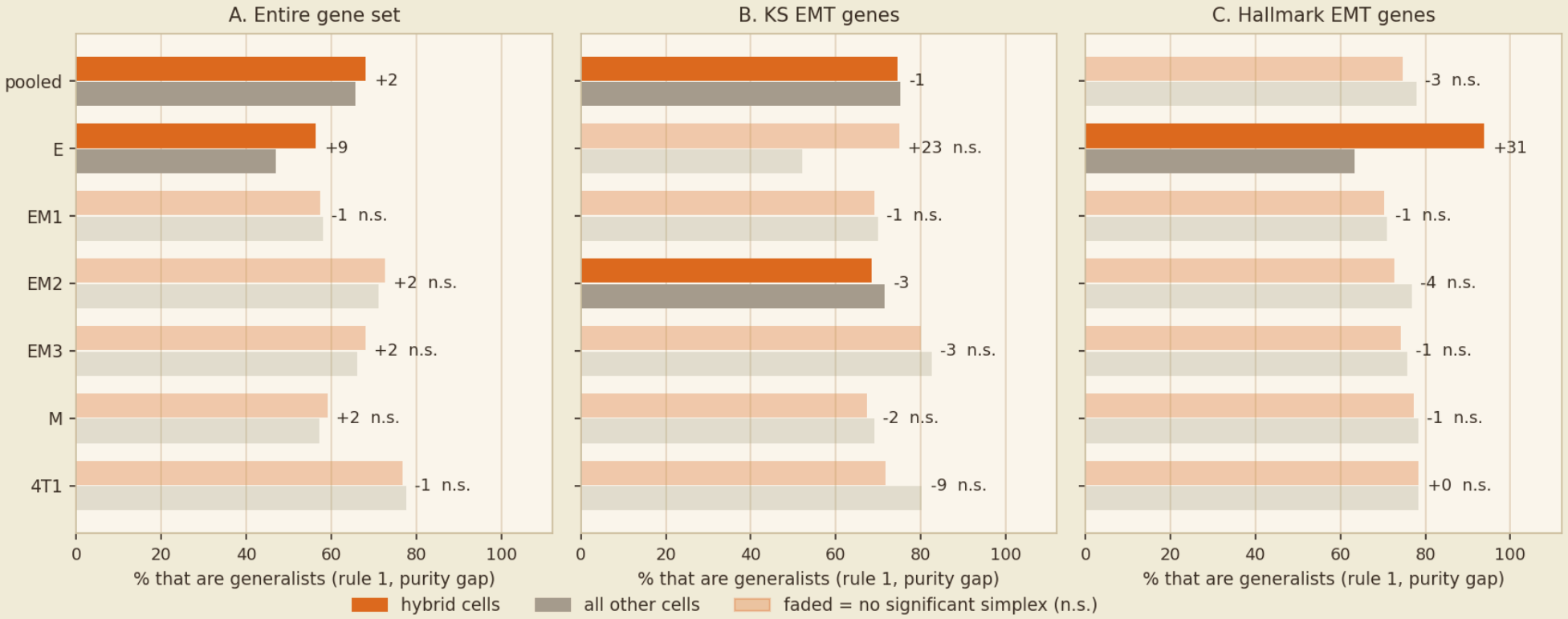
V3 inflammatory / stress response — Inflammatory Response +0.93 SD; 6% S/G2M; 36% Early-Hybrid; 100% 4T1; Cxcl1 Serpine2 Spp1

V4 UV Response Dn — UV Response Dn +0.78 SD; 10% S/G2M; 30% Late-Hybrid; 100% 4T1; Fn1 Serpine2 Jun

Hallmark EMT set left out when naming (circular: fitted on those genes).

Question 2 · 4T1 clones: are hybrid cells more often generalists than other cells?

Orange = hybrid cells, grey = all other cells; one pair of bars per unit; the number = orange minus grey (percentage points)



Reading

- A. Entire gene set (2 significant): hybrids 62% vs others 56% generalist under rule 1 (rule 2: 90% vs 84%) → hybrids are slightly more often generalists.
- B. KS EMT genes (2 significant): hybrids 72% vs others 73% generalist under rule 1 (rule 2: 85% vs 79%) → no real difference.
- C. Hallmark EMT genes (1 significant): hybrids 94% vs others 63% generalist under rule 1 (rule 2: 94% vs 85%) → hybrids are clearly more often generalists.

Rule 3 (weight > 0.95) is not drawn: it calls about 100% of every group generalist, so it cannot show a difference.

MCF10A/HuMEC

What the dataset is, and what to keep in mind when reading its slides

What the dataset means

Human mammary cells (MCF10A, HuMEC) grown as colonies so EMT rises with distance from the colony edge; Mock vs TGF-β. Induced EMT in culture, no replicates.

Units in the main analysis

MCF10A, MCF10A_Mock, MCF10A_TGFB, HuMEC, HuMEC_Mock, HuMEC_TGFB

Gene sets

A. all genes (11,000 to 14,500); B. KS EMT genes (Tan 2014; about 160 to 190 present); C. Hallmark Epithelial-Mesenchymal-Transition set (200 genes, those present in the data).

Keep in mind

- Colony position (inner vs outer) is an independent EMT coordinate; hybrids = Early- + Late-Hybrid (KS-score quartiles).
- Cleaning keeps only 45% of cells (HuMEC 35 to 40%), so cleaned and light-QC fits differ.
- No replicates: one experiment per line and arm.

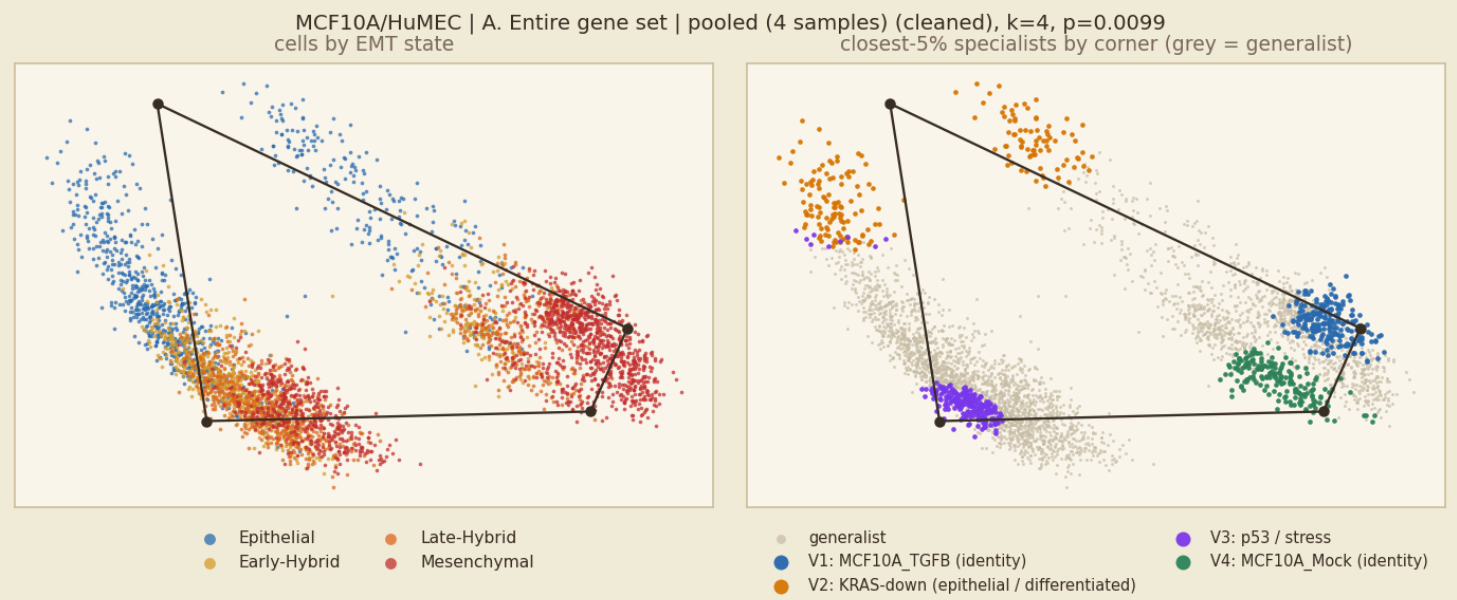
This dataset in numbers (main fits)

- Entire gene set: 7 of 7 fits significant; hybrids 61% / 88% / 100% generalist (median, rules 1/2/3).
- KS EMT genes: 5 of 7 fits significant; hybrids 64% / 91% / 100% generalist (median, rules 1/2/3).
- Hallmark EMT genes: 6 of 7 fits significant; hybrids 69% / 90% / 100% generalist (median, rules 1/2/3).

Question 1 slides come first, then the Question 2 slides for this dataset.

MCF10A/HuMEC: A. Entire gene set, pooled fit

Fit shown: cleaned · k = 4 · p = 0.0099 · 13,654 cells · 100 shuffles l comparison, light QC: k=6, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	14%	38%
2 closest 5% of cells to each corner	84%	77%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 45% of all cells here; 16% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=4, p=0.0099) whose corners mix samples (no corner is 70% or more one sample), so it is not a pure sample-identity simplex.

Corners are programmes: EMT / matrix; KRAS-down (epithelial / differentiated); p53 / stress; MYC / growth (translation).

Hybrid vs other cells: 14% of hybrid cells and 38% of other cells are generalists (rule 1; rule 2: 84% vs 77%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 identity corner: MCF10A_TGFB (programme: EMT / matrix) — Epithelial Mesenchymal Transition +1.96 SD; 21% S/G2M; 98% Mesenchymal; 100% MCF10A_TGFB; FN1 KRT81 CXCL3

V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.94 SD; 0% S/G2M; 100% Epithelial; SPRR1A SBSN PI3

V3 p53 / stress — p53 Pathway +1.0 SD; 55% S/G2M; 46% Late-Hybrid; CALML3 S100A9 FGFBP1

V4 identity corner: MCF10A_Mock (programme: MYC / growth (translation)) — Myc Targets V1 +1.86 SD; 96% S/G2M; 43% Late-Hybrid; 98% MCF10A_Mock; KRT81 MT1E MT2A

MCF10A/HuMEC: A. Entire gene set, all units at a glance

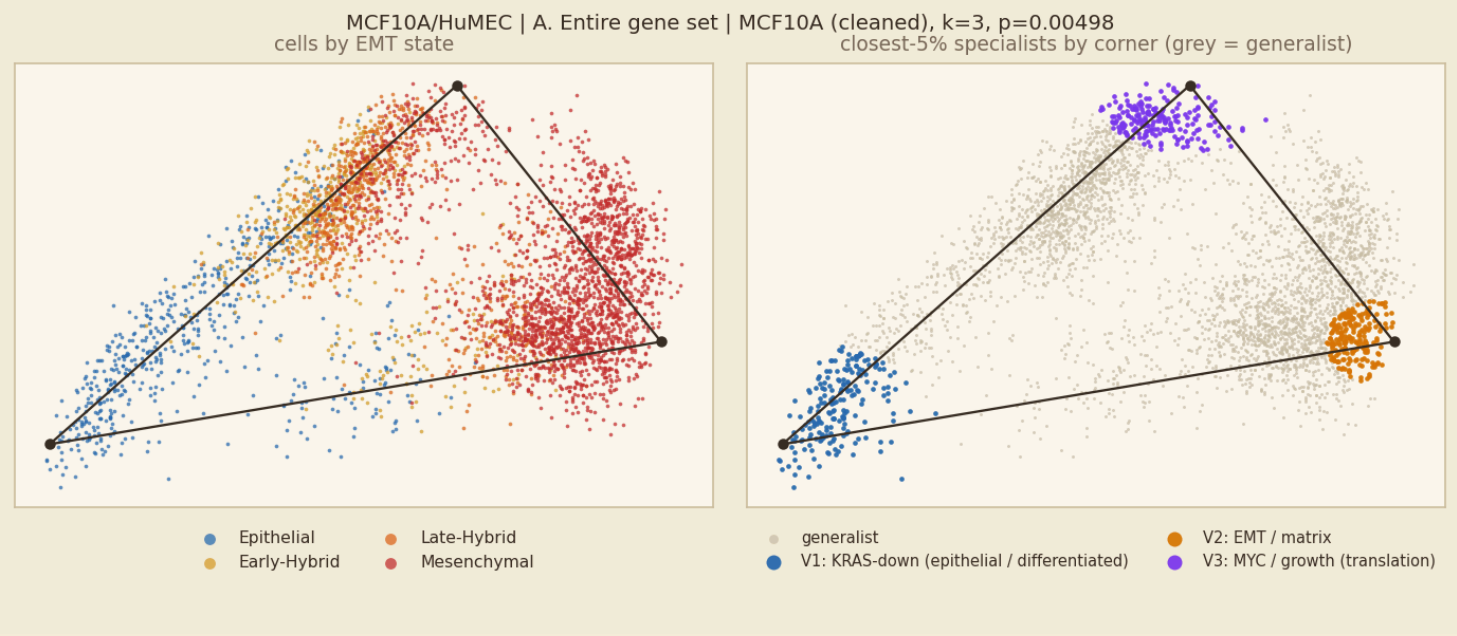
Main fit of each unit: 6 of 6 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
MCF10A	cleaned	3	0.00498 *	0.00498	32%	47 / 94 / 100%	40 / 36 / 32%
MCF10A_Mock	cleaned	3	0.00498 *	0.00498	54%	61 / 90 / 99%	59 / 57 / 55%
MCF10A_TGFB	cleaned	3	0.00498 *	0.00498	16%	76 / 90 / 100%	18 / 17 / 16%
HuMEC	cleaned	4	0.00498 *	0.00498	54%	67 / 85 / 100%	55 / 57 / 54%
HuMEC_Mock	cleaned	3	0.00498 *	0.0896	56%	54 / 88 / 99%	53 / 58 / 56%
HuMEC_TGFB	cleaned	4	0.00498 *	0.746	52%	74 / 85 / 100%	55 / 55 / 52%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

MCF10A/HuMEC: A. Entire gene set, unit MCF10A

Fit shown: cleaned · k = 3 · p = 0.00498 · 4,000 cells · 200 shuffles l comparison, light QC: k=3, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	47%	34%
2 closest 5% of cells to each corner	94%	81%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 32% of all cells here; 58% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MCF10A alone, cells fill a 3-corner simplex (p=0.00498). 6 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); EMT / matrix; MYC / growth (translation).

Hybrid vs other cells: 47% of hybrid cells and 34% of other cells are generalists (rule 1; rule 2: 94% vs 81%), so hybrids are clearly more often generalists.

Corners and their tasks

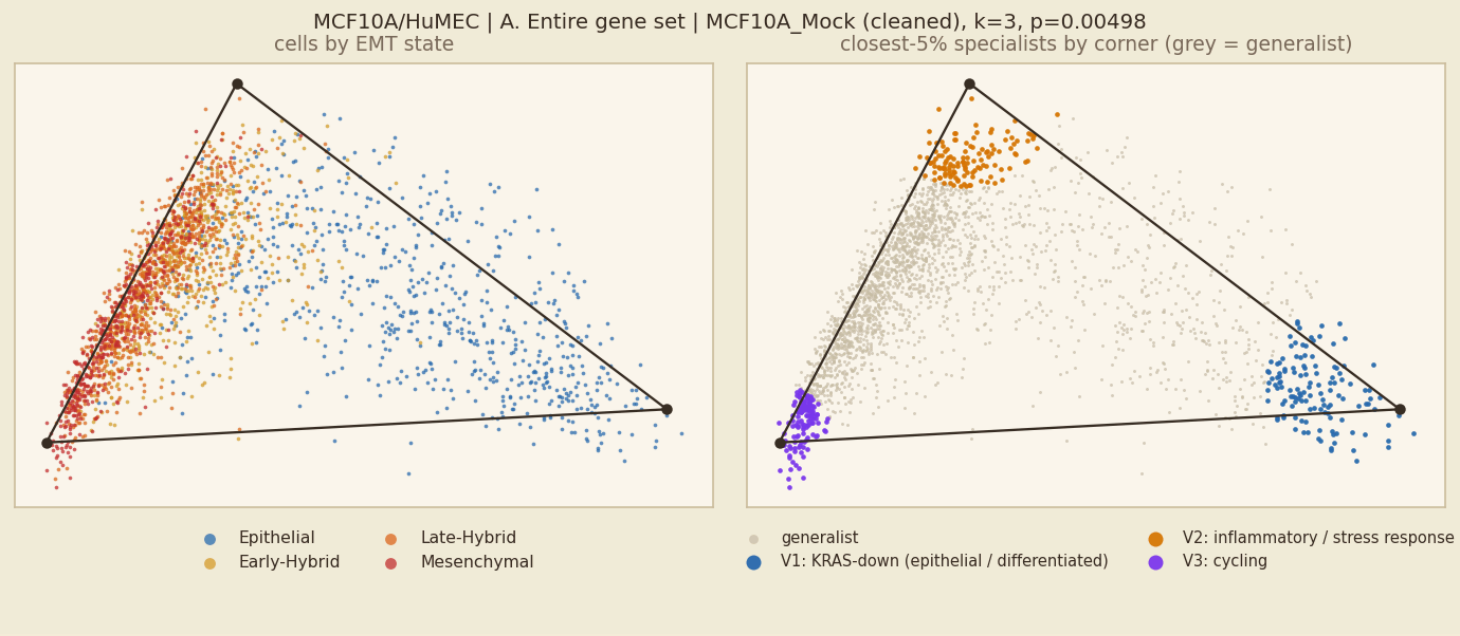
V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.86 SD; 0% S/G2M; 100% Epithelial; 100% MCF10A_Mock; PI3 SPRR1A SPRR1B

V2 EMT / matrix — Epithelial Mesenchymal Transition +1.53 SD; 27% S/G2M; 100% Mesenchymal; 100% MCF10A_TGFB; FN1 CXCL3 CXCL1

V3 MYC / growth (translation) — Myc Targets V1 +1.79 SD; 100% S/G2M; 60% Mesenchymal; 98% MCF10A_Mock; PTTG1 HMGB2 CCNB1

MCF10A/HuMEC: A. Entire gene set, unit MCF10A_Mock

Fit shown: cleaned · k = 3 · p = 0.00498 · 2,536 cells · 200 shuffles | comparison, light QC: k=3, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	61%	50%
2 closest 5% of cells to each corner	90%	80%
3 weight > 0.95 on one corner	99%	97%

Hybrids are 54% of all cells here; 85% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MCF10A_Mock alone, cells fill a 3-corner simplex (p=0.00498). 6 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); inflammatory / stress response; cycling.

Hybrid vs other cells: 61% of hybrid cells and 50% of other cells are generalists (rule 1; rule 2: 90% vs 80%), so hybrids are clearly more often generalists.

Corners and their tasks

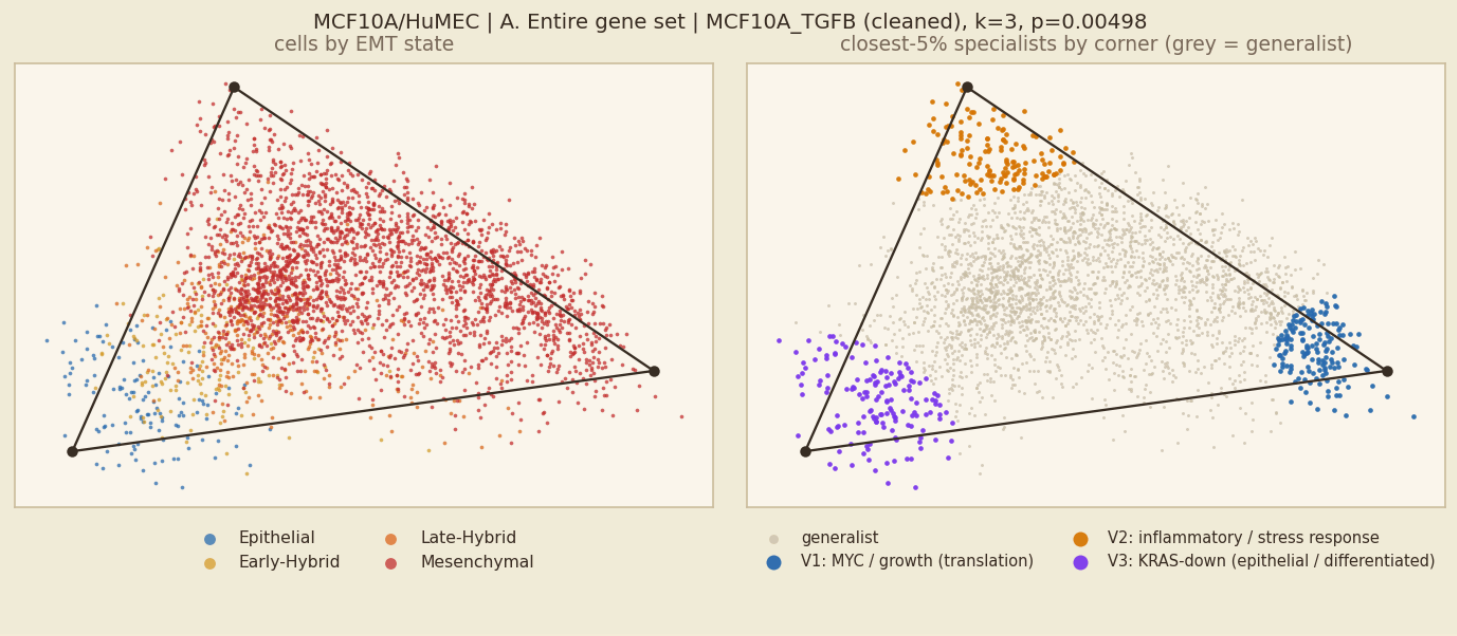
V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.5 SD; 0% S/G2M; 100% Epithelial; 100% MCF10A_Mock; SPRR1A SBSN PI3

V2 inflammatory / stress response — TNF-alpha Signaling via NF-kB +0.67 SD; 20% S/G2M; 39% Late-Hybrid; 100% MCF10A_Mock; MIR205HG TGFBI NDRG1

V3 cycling — G2-M Checkpoint +1.85 SD; 100% S/G2M; 61% Mesenchymal; 100% MCF10A_Mock; CCNB1 STMN1 HMGB2

MCF10A/HuMEC: A. Entire gene set, unit MCF10A_TGFB

Fit shown: cleaned · k = 3 · p = 0.00498 · 3,127 cells · 200 shuffles l comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	76%	67%
2 closest 5% of cells to each corner	90%	84%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 16% of all cells here; 87% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MCF10A_TGFB alone, cells fill a 3-corner simplex (p=0.00498). 6 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: MYC / growth (translation); inflammatory / stress response; KRAS-down (epithelial / differentiated).

Hybrid vs other cells: 76% of hybrid cells and 67% of other cells are generalists (rule 1; rule 2: 90% vs 84%), so hybrids are clearly more often generalists.

Corners and their tasks

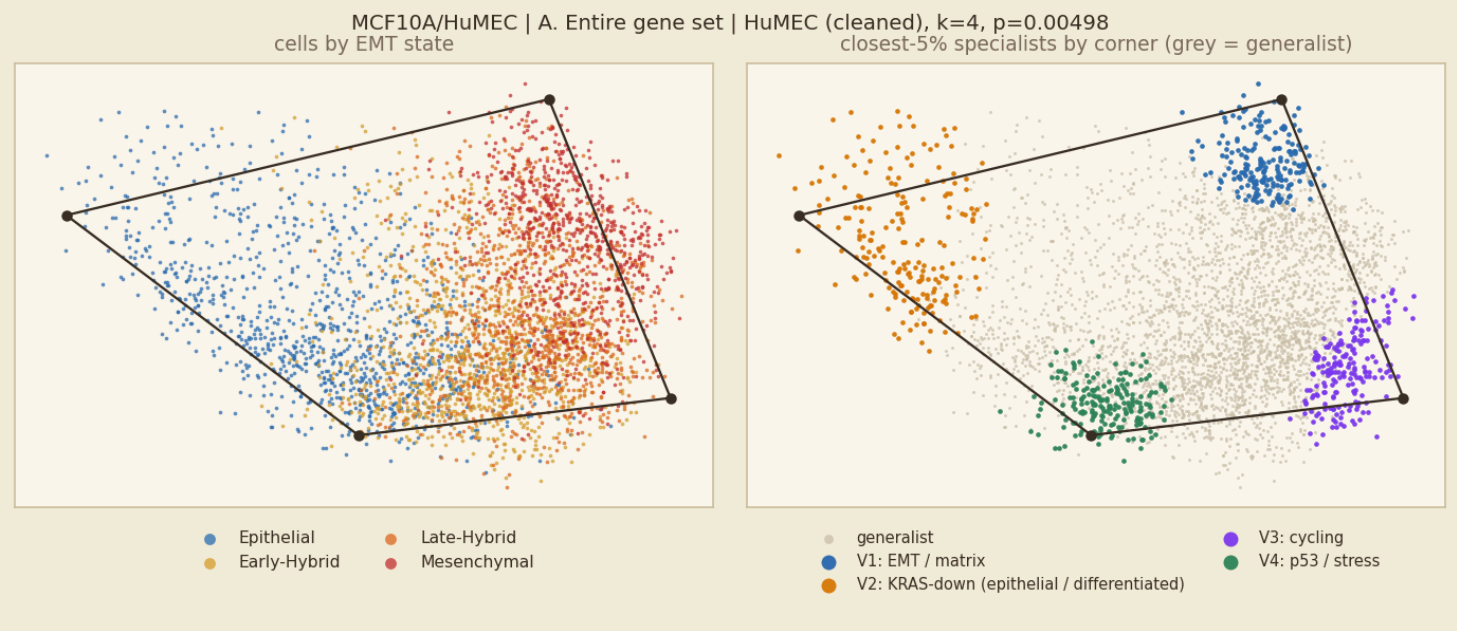
V1 MYC / growth (translation) — Myc Targets V1 +2.07 SD; 100% S/G2M; 100% Mesenchymal; 100% MCF10A_TGFB; PTTG1 CCNB1 CENPF

V2 inflammatory / stress response — TNF-alpha Signaling via NF-kB +0.96 SD; 5% S/G2M; 99% Mesenchymal; 100% MCF10A_TGFB; CXCL3 FN1 TGM2

V3 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.35 SD; 0% S/G2M; 69% Epithelial; 100% MCF10A_TGFB; FXYD3 SAA1 KRT16

MCF10A/HuMEC: A. Entire gene set, unit HuMEC

Fit shown: cleaned · k = 4 · p = 0.00498 · 4,000 cells · 200 shuffles l comparison, light QC: k=4, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	67%	63%
2 closest 5% of cells to each corner	85%	74%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 54% of all cells here; 48% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside HuMEC alone, cells fill a 4-corner simplex (p=0.00498). 6 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: EMT / matrix; KRAS-down (epithelial / differentiated); cycling; p53 / stress.

Hybrid vs other cells: 67% of hybrid cells and 63% of other cells are generalists (rule 1; rule 2: 85% vs 74%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.72 SD; 16% S/G2M; 70% Mesenchymal; 100% HuMEC_TGFB; FN1 LAMC2 LAMA3

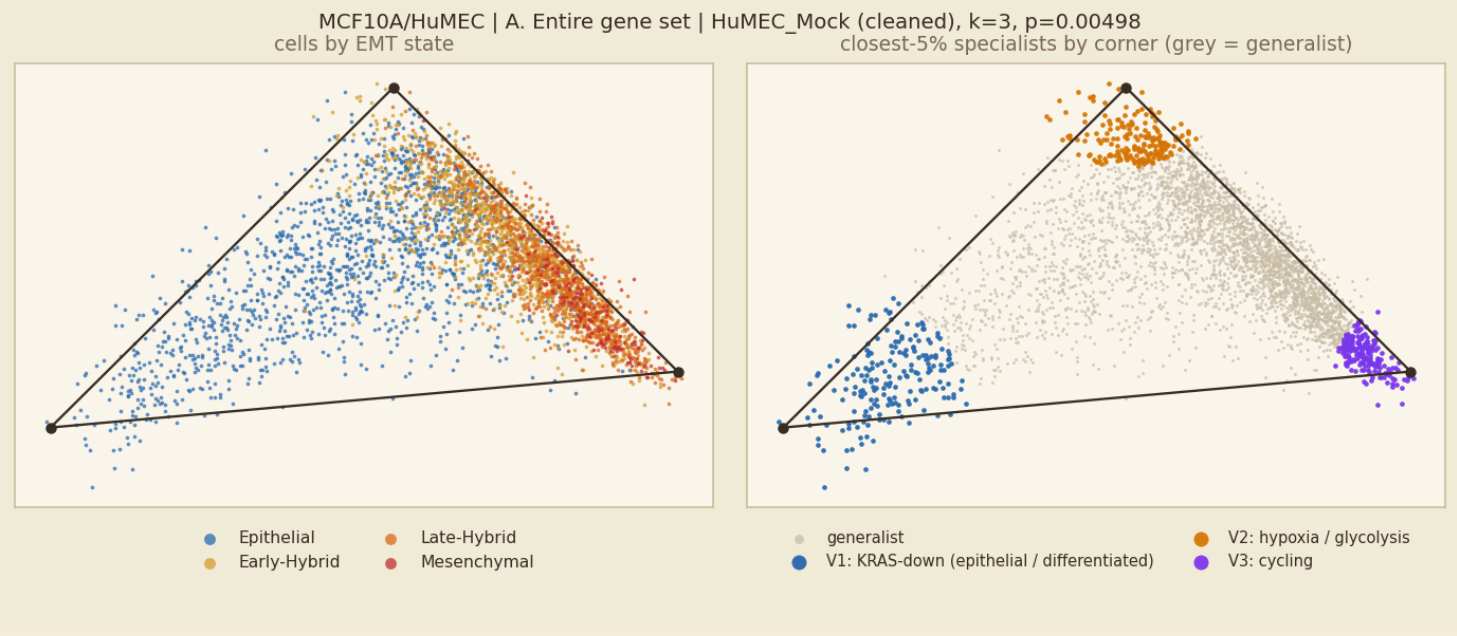
V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.25 SD; 0% S/G2M; 100% Epithelial; 70% HuMEC_Mock; SPRR1A SBSN CLIC3

V3 cycling — G2-M Checkpoint +2.19 SD; 100% S/G2M; 51% Late-Hybrid; 60% HuMEC_Mock; HMGB2 UBE2C CCNB1

V4 p53 / stress — p53 Pathway +0.95 SD; 14% S/G2M; 44% Early-Hybrid; 96% HuMEC_Mock; CKB CXCL14 C10orf99

MCF10A/HuMEC: A. Entire gene set, unit HuMEC_Mock

Fit shown: cleaned · k = 3 · p = 0.00498 · 3,889 cells · 200 shuffles | comparison, light QC: k=4, p=0.0896



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	54%	60%
2 closest 5% of cells to each corner	88%	81%
3 weight > 0.95 on one corner	99%	99%

Hybrids are 56% of all cells here; 92% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside HuMEC_Mock alone, cells fill a 3-corner simplex (p=0.00498). 6 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); hypoxia / glycolysis; cycling.

Hybrid vs other cells: 54% of hybrid cells and 60% of other cells are generalists (rule 1; rule 2: 88% vs 81%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

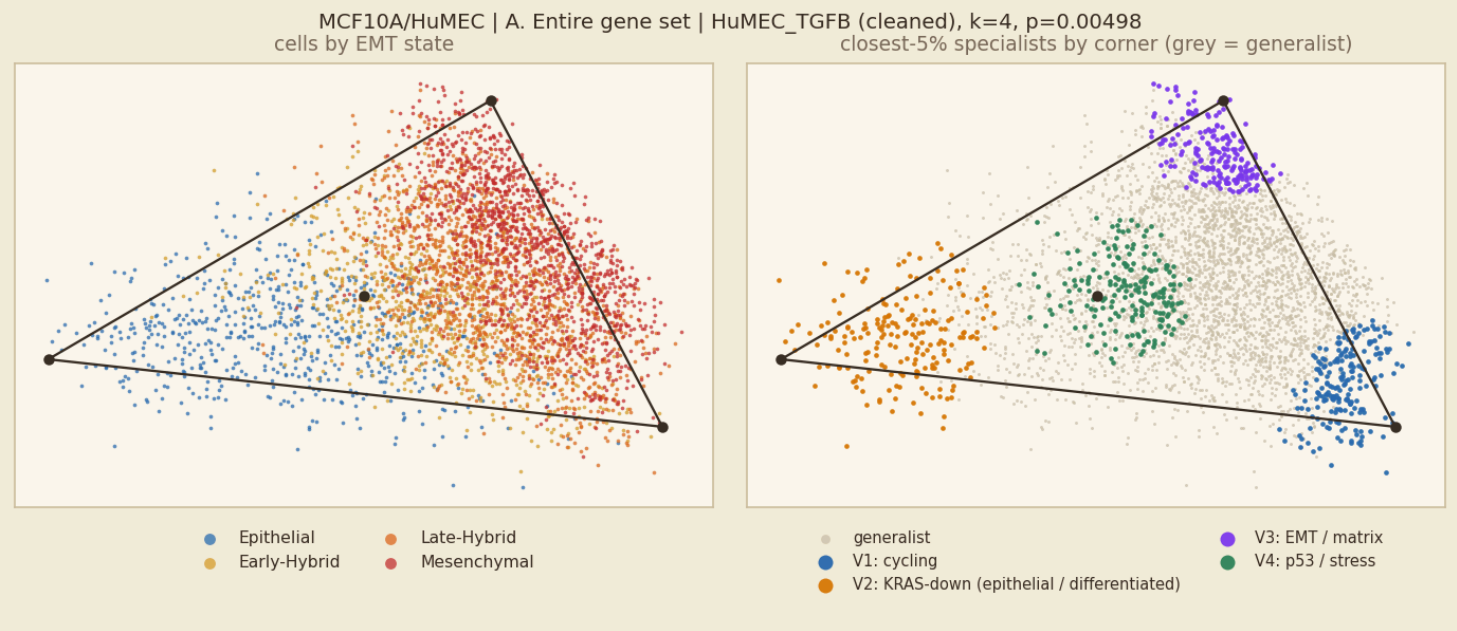
V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.25 SD; 0% S/G2M; 100% Epithelial; 100% HuMEC_Mock; SPRR1A SBSN CLIC3

V2 hypoxia / glycolysis — Hypoxia +0.73 SD; 7% S/G2M; 47% Epithelial; 100% HuMEC_Mock; CXCL14 C10orf99 CKB

V3 cycling — G2-M Checkpoint +2.1 SD; 99% S/G2M; 51% Late-Hybrid; 100% HuMEC_Mock; MT2A TUBA1B HMGB2

MCF10A/HuMEC: A. Entire gene set, unit HuMEC_TGFB

Fit shown: cleaned · k = 4 · p = 0.00498 · 4,000 cells · 200 shuffles l comparison, light QC: k=4, p=0.746



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	74%	65%
2 closest 5% of cells to each corner	85%	75%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 52% of all cells here; 44% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside HuMEC_TGFB alone, cells fill a 4-corner simplex (p=0.00498). 6 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: cycling; KRAS-down (epithelial / differentiated); EMT / matrix; p53 / stress.

Hybrid vs other cells: 74% of hybrid cells and 65% of other cells are generalists (rule 1; rule 2: 85% vs 75%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 cycling — G2-M Checkpoint +2.25 SD; 100% S/G2M; 48% Late-Hybrid; 100% HuMEC_TGFB; UBE2C HMGB2 CCNB1

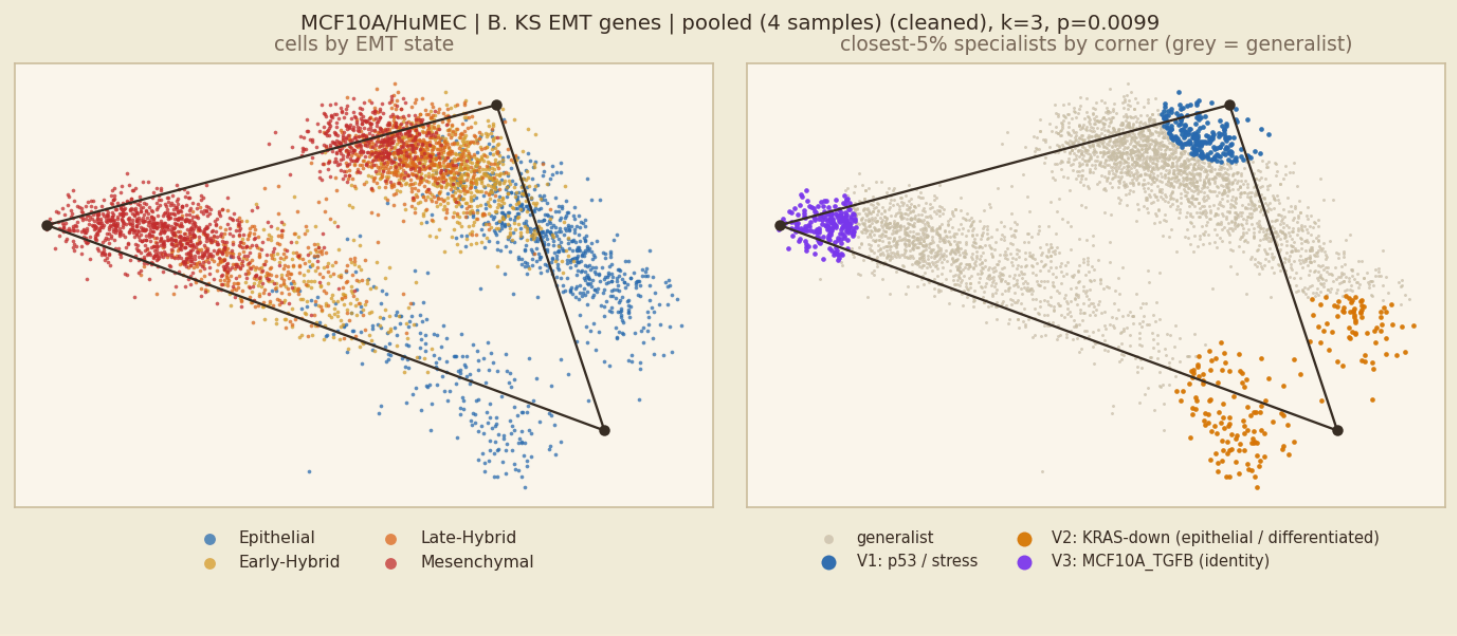
V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.12 SD; 0% S/G2M; 96% Epithelial; 100% HuMEC_TGFB; SPRR1A KRT6C PI3

V3 EMT / matrix — Epithelial Mesenchymal Transition +1.6 SD; 19% S/G2M; 80% Mesenchymal; 100% HuMEC_TGFB; FN1 LAMC2 MT2A

V4 p53 / stress — p53 Pathway +0.91 SD; 20% S/G2M; 40% Early-Hybrid; 100% HuMEC_TGFB; KRT15 GLUL NEAT1

MCF10A/HuMEC: B. KS EMT genes, pooled fit

Fit shown: cleaned · k = 3 · p = 0.0099 · 13,654 cells · 100 shuffles | comparison, light QC: k=4, p=0.119



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	34%	34%
2 closest 5% of cells to each corner	91%	80%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 45% of all cells here; 74% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=3, p=0.0099) whose corners mix samples (no corner is 70% or more one sample), so it is not a pure sample-identity simplex.

Corners are programmes: p53 / stress; KRAS-down (epithelial / differentiated); TGF-β response.

Hybrid vs other cells: 34% of hybrid cells and 34% of other cells are generalists (rule 1; rule 2: 91% vs 80%), so no real difference.

Corners and their tasks

V1 p53 / stress — p53 Pathway +1.0 SD; 37% S/G2M; 47% Early-Hybrid; KRT15 PYCARD FXYD3

V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.43 SD; 1% S/G2M; 100% Epithelial; SLPI TACSTD2 S100P

V3 identity corner: MCF10A_TGFB (programme: TGF-β response) — TGF-beta Signaling +1.73 SD; 63% S/G2M; 100% Mesenchymal; 97% MCF10A_TGFB; SERPINE1 VIM EMP3

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

MCF10A/HuMEC: B. KS EMT genes, all units at a glance

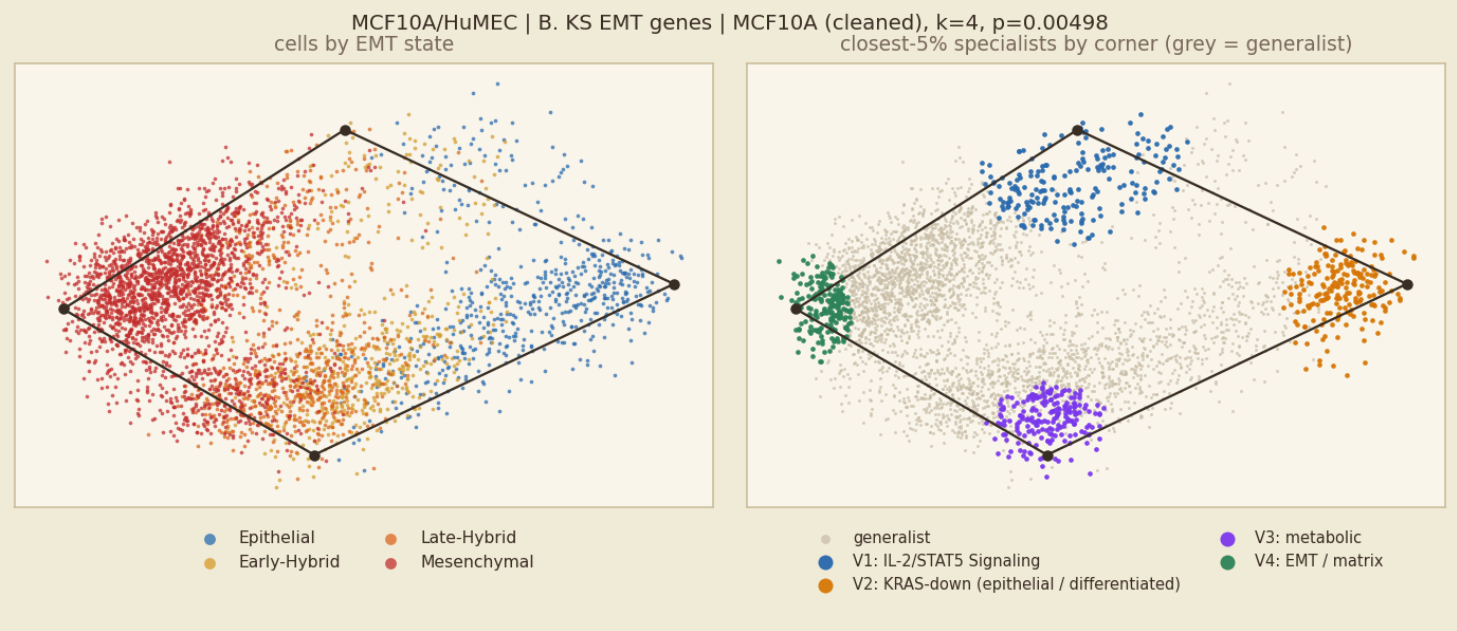
Main fit of each unit: 4 of 6 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
MCF10A	cleaned	4	0.00498 *	0.0149	32%	44 / 76 / 100%	28 / 31 / 32%
MCF10A_Mock	cleaned	3	0.00498 *	0.289	54%	69 / 91 / 100%	62 / 58 / 54%
MCF10A_TGFB	cleaned	4	0.393	0.0448	16%	73 / 82 / 100%	18 / 16 / 16%
HuMEC	cleaned	3	0.0746	0.00498	54%	71 / 91 / 100%	61 / 57 / 54%
HuMEC_Mock	cleaned	3	0.00498 *	0.239	56%	64 / 86 / 100%	57 / 57 / 56%
HuMEC_TGFB	cleaned	3	0.00498 *	0.00995	52%	69 / 91 / 100%	54 / 55 / 52%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

MCF10A/HuMEC: B. KS EMT genes, unit MCF10A

Fit shown: cleaned · k = 4 · p = 0.00498 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.0149



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	44%	55%
2 closest 5% of cells to each corner	76%	82%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 32% of all cells here; 30% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MCF10A alone, cells fill a 4-corner simplex (p=0.00498). 4 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: IL-2/STAT5 Signaling; KRAS-down (epithelial / differentiated); metabolic; EMT / matrix.

Hybrid vs other cells: 44% of hybrid cells and 55% of other cells are generalists (rule 1; rule 2: 76% vs 82%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 IL-2/STAT5 Signaling — IL-2/STAT5 Signaling +1.24 SD; 6% S/G2M; 36% Late-Hybrid; 100% MCF10A_TGFB; TACSTD2 IRF6 CDH3

V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.87 SD; 1% S/G2M; 100% Epithelial; 98% MCF10A_Mock; S100P FXYD3 SLPI

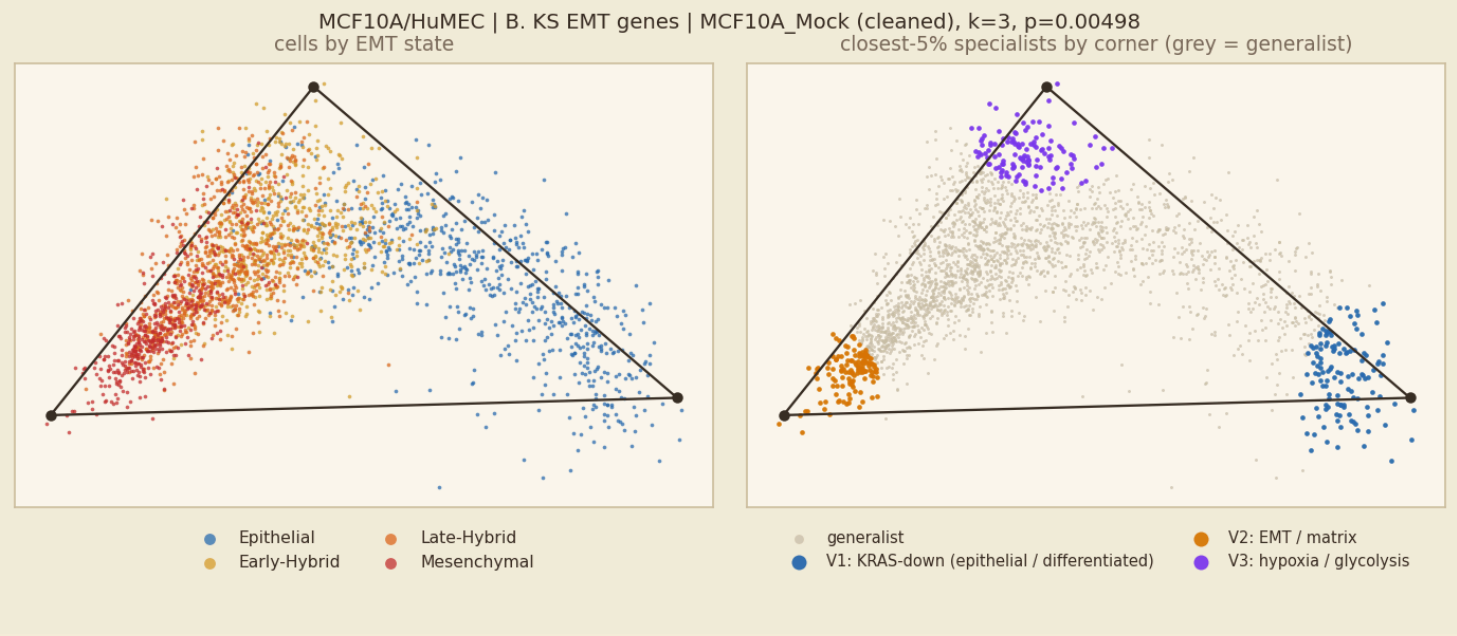
V3 metabolic — Oxidative Phosphorylation +0.91 SD; 71% S/G2M; 46% Late-Hybrid; 100% MCF10A_Mock; KRT15 S100A14 HES1

V4 EMT / matrix — Epithelial Mesenchymal Transition +1.25 SD; 68% S/G2M; 100% Mesenchymal; 100% MCF10A_TGFB; SERPINE1 VIM TPM2

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

MCF10A/HuMEC: B. KS EMT genes, unit MCF10A_Mock

Fit shown: cleaned · k = 3 · p = 0.00498 · 2,536 cells · 200 shuffles | comparison, light QC: k=4, p=0.289



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	69%	50%
2 closest 5% of cells to each corner	91%	77%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 54% of all cells here; 88% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MCF10A_Mock alone, cells fill a 3-corner simplex (p=0.00498). 4 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); EMT / matrix; hypoxia / glycolysis.

Hybrid vs other cells: 69% of hybrid cells and 50% of other cells are generalists (rule 1; rule 2: 91% vs 77%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.5 SD; 0% S/G2M; 100% Epithelial; 100% MCF10A_Mock; S100P TACSTD2 FXYD3

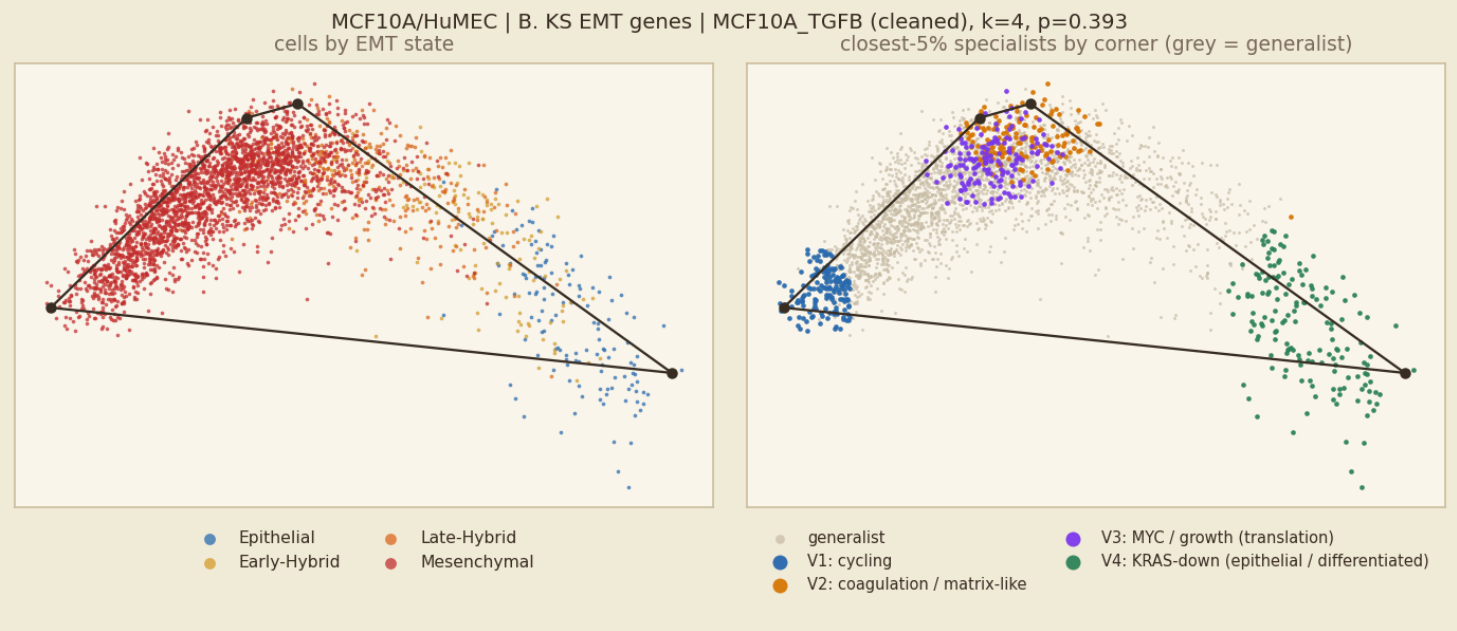
V2 EMT / matrix — Epithelial Mesenchymal Transition +1.34 SD; 82% S/G2M; 90% Mesenchymal; 100% MCF10A_Mock; VIM EMP3 TPM2

V3 hypoxia / glycolysis — Hypoxia +0.94 SD; 35% S/G2M; 51% Early-Hybrid; 100% MCF10A_Mock; KRT15 FXYD3 SFN

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

MCF10A/HuMEC: B. KS EMT genes, unit MCF10A_TGFB

Fit shown: cleaned · k = 4 · p = 0.393 · 3,127 cells · 200 shuffles | comparison, light QC: k=4, p=0.0448



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	73%	64%
2 closest 5% of cells to each corner	82%	79%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 16% of all cells here; 51% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside MCF10A_TGFB: the fit (k=4, p=0.393) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 73% of hybrid cells and 64% of other cells are generalists (rule 1; rule 2: 82% vs 79%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 cycling — G2-M Checkpoint +0.84 SD; 75% S/G2M; 100% Mesenchymal; 100% MCF10A_TGFB; SERPINE1 TPM2 MYL9

V2 coagulation / matrix-like — Coagulation +1.28 SD; 13% S/G2M; 94% Mesenchymal; 100% MCF10A_TGFB; S100A14 SFN CD9

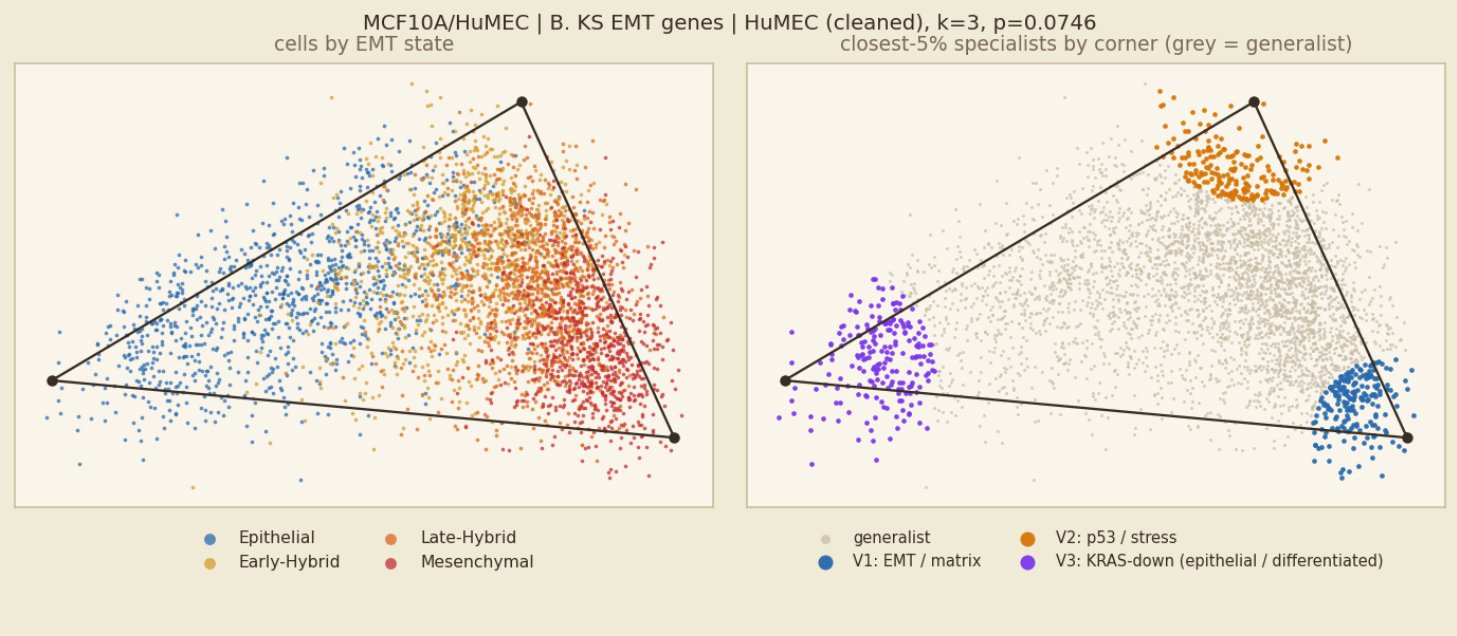
V3 MYC / growth (translation) — Myc Targets V2 +0.52 SD; 38% S/G2M; 76% Mesenchymal; 100% MCF10A_TGFB; S100A14 SLPI ITGB4

V4 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.56 SD; 0% S/G2M; 74% Epithelial; 100% MCF10A_TGFB; FXYD3 TACSTD2 SLPI

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

MCF10A/HuMEC: B. KS EMT genes, unit HuMEC

Fit shown: cleaned · k = 3 · p = 0.0746 · 4,000 cells · 200 shuffles l comparison, light QC: k=3, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	71%	52%
2 closest 5% of cells to each corner	91%	78%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 54% of all cells here; 87% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside HuMEC: the fit (k=3, p=0.0746) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 71% of hybrid cells and 52% of other cells are generalists (rule 1; rule 2: 91% vs 78%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.74 SD; 42% S/G2M; 88% Mesenchymal; 99% HuMEC_TGFB; VIM KRT7 EMP3

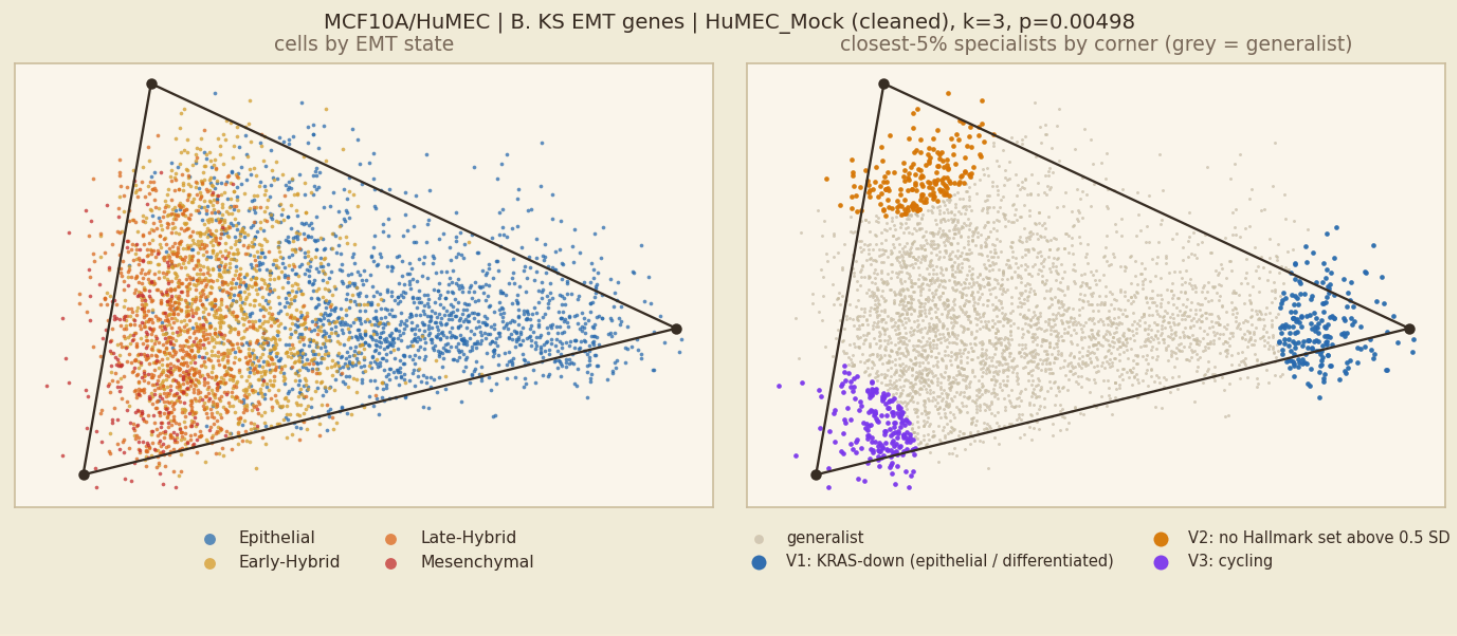
V2 p53 / stress — p53 Pathway +0.52 SD; 62% S/G2M; 56% Early-Hybrid; 77% HuMEC_Mock; KRT15 HES1 TPM2

V3 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.13 SD; 0% S/G2M; 100% Epithelial; 74% HuMEC_Mock; SLPI DSC2 RAB25

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

MCF10A/HuMEC: B. KS EMT genes, unit HuMEC_Mock

Fit shown: cleaned · k = 3 · p = 0.00498 · 3,889 cells · 200 shuffles | comparison, light QC: k=4, p=0.239



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	64%	59%
2 closest 5% of cells to each corner	86%	83%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 56% of all cells here; 92% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside HuMEC_Mock alone, cells fill a 3-corner simplex (p=0.00498). 4 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); no Hallmark set above 0.5 SD; cycling.

Hybrid vs other cells: 64% of hybrid cells and 59% of other cells are generalists (rule 1; rule 2: 86% vs 83%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.1 SD; 0% S/G2M; 100% Epithelial; 100% HuMEC_Mock; SLPI CLDN4 DSC2

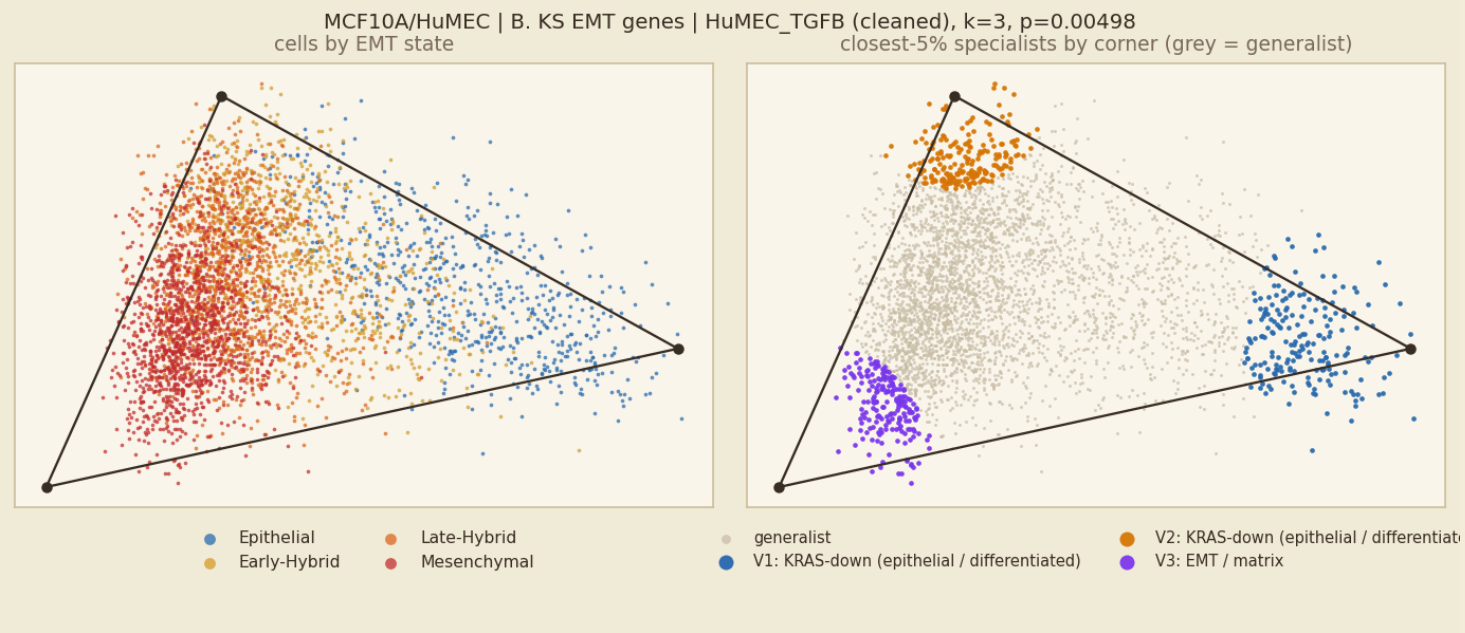
V2 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 46% S/G2M; 55% Early-Hybrid; 100% HuMEC_Mock; KRT15 VIM TPM2

V3 cycling — E2F Targets +1.26 SD; 83% S/G2M; 55% Late-Hybrid; 100% HuMEC_Mock; VIM EMP3 KRT7

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

MCF10A/HuMEC: B. KS EMT genes, unit HuMEC_TGFB

Fit shown: cleaned · k = 3 · p = 0.00498 · 4,000 cells · 200 shuffles | comparison, light QC: k=4, p=0.00995



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	69%	62%
2 closest 5% of cells to each corner	91%	79%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 52% of all cells here; 93% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside HuMEC_TGFB alone, cells fill a 3-corner simplex (p=0.00498). 4 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); EMT / matrix.

Hybrid vs other cells: 69% of hybrid cells and 62% of other cells are generalists (rule 1; rule 2: 91% vs 79%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.12 SD; 0% S/G2M; 98% Epithelial; 100% HuMEC_TGFB; SLPI DSC2 TACSTD2

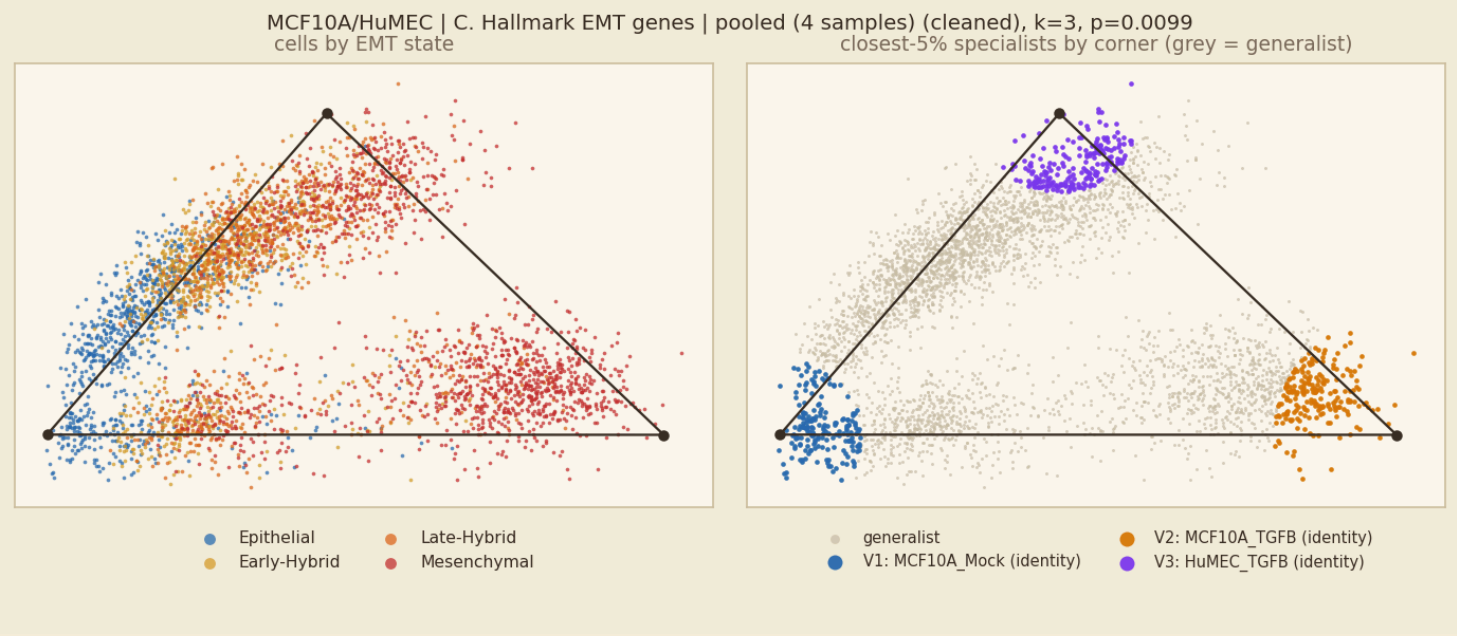
V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +0.7 SD; 66% S/G2M; 44% Early-Hybrid; 100% HuMEC_TGFB; KRT15 HES1 TPM2

V3 EMT / matrix — Epithelial Mesenchymal Transition +1.5 SD; 44% S/G2M; 92% Mesenchymal; 100% HuMEC_TGFB; KRT7 TUBA1A SERPINE1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

MCF10A/HuMEC: C. Hallmark EMT genes, pooled fit

Fit shown: cleaned · k = 3 · p = 0.0099 · 13,654 cells · 100 shuffles | comparison, light QC: k=5, p=0.0099



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	50%	36%
2 closest 5% of cells to each corner	93%	78%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 45% of all cells here; 68% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=3, p=0.0099), but 3 of 3 corners are almost entirely one sample (MCF10A_Mock, MCF10A_TGFB, HuMEC_TGFB). It separates samples; it is not a shared EMT trade-off.

Each corner is essentially one sample. The programme names (Hallmark) only describe what distinguishes that sample, and the EMT-state colours follow the samples. These corners are not tasks.

Hybrid vs other cells: 50% of hybrid cells and 36% of other cells are generalists (rule 1; rule 2: 93% vs 78%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 identity corner: MCF10A_Mock (programme: KRAS-down (epithelial / differentiated)) — KRAS Signaling Dn +0.81 SD; 8% S/G2M; 91% Epithelial; 80% MCF10A_Mock; ID2 QSOX1 MFAP5

V2 identity corner: MCF10A_TGFB (programme: UV Response Dn) — UV Response Dn +1.95 SD; 39% S/G2M; 99% Mesenchymal; 100% MCF10A_TGFB; FN1 CXCL1 TGFBI

V3 identity corner: HuMEC_TGFB (programme: coagulation / matrix-like) — Coagulation +1.19 SD; 47% S/G2M; 45% Mesenchymal; 85% HuMEC_TGFB; TFPI2 SERPINE2 VIM

Hallmark EMT set left out when naming (circular: fitted on those genes).

MCF10A/HuMEC: C. Hallmark EMT genes, all units at a glance

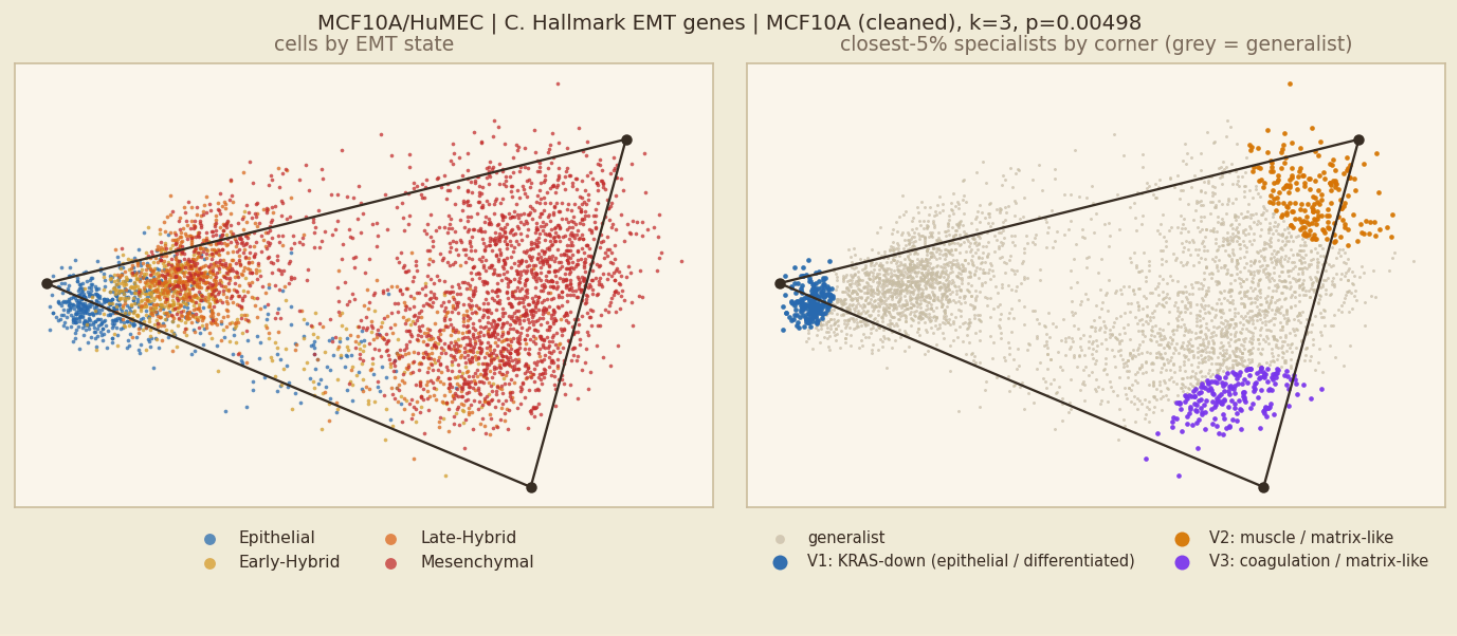
Main fit of each unit: 5 of 6 units have a significant simplex

unit	fit shown	k	p	light-QC p	hybrid % of cells	hybrids→generalist (1/2/3)	generalists→hybrid (1/2/3)
MCF10A	cleaned	3	0.00498 *	0.00498	32%	24 / 95 / 100%	18 / 36 / 32%
MCF10A_Mock	cleaned	4	0.0249 *	0.617	54%	67 / 85 / 100%	57 / 58 / 54%
MCF10A_TGFB	cleaned	3	0.0249 *	0.0348	16%	72 / 87 / 100%	17 / 16 / 16%
HuMEC	cleaned	4	0.0647	0.284	54%	73 / 86 / 100%	58 / 58 / 54%
HuMEC_Mock	cleaned	3	0.00498 *	0.791	56%	74 / 88 / 100%	62 / 58 / 56%
HuMEC_TGFB	cleaned	3	0.0348 *	0.00995	52%	77 / 93 / 100%	60 / 56 / 52%

* $p \le 0.05$ for the fit shown. Rules: 1 = gap, 2 = closest 5%, 3 = weight > 0.95. One slide per unit follows; non-significant units are included for completeness and their corners should not be read as tasks.

MCF10A/HuMEC: C. Hallmark EMT genes, unit MCF10A

Fit shown: cleaned · k = 3 · p = 0.00498 · 4,000 cells · 200 shuffles | comparison, light QC: k=3, p=0.00498



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	24%	50%
2 closest 5% of cells to each corner	95%	80%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 32% of all cells here; 82% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MCF10A alone, cells fill a 3-corner simplex (p=0.00498). 5 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); muscle / matrix-like; coagulation / matrix-like.

Hybrid vs other cells: 24% of hybrid cells and 50% of other cells are generalists (rule 1; rule 2: 95% vs 80%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.44 SD; 2% S/G2M; 99% Epithelial; 100% MCF10A_Mock; MFAP5 ECM1 CAPG

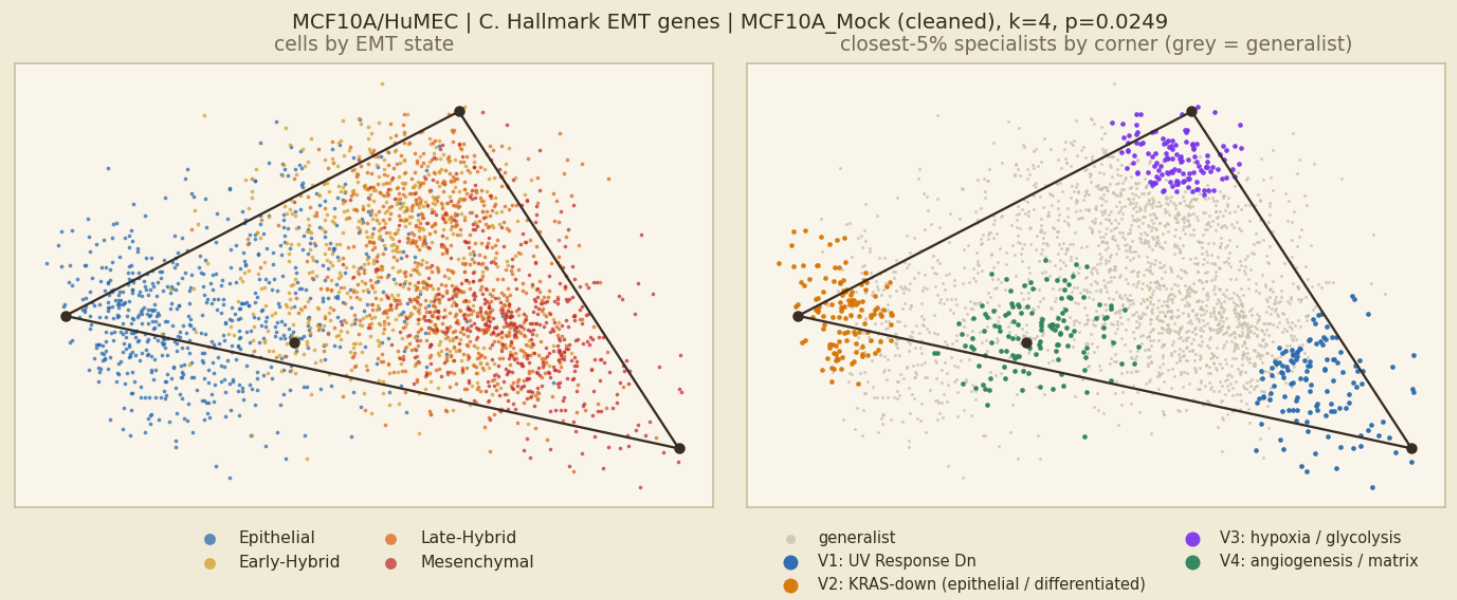
V2 muscle / matrix-like — Myogenesis +1.29 SD; 61% S/G2M; 100% Mesenchymal; 100% MCF10A_TGFB; TAGLN FN1 CXCL1

V3 coagulation / matrix-like — Coagulation +1.3 SD; 7% S/G2M; 71% Mesenchymal; 100% MCF10A_TGFB; FN1 LAMA3 LAMC2

Hallmark EMT set left out when naming (circular: fitted on those genes).

MCF10A/HuMEC: C. Hallmark EMT genes, unit MCF10A_Mock

Fit shown: cleaned · k = 4 · p = 0.0249 · 2,536 cells · 200 shuffles l comparison, light QC: k=3, p=0.617



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	67%	60%
2 closest 5% of cells to each corner	85%	73%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 54% of all cells here; 50% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MCF10A_Mock alone, cells fill a 4-corner simplex (p=0.0249). 5 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: UV Response Dn; KRAS-down (epithelial / differentiated); hypoxia / glycolysis; angiogenesis / matrix.

Hybrid vs other cells: 67% of hybrid cells and 60% of other cells are generalists (rule 1; rule 2: 85% vs 73%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 UV Response Dn — UV Response Dn +0.97 SD; 59% S/G2M; 78% Mesenchymal; 100% MCF10A_Mock; DKK1 VIM EMP3

V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.21 SD; 1% S/G2M; 100% Epithelial; 100% MCF10A_Mock; CD59 ECM1 MFAP5

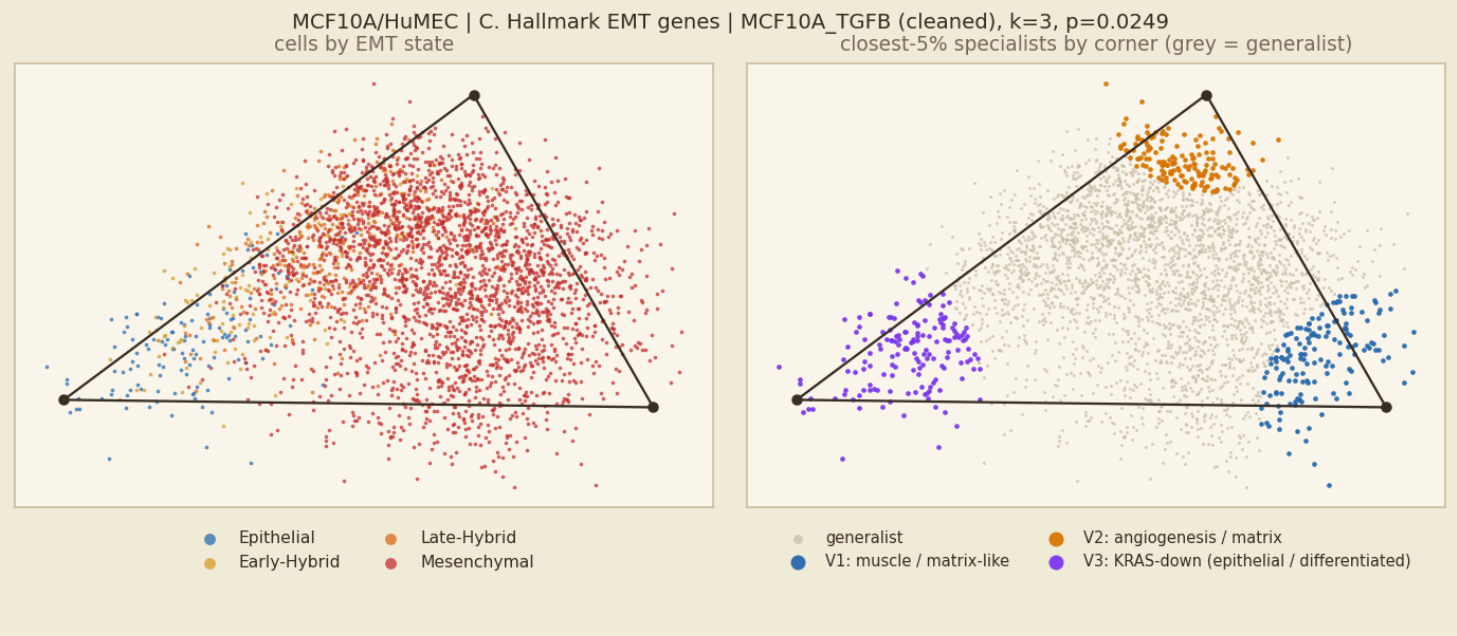
V3 hypoxia / glycolysis — Hypoxia +1.13 SD; 57% S/G2M; 48% Late-Hybrid; 100% MCF10A_Mock; TGFBI DST TNFRSF12A

V4 angiogenesis / matrix — Angiogenesis +0.54 SD; 25% S/G2M; 50% Epithelial; 100% MCF10A_Mock; TIMP1 ID2 SDC1

Hallmark EMT set left out when naming (circular: fitted on those genes).

MCF10A/HuMEC: C. Hallmark EMT genes, unit MCF10A_TGFB

Fit shown: cleaned · k = 3 · p = 0.0249 · 3,127 cells · 200 shuffles l comparison, light QC: k=4, p=0.0348



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	72%	67%
2 closest 5% of cells to each corner	87%	85%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 16% of all cells here; 83% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MCF10A_TGFB alone, cells fill a 3-corner simplex (p=0.0249). 5 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: muscle / matrix-like; angiogenesis / matrix; KRAS-down (epithelial / differentiated).

Hybrid vs other cells: 72% of hybrid cells and 67% of other cells are generalists (rule 1; rule 2: 87% vs 85%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 muscle / matrix-like — Myogenesis +0.94 SD; 62% S/G2M; 100% Mesenchymal; 100% MCF10A_TGFB; TAGLN TPM1 CXCL1

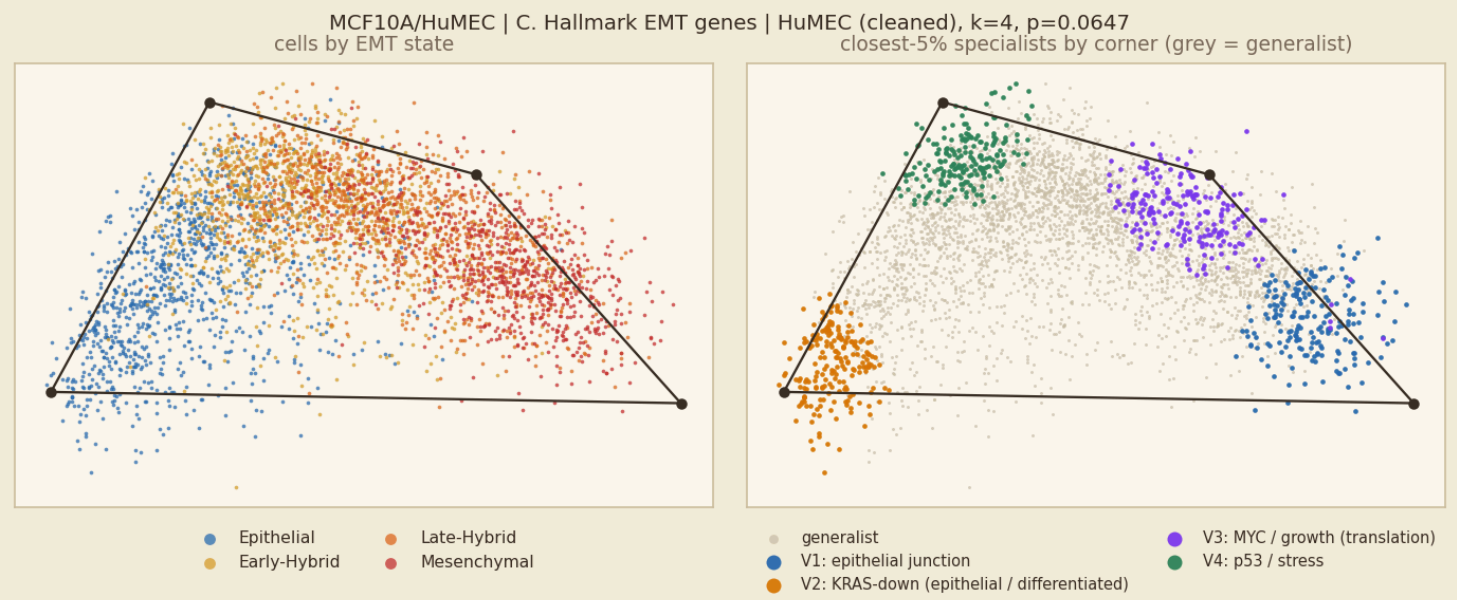
V2 angiogenesis / matrix — Angiogenesis +1.05 SD; 8% S/G2M; 93% Mesenchymal; 100% MCF10A_TGFB; TGM2 FBLN5 LOX

V3 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +2.13 SD; 11% S/G2M; 59% Epithelial; 100% MCF10A_TGFB; COL1A1 PDLIM4 CAPG

Hallmark EMT set left out when naming (circular: fitted on those genes).

MCF10A/HuMEC: C. Hallmark EMT genes, unit HuMEC

Fit shown: cleaned · k = 4 · p = 0.0647 · 4,000 cells · 200 shuffles l comparison, light QC: k=4, p=0.284



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	73%	61%
2 closest 5% of cells to each corner	86%	73%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 54% of all cells here; 48% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside HuMEC: the fit (k=4, p=0.0647) is no more triangular than shuffled data, so its corners should not be read as tasks. 5 of 6 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 73% of hybrid cells and 61% of other cells are generalists (rule 1; rule 2: 86% vs 73%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 epithelial junction — Apical Junction +1.7 SD; 24% S/G2M; 76% Mesenchymal; 100% HuMEC_TGFB; FN1 LAMC2 LAMA3

V2 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.86 SD; 0% S/G2M; 99% Epithelial; 87% HuMEC_Mock; QSOX1 ECM1 CAPG

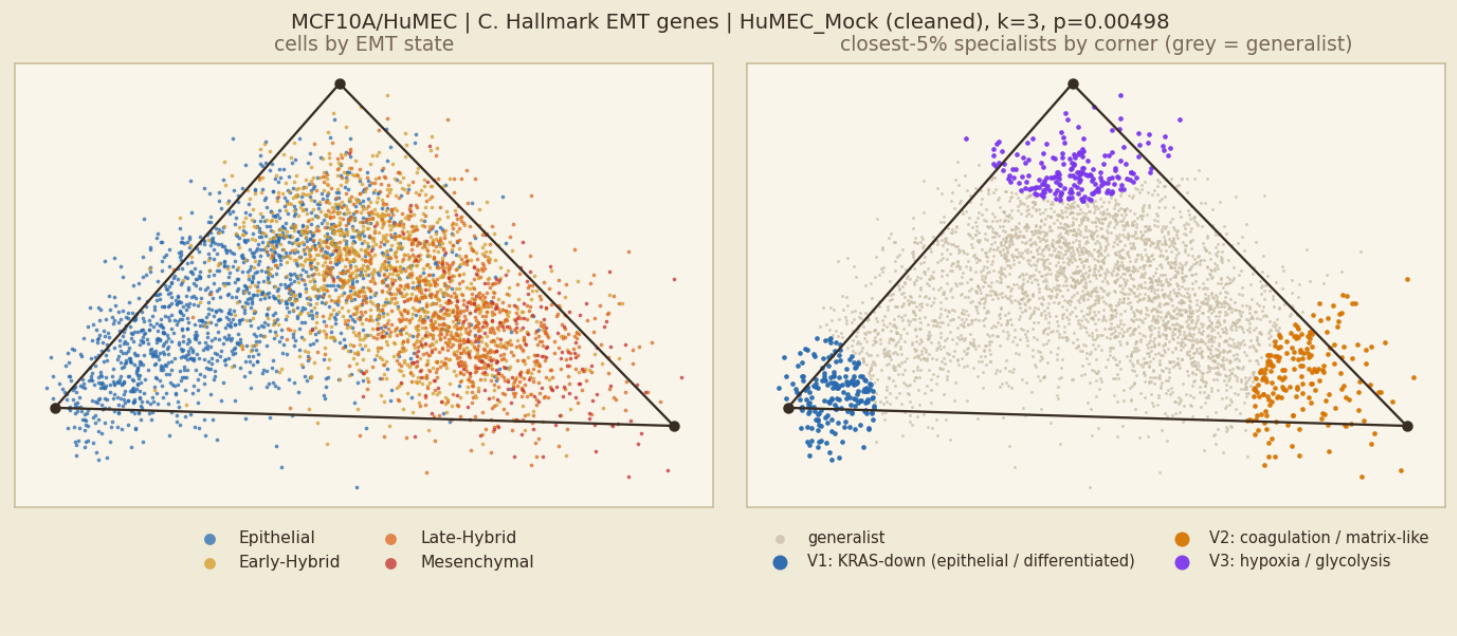
V3 MYC / growth (translation) — Myc Targets V1 +0.89 SD; 72% S/G2M; 52% Late-Hybrid; 80% HuMEC_TGFB; TFPI2 VIM EMP3

V4 p53 / stress — p53 Pathway +0.96 SD; 34% S/G2M; 46% Early-Hybrid; 94% HuMEC_Mock; DST IGFBP2 HTRA1

Hallmark EMT set left out when naming (circular: fitted on those genes).

MCF10A/HuMEC: C. Hallmark EMT genes, unit HuMEC_Mock

Fit shown: cleaned · k = 3 · p = 0.00498 · 3,889 cells · 200 shuffles | comparison, light QC: k=4, p=0.791



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	74%	56%
2 closest 5% of cells to each corner	88%	81%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 56% of all cells here; 91% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside HuMEC_Mock alone, cells fill a 3-corner simplex (p=0.00498). 5 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); coagulation / matrix-like; hypoxia / glycolysis.

Hybrid vs other cells: 74% of hybrid cells and 56% of other cells are generalists (rule 1; rule 2: 88% vs 81%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.82 SD; 0% S/G2M; 100% Epithelial; 100% HuMEC_Mock; QSOX1 MFAP5 ECM1

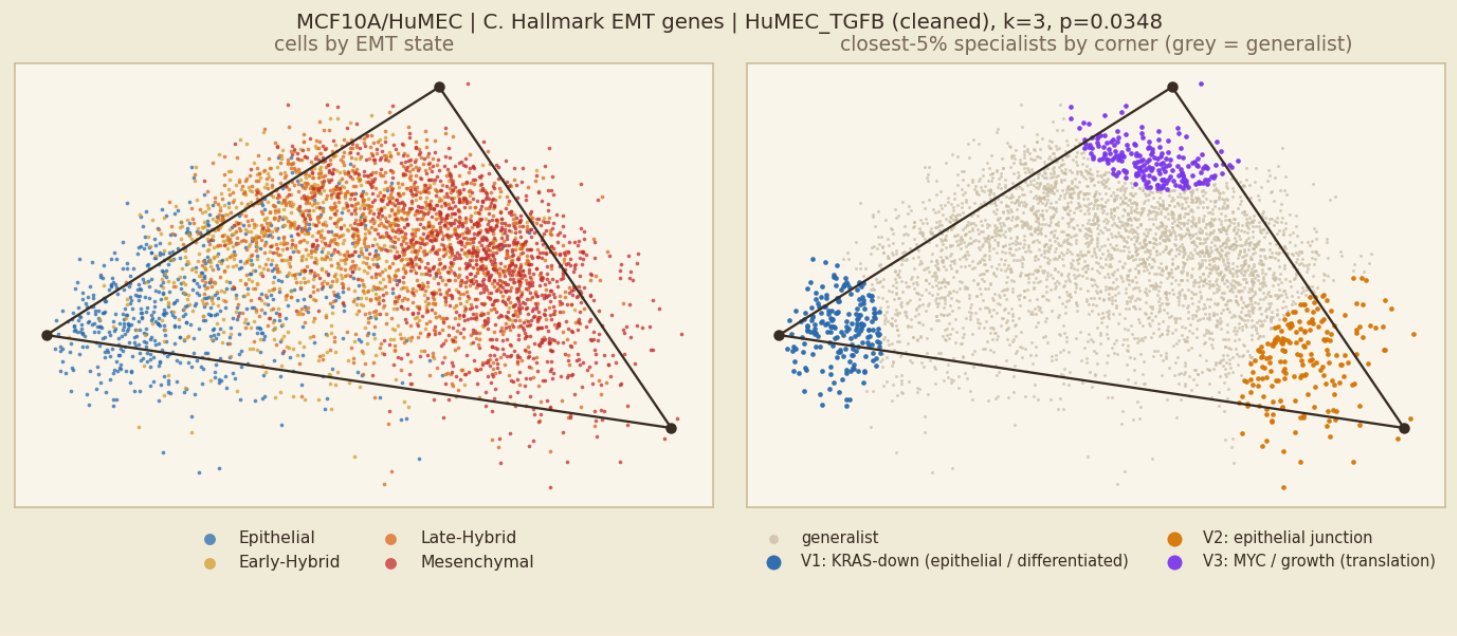
V2 coagulation / matrix-like — Coagulation +1.7 SD; 59% S/G2M; 53% Late-Hybrid; 100% HuMEC_Mock; FN1 TFPI2 VIM

V3 hypoxia / glycolysis — Hypoxia +0.92 SD; 24% S/G2M; 43% Early-Hybrid; 100% HuMEC_Mock; DST IGFBP2 HTRA1

Hallmark EMT set left out when naming (circular: fitted on those genes).

MCF10A/HuMEC: C. Hallmark EMT genes, unit HuMEC_TGFB

Fit shown: cleaned · k = 3 · p = 0.0348 · 4,000 cells · 200 shuffles l comparison, light QC: k=3, p=0.00995



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	77%	56%
2 closest 5% of cells to each corner	93%	77%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 52% of all cells here; 83% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside HuMEC_TGFB alone, cells fill a 3-corner simplex (p=0.0348). 5 of 6 units of this dataset have a significant simplex for this gene set.

Corners are programmes: KRAS-down (epithelial / differentiated); epithelial junction; MYC / growth (translation).

Hybrid vs other cells: 77% of hybrid cells and 56% of other cells are generalists (rule 1; rule 2: 93% vs 77%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +1.7 SD; 8% S/G2M; 94% Epithelial; 100% HuMEC_TGFB; MFAP5 ECM1 QSOX1

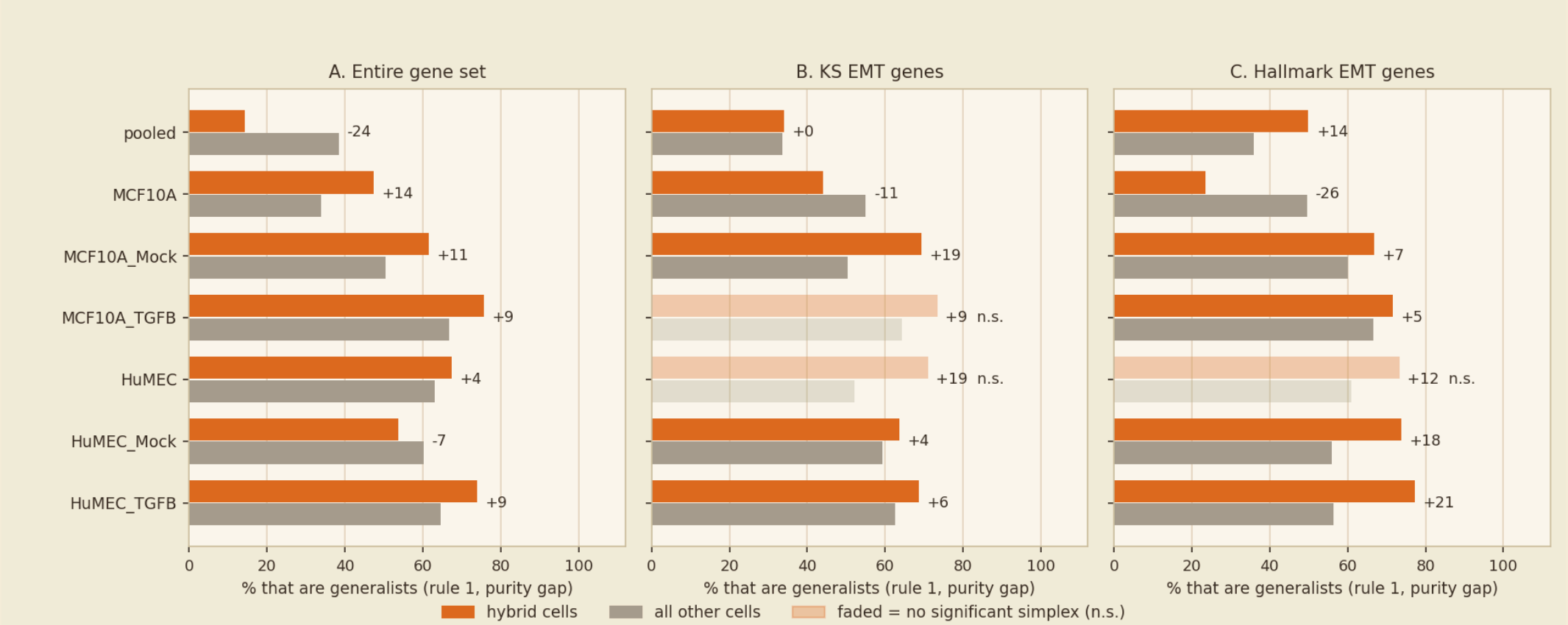
V2 epithelial junction — Apical Junction +1.56 SD; 22% S/G2M; 80% Mesenchymal; 100% HuMEC_TGFB; FN1 LAMA3 LAMC2

V3 MYC / growth (translation) — Myc Targets V1 +0.71 SD; 74% S/G2M; 48% Mesenchymal; 100% HuMEC_TGFB; TFPI2 VIM TIMP1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Question 2 · MCF10A/HuMEC: are hybrid cells more often generalists than other cells?

Orange = hybrid cells, grey = all other cells; one pair of bars per unit; the number = orange minus grey (percentage points)



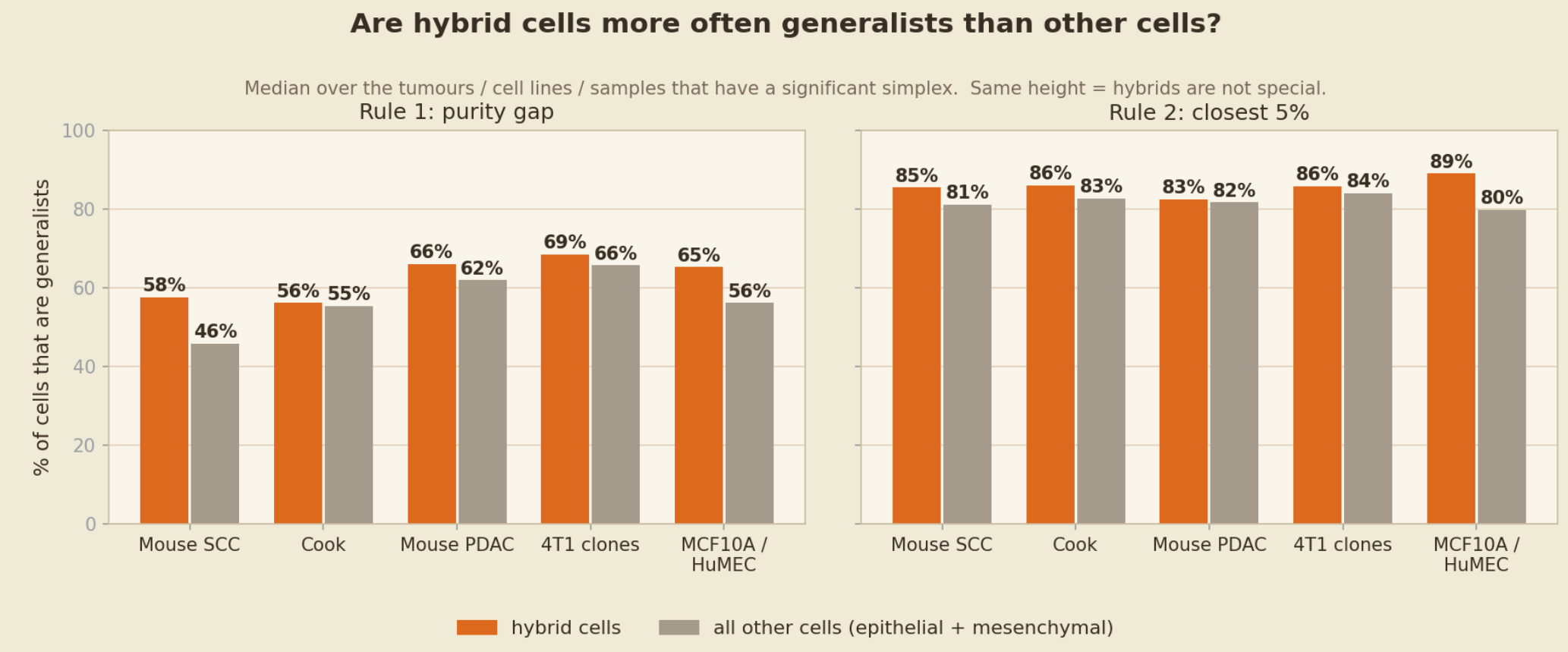
Reading

- A. Entire gene set (7 significant): hybrids 61% vs others 60% generalist under rule 1 (rule 2: 88% vs 80%) → hybrids are clearly more often generalists.
- B. KS EMT genes (5 significant): hybrids 64% vs others 55% generalist under rule 1 (rule 2: 91% vs 80%) → hybrids are slightly more often generalists.
- C. Hallmark EMT genes (6 significant): hybrids 69% vs others 56% generalist under rule 1 (rule 2: 90% vs 79%) → hybrids are clearly more often generalists.

Rule 3 (weight > 0.95) is not drawn: it calls about 100% of every group generalist, so it cannot show a difference.

What it all says: Question 2

Are hybrid cells more often generalists than other cells? (all five datasets)



Most cells are generalists, hybrid or not. Under rule 1 a median 62% of hybrid cells and 56% of other cells are generalists; under rule 2 86% vs 81%.

Hybrids lean slightly toward the interior. Median difference, hybrids minus other cells (rule 1): Mouse SCC +10, MCF10A/HuMEC +6, Mouse PDAC +5, 4T1 clones +2, Cook +0 points. The tilt is clearest in Mouse SCC and absent in Cook.

No hybrid corner. In no dataset do hybrid cells pile up at a corner of their own: hybrids are mixtures in the interior about as often as, or slightly more often than, other cells.

What it all says: Question 1

Is there a simplex, and are its corners tasks? (main fits)

Significance depends on the gene set and the level. Main fits significant: A. Entire gene set 26 of 38; B. KS EMT genes 16 of 38; C. Hallmark EMT genes 19 of 38. Hallmark behaves like KS rather than like the entire gene set.

Pooled simplices mostly separate samples. 53% of the corners of significant pooled simplices are at least 70% one sample, tumour, line or clone.

Within one unit the corners are programmes, not epithelial / hybrid / mesenchymal states. Most frequent corner names in significant within-unit fits: KRAS-down (epithelial / differentiated) (23); MYC / growth (translation) (20); EMT / matrix (16); metabolic (16); angiogenesis / matrix (11); cycling (10). 14% of corners are hybrid-dominated (≥ 60% of their specialists), mainly cycling or MYC/growth corners; 14 of 19 'EMT / matrix' corners are mesenchymal or late-EMT.

So: a trade-off structure exists inside single units (proliferation, growth, differentiation, EMT / matrix), but EMT states do not form their own corners and no dedicated hybrid corner is seen.

Caveats

What these results do not show

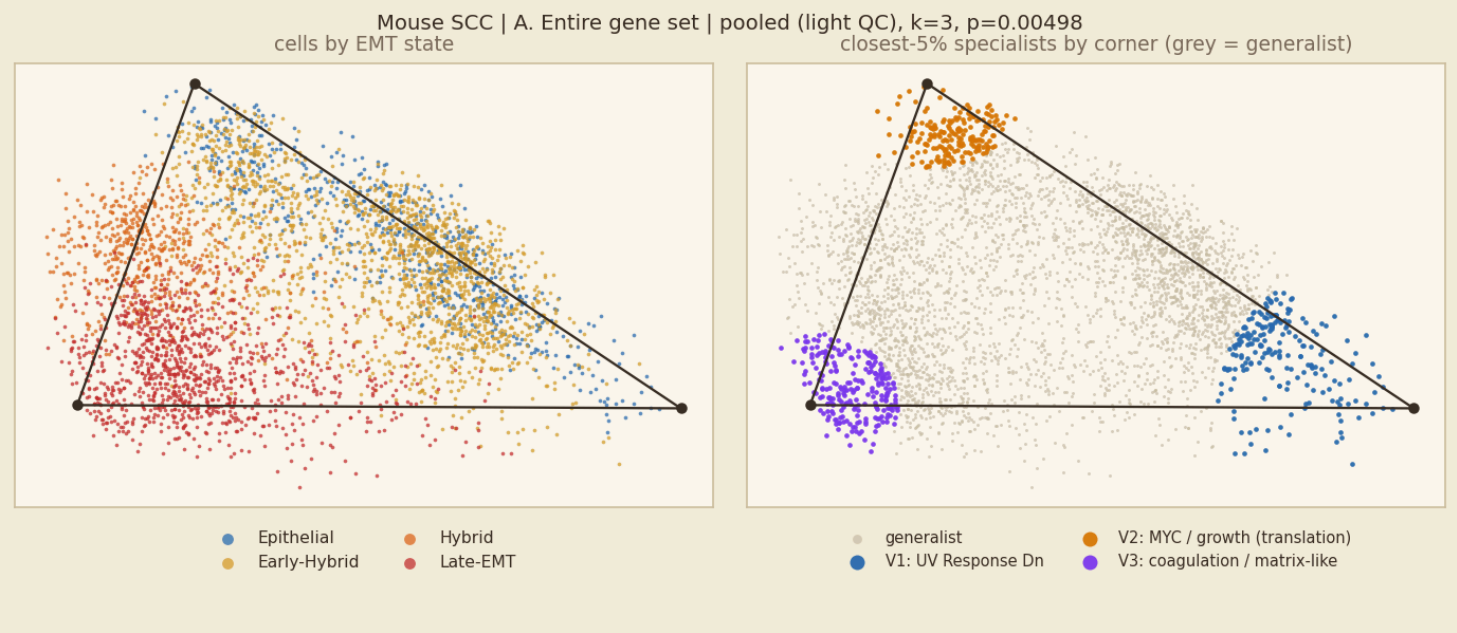
- Many fits per dataset (gene sets, k, arms); no multiplicity correction. p cannot go below about 0.005 (200 shuffles) or 0.01 (100 shuffles).
- Hallmark pooled, cleaned-pooled and SCC scRNA-only KS / Hallmark pooled fits used k = 3 to 5 and 100 shuffles.
- Hybrid labels for PDAC, 4T1, GSE114687 and Cook are cut from the same KS score used for the KS-gene fits (circular); Cook bins are pooled across lines.
- Within-unit fits are capped at 4,000 random cells; corner names are automatic (top Hallmark set) and should be sanity-checked against biology.
- Design limits: one tumour per clone (4T1), two mice (PDAC), no replicates (GSE114687), tumour / assay / state tangled (SCC). Nothing here shows that EMT is or is not a Pareto trade-off in tumours.
- Files: main_fits.csv (the fits shown), best_fits.csv (all arms), corner_annotations.csv; scripts analyze.py, corners.py, build_explainer_ppt.py in docs/emt_single_cell/best_simplex/.

Appendix

A1. Mouse SCC: the all-cells pooled fit (with multiome nuclei) and the four multiome tumours. A2. Light QC versus cleaned arm, per unit, for Cook, PDAC, 4T1 and GSE114687.

Appendix A1: Mouse SCC: A. Entire gene set, pooled fit

Fit shown: light QC · k = 3 · p = 0.00498 · 23,959 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	74%	41%
2 closest 5% of cells to each corner	89%	79%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 57% of all cells here; 73% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

A significant simplex (k=3, p=0.00498) whose corners mix samples (no corner is 70% or more one sample), so it is not a pure sample-identity simplex.

Corners are programmes: UV Response Dn; MYC / growth (translation); coagulation / matrix-like.

Hybrid vs other cells: 74% of hybrid cells and 41% of other cells are generalists (rule 1; rule 2: 89% vs 79%), so hybrids are clearly more often generalists.

Corners and their tasks

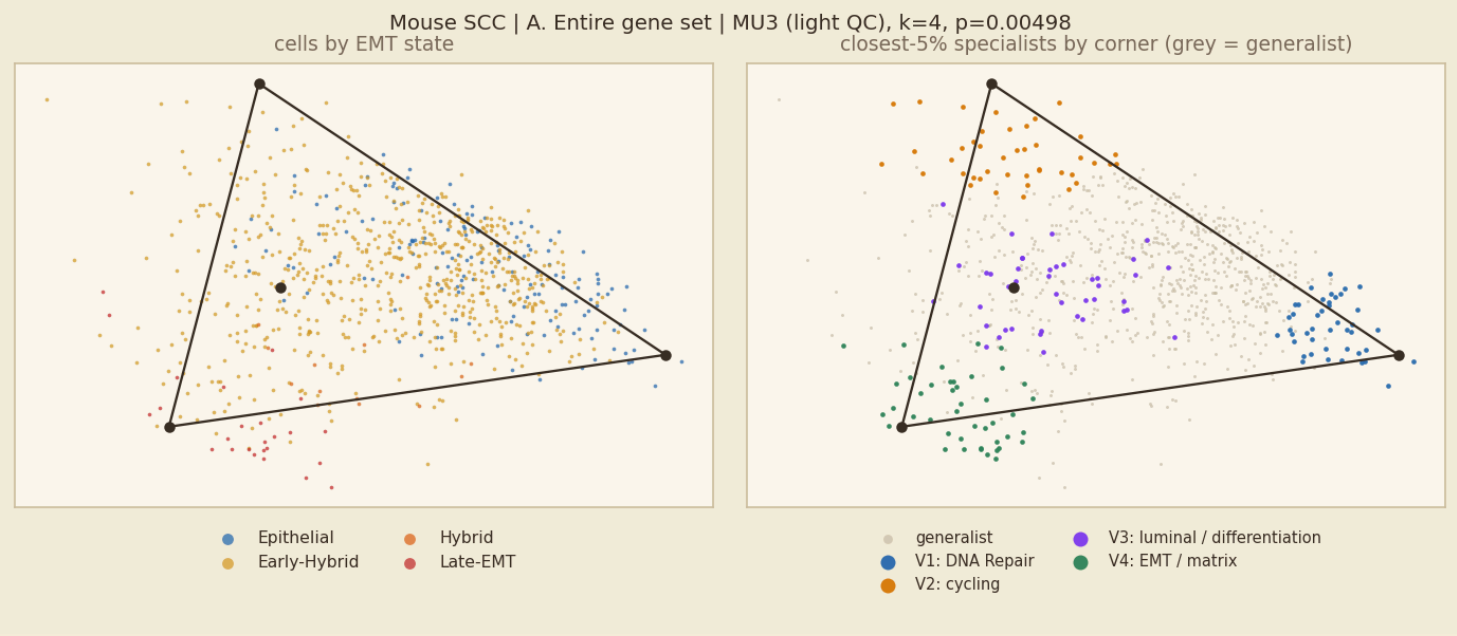
V1 UV Response Dn — UV Response Dn +1.34 SD; 24% S/G2M; 57% Early-Hybrid; Frmd4b Adgrl3 Maml2

V2 MYC / growth (translation) — Myc Targets V1 +1.8 SD; 75% S/G2M; 62% Early-Hybrid; Krt14 Krt5 Krt6a

V3 coagulation / matrix-like — Coagulation +1.42 SD; 9% S/G2M; 98% Late-EMT; 60% SC2b1; Col3a1 Col1a1 Col1a2

Appendix A1: Mouse SCC: A. Entire gene set, unit MU3

Fit shown: light QC · k = 4 · p = 0.00498 · 852 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	75%	52%
2 closest 5% of cells to each corner	83%	69%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 78% of all cells here; 50% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MU3 alone, cells fill a 4-corner simplex (p=0.00498). 4 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: DNA Repair; cycling; luminal / differentiation; EMT / matrix.

Hybrid vs other cells: 75% of hybrid cells and 52% of other cells are generalists (rule 1; rule 2: 83% vs 69%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 DNA Repair — DNA Repair +1.67 SD; 88% S/G2M; 65% Epithelial; 100% MU3; H2afz Hmgb2 Ly6c1

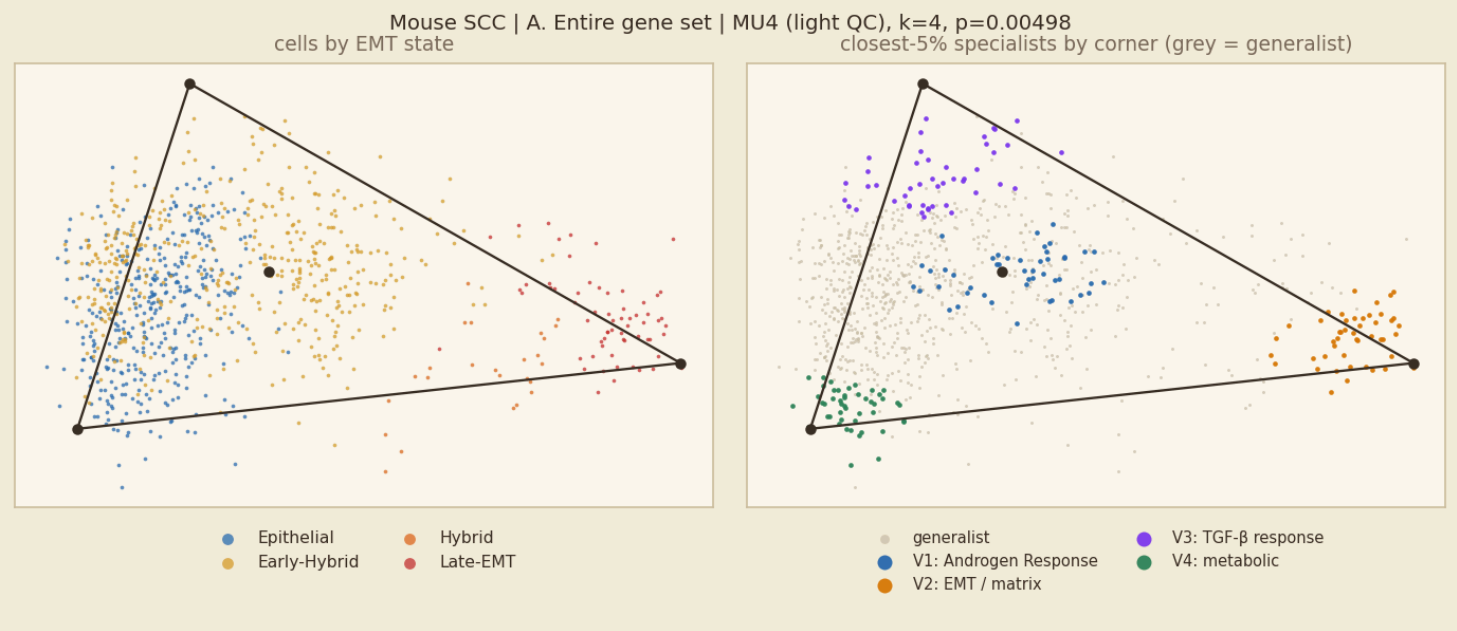
V2 cycling — Mitotic Spindle +1.37 SD; 65% S/G2M; 93% Early-Hybrid; 100% MU3; Grip1 Jmjd1c Pkp4

V3 luminal / differentiation — Estrogen Response Early +1.15 SD; 9% S/G2M; 79% Early-Hybrid; 100% MU3; Slpi Krt16 Aqp3

V4 EMT / matrix — Epithelial Mesenchymal Transition +2.64 SD; 7% S/G2M; 53% Early-Hybrid; 100% MU3; Fn1 Col5a2 Col1a1

Appendix A1: Mouse SCC: A. Entire gene set, unit MU4

Fit shown: light QC · k = 4 · p = 0.00498 · 939 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	62%	52%
2 closest 5% of cells to each corner	84%	75%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 51% of all cells here; 26% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MU4 alone, cells fill a 4-corner simplex (p=0.00498). 4 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: Androgen Response; EMT / matrix; TGF-β response; metabolic.

Hybrid vs other cells: 62% of hybrid cells and 52% of other cells are generalists (rule 1; rule 2: 84% vs 75%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 Androgen Response — Androgen Response +1.22 SD; 53% S/G2M; 81% Early-Hybrid; 100% MU4; Cxcl3 Csf3 Areg

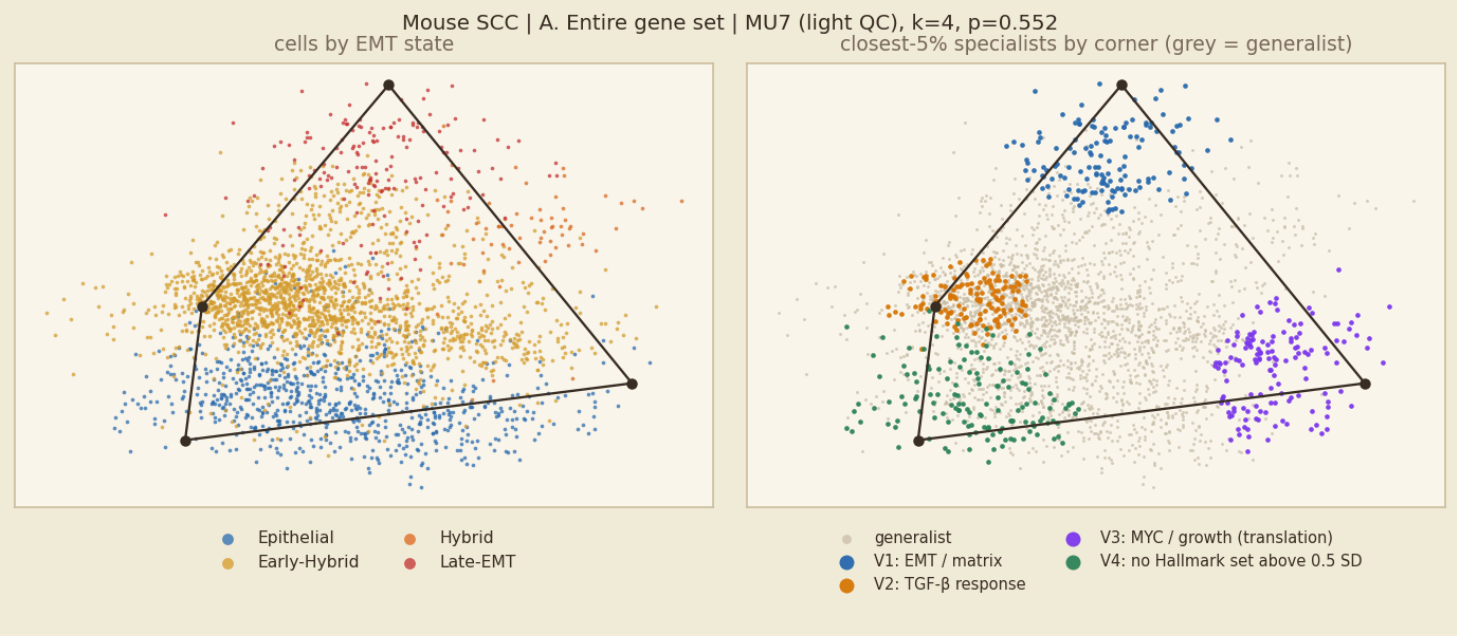
V2 EMT / matrix — Epithelial Mesenchymal Transition +2.81 SD; 11% S/G2M; 91% Late-EMT; 100% MU4; Dcn Col1a2 Spp1

V3 TGF-β response — TGF-beta Signaling +0.77 SD; 26% S/G2M; 62% Early-Hybrid; 100% MU4; Kcnq5 Grip1 Adgrl3

V4 metabolic — Oxidative Phosphorylation +1.76 SD; 98% S/G2M; 91% Epithelial; 100% MU4; Hmgb2 Tubb5 H2afz

Appendix A1: Mouse SCC: A. Entire gene set, unit MU7

Fit shown: light QC · k = 4 · p = 0.552 · 2,801 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight – second ≥ 0.35)	68%	73%
2 closest 5% of cells to each corner	84%	73%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 66% of all cells here; 40% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside MU7: the fit (k=4, p=0.552) is no more triangular than shuffled data, so its corners should not be read as tasks. 4 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 68% of hybrid cells and 73% of other cells are generalists (rule 1; rule 2: 84% vs 73%), so hybrids are less often generalists (they sit at corners). (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 EMT / matrix — Epithelial Mesenchymal Transition +1.74 SD; 26% S/G2M; 70% Late-EMT; 100% MU7; Brinp3 Col1a1 Col1a2

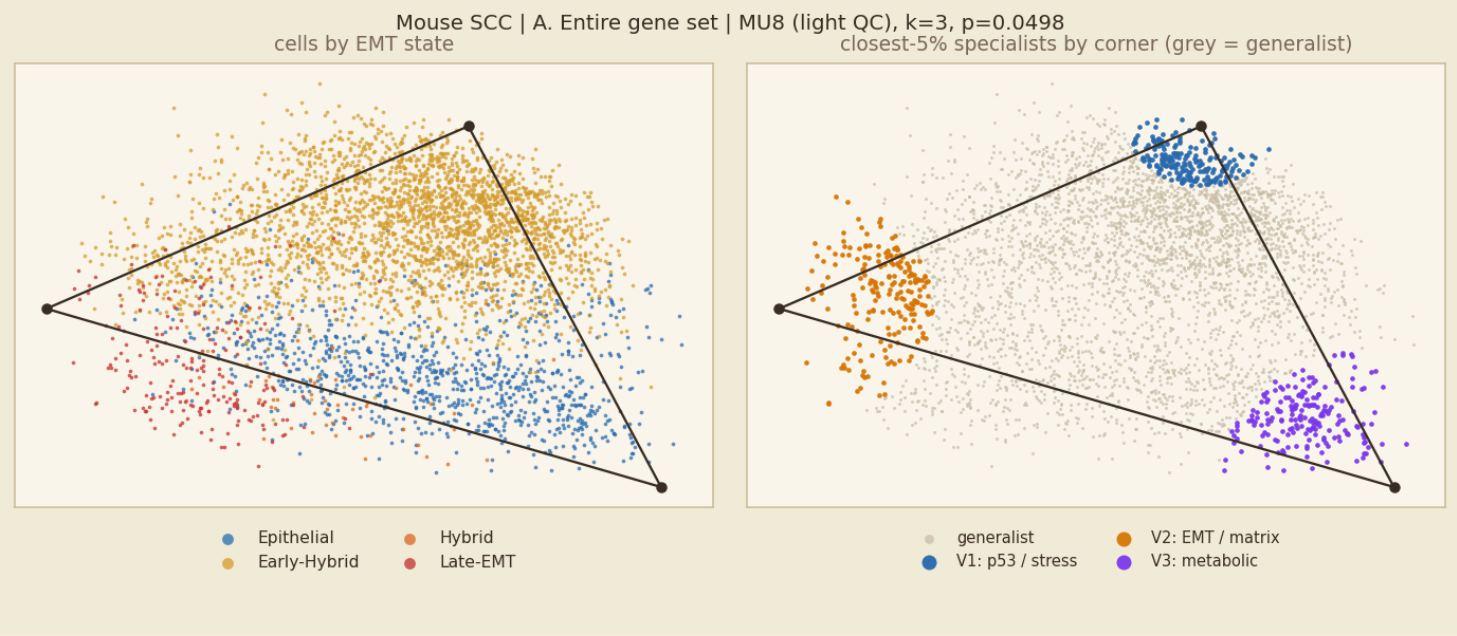
V2 TGF-β response — TGF-beta Signaling +1.23 SD; 5% S/G2M; 99% Early-Hybrid; 100% MU7; Macrod2 Ccnd2 Thbs1

V3 MYC / growth (translation) — Myc Targets V1 +2.37 SD; 92% S/G2M; 60% Early-Hybrid; 100% MU7; Ly6a H2afz Krt6a

V4 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 26% S/G2M; 78% Epithelial; 100% MU7; Arap2 2610035D17Rik Grip1

Appendix A1: Mouse SCC: A. Entire gene set, unit MU8

Fit shown: light QC · k = 3 · p = 0.0498 · 5,308 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	48%	58%
2 closest 5% of cells to each corner	88%	77%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 70% of all cells here; 72% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MU8 alone, cells fill a 3-corner simplex (p=0.0498). 4 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: p53 / stress; EMT / matrix; metabolic.

Hybrid vs other cells: 48% of hybrid cells and 58% of other cells are generalists (rule 1; rule 2: 88% vs 77%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

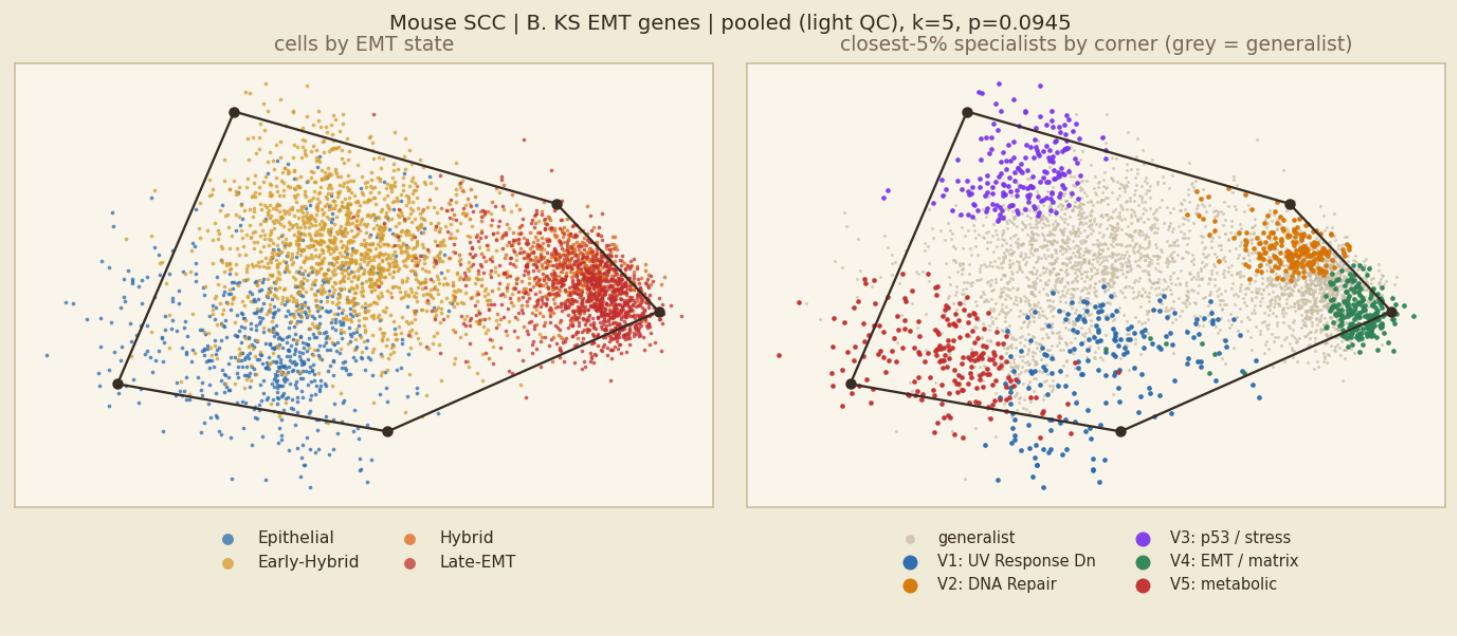
V1 p53 / stress — p53 Pathway +1.0 SD; 34% S/G2M; 100% Early-Hybrid; 100% MU8; Nrg1 Areg Mmp10

V2 EMT / matrix — Epithelial Mesenchymal Transition +1.64 SD; 9% S/G2M; 60% Early-Hybrid; 100% MU8; Col5a2 Col12a1 Hmcn1

V3 metabolic — Oxidative Phosphorylation +1.32 SD; 79% S/G2M; 99% Epithelial; 100% MU8; Epcam Krt17 S100a14

Appendix A1: Mouse SCC: B. KS EMT genes, pooled fit

Fit shown: light QC · k = 5 · p = 0.0945 · 23,959 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	84%	61%
2 closest 5% of cells to each corner	77%	72%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 57% of all cells here; 14% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex: the pooled fit (k=5, p=0.0945) is no more triangular than shuffled data, so its corners should not be read as tasks.

Hybrid vs other cells: 84% of hybrid cells and 61% of other cells are generalists (rule 1; rule 2: 77% vs 72%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 UV Response Dn — UV Response Dn +1.16 SD; 29% S/G2M; 53% Early-Hybrid; Rabgap1l Ank3 Myo1d

V2 DNA Repair — DNA Repair +1.17 SD; 68% S/G2M; 90% Hybrid; Lgals1 Krt8 Vim

V3 p53 / stress — p53 Pathway +1.1 SD; 30% S/G2M; 96% Early-Hybrid; Slpi Aqp3 Areg

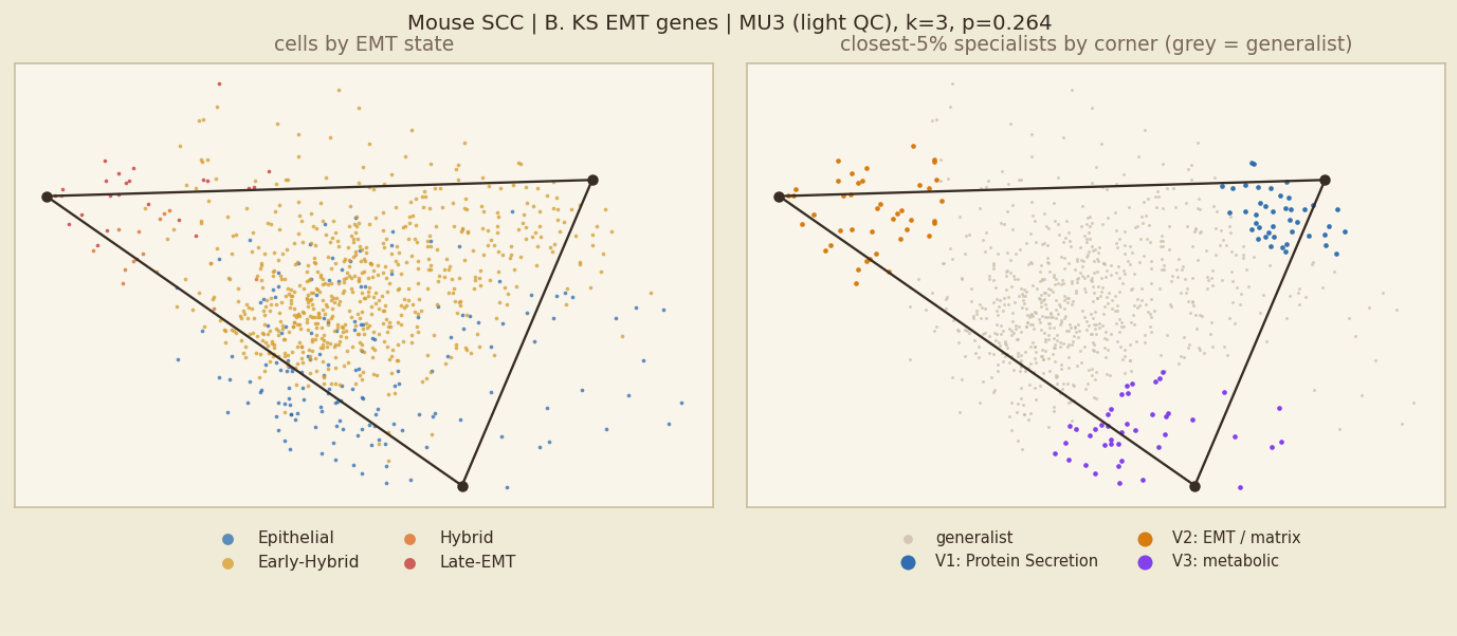
V4 EMT / matrix — Epithelial Mesenchymal Transition +1.83 SD; 12% S/G2M; 96% Late-EMT; Col5a2 Sparc Fstl1

V5 metabolic — Cholesterol Homeostasis +0.75 SD; 48% S/G2M; 87% Epithelial; Epcam Tacstd2 S100a14

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Appendix A1: Mouse SCC: B. KS EMT genes, unit MU3

Fit shown: light QC · k = 3 · p = 0.264 · 852 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	80%	58%
2 closest 5% of cells to each corner	89%	70%
3 weight > 0.95 on one corner	100%	98%

Hybrids are 78% of all cells here; 79% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside MU3: the fit (k=3, p=0.264) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 80% of hybrid cells and 58% of other cells are generalists (rule 1; rule 2: 89% vs 70%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 Protein Secretion — Protein Secretion +0.86 SD; 12% S/G2M; 98% Early-Hybrid; 100% MU3; Slpi Aqp3 Sdc1

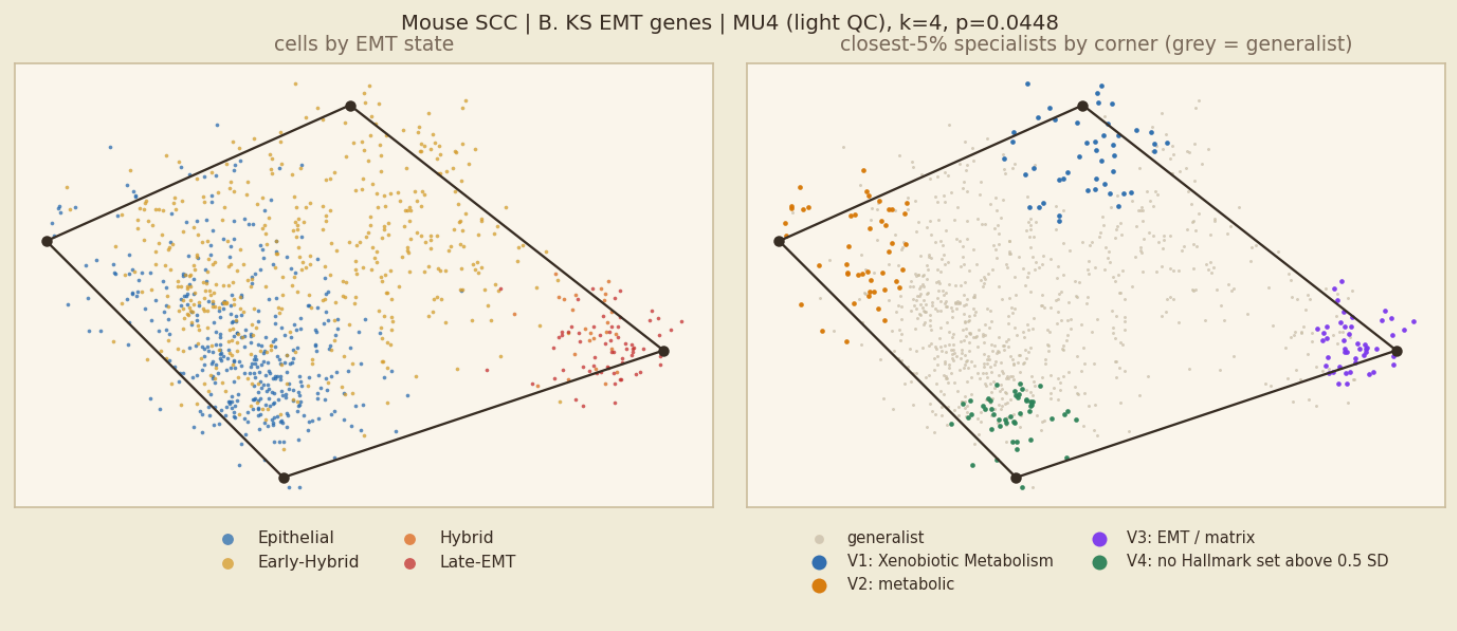
V2 EMT / matrix — Epithelial Mesenchymal Transition +2.43 SD; 14% S/G2M; 47% Late-EMT; 100% MU3; Col5a2 Fstl1 Cald1

V3 metabolic — Cholesterol Homeostasis +0.9 SD; 65% S/G2M; 79% Epithelial; 100% MU3; Epcam S100a14 Gpx2

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Appendix A1: Mouse SCC: B. KS EMT genes, unit MU4

Fit shown: light QC · k = 4 · p = 0.0448 · 939 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	67%	64%
2 closest 5% of cells to each corner	81%	79%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 51% of all cells here; 32% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MU4 alone, cells fill a 4-corner simplex (p=0.0448). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: Xenobiotic Metabolism; metabolic; EMT / matrix; no Hallmark set above 0.5 SD.

Hybrid vs other cells: 67% of hybrid cells and 64% of other cells are generalists (rule 1; rule 2: 81% vs 79%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 Xenobiotic Metabolism — Xenobiotic Metabolism +1.15 SD; 38% S/G2M; 100% Early-Hybrid; 100% MU4; Areg Aqp3 Slpi

V2 metabolic — Cholesterol Homeostasis +1.05 SD; 51% S/G2M; 53% Epithelial; 100% MU4; Tacstd2 Dsc2 Epcam

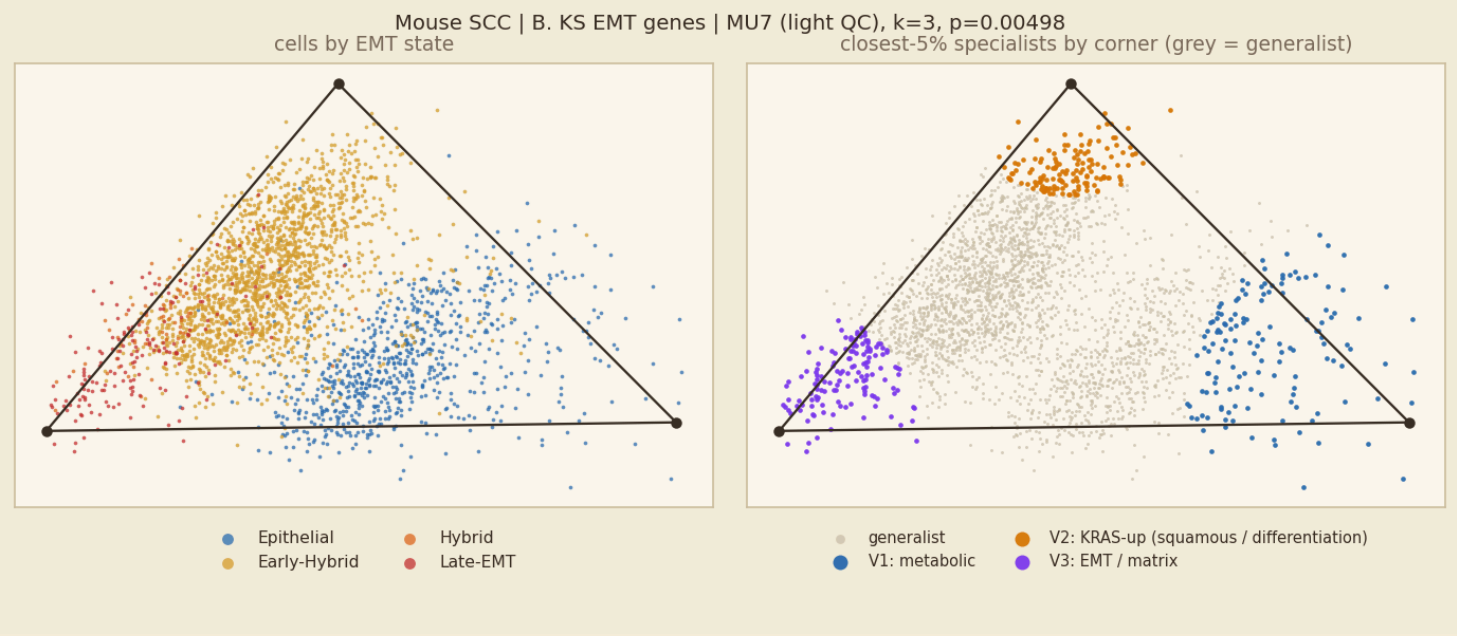
V3 EMT / matrix — Epithelial Mesenchymal Transition +2.48 SD; 26% S/G2M; 72% Late-EMT; 100% MU4; Vim Lgals1 Col5a2

V4 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 43% S/G2M; 83% Epithelial; 100% MU4; Fxyd3 Map7 Myo1d

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Appendix A1: Mouse SCC: B. KS EMT genes, unit MU7

Fit shown: light QC · k = 3 · p = 0.00498 · 2,801 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	72%	78%
2 closest 5% of cells to each corner	90%	76%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 66% of all cells here; 92% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MU7 alone, cells fill a 3-corner simplex (p=0.00498). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: metabolic; KRAS-up (squamous / differentiation); EMT / matrix.

Hybrid vs other cells: 72% of hybrid cells and 78% of other cells are generalists (rule 1; rule 2: 90% vs 76%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 metabolic — Cholesterol Homeostasis +0.79 SD; 37% S/G2M; 94% Epithelial; 100% MU7; Epcam Tacstd2 Cdh1

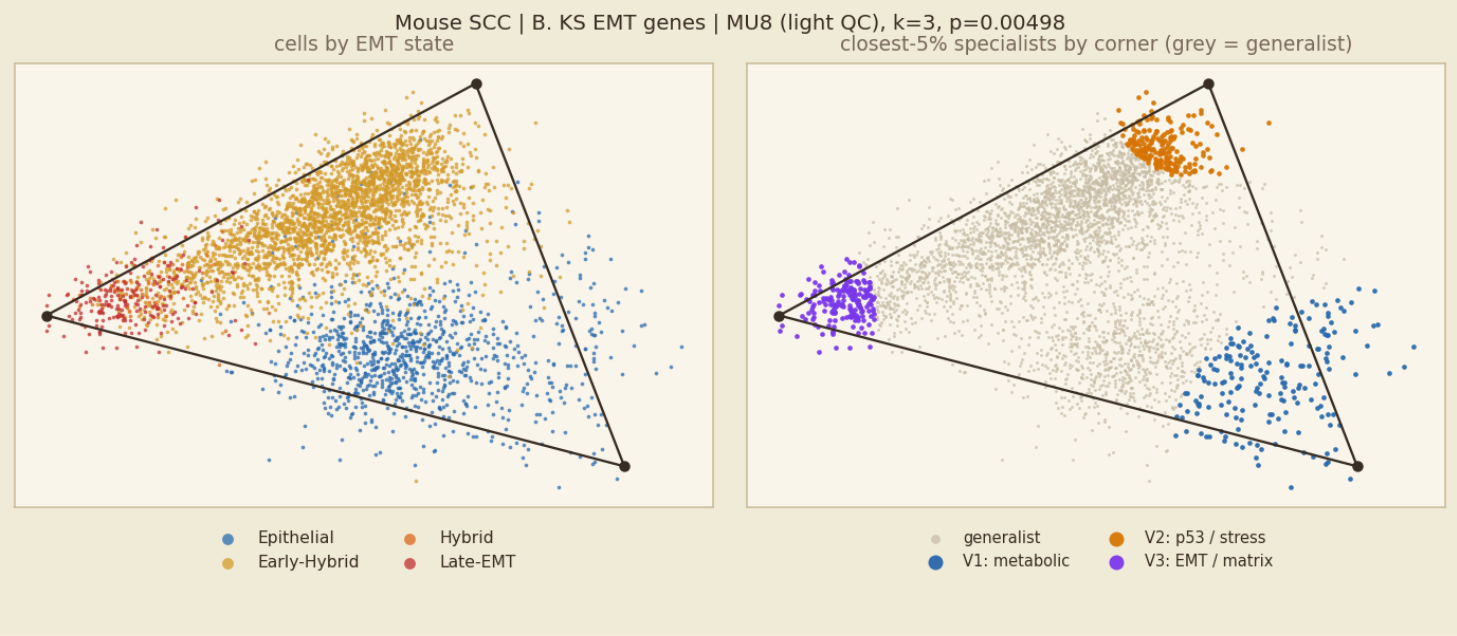
V2 KRAS-up (squamous / differentiation) — KRAS Signaling Up +0.94 SD; 34% S/G2M; 99% Early-Hybrid; 100% MU7; Areg Aqp3 Itgb4

V3 EMT / matrix — Epithelial Mesenchymal Transition +2.05 SD; 35% S/G2M; 69% Late-EMT; 100% MU7; Col5a2 Fstl1 Vim

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Appendix A1: Mouse SCC: B. KS EMT genes, unit MU8

Fit shown: light QC · k = 3 · p = 0.00498 · 5,308 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	61%	68%
2 closest 5% of cells to each corner	90%	73%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 70% of all cells here; 92% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MU8 alone, cells fill a 3-corner simplex (p=0.00498). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: metabolic; p53 / stress; EMT / matrix.

Hybrid vs other cells: 61% of hybrid cells and 68% of other cells are generalists (rule 1; rule 2: 90% vs 73%), so hybrids are less often generalists (they sit at corners).

Corners and their tasks

V1 metabolic — Cholesterol Homeostasis +1.19 SD; 45% S/G2M; 98% Epithelial; 100% MU8; Tacstd2 Epcam Cdh1

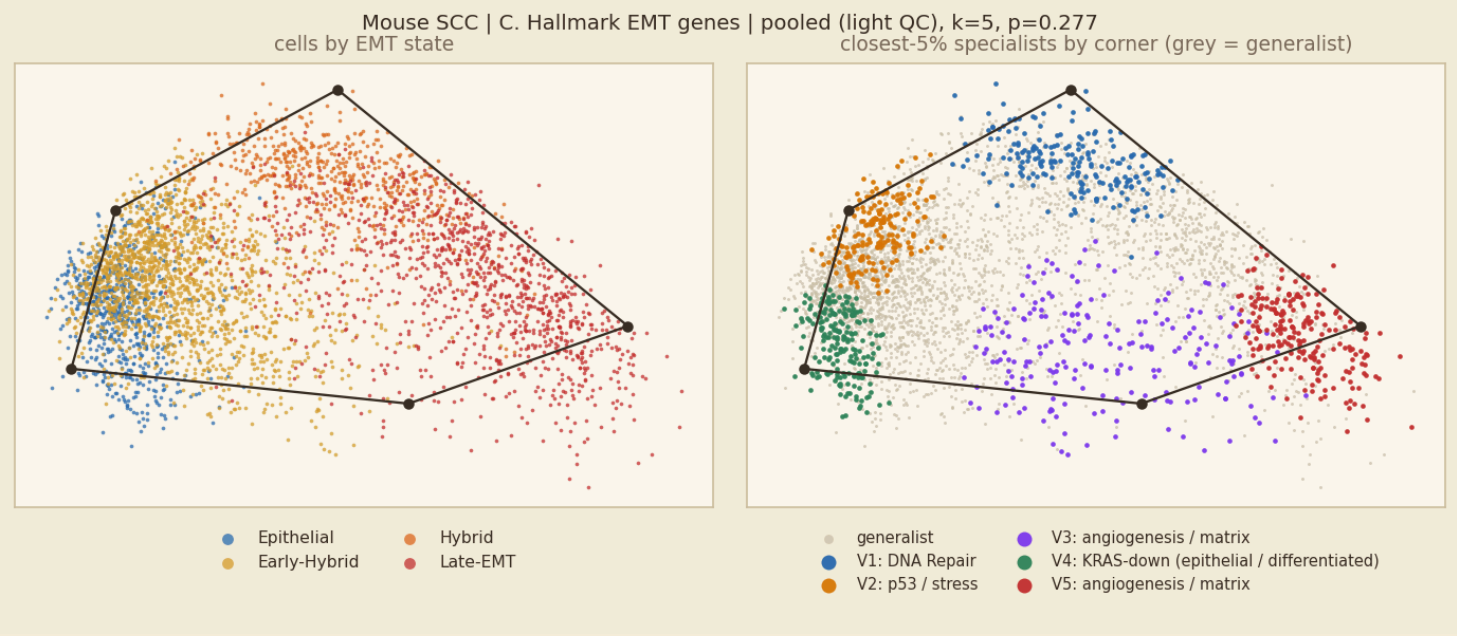
V2 p53 / stress — p53 Pathway +0.96 SD; 33% S/G2M; 100% Early-Hybrid; 100% MU8; Areg Aqp3 Sfn

V3 EMT / matrix — Epithelial Mesenchymal Transition +2.0 SD; 17% S/G2M; 64% Late-EMT; 100% MU8; Col5a2 Vim Fstl1

KS genes overlap the EMT set: an 'EMT / matrix' label is partly expected.

Appendix A1: Mouse SCC: C. Hallmark EMT genes, pooled fit

Fit shown: light QC · k = 5 · p = 0.277 · 23,959 cells · 100 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	80%	66%
2 closest 5% of cells to each corner	78%	71%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 57% of all cells here; 13% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex: the pooled fit (k=5, p=0.277) is no more triangular than shuffled data, so its corners should not be read as tasks.

Hybrid vs other cells: 80% of hybrid cells and 66% of other cells are generalists (rule 1; rule 2: 78% vs 71%), so hybrids are clearly more often generalists. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 DNA Repair — DNA Repair +1.3 SD; 67% S/G2M; 87% Hybrid; Lgals1 Timp1 Vim

V2 p53 / stress — p53 Pathway +1.12 SD; 46% S/G2M; 92% Early-Hybrid; Areg Cxcl1 Ecm1

V3 angiogenesis / matrix — Angiogenesis +1.22 SD; 13% S/G2M; 50% Early-Hybrid; Fn1 Col5a2 Spp1

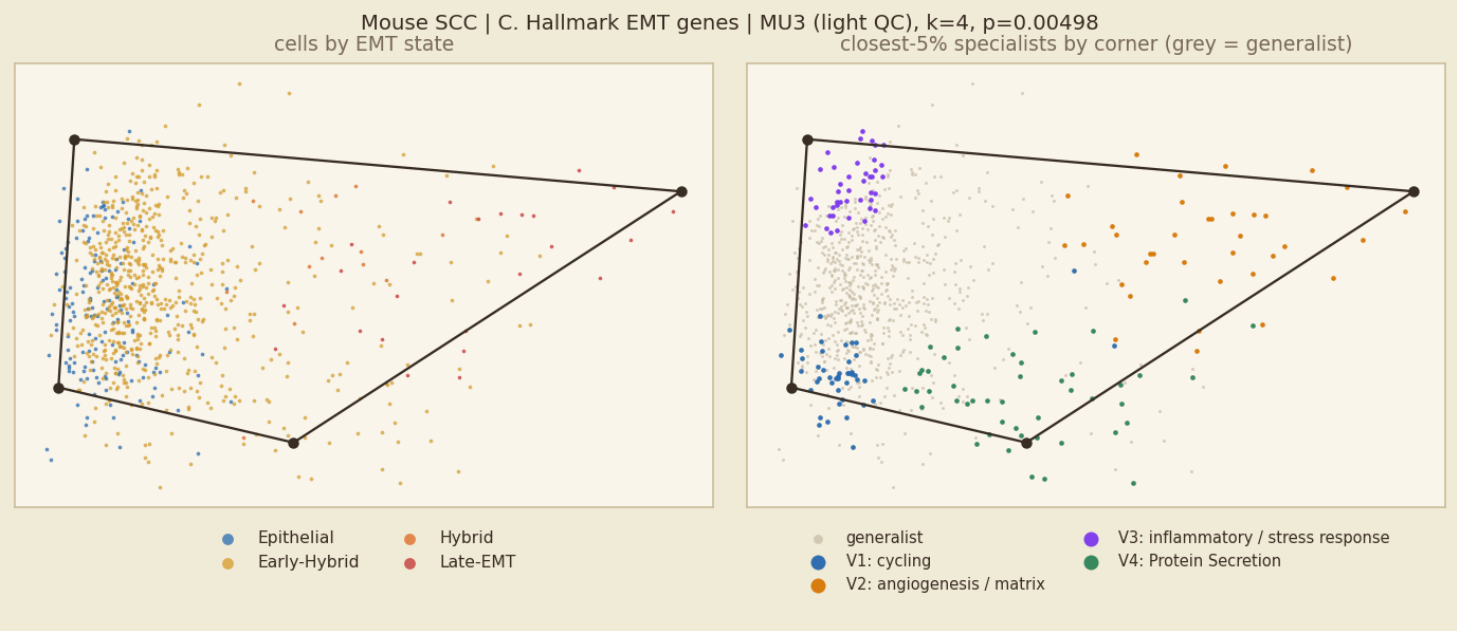
V4 KRAS-down (epithelial / differentiated) — KRAS Signaling Dn +0.61 SD; 44% S/G2M; 83% Epithelial; Sfrp1 Lama3 Cadm1

V5 angiogenesis / matrix — Angiogenesis +1.54 SD; 9% S/G2M; 98% Late-EMT; Col3a1 Col1a1 Col1a2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Appendix A1: Mouse SCC: C. Hallmark EMT genes, unit MU3

Fit shown: light QC · k = 4 · p = 0.00498 · 852 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	67%	52%
2 closest 5% of cells to each corner	83%	69%
3 weight > 0.95 on one corner	100%	98%

Hybrids are 78% of all cells here; 50% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MU3 alone, cells fill a 4-corner simplex (p=0.00498). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: cycling; angiogenesis / matrix; inflammatory / stress response; Protein Secretion.

Hybrid vs other cells: 67% of hybrid cells and 52% of other cells are generalists (rule 1; rule 2: 83% vs 69%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 cycling — no Hallmark set above 0.5 SD; 65% S/G2M; 63% Epithelial; 100% MU3; Jun Mcm7 Plod1

V2 angiogenesis / matrix — Angiogenesis +2.54 SD; 9% S/G2M; 51% Late-EMT; 100% MU3; Dcn Col5a2 Col1a1

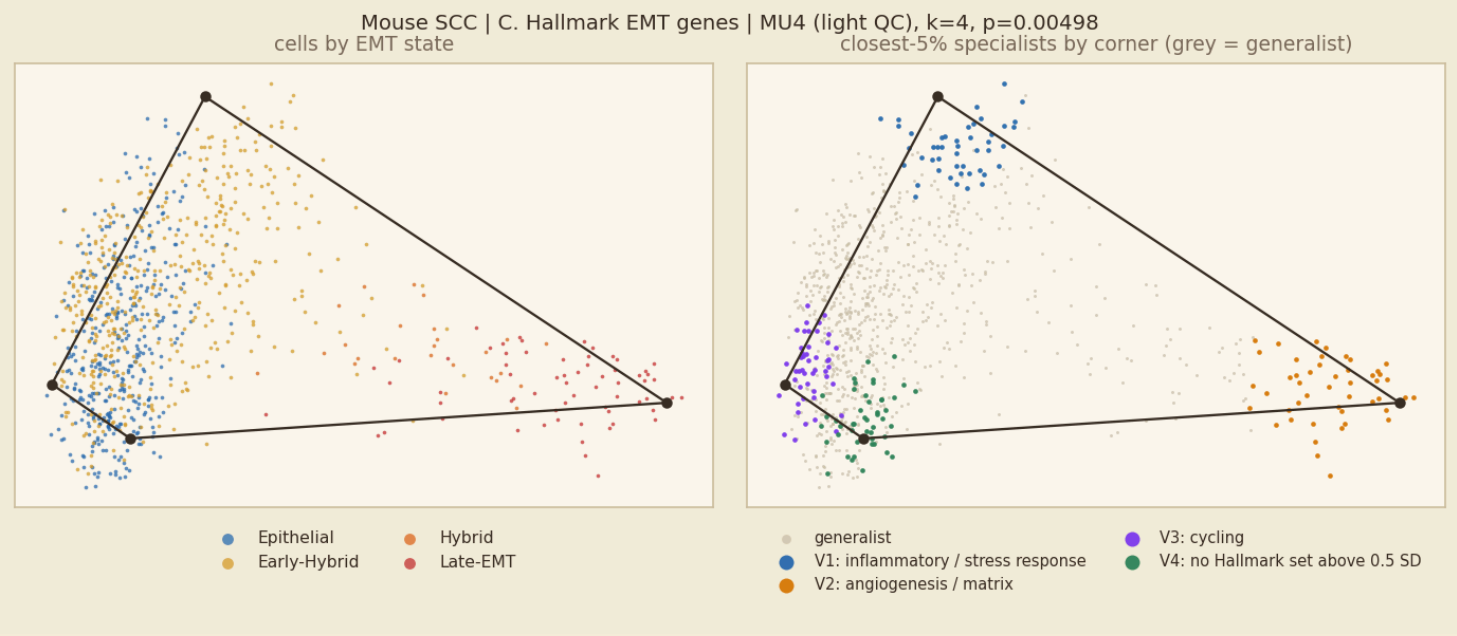
V3 inflammatory / stress response — Inflammatory Response +1.09 SD; 51% S/G2M; 84% Early-Hybrid; 100% MU3; Areg Cxcl1 Lama3

V4 Protein Secretion — Protein Secretion +1.16 SD; 0% S/G2M; 98% Early-Hybrid; 100% MU3; Spp1 Col6a2 Ecm1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Appendix A1: Mouse SCC: C. Hallmark EMT genes, unit MU4

Fit shown: light QC · k = 4 · p = 0.00498 · 939 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	65%	61%
2 closest 5% of cells to each corner	86%	74%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 51% of all cells here; 36% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MU4 alone, cells fill a 4-corner simplex (p=0.00498). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: inflammatory / stress response; angiogenesis / matrix; cycling; no Hallmark set above 0.5 SD.

Hybrid vs other cells: 65% of hybrid cells and 61% of other cells are generalists (rule 1; rule 2: 86% vs 74%), so hybrids are slightly more often generalists.

Corners and their tasks

V1 inflammatory / stress response — Inflammatory Response +1.37 SD; 36% S/G2M; 85% Early-Hybrid; 100% MU4; Areg Vim Cxcl1

V2 angiogenesis / matrix — Angiogenesis +2.94 SD; 9% S/G2M; 94% Late-EMT; 100% MU4; Col1a2 Dcn Spp1

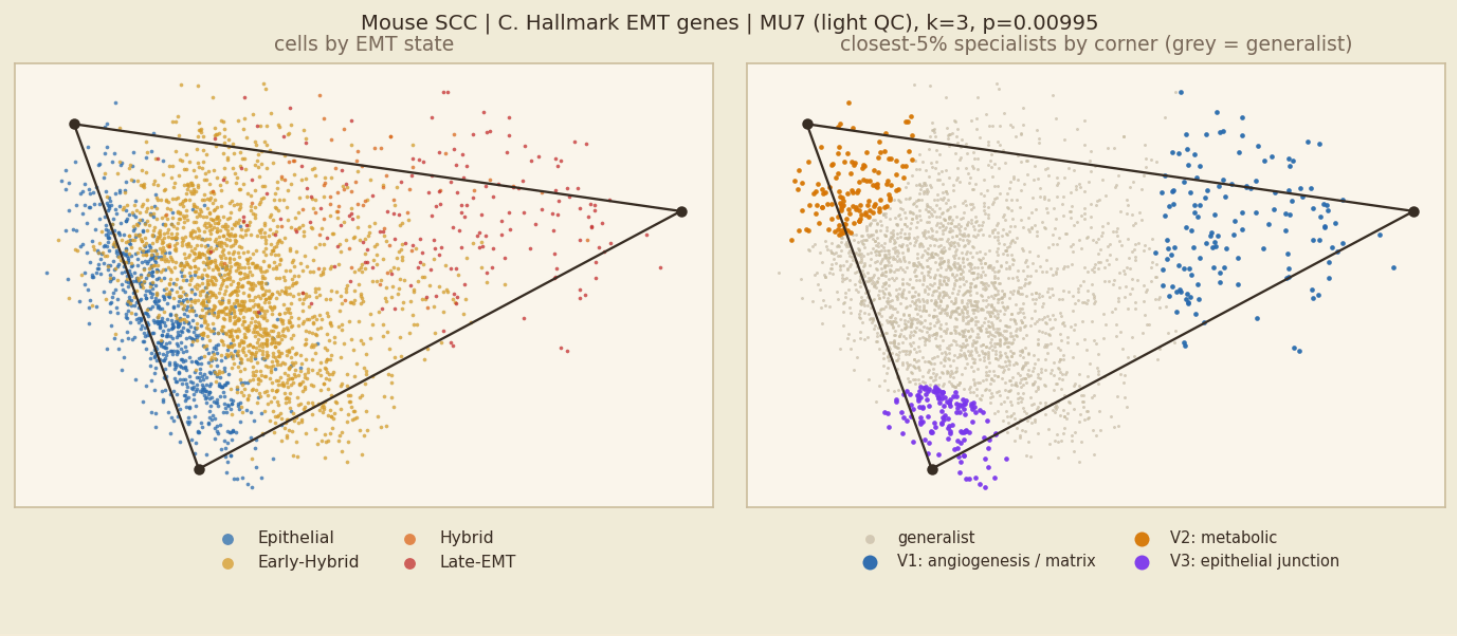
V3 cycling — no Hallmark set above 0.5 SD; 64% S/G2M; 55% Epithelial; 100% MU4; Igfbp3 Areg Pthlh

V4 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 49% S/G2M; 89% Epithelial; 100% MU4; Thbs1 Timp3 Tnc

Hallmark EMT set left out when naming (circular: fitted on those genes).

Appendix A1: Mouse SCC: C. Hallmark EMT genes, unit MU7

Fit shown: light QC · k = 3 · p = 0.00995 · 2,801 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	67%	54%
2 closest 5% of cells to each corner	95%	65%
3 weight > 0.95 on one corner	100%	99%

Hybrids are 66% of all cells here; 84% of cells lie inside the simplex.

SIGNIFICANT permutation test, p ≤ 0.05

In plain words

Inside MU7 alone, cells fill a 3-corner simplex (p=0.00995). 3 of 8 units of this dataset have a significant simplex for this gene set.

Corners are programmes: angiogenesis / matrix; metabolic; epithelial junction.

Hybrid vs other cells: 67% of hybrid cells and 54% of other cells are generalists (rule 1; rule 2: 95% vs 65%), so hybrids are clearly more often generalists.

Corners and their tasks

V1 angiogenesis / matrix — Angiogenesis +2.15 SD; 28% S/G2M; 80% Late-EMT; 100% MU7; Col1a2 Col1a1 Col3a1

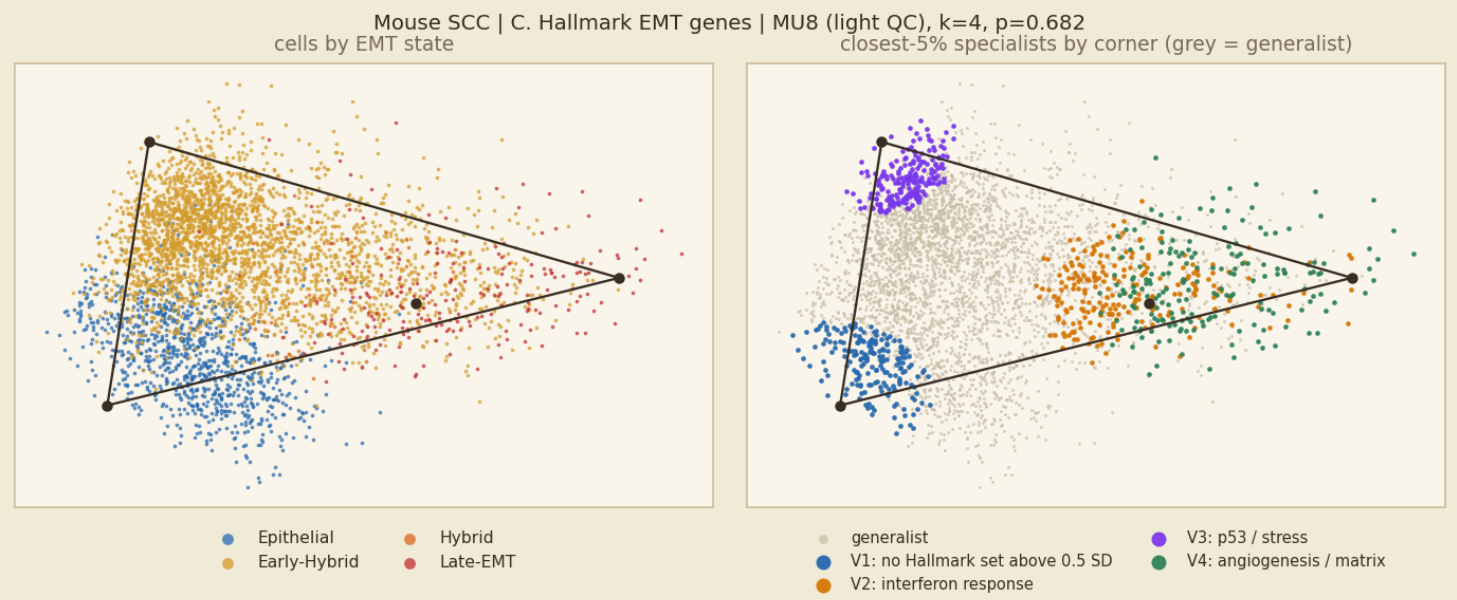
V2 metabolic — Cholesterol Homeostasis +0.52 SD; 30% S/G2M; 61% Epithelial; 100% MU7; Spp1 Areg Sdc1

V3 epithelial junction — Apical Junction +0.88 SD; 13% S/G2M; 96% Epithelial; 100% MU7; Lama3 Adam12 Lamc2

Hallmark EMT set left out when naming (circular: fitted on those genes).

Appendix A1: Mouse SCC: C. Hallmark EMT genes, unit MU8

Fit shown: light QC · k = 4 · p = 0.682 · 5,308 cells · 200 shuffles



Left: cells coloured by EMT state (blue epithelial → red mesenchymal; hybrids are the two middle colours). Right: the 5% of cells nearest each corner coloured by corner, all other cells grey = generalists.

rule	% of hybrid cells that are generalists	% of other cells that are generalists
1 gap (top weight − second ≥ 0.35)	65%	64%
2 closest 5% of cells to each corner	84%	72%
3 weight > 0.95 on one corner	100%	100%

Hybrids are 70% of all cells here; 37% of cells lie inside the simplex.

NOT SIGNIFICANT $p > 0.05$

In plain words

No simplex inside MU8: the fit (k=4, p=0.682) is no more triangular than shuffled data, so its corners should not be read as tasks. 3 of 8 units of this dataset have a significant simplex for this gene set.

Hybrid vs other cells: 65% of hybrid cells and 64% of other cells are generalists (rule 1; rule 2: 84% vs 72%), so no real difference. (No significant simplex, so this is descriptive only.)

Corners and their tasks (shown for completeness only)

V1 no Hallmark set above 0.5 SD — no Hallmark set above 0.5 SD; 52% S/G2M; 93% Epithelial; 100% MU8; Lama3 Lamc2 Sdc1

V2 interferon response — Interferon Alpha Response +1.07 SD; 38% S/G2M; 98% Early-Hybrid; 100% MU8; Timp3 Thbs1 Col12a1

V3 p53 / stress — p53 Pathway +0.94 SD; 43% S/G2M; 100% Early-Hybrid; 100% MU8; Areg Igfbp3 Ecm1

V4 angiogenesis / matrix — Angiogenesis +1.97 SD; 9% S/G2M; 77% Late-EMT; 100% MU8; Col5a2 Dcn Fn1

Hallmark EMT set left out when naming (circular: fitted on those genes).

Appendix A2. Cook: light QC vs cleaned

p of each arm (* = p ≤ 0.05); the cleaned arm is the one used in the main slides

unit	A. light QC	A. cleaned	B. light QC	B. cleaned	C. light QC	C. cleaned
pooled (4 lines)	k=3, 0.00498 *	k=3, 0.0099 *	k=3, 0.00498 *	k=4, 0.0099 *	k=4, 0.0099 *	k=4, 0.0099 *
A549	k=4, 0.0647	k=3, 0.00995 *	k=5, 0.169	k=4, 0.716	k=3, 0.259	k=3, 0.0448 *
DU145	k=3, 0.0995	k=3, 0.254	k=4, 0.9	k=3, 0.95	k=3, 0.0199 *	k=3, 0.0249 *
MCF7	k=5, 0.0896	k=3, 0.463	k=3, 0.0149 *	k=3, 0.0299 *	k=4, 0.891	k=4, 0.368
OVCA420	k=5, 0.453	k=3, 0.756	k=3, 0.418	k=4, 0.448	k=3, 0.00498 *	k=3, 0.00995 *

Appendix A2. Mouse PDAC: light QC vs cleaned

p of each arm (* = p ≤ 0.05); the cleaned arm is the one used in the main slides

unit	A. light QC	A. cleaned	B. light QC	B. cleaned	C. light QC	C. cleaned
pooled (in vivo)	k=5, 0.0796	k=4, 0.0396 *	k=4, 0.97	k=4, 0.931	k=3, 0.129	k=3, 0.208
M1	k=4, 0.00498 *	k=4, 0.00498 *	k=3, 1	k=4, 0.985	k=3, 0.234	k=3, 0.493
M1-PT	k=3, 0.00498 *	k=3, 0.0199 *	k=4, 0.00498 *	k=4, 0.154	k=4, 0.726	k=4, 0.642
M1-SS	k=3, 0.00498 *	k=3, 0.00498 *	k=4, 0.448	k=4, 0.00498 *	k=3, 0.00995 *	k=3, 0.00498 *
M1-Liver	k=4, 0.00498 *	k=4, 0.00498 *	k=4, 0.945	k=4, 0.97	k=3, 0.204	k=4, 0.443
M1-Met	k=4, 0.0249 *	k=4, 0.00995 *	k=4, 0.343	k=4, 0.597	k=3, 0.204	k=4, 0.0448 *
M2	k=4, 0.0498 *	k=3, 0.00498 *	k=4, 0.363	k=4, 0.0249 *	k=3, 0.124	k=4, 0.00995 *
M2-PT	k=4, 0.00498 *	k=3, 0.00498 *	k=3, 0.99	k=3, 0.622	k=4, 0.0348 *	k=4, 0.0299 *
M2-Liver	k=4, 0.00498 *	k=3, 0.00498 *	k=4, 0.0796	k=4, 0.746	k=4, 0.0249 *	k=3, 0.164
M2-Met	k=3, 0.0647	k=3, 0.00498 *	k=3, 0.925	k=4, 0.0398 *	k=3, 0.0846	k=3, 0.537

Appendix A2. 4T1 clones: light QC vs cleaned

p of each arm (* = p ≤ 0.05); the cleaned arm is the one used in the main slides

unit	A. light QC	A. cleaned	B. light QC	B. cleaned	C. light QC	C. cleaned
pooled (6 clones)	k=4, 0.00498 *	k=3, 0.00995 *	k=5, 0.174	k=4, 0.0299 *	k=4, 0.0198 *	k=4, 0.119
E	k=3, 0.00995 *	–	k=3, 0.348	–	k=3, 0.0149 *	–
EM1	k=3, 1	k=3, 0.622	k=4, 0.00498 *	k=4, 0.219	k=4, 0.00498 *	k=4, 0.403
EM2	k=3, 0.642	k=3, 0.627	k=3, 0.279	k=3, 0.0149 *	k=3, 0.209	k=4, 0.139
EM3	k=4, 0.95	k=4, 0.403	k=4, 0.0746	k=4, 0.458	k=3, 0.134	k=3, 0.124
M	k=4, 0.00995 *	k=3, 0.498	k=4, 0.174	k=3, 0.164	k=3, 0.408	k=4, 0.398
4T1	k=4, 0.00995 *	k=4, 0.0597	k=4, 0.199	k=4, 0.0597	k=4, 0.905	k=4, 0.562

Appendix A2. MCF10A/HuMEC: light QC vs cleaned

p of each arm (* = p ≤ 0.05); the cleaned arm is the one used in the main slides

unit	A. light QC	A. cleaned	B. light QC	B. cleaned	C. light QC	C. cleaned
pooled (4 samples)	k=6, 0.00498 *	k=4, 0.0099 *	k=4, 0.119	k=3, 0.0099 *	k=5, 0.0099 *	k=3, 0.0099 *
MCF10A	k=3, 0.00498 *	k=3, 0.00498 *	k=4, 0.0149 *	k=4, 0.00498 *	k=3, 0.00498 *	k=3, 0.00498 *
MCF10A_Mock	k=3, 0.00498 *	k=3, 0.00498 *	k=4, 0.289	k=3, 0.00498 *	k=3, 0.617	k=4, 0.0249 *
MCF10A_TGFB	k=4, 0.00498 *	k=3, 0.00498 *	k=4, 0.0448 *	k=4, 0.393	k=4, 0.0348 *	k=3, 0.0249 *
HuMEC	k=4, 0.00498 *	k=4, 0.00498 *	k=3, 0.00498 *	k=3, 0.0746	k=4, 0.284	k=4, 0.0647
HuMEC_Mock	k=4, 0.0896	k=3, 0.00498 *	k=4, 0.239	k=3, 0.00498 *	k=4, 0.791	k=3, 0.00498 *
HuMEC_TGFB	k=4, 0.746	k=4, 0.00498 *	k=4, 0.00995 *	k=3, 0.00498 *	k=3, 0.00995 *	k=3, 0.0348 *